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(54) **ANTIBODIES THAT SPECIFICALLY BIND TO TL1a AND METHODS OF TREATING GASTROINTESTINAL OR LUNG DISEASES**

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None

See application file for complete search history.

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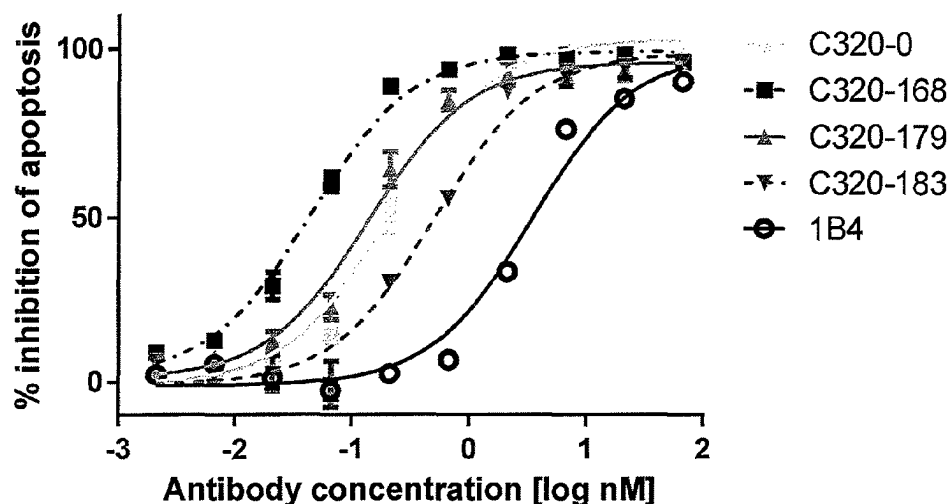
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(57) **ABSTRACT**

The disclosure provides TNF-like ligand 1a (TL1a)-binding proteins comprising an antigen binding domain of an antibody which binds specifically to TL1a and inhibits interaction of TL1a and Death Receptor 3 (DR3) and which does not inhibit the interaction of TL1a and Decoy Receptor 3 (DcR3). The disclosure also provides uses of the TL1a-binding proteins.

19 Claims, 34 Drawing Sheets

Specification includes a Sequence Listing.



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C336	EVQLVQSGAEVKKPGASVKVSCKAS	GYTFTGYYMH	WVRQAPGQGLEWMG	CDR2	WINPNSGGTNYA
C334	QMQLVQSGAEVKKPGSSVKVSCKAS	GGTFSSYTIS	WVRQAPGQGLEWMG		GIIPIFGTANHA
C333	QVQLVQSGAEVKKPGSSVKVSCKAS	GGTFSSYTIS	WVRQAPGQGLEWMG		GIIPIFGTNYA
C323	QMQLVQSGAEVKKPGSSVKVSCKAS	GGTFSSYTIS	WVRQAPGQGLEWMG		GIIPIFGTNYA
C321	QVQLVQSGAEVKKPGASVKVSCKAS	GYTFTSYGIS	WVRQAPGQGLEWMG		WISAYNGNTNYA
C320	QVQLVQSGAEVKKPGASVKVSCKAS	GYTFTSYDIN	WVRQATGQGLEWMG		WMNPNSGNTGYA
C319	QLQLQESGPGLVKPSGTLSTLCAVS	GGSISSRNWWS	WVRQSPGKGLEWIG		EIY HSDITNYN

C336	QKFGG	RVTMTRDTSISTAYMELSLRSDDTAVYYCAS	GGQTHL	DV	WGQGTITVTVSS	SEQ ID NO: 2
C334	QSFQG	RVTITADESTSTAYMELSSLRSEDTAIYYCST	NSYSSSWYDAF	DI	WGQGTMTVTVSS	SEQ ID NO: 10
C333	QRFQG	RVTITADESTSTAYMELSSLRSEDTAIYYCST	NSYSSSWYDAF	DI	WGQGTMTVTVSS	SEQ ID NO: 18
C323	QRFQG	RVTITADESTSTAYMELSSLRSEDTAIYYCST	NSYSSSWYDAF	DI	WGQGTMTVTVSS	SEQ ID NOT 26
C321	QKLQG	RVTMTTDTSTSTAYMELSLRSDDTAVYYCAR	DSHIYDILTGY	DY	WGQGTITVTVSS	SEQ ID NO: 34
C320	QKFGG	RVTMTRNSTISITAYMELSSLRSEDTAIYYCAR	EYEDTASF	EY	WGQGTITVTVSS	SEQ ID NO: 42
C319	PSLKS	RVTISVDKSKNQFSLKLSVTTAADTAIYYCAK	DGEAGGTYIDAFDV		WGQGTMTVTVSS	SEQ ID NO: 50

FIGURE 1A

C336	DIQMTQSPSSLASVGDRTITC	QASQ	DITDYLN	WYQQRPQKAPKLLIY	CDR2
C334	DVVMTQSPAFLSVSPGERATLSC	RASQ	SISNNLA	WYQQMPGQAPRLLLY	DASNLET
C333	EIVLTQSPATLSVSPGERATLSC	RASQ	STNNLA	WYQQLPGQAPRLLLY	DASTRAT
C323	AIQLTQSPSSLASVGDRTITC	RASQ	GIGSALA	WYQQKPGKAPKLLIY	DASTRAT
C321	DVVMTQTPLSSPVTILGQPASISQ	HSSQSLVHSDGNTYLS		WLQQRPGQPPRLLLY	DASSLQS
C320	QSVLTQPPPSVSGA PGQRTITSC	AGSSSDIG	AGLGVH	WYQQLPGTAPKLLIY	KISNRFS
C319	QSALTQPRSVSGS PQQSVTISC	TGTSSDVG	IYNYVS	WYQQHPGKAPKLLIY	GYNNRPS
					DVSRPFS
C336	GVPSRFSGSGGYFTFTISSLQPEDFATYYC	QQYDNLQ	IT	FGQGTTRLEIKR	SEQ ID NO: 6
C334	DIPARFSGSGGSEFTLTISGLQADFAVYYC	QQYNNWP	LT	FGGGTKLEIKR	SEQ ID NO: 14
C333	DIPARFSGTSGSEFTLTISGLQADFAVYYC	QQYNNWP	LT	FGGGTKVDIKR	SEQ ID NO: 22
C323	GVPSRYSGSGGTDTLTISGLQPEDFATYYC	QQFSYYP	LT	FGGGTKLEIKR	SEQ ID NO: 30
C321	GVPRFSGSGAGTDTLTKISRVEAEDVGVIYC	MQDTQFP	QT	FGQGTKVEIKR	SEQ ID NO: 38
C320	GVPRFSGSKGTSASLAI TGLLPEDEGDYYC	QSYDGTLSAL		FGGTKLTVLG	SEQ ID NO: 46
C319	GVPRFSGSKSDNTASLTISGLQAEDEADYYC	YSVAGTYTSL		FGGTKVTVLG	SEQ ID NO: 54

FIGURE 1B

C320	QVQLVQSGAEVKKPGASVKVCKAS	GYTFTSYDIN	WVRQATGQGLEWMG	WMNPNSGNTGYAQKFQG
C320-3	QVQLVQSGAEVKKPGASVKVCKAS	GYTFTSYDIN	WVRQAPGQGLEWMG	WMNPNSGNTGYAQKFQG
C320-90	QVQLVQSGAEVKKPGASVKVCKAS	GYTFTSYDIN	WVRQATGQGLEWMG	WMNPNSGNTGYAQKFQG
C320-103	QVQLVQSGAEVKKPGASVKVCKAS	GYTFTSYDIN	WVRQATGQGLEWMG	WMNPNSGNTGYAQKFQG
C320-114	QVQLVQSGAEVKKPGASVKVCKAS	GYTFTSYDIN	WVRQATGQGLEWMG	WMNPNSGNTGYAQKFQG
C320-115	QVQLVQSGAEVKKPGASVKVCKAS	GYTFTSYDIN	WVRQATGQGLEWMG	WMNPNSGNTGYAQKFQG
C320-129	QVQLVQSGAEVKKPGASVKVCKAS	GYTFTSYDIN	WVRQATGQGLEWMG	WMNPNSGNTGYAQKFQG
C320-130	QVQLVQSGAEVKKPGASVKVCKAS	GYTFTSYDIN	WVRQATGQGLEWMG	WMNPNSGNTGYAQKFQG
CONSENSUS	QVQLVQSGAEVKKPGASVKVCKAS	GYTFTSYDIN	WVRQAXGQGLEWMG	WMNPNSGNTGYAQKFQG
			T	M
			P	L

C320	RVTMTRNTSISTAYMELSSLRSED	EVPDTASFXY	WGQGTILVTVSS	SEQ ID NO: 42
C320-3	RVTMTRNTSISTAYMELSSLRSED	EVPDTASFXY	WGQGTILVTVSS	SEQ ID NO: 58
C320-90	RVTMTRNTSISTAYMELSSLRSED	EVPDDASFXY	WGQGTILVTVSS	SEQ ID NO: 66
C320-103	RVTMTRNTSISTAYMELSSLRSED	EVPDTAAFXY	WGQGTILVTVSS	SEQ ID NO: 70
C320-114	RVTMTRNTSISTAYMELSSLRSED	EVPDTASFXY	WGQGTILVTVSS	SEQ ID NO: 74
C320-115	RVTMTRNTSISTAYMELSSLRSED	EVPDTASFXY	WGQGTILVTVSS	SEQ ID NO: 78
C320-129	RVTMTRNTSISTAYMELSSLRSED	EVPDTASFXY	WGQGTILVTVSS	SEQ ID NO: 86
C320-130	RVTMTRNTSISTAYMELSSLRSED	EVPDTASFXY	WGQGTILVTVSS	SEQ ID NO: 90
CONSENSUS	RVTMTRNTSISTAYMELSSLRSED	EVPXXAXFXY	WGQGTILVTVSS	SEQ ID NO: 94
		ET S L		
		DD A D		
		E		

FIGURE 1C

C320	QVQLV-QSGAEVKKPGASVKVSCKAS	<u>CDR1</u> GYTFTSYDIN	WVRQATGQGLEWMG	<u>CDR2</u> WMNPNSGNTGYAOKFQG
1TZG	...-...R..S..T..		...P.R.....	
1RHH	E...LE.....S..Q.....		...P.H.....	
2DD8	...Q-.....S.....		...P.....	
2JB5	...-.....S.....		...P.....	
3FKU	...-.....S.....TS.		...P...L.	
3GBM	E...-E.....S.....		...P...P.	
3LMJ	...V-.....R.....V.		...P.K.....	...K
3P30	...-.....A.S..R.....		...P...PQ.	
3IYW	E...-.....R.....T.....		...P.....	
Consensus	XVQXXXSGAEVXXPGXSXXVSCXXS	<u>GYTFTSYDIN</u>	WVRQAXGXGXXWXG	<u>WMNPNSGNTGYAOKFQG</u>
	Q LV-Q KK A VK KA		T Q LE M	
	E VLLE RR S TT TS		P R PQ L	
	Q A Q V		H K	
C320	RVTMTRNTSISTAYMELSLRSEDTAIVYCAR	<u>CDR3</u> EVPDTASFEY	WGQGTLVTVSS	SEQ ID NO: 42
1TZG	..I.I.ADR.T...L..N...P.....			SEQ ID NO: 154
1RHH	...F.ADQAT...TN..D.....			SEQ ID NO: 155
2DD8	...I.TDE.T.....TN.....		...T.....	SEQ ID NO: 156
2JB5	...I.ADE.T.....D.R.....			SEQ ID NO: 157
3FKU	...I.ADQ.TR...D.R.....			SEQ ID NO: 158
3GBM	...I.ADDFAG.V.....M...K.....		...K..T.....	SEQ ID NO: 159
3LMJ	..S..ED..TN.....M...T.....			SEQ ID NO: 160
3P30	..L.I.AD..TN...L...YD...T.....			SEQ ID NO: 161
3IYW	D..T..V.....			SEQ ID NO: 234
Consensus	RXXXTXXXXXXXTXXXXLRLRXXDXTAXVYCX	<u>EVPDTASFEY</u>	WGXGTXVTVSS	SEQ ID NO: 162
	VTM RNTSIS A ME SS SE V AR		Q L	
	ISI ADRATR V LD NN PD M TK		K T	
	L F T QFAG T Y T			
	E E N R			
	D			

FIGURE 1D

C320	QSVLTQPPSVSGAPGQRTVISC	AGSSSDIGAGLG ¹ VH	WYQQLPGTAPKLLIY	CDR2
C320-5	QSVLTQPPSVSGAPGQRTVISC	AGSSSDIGAGLG ¹ VH	WYQQLPGTAPKLLIY	<u>GYNRP</u> S
C320-120	QSVLTQPPSVSGAPGQRTVISC	TGSSSDIGAGLG ¹ VH	WYQQLPGTAPKLLIY	<u>GYNRP</u> S
CONSENSUS	QSVLTQPPSVSGAPGQRTVISC	XGSSSDIGAGLG ¹ VH	WYQQLPGTAPKLLIY	<u>GYNRP</u> S
		A		
		T		
C320	GVPDRFSGSKSGTSASLITGLLPEDEGDYIC	QSYDGLSAL	FGGGTKLTVLG	SEQ ID NO: 46
C320-5	GVPDRFSGSKSGTSASLITGLLPEDEGDYIC	<u>QSYDGLSAL</u>	FGGGTKLTVLG	SEQ ID NO: 62
C320-120	GVPDRFSGSKSGTSASLITGLLPEDEGDYIC	<u>QSYDGLSAL</u>	FGGGTKLTVLG	SEQ ID NO: 82
CONSENSUS	GVPDRFSGSKSGTSASLITGLLPEDEGDYIC	<u>QSYDGLSAL</u>	FGGGTKLTVLG	SEQ ID NO: 95
	A			
	T			

FIGURE 1F

C320	QSVLTQPP-SVSGAPGQRTVISC	CDR1 <u>AGSSSDIGAGLGVH</u>	CDR2 <u>GYNRPSP</u>
1RHH	EL...S.GTL.LSA.E.A.L.	K.Q..R....	
1TZGL	EI...S.GTQ.LS.E.A.L.	R.Q..R....	
2DD8	SYE.....V...KTAR.T.	K.Q..V.VV..	
2JB5	DIA...A...S...SI....	H..K...M....	
3FKU	.PG.....KGLR.TA.LT.	L.HQ.HP...S..	
3GBM	A...K....		
3LMJ	S...T....		
3P30	T....		
3IYW	S...T....		
CONSENSUS	XXLTOXXXXXXXSXXXXXXXXXC	<u>WXQXXGXXPLXXY</u>	<u>GYNRPSP</u>
	QSV PP-SV GAPGQRTVIS	Y LP TA K LI	
	ELE SAGTL LSARETARLT	L KQ QP R VV	
	SIA S Q VGL KSI	R K V MS	
	DYG KT K	H H	
	P A		

C320	GVPDRFSGSKGTSASLAITGLLPEDEGYVC	CDR3 <u>QSYDGTLSAL</u>	FGGGTKLTVLG SEQ ID NO: 46
1RHH	I.....C.G...DFT.T.GR.E...LAV...	Q....EIKR. SEQ ID NO: 164	
1TZGL	.A.....G...DFT.T.SR.E...FAV...	Q....VE.KR. SEQ ID NO: 165	
2DD8	I.E.....N..NT.T.T.SRVEAG..A....	T...V.... SEQ ID NO: 166	
2JB5	.SN.....NT...T.S..QA..A....	 SEQ ID NO: 167	
3FKU	ISE...A.R.NT...T...Q...A....	 SEQ ID NO: 168	
3GBM	I.....T.G...QTG..AN....	 SEQ ID NO: 169	
3LMJ	S...QS..A....	 SEQ ID NO: 170	
3P30	S...RS..A....	A..... SEQ ID NO: 171	
3IYW	S...RS..A....	A..... SEQ ID NO: 172	
CONSENSUS	XXPDRFXGXKXXXXXXSXXXXXXXXXXDYVC	<u>QSYDGTLSAL</u>	FXGGTXXXXXG SEQ ID NO: 173
	GVP S S GTSA L ITGLLP DEG		G KLTVL
	IAE A G NFFT T GRVEAG LAV		Q VEIKR
	SN N G S QT F N		T
	R RS		A

FIGURE 1G

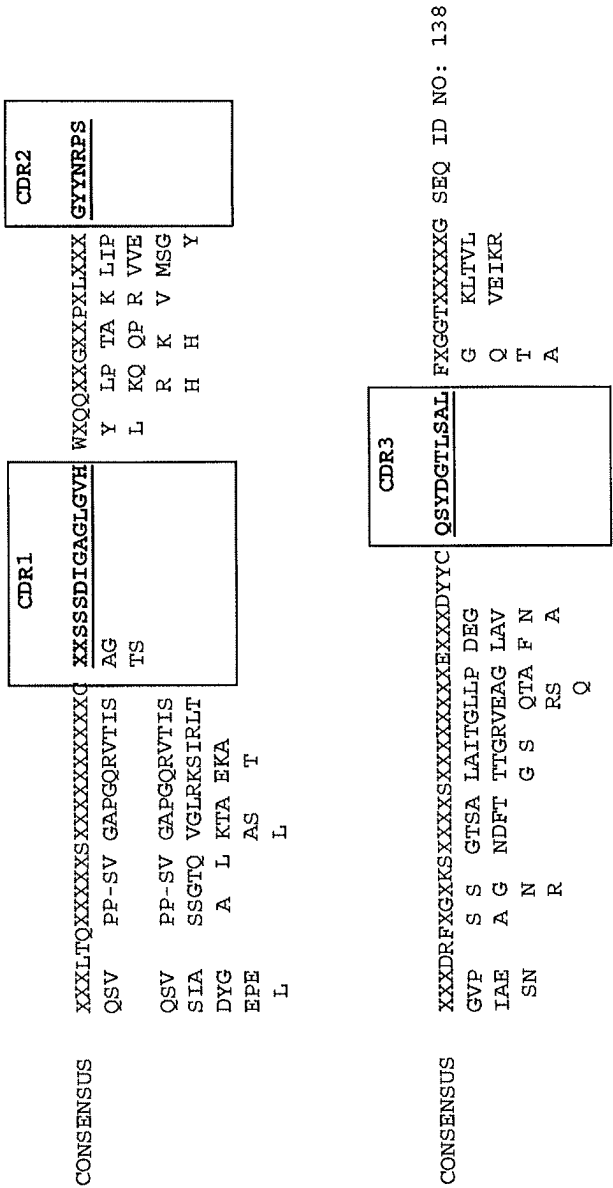


FIGURE 1H

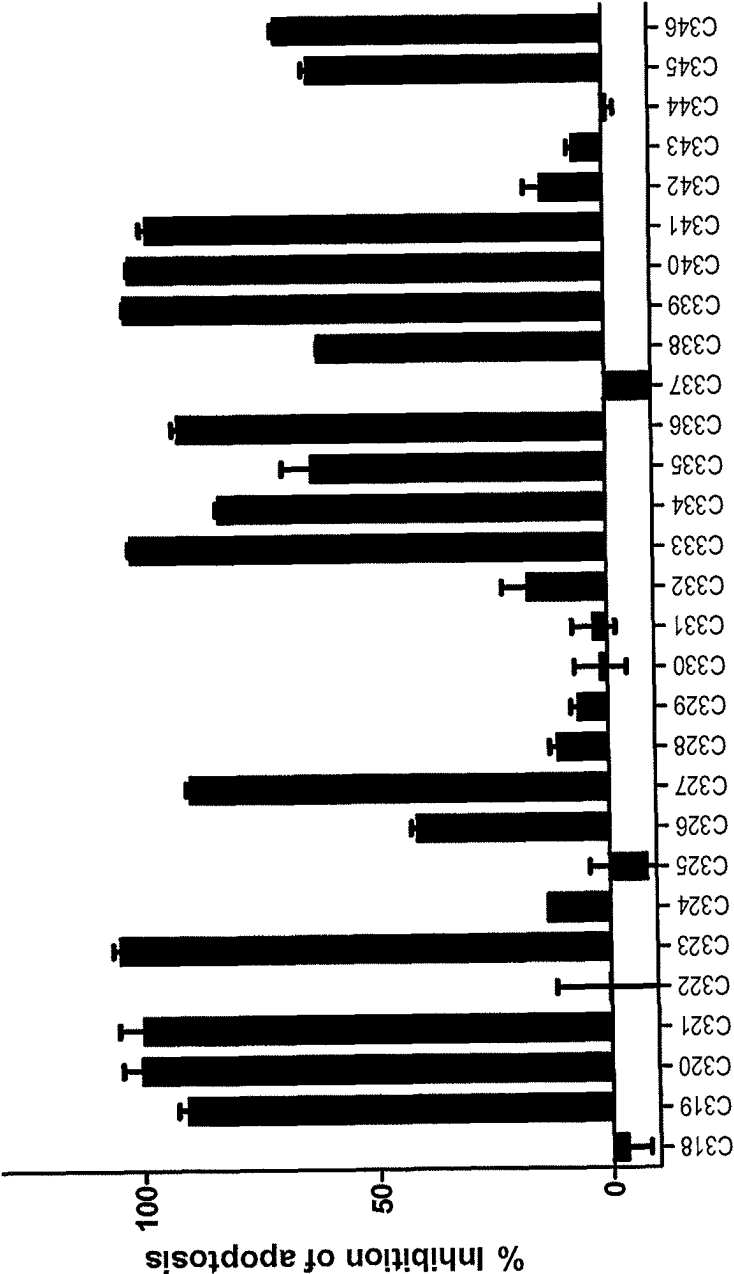


FIGURE 2

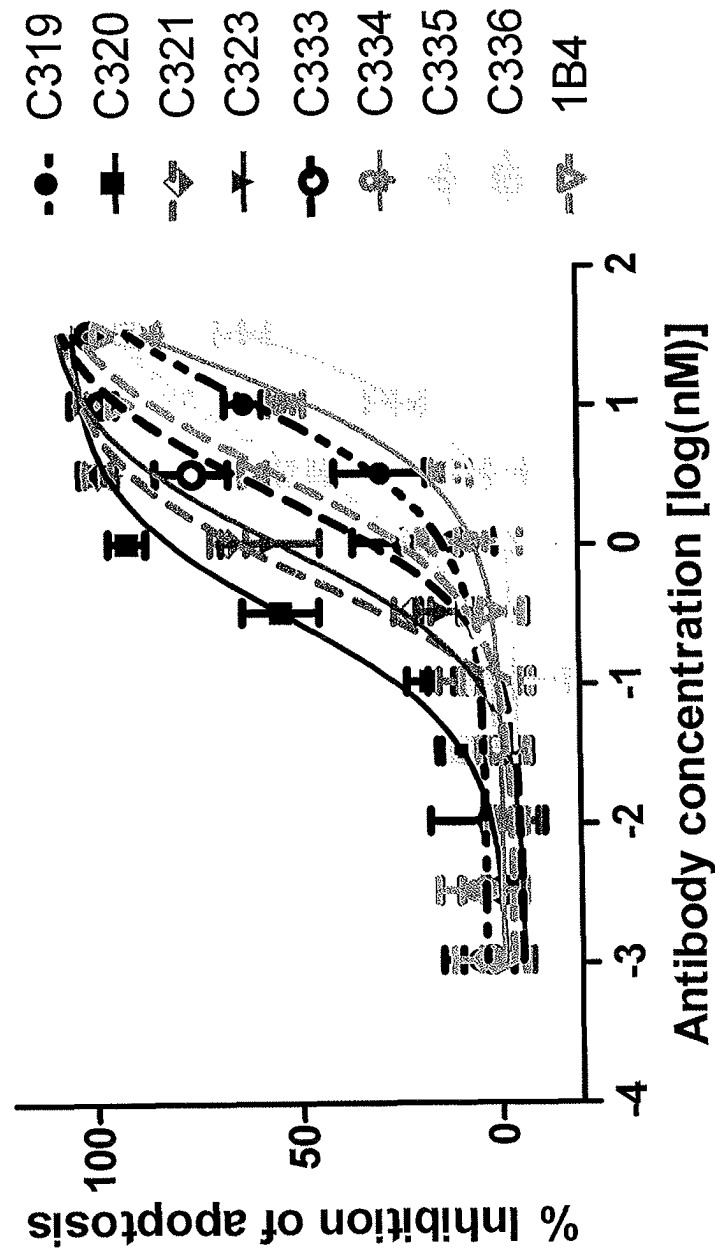


FIGURE 3

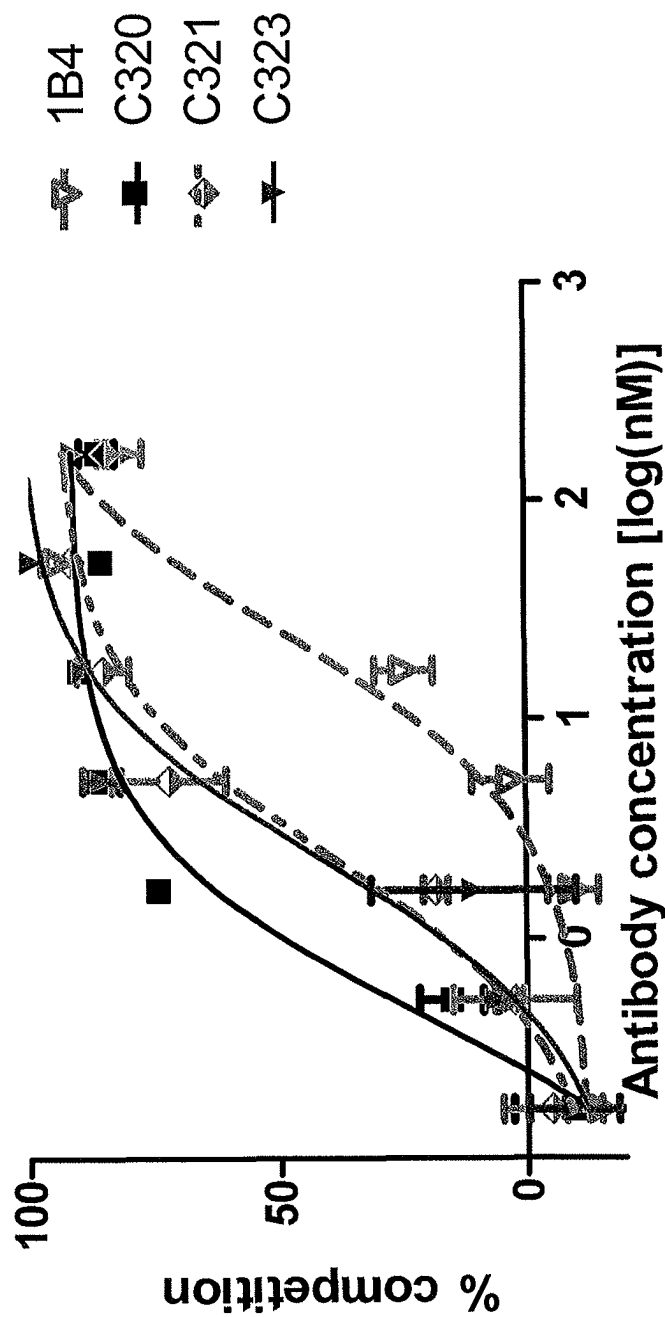


FIGURE 4A

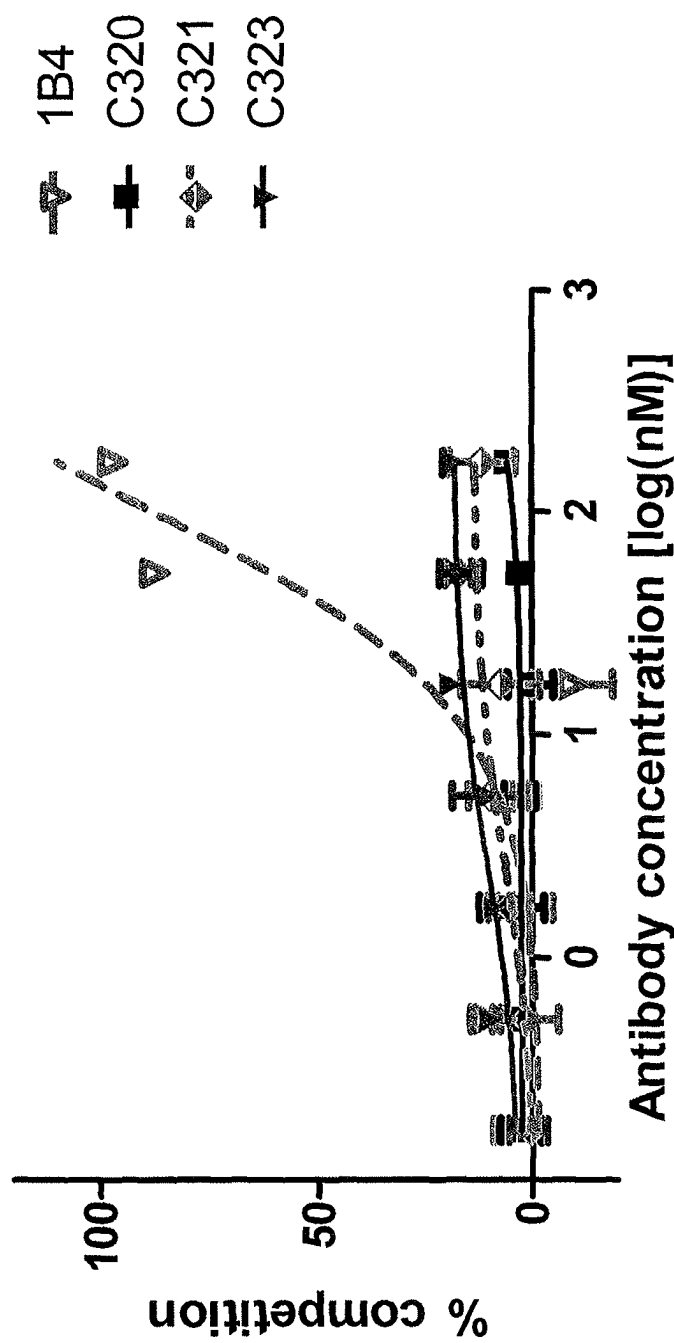


FIGURE 4B

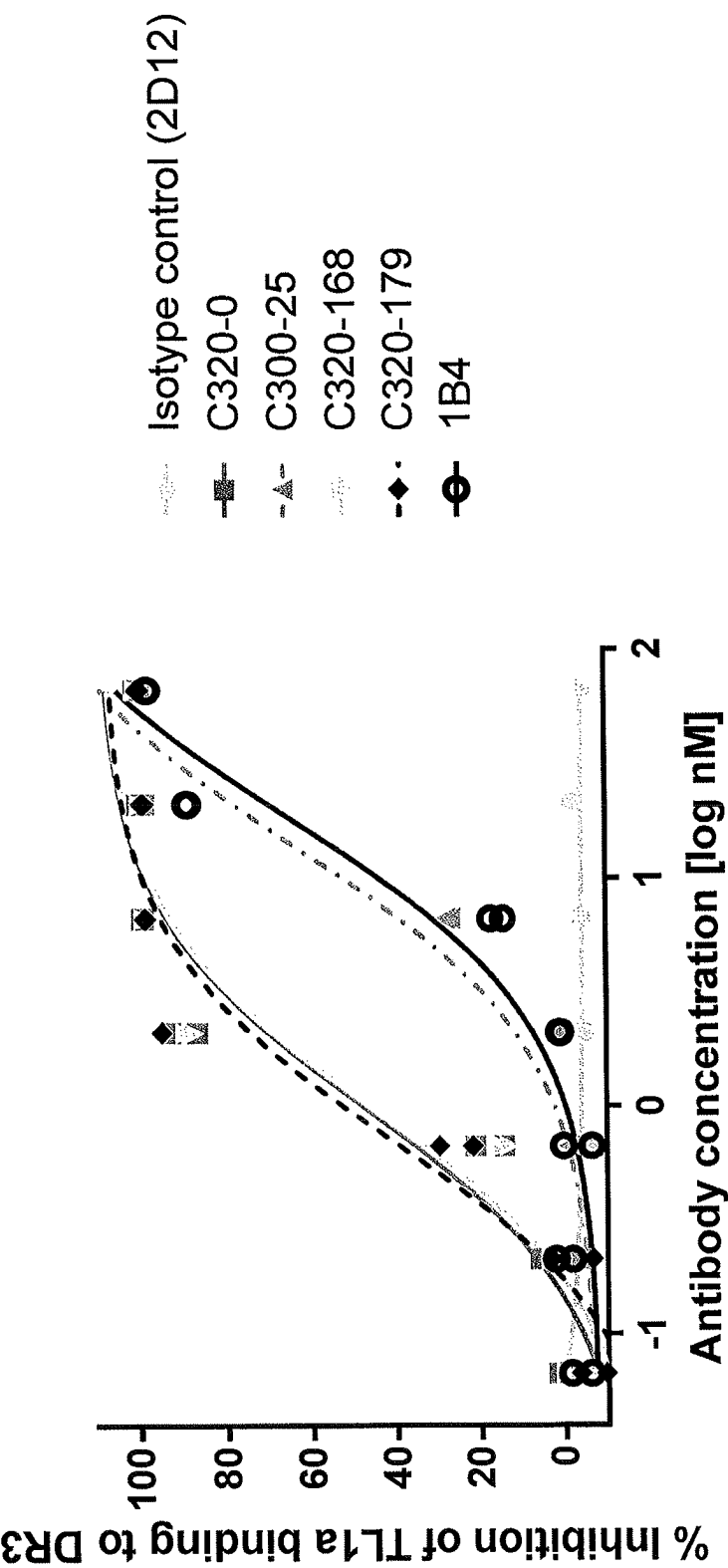


Figure 4C

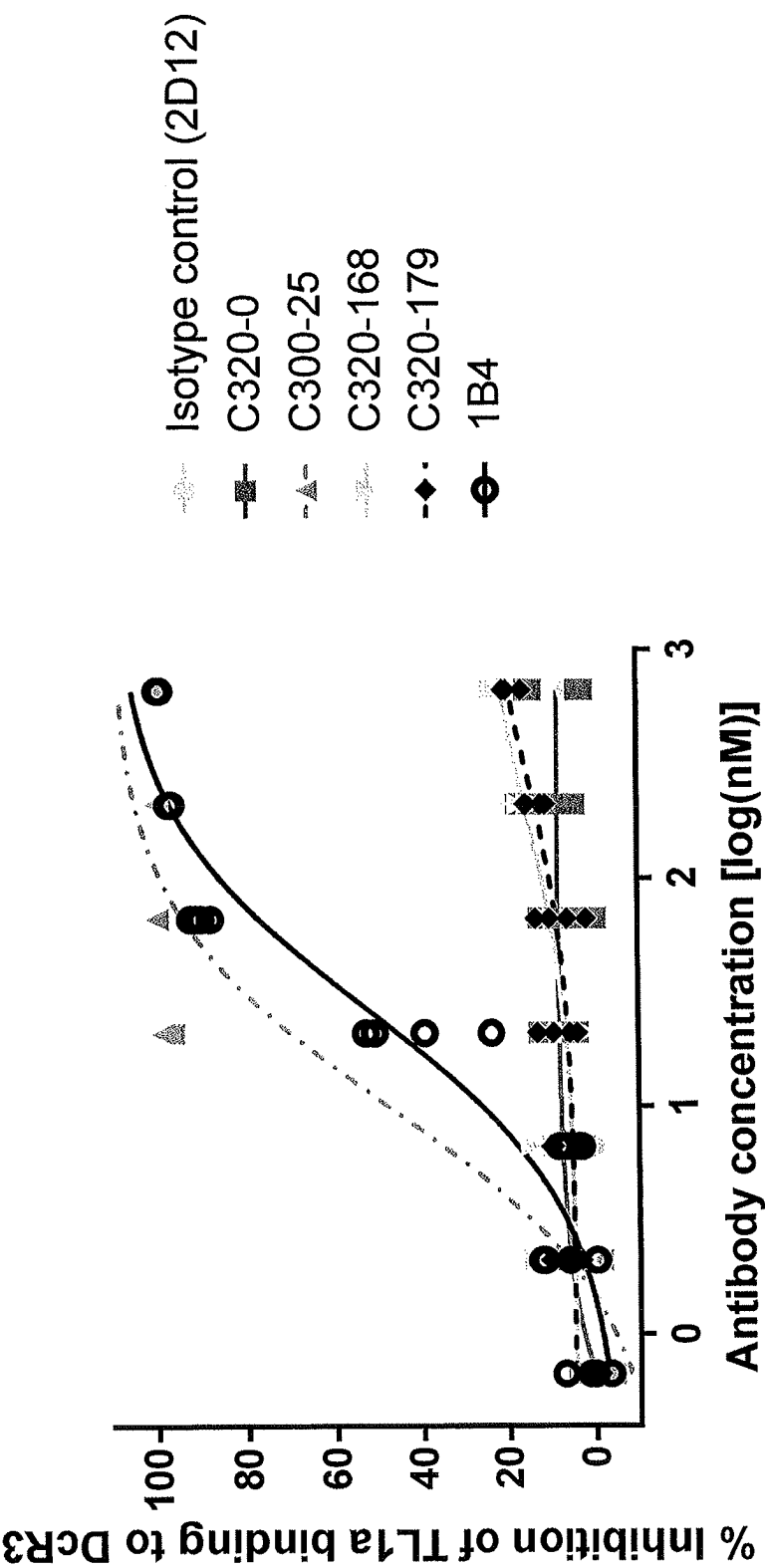


FIGURE 4D

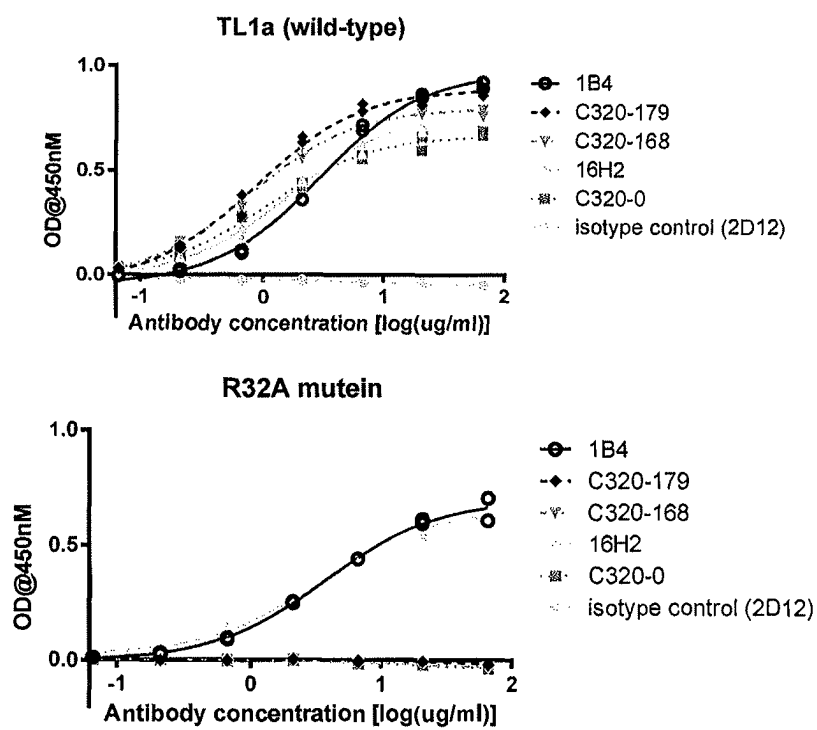


FIGURE 5A

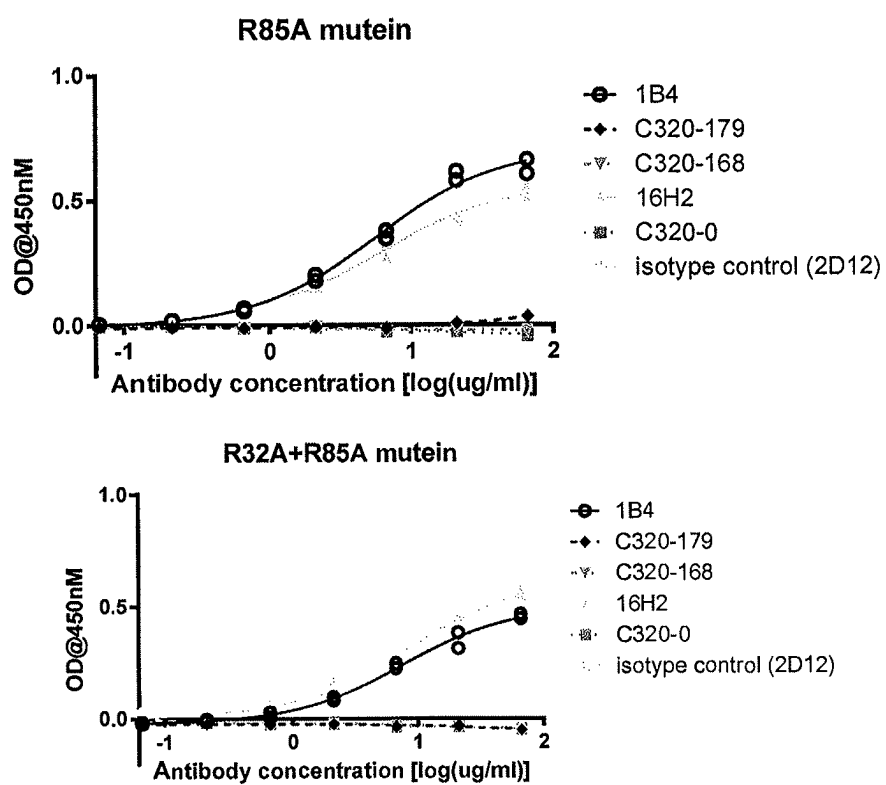


FIGURE 5B

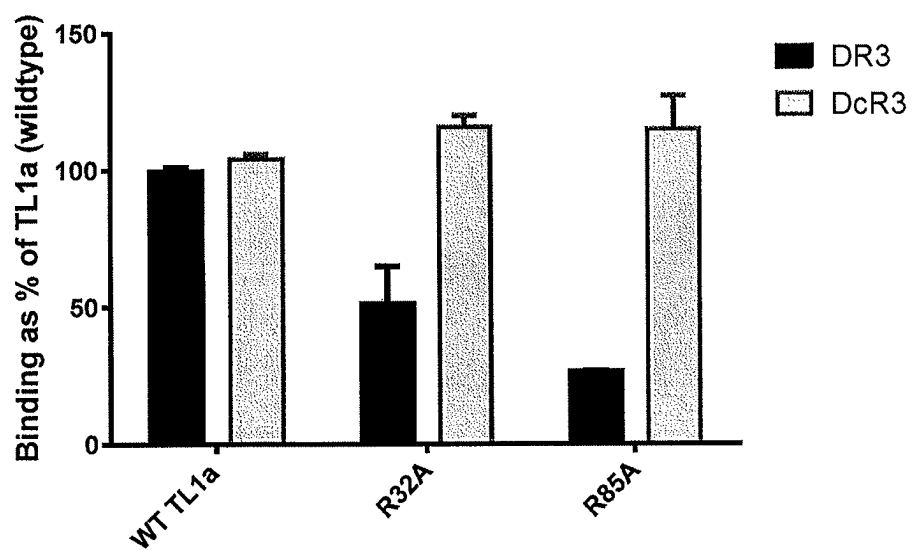


FIGURE 5C

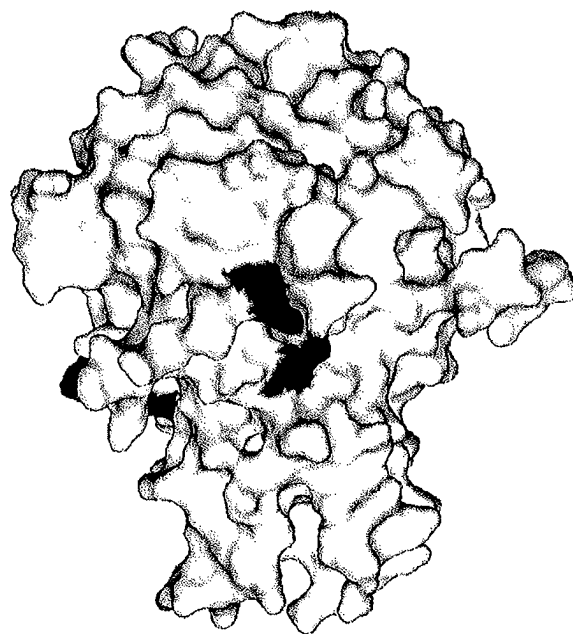


FIGURE 5D

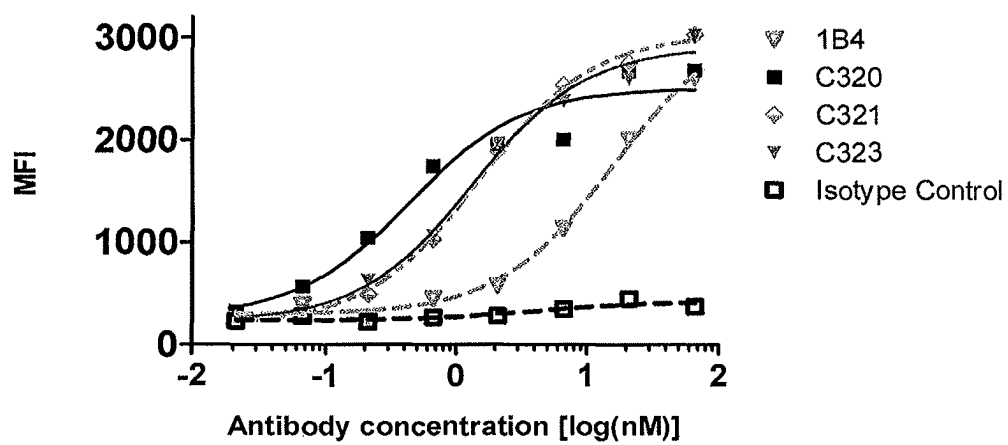


FIGURE 6

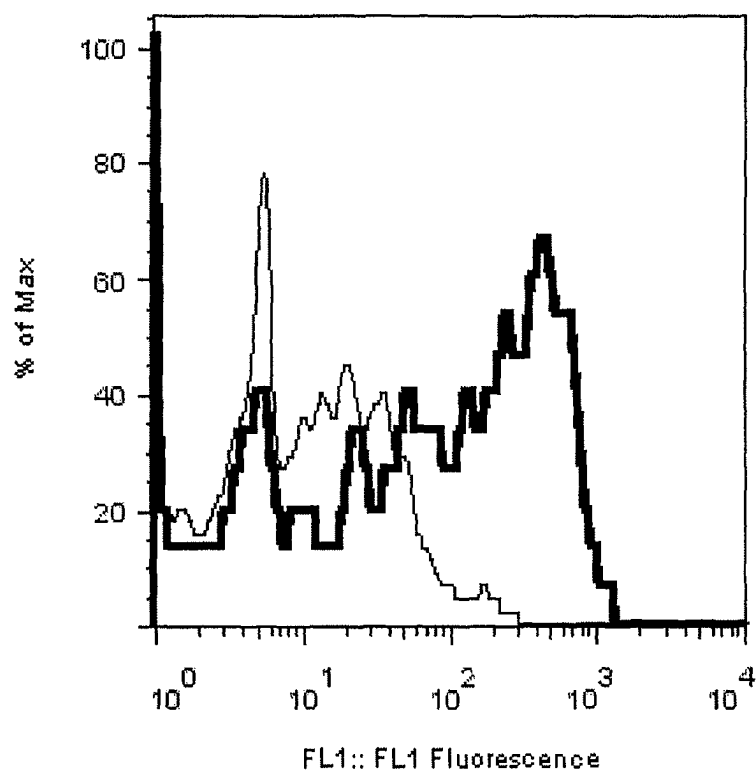


FIGURE 7A

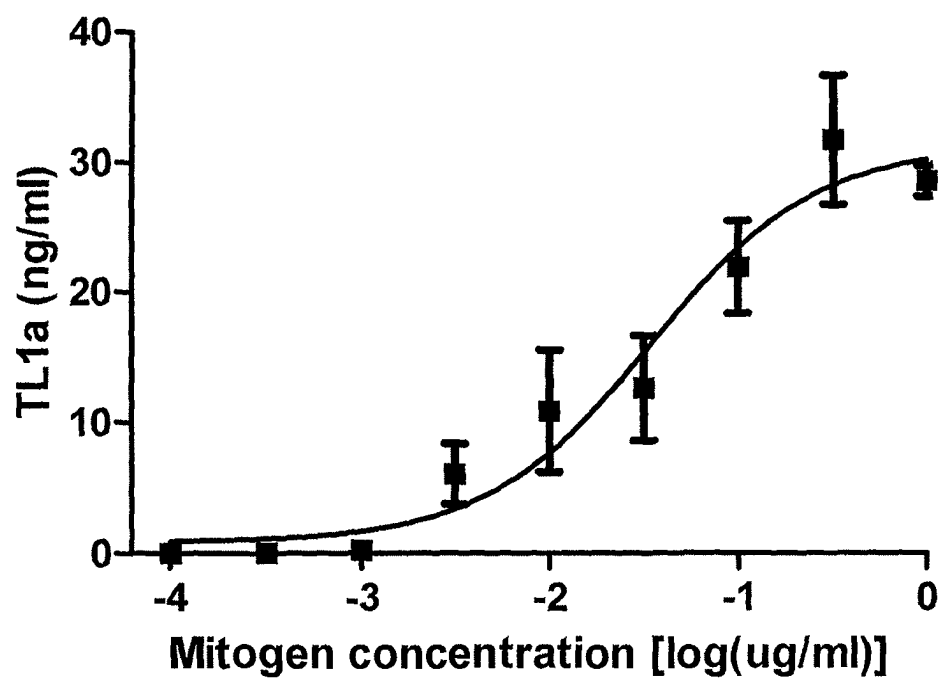


FIGURE 7B

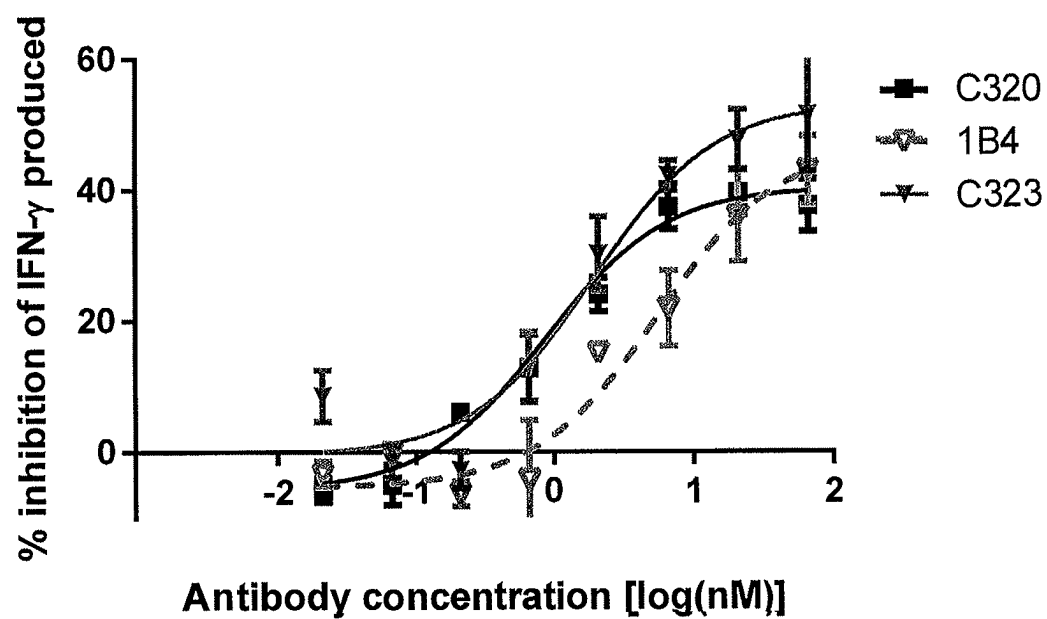


FIGURE 8A

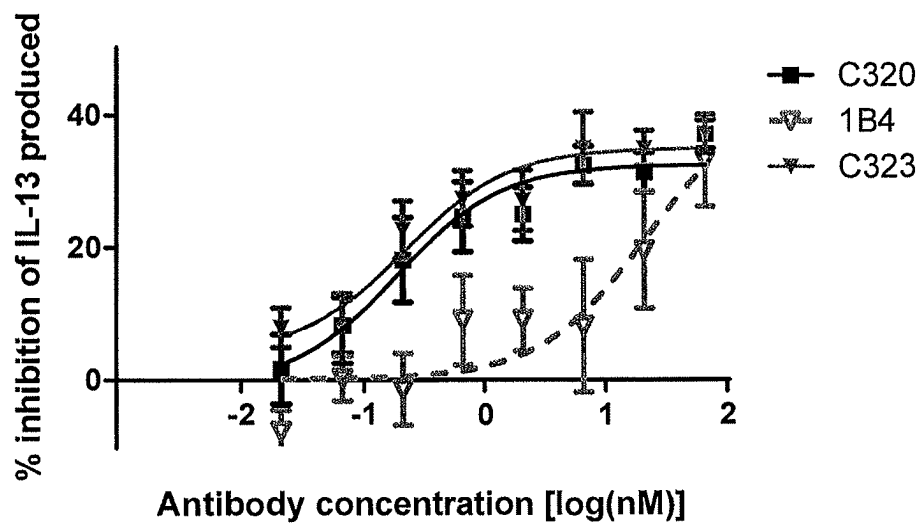


FIGURE 8B

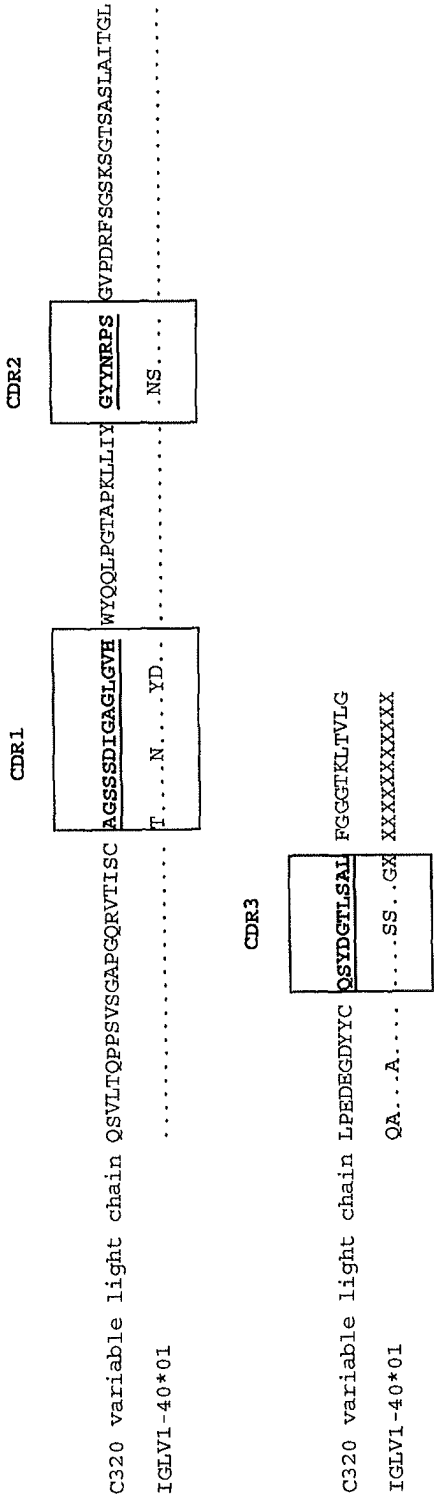


FIGURE 9A

C320	QVQLVQSGAEVKKPGASVKVCKAS	CDR1	WVROQTGGQLEWNG	CDR2	WMPNSGNTGYAOKFQG	RVTMTNTISISTAY
C320-162		GYTFTSYDIN	P.....	L.....	A.....	
C320-163			P.....	L.....	ADR.T.....	
C320-164			P.....	L.....		
C320-165			P.....	L.....		
C320-166			P.....	L.....		
C320-167			P.....	L.....		
C320-168			P.....	L.....		
C320-169			P.....	L.....	ADR.T.....	
C320-170			P.....		ADR.T.....	
C320-171			P.....		ADR.T.....	
C320-172			P.....		ADR.T.....	
C320-179			P.....	L.....	ADR.T.....	
C320-183			P.....	L.....	ADR.T.....	
Consensus			X.....	X.....	XXXX.....	
			T.....	M.....	RNTSI.....	
			P.....	L.....	ADRAT.....	
C320	MELSSLRSEDIAVYYCAR	CDR3	WGQGITLVTVSS			
C320-162	E.....	EVPDTASFEY		SEQ ID NO: 175		
C320-163				SEQ ID NO: 176		
C320-164				SEQ ID NO: 177		
C320-165				SEQ ID NO: 178		
C320-166				SEQ ID NO: 179		
C320-167				SEQ ID NO: 180		
C320-168				SEQ ID NO: 181		
C320-169				SEQ ID NO: 182		
C320-170				SEQ ID NO: 183		
C320-171				SEQ ID NO: 184		
C320-172				SEQ ID NO: 185		
C320-179				SEQ ID NO: 186		
C320-183				SEQ ID NO: 187		
Consensus				SEQ ID NO: 173		
	X.X.....					
	D S					
	E A					

FIGURE 9B

C320	QSVLTQPPSVSGAPGQRTISC	CDR1	AGSSSDICAGLGVH	MYOQLPGTAPKLLIY	CDR2	GYNRP	GVPDRFSGSKSGTSASIAITGL
C320-162							T.....
C320-163							T.....
C320-164			S.....				T.....
C320-165			TS.....				T.....
C320-166			T.....				T.....
C320-167							T.....
C320-168			TS.....				T.....
C320-169							T.....
C320-170			TS.....				T.....
C320-171			TS.....	P.....			T.....
C320-172			TS.....	E.....			T.....
C320-179			TS.....	E.....			T.....
C320-183			TS.....	G.....			T.....
Consensus			XX.....	X.....			X.....
			AG.....	Y.....			A.....
			TS.....	P.....			T.....
				E.....			
				G.....			
C320	LPEDEGDYIC	CDR3	QSYDGLSAL	FGGTKLTVLG			
C320-162							SEQ ID NO: 188
C320-163							SEQ ID NO: 189
C320-164							SEQ ID NO: 190
C320-165							SEQ ID NO: 191
C320-166							SEQ ID NO: 192
C320-167							SEQ ID NO: 193
C320-168							SEQ ID NO: 194
C320-169							SEQ ID NO: 195
C320-170							SEQ ID NO: 196
C320-171							SEQ ID NO: 197
C320-172							SEQ ID NO: 198
C320-179							SEQ ID NO: 199
C320-183							SEQ ID NO: 200
Consensus							SEQ ID NO: 174

FIGURE 9C

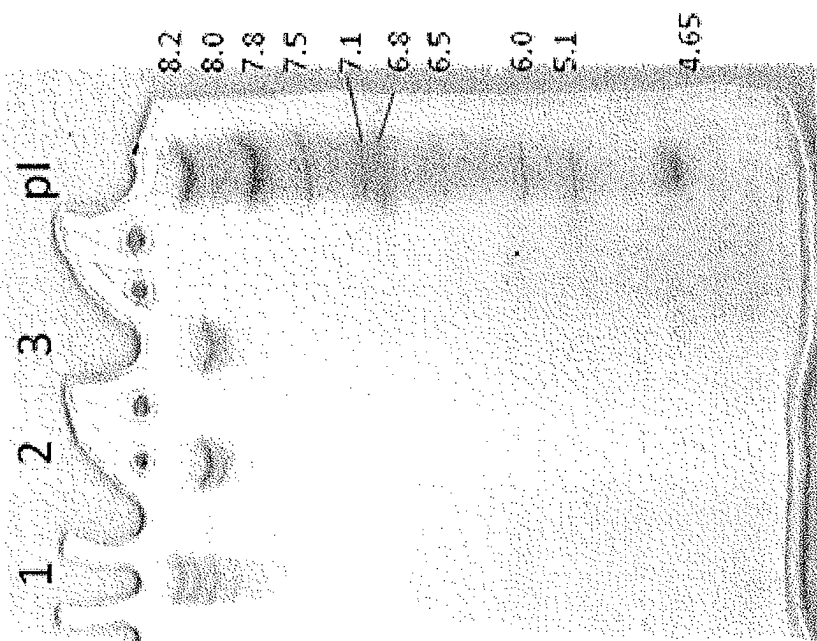


FIGURE 9D

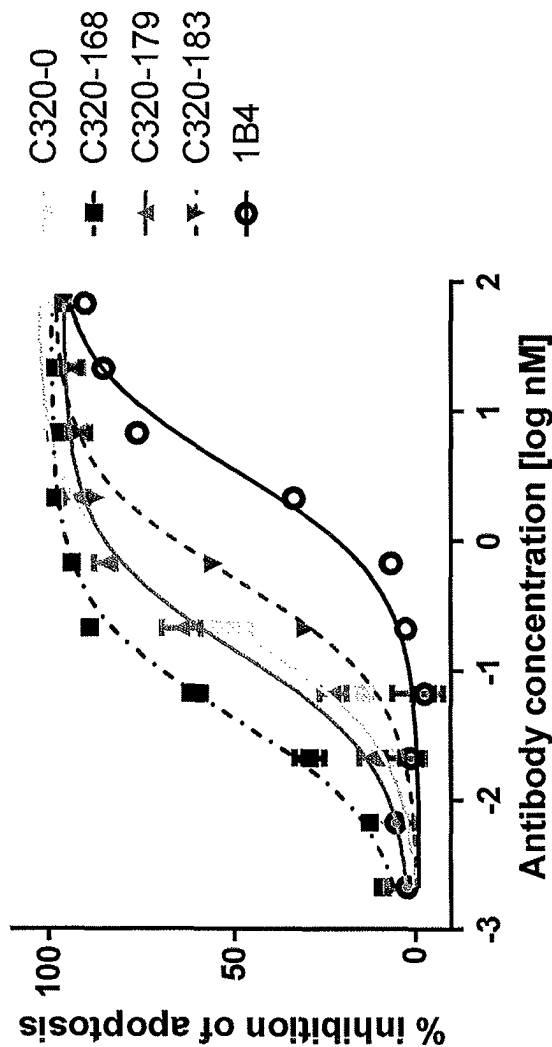


FIGURE 9E

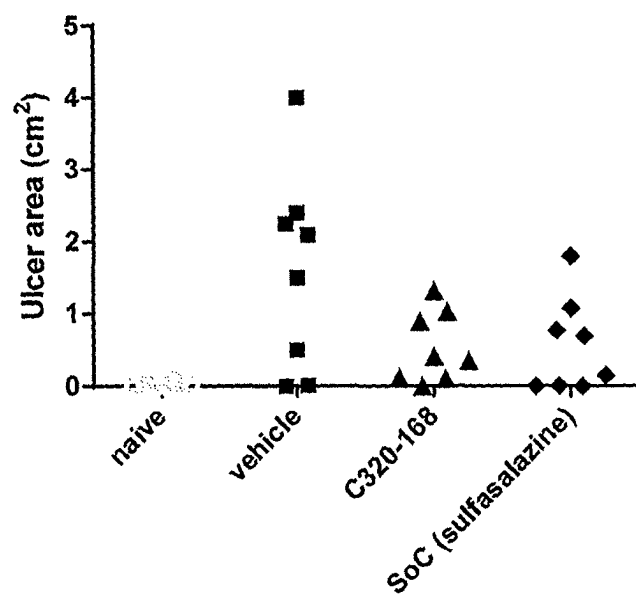


FIGURE 10A

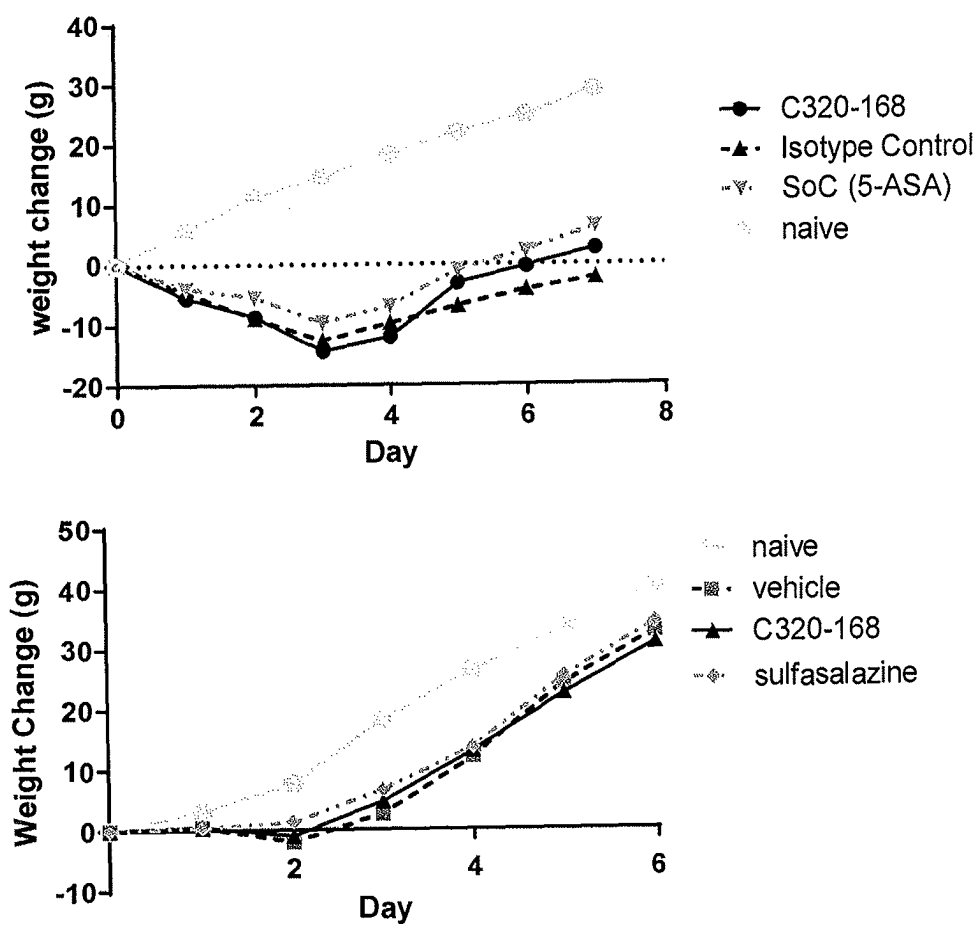


FIGURE 10B

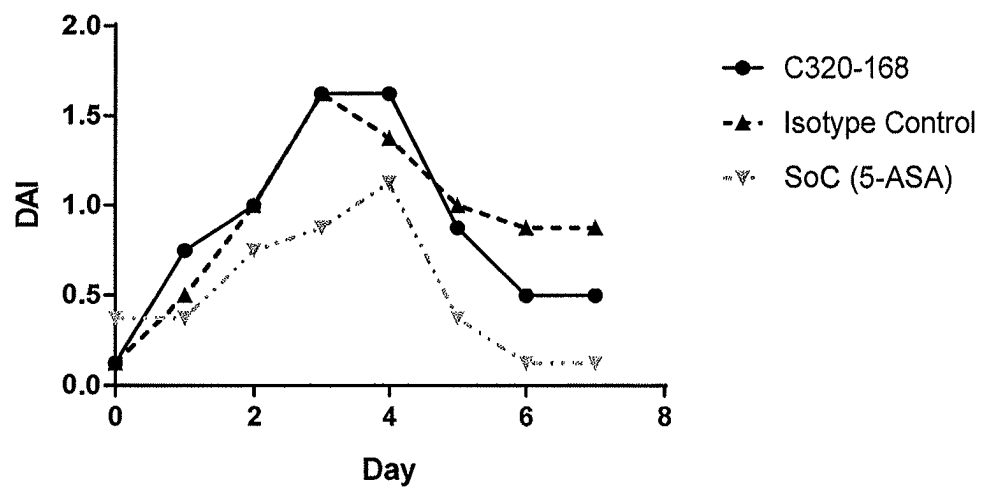


FIGURE 10C

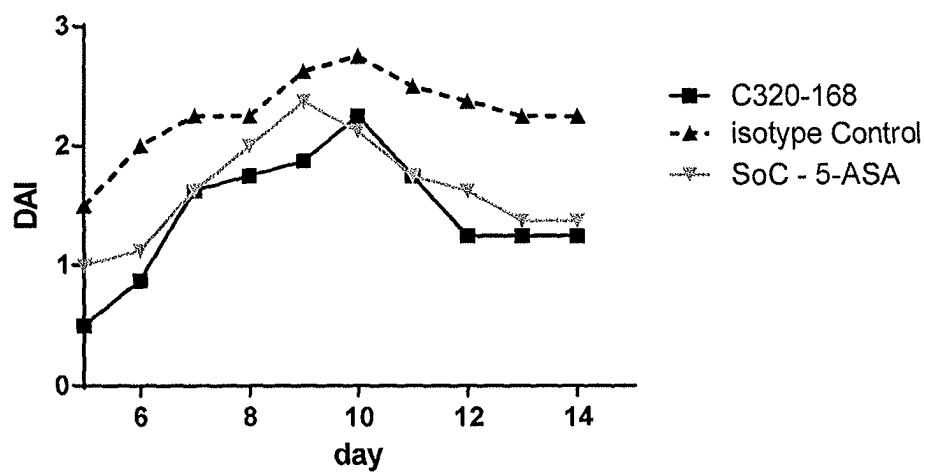


FIGURE 11A

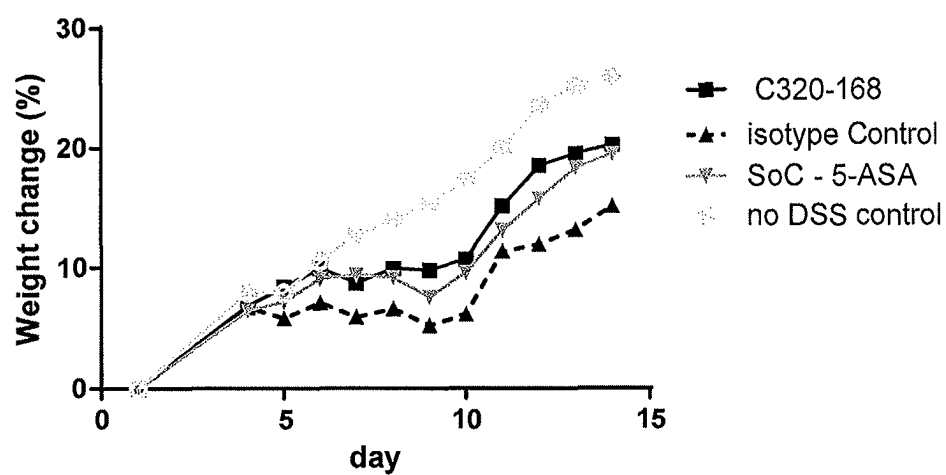


FIGURE 11B

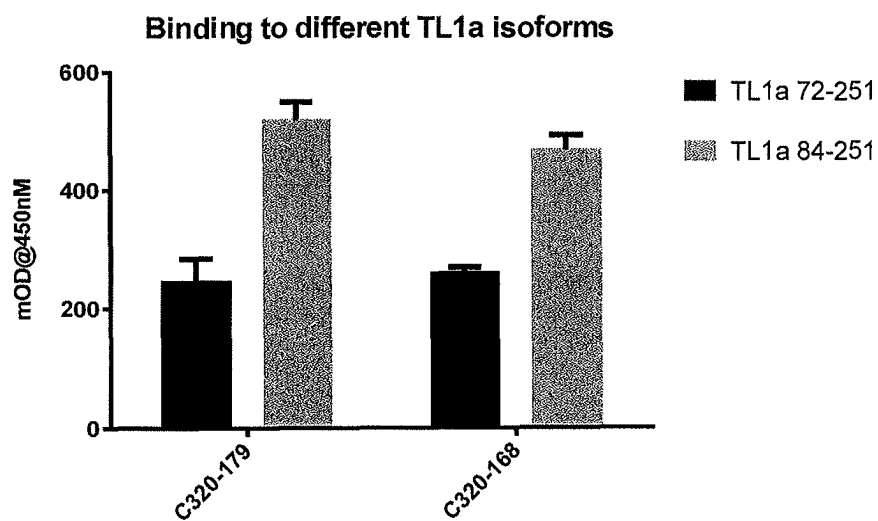


FIGURE 12

ANTIBODIES THAT SPECIFICALLY BIND TO TL1a AND METHODS OF TREATING GASTROINTESTINAL OR LUNG DISEASES

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/031,615 filed on Sep. 24, 2020, which is a divisional of U.S. patent application Ser. No. 15/206,493 filed on Jul. 11, 2016 (now U.S. Pat. No. 10,822,422; issued on Nov. 3, 2020), which is a continuation of U.S. patent application Ser. No. 14/228,367 filed on Mar. 28, 2014 (now abandoned), which is a continuation of International Application No. PCT/AU2012/001161 filed on Sep. 28, 2012, and claims priority to U.S. Provisional Application No. 61/541,590 filed on Sep. 30, 2011 and Australian Patent Application No. 2011904042 filed on Sep. 30, 2011. Each of these applications is incorporated by reference herein, in its entirety and for all purposes.

REFERENCE TO A SEQUENCE LISTING

This application includes a Sequence Listing submitted electronically as a text file named, 2873 2720005 Seglisting ST26, created on Apr. 12, 2023, with a size of 362,271 bytes. The Sequence Listing is incorporated by reference herein.

FIELD OF THE INVENTION

The disclosure relates to proteins that bind to TL1a and uses thereof, e.g., in therapy, prophylaxis, diagnosis or prognosis.

BACKGROUND OF THE INVENTION

TNF-like ligand 1a (TL1a, syn. TNF superfamily member 15 (TNFSF15); TL1 and VEGI) is a member of the tumor necrosis factor superfamily, which is expressed by antigen presenting cells (including dendritic cells, B cells and macrophages), CD4⁺ and CD8⁺ T cells and endothelial cells and can be expressed on the cell surface or secreted as a soluble cytokine. The receptor for TL1a, Death Receptor 3 (DR3) is expressed by a variety of cells, including CD4⁺ and CD8⁺ T cells, NK cells, NKT cells and FOXP3⁺ regulatory T (Treg) cells.

TL1a can also bind a decoy receptor (DcR3), which is a competitive inhibitor of DR3. DcR3 also acts as a decoy receptor for Fas-ligand (Fas-L) and lymphotoxin-like inducible protein that competes with glycoprotein D for binding herpesvirus entry mediator on T-cells (LIGHT). Accordingly, DcR3 is an important regulator of several signal transduction pathways.

The TL1a/DR3 signaling pathway has been implicated in several biological systems, which are associated with human diseases. For example, TL1a has been shown to play a role in immunity and in angiogenesis.

Using mice deficient in TL1a and/or DR3, researchers have also shown that inhibiting this pathway can provide prophylactic or therapeutic benefit in several immune-mediated conditions, such as, experimental autoimmune encephalomyelitis (EAE; a model of multiple sclerosis), colitis, inflammatory bowel disease, asthma and arthritis. TL1a has also been shown to promote formation of foam cells and atherosclerotic plaques.

It will be apparent to the skilled artisan from the foregoing that TL1a plays an important role in biological processes

involved in several important human diseases. Accordingly, compounds that inhibit TL1a activity are desirable, e.g., for their therapeutic, prophylactic, diagnostic and prognostic uses.

SUMMARY OF THE INVENTION

The inventors have produced TL1a-binding proteins comprising antigen binding domains of antibodies which are capable of specifically binding to TL1a and inhibiting interaction of TL1a and DR3 (thereby neutralizing TL1a activity(ies)) without inhibiting interaction of TL1a and DcR3. Without being bound by any theory or mode of action, the inventors reasoned that such TL1a-binding proteins may be capable of reducing or preventing signaling of TL1a through DR3 without significantly disturbing the homeostatic interaction of DcR3 and TL1a. This preserves the natural antagonistic effects of DcR3 on TL1a-DR3 interactions, which may be advantageous because DcR3 also regulates the amount of free Fas-L and LIGHT available for binding to their receptors (Fas and H-VEM, respectively). Since Fas-mediated killing plays a role in cancer surveillance, potential downstream consequences of increasing the amount of DcR3 to bind to Fas-L could include increased susceptibility to cancer. Again, without being bound by theory or mode of action, proteins that specifically inhibit interaction of TL1a and DR3, but not DcR3, could be advantageous in treating disease but without compromising safety.

A subclass of the TL1a-binding proteins identified by the inventors was also found to inhibit or prevent apoptosis of TF-1 cells induced by human TL1a at low concentrations, i.e., the antibodies had a low effective concentration or EC₅₀. TL1a-binding proteins capable of inhibiting or preventing TL1a activity (e.g., TL1a-induced apoptosis of TF-1 cells) are sometimes referred to herein as highly potent TL1a-binding proteins.

The inventors have also identified a region of TL1a which is bound by a highly potent TL1a-binding protein which binds specifically to TL1a and inhibits interaction of TL1a with DR3 without inhibiting the ability of TL1a to interact with DcR3.

The TL1a-binding proteins identified by the inventors form the basis for various therapeutic/prophylactic/diagnostic/prognostic uses. This is demonstrated by the inventors' use of a TL1a-binding protein of the disclosure to treat accepted models of colitis, with the protein showing efficacy at least equal to the current standard of care for this condition.

Accordingly, the disclosure provides an isolated or recombinant TL1a-binding protein comprising an antigen binding domain of an antibody, wherein the antigen binding domain specifically binds to TL1a and, wherein the TL1a-binding protein inhibits interaction of TL1a and DR3 and does not inhibit interaction of TL1a and DcR3.

In one example, the TL1a-binding protein does not detectably reduce interaction of TL1a and DcR3. For example, the effect of the TL1a-binding protein on interaction of TL1a and DcR3 is assessed using a competition enzyme linked immunosorbent assay (ELISA). For example, the TL1a-binding protein is incubated with TL1a (e.g., human TL1a) and then contacted with a polypeptide comprising DcR3 (e.g., human DcR3 (hDcR3)) fused to an Fc region of an antibody ("DcR3/Fc") and the level of bound TL1a is detected. In one example, the level of bound TL1a in the presence or absence of the protein is not significantly different and/or is insufficiently different to permit calculation of an EC₅₀.

In one example, the level of inhibition of interaction of TL1a and DcR3 (or DcR3/Fc) in the presence of the TL1a-binding protein expressed as a percentage of the level of binding in the absence of the protein is 25% or less, or 22% or less, or 20% or less, or 18% or less, or 15% or less, or 12% or less, or 10% or less, or 7% or less, or 5% or less.

In one example, the ability of a TL1a-binding protein to inhibit interaction of TL1a and DR3 or DcR3 is assessed by immobilizing DcR3/Fc or a polypeptide comprising DR3 (e.g., human DR3 (hDR3)) fused to a Fc region of an antibody (DR3/Fc) on a solid or semi-solid surface (e.g., an ELISA plate) at a concentration of about 2 $\mu\text{g}/\text{mL}$. The TL1a-binding protein is then contacted with biotinylated human TL1a (at a concentration of about 1 $\mu\text{g}/\text{mL}$) for about 30 minutes then added to the immobilized DcR3/Fc or DR3/Fc. Following washing, bound TL1a is detected. To determine percentage binding or inhibition, data are normalized by expression as a percentage of maximum binding of TL1a to the immobilized DcR3/Fc or DR3/Fc in the absence of a TL1a-binding protein. By calculating the level of inhibition at multiple concentrations of the TL1a-binding protein, an EC_{50} can be determined.

In one example, the TL1a-binding protein inhibits interaction of TL1a and DR3 (or DR3/Fc) but not TL1a and DcR3 (or DcR3/Fc).

For example, the TL1a-binding protein inhibits interaction of DR3/Fc and TL1a with an EC_{50} of from about 20 nM to about 10 fM, or an EC_{50} of 20 nM or less, such as, 15 nM or less, for example, 11 nM or less, for example 5 nM or less. In one example, the EC_{50} is 5 nM or less. For example, the EC_{50} is 3 nM or less. For example, the EC_{50} is 2.5 nM or less. For example, the EC_{50} is 1 nM or less. For example, the EC_{50} is 0.5 nM or less. In one example, the EC_{50} is assessed using a competition enzyme linked immunosorbent assay (ELISA). For example, various concentrations of the TL1a-binding protein are incubated with TL1a (e.g., human TL1a) (e.g., about 1 $\mu\text{g}/\text{mL}$ of TL1a) and then contacted with the DR3/Fc (e.g., about 2 $\mu\text{g}/\text{mL}$ of the DR3/Fc) and the level of bound TL1a is detected. The concentration of protein at which half maximal inhibition of binding to TL1a is detected is considered the EC_{50} .

In one example, the TL1a-binding protein neutralizes TL1a activity in or on a cell by interfering with TL1a and DR3 interactions.

In one example, the TL1a-binding protein binds to the extracellular domain of TL1a, such as the extracellular domain of human TL1a.

In one example, the TL1a-binding protein binds to human TL1a produced by mammalian cells, such as human cells.

Exemplary TL1a-binding proteins of the disclosure reduce the level of apoptosis of TF-1 cells cultured in the presence of human TL1a, such as human TL1a produced by mammalian cells (e.g., human cells) (e.g., about 100 ng human TL1a per mL of culture) and cycloheximide. For example, about 7×10^4 to 8×10^4 TF-1 cells (e.g., 7.5×10^4 cells) are contacted with about 1 μg human TL1a per mL of culture and cycloheximide. For example, the TL1a-binding protein reduces the level of apoptosis of the TF-1 cells with an EC_{50} (i.e., a concentration of the TL1a-binding protein that achieves 50% of the maximum inhibition of TL1a-induced apoptosis of TF-1 cells achieved by the TL1a-binding protein) of 25 nM or less. In one example, the EC_{50} is 5 nM or less. In one example, the EC_{50} is 2 nM or less. In one example, the EC_{50} is 1.5 nM or less or 1.2 nM or less or 1.1 nM or less. In one example, the EC_{50} is 1 nM or less. In one example, the EC_{50} is 0.75 nM or less. In one example, the EC_{50} is 0.3 nM or less. In one example, the EC_{50} is 0.1

nM or less. In one example, the EC_{50} is from about 1.5 nM to about 10 fM, such as from about 1 nM to about 50 fM, for example, from about 1 nM to about 100 fM.

In one example, the TL1a-binding protein binds to TL1a on the surface of a cell with an EC_{50} (i.e., a concentration of the TL1a-binding protein that achieves 50% of the maximum binding to the cell achieved by the TL1a-binding protein) of about 10 nM or less, e.g., as determined using flow cytometry. In one example, the flow cytometry is performed with about 2×10^5 to 3×10^5 cells (e.g., 2.5×10^5 cells). In one example, the EC_{50} is 5 nM or less. In one example, the EC_{50} is 2 nM or less. In one example, the EC_{50} is from about 10.0 nM or 5.0 nM or 1.0 nM or 0.5 nM or 0.1 nM to about 10 fM.

The disclosure also provides an isolated or recombinant TL1a-binding protein comprising an antigen binding domain of an antibody, wherein the antigen binding domain binds specifically to TL1a and inhibits the interaction of biotinylated TL1a and DR3/Fc with an EC_{50} of about 2.5 nM or less, such as 1 nM or less or from about 2.5 nM, or about 1.0 nM, or about 0.5 nM, or about 0.1 nM to about 10 fM in a competition ELISA,

wherein the DR3/Fc is immobilized on a solid or semi-solid substrate (e.g., a solid substrate such as an ELISA plate) at a concentration of about 2 $\mu\text{g}/\text{mL}$, and wherein the biotinylated TL1a is contacted for about 30 minutes at a concentration of about 1 $\mu\text{g}/\text{mL}$ with the TL1a binding protein at a concentration range of from about 10 $\mu\text{g}/\text{mL}$ to about 0.01 $\mu\text{g}/\text{mL}$ and is then contacted to the immobilized DR3/Fc,

and wherein the TL1a-binding protein does not detectably reduce interaction of biotinylated TL1a and DcR3/Fc in a competition ELISA compared to the level of the binding of biotinylated TL1a to DcR3/Fc in the absence of the TL1a-binding protein, wherein the DcR3/Fc is immobilized on a solid or semi-solid substrate (e.g., a solid substrate such as an ELISA plate) at a concentration of about 2 $\mu\text{g}/\text{mL}$, wherein the biotinylated TL1a is contacted for about 30 minutes at a concentration of about 1 $\mu\text{g}/\text{mL}$ with the TL1a-binding protein at a concentration of from about 10 $\mu\text{g}/\text{mL}$ to 0.1 $\mu\text{g}/\text{mL}$ and is then contacted to the immobilized DcR3/Fc.

The disclosure also provides an isolated or recombinant TL1a-binding protein comprising an antigen binding domain of an antibody, wherein the antigen binding domain binds specifically to TL1a and inhibits the interaction of biotinylated TL1a and DR3/Fc with an EC_{50} of about 2.5 nM or less, such as 1 nM or less or from about 2.5 nM, or about 1.0 nM, or about 0.5 nM, or about 0.1 nM to about 10 fM in a competition ELISA,

wherein the DR3/Fc is immobilized on a solid or semi-solid substrate (e.g., a solid substrate such as an ELISA plate) at a concentration of about 2 $\mu\text{g}/\text{mL}$, and wherein the biotinylated TL1a is contacted for about 30 minutes at a concentration of about 1 $\mu\text{g}/\text{mL}$ with the TL1a binding protein at a concentration range of from about 10 $\mu\text{g}/\text{mL}$ to about 0.01 $\mu\text{g}/\text{mL}$ and is then contacted to the immobilized DR3/Fc,

and wherein the TL1a-binding protein does not detectably reduce interaction of biotinylated TL1a and DcR3/Fc in a competition ELISA compared to the level of the binding of biotinylated TL1a to DcR3/Fc in the absence of the TL1a-binding protein, and wherein the DcR3/Fc is immobilized on a solid or semi-solid substrate (e.g., a solid substrate such as an ELISA plate) at a concentration of about 2 $\mu\text{g}/\text{mL}$, wherein the biotinylated TL1a is contacted for about 30 minutes at a concen-

tration of about 1 $\mu\text{g/mL}$ with the TL1a-binding protein at a concentration of from about 10 $\mu\text{g/mL}$ to 0.1 $\mu\text{g/mL}$ and is then contacted to the immobilized DcR3/Fc. and wherein the TL1a-binding protein reduces the level of apoptosis of TF-1 cells cultured in the presence of human TL1a produced by human cells and cycloheximide with an EC_{50} (i.e., a concentration of the TL1a-binding protein that achieves 50% of the maximum inhibition of TL1a-induced apoptosis of TF-1 cells achieved by the TL1a-binding protein) of 25 nM or less, or from about 1.5 nM or 1.0 nM or 0.5 nM or 0.1 nM or 0.05 nM to about 10 fM

wherein about 7.5×10^4 TF-1 cells are contacted with about 100 ng human TL1a per mL of culture and about 10 $\mu\text{g/mL}$ cycloheximide with the TL1a-binding protein at a concentration of about 5 $\mu\text{g/mL}$ or less for about 4 to 5 hours.

In one example, the TL1a is biotinylated at one site, i.e., the biotin is linked to only one amino acid in TL1a.

In one example, the TL1a-binding protein does not detectably reduce interaction of the biotinylated TL1a and DcR3/Fc in the competition ELISA compared to the level of the binding of biotinylated TL1a to DcR3/Fc in the absence of the TL1a-binding protein, wherein the biotinylated TL1a is contacted for about 30 minutes at a concentration of about 1 $\mu\text{g/mL}$ with the TL1a-binding protein at a concentration of about 100 $\mu\text{g/mL}$ and is then contacted to the immobilized DcR3/Fc.

In one example, the TL1a-binding protein does not detectably reduce interaction of the biotinylated TL1a and DcR3/Fc in the competition ELISA compared to the level of the binding of biotinylated TL1a to DcR3/Fc in the absence of the TL1a-binding protein, wherein the biotinylated TL1a is contacted for about 30 minutes at a concentration of about 1 $\mu\text{g/mL}$ with the TL1a-binding protein at a concentration of about 10 $\mu\text{g/mL}$ and is then contacted to the immobilized DcR3/Fc.

In one example, the TL1a-binding protein reduces the level of apoptosis of the TF-1 cells with an EC_{50} of 22 nM or less. In one example, the EC_{50} is 10 nM or less. In one example, the EC_{50} is 5 nM or less. In one example, the EC_{50} is 2 nM or less. In one example, the EC_{50} is 1.5 nM or less or 1.2 nM or less or 1.1 nM or less. In one example, the EC_{50} is 1 nM or less. In one example, the EC_{50} is 0.75 nM or less. In one example, the EC_{50} is 0.3 nM or less. In one example, the EC_{50} is 0.1 nM or less.

The disclosure additionally, or alternatively, provides an isolated or recombinant TL1a-binding protein comprising an antigen binding domain of an antibody, wherein the antigen binding domain specifically binds to TL1a and, wherein the TL1a-binding protein inhibits interaction of TL1a and DR3 and does not inhibit interaction of TL1a and DcR3, and wherein the TL1a-binding protein binds a mutant form of soluble human TL1a comprising a sequence set forth in SEQ ID NO: 202 in which the arginine at position 32 has been substituted with alanine and/or the arginine at position 85 has been substituted with alanine at a level that is at least 75% lower than the level with which the TL1a-binding protein binds to soluble human TL1a comprising a sequence set forth in SEQ ID NO: 202.

In one example, the mutant form of soluble human TL1a is immobilized on a solid or semi-solid substrate (e.g., a solid substrate such as an ELISA plate) at a concentration of about 1 $\mu\text{g/mL}$, and wherein the TL1a binding protein at a concentration range of from about 10 $\mu\text{g/mL}$ to about 0.01 $\mu\text{g/mL}$ is then contacted to the immobilized mutant TL1a.

In one example, the TL1a-binding protein binds a mutant form of soluble human TL1a comprising a sequence set forth in SEQ ID NO: 202 in which the arginine at position 32 has been substituted with alanine and/or the arginine at position 85 has been substituted with alanine at a level that is no greater than 25% of the level with which the protein binds to soluble human TL1a comprising a sequence set forth in SEQ ID NO: 202. For example, the level of binding of the TL1a-binding protein to the mutant form of soluble human TL1a is no greater than 25% of the level with which the protein binds to soluble human TL1a, when the TL1a-binding protein is tested at a concentration of 10 $\mu\text{g/mL}$.

The disclosure additionally, or alternatively, provides an isolated or recombinant TL1a-binding protein comprising an antigen binding domain of an antibody, wherein the antigen binding domain specifically binds to TL1a and, wherein the TL1a-binding protein inhibits interaction of TL1a and DR3 and does not inhibit interaction of TL1a and DcR3, and wherein the TL1a-binding protein binds a mutant form of soluble human TL1a comprising a sequence set forth in SEQ ID NO: 202 in which the arginine at position 32 has been substituted with alanine and/or the arginine at position 85 has been substituted with alanine at a level that is at least 75% lower than the level with which the protein binds to soluble human TL1a comprising a sequence set forth in SEQ ID NO: 202,

wherein the mutant form of TL1a is immobilized on a solid or semi-solid substrate at a concentration of about 1 $\mu\text{g/mL}$, and wherein the TL1a binding protein at a concentration of 10 $\mu\text{g/mL}$ is then contacted to the immobilized mutant TL1a.

In one example, the mutant form of soluble human TL1a comprises a sequence set forth in SEQ ID NO: 202 in which the arginine at position 32 has been substituted with alanine.

In one example, the mutant form of soluble human TL1a comprises a sequence set forth in SEQ ID NO: 202 in which the arginine at position 85 has been substituted with alanine.

In one example, the mutant form of soluble human TL1a comprises a sequence set forth in SEQ ID NO: 202 in which the arginine at position 32 has been substituted with alanine and in which the arginine at position 85 has been substituted with alanine.

In one example, the level of binding of the TL1a-binding protein to the mutant form of soluble human TL1a is at least 80% or 85% or 90% or 95% lower than the level with which the protein binds to soluble human TL1a comprising a sequence set forth in SEQ ID NO: 202.

In one example, the TL1a-binding protein does not detectably bind to the mutant form of soluble human TL1a.

In one example, the binding of the TL1a-binding protein to soluble human TL1a or a mutant form thereof is assessed using Surface Plasmon Resonance. For example, the soluble human TL1a or a mutant form thereof is immobilized (e.g., at a concentration of about 1 $\mu\text{g/mL}$) and the TL1a-binding protein (e.g., at a concentration of about 500 ng/mL) contacted to the immobilized soluble human TL1a or a mutant form thereof and binding detected by Surface Plasmon Resonance. By comparing the level of binding to the soluble human TL1a or a mutant form thereof a comparison can be made to determine a TL1a-binding protein that binds at a level that is at least 75% lower than the level with which the protein binds to soluble human TL1a.

In one example, the binding of the TL1a-binding protein to soluble human TL1a or a mutant form thereof is assessed using ELISA. For example, the soluble human TL1a or a mutant form thereof is immobilized (e.g., at a concentration of about 1 $\mu\text{g/mL}$) and the TL1a-binding protein (e.g., at a

concentration of about 10 µg/mL) contacted to the immobilized soluble human TL1a or a mutant form thereof and binding detected by ELISA (e.g., using standard methods in the art).

In one example, the TL1a-binding protein binds to an epitope within TL1a comprising residues corresponding to arginine at position 32 of SEQ ID NO: 202 and the arginine at position 85 of SEQ ID NO: 202. In one example, the epitope is a conformational epitope.

In one example, the TL1a-binding protein binds at least at amino acid residues arginine at position 32 and arginine at position 85 of a human TL1a which comprises an amino acid sequence as set forth in SEQ ID NO:202.

Exemplary TL1a-binding proteins having the binding characteristics set forth in the foregoing paragraphs will be apparent to the skilled artisan from the description herein and include those comprising the following pairs of V_H and V_L :

- (i) a V_H comprising a sequence set forth in SEQ ID NO: 94 and a V_L comprising a sequence set forth in SEQ ID NO: 95;
- (ii) a V_H comprising a sequence set forth in SEQ ID NO: 137 and a V_L comprising a sequence set forth in SEQ ID NO: 138;
- (iii) a V_H comprising a sequence set forth in SEQ ID NO: 162 and a V_L comprising a sequence set forth in SEQ ID NO: 172; or
- (iv) a V_H comprising a sequence set forth in SEQ ID NO: 173 and a V_L comprising a sequence set forth in SEQ ID NO: 174.

V_H and V_L falling within the foregoing sequences will be apparent to the skilled person from the description herein and are to be taken to apply mutatis mutandis to the present example of the disclosure.

In one example, a TL1a-binding protein of the disclosure inhibits interaction of TL1a from human, cynomolgus monkey or rhesus monkey and DR3. Such TL1a-binding proteins are useful for characterization in animal models of human disease.

In one example, a TL1a-binding protein of the disclosure does not detectably inhibit interaction of TL1a from mouse, pig, rabbit or guinea pig and DR3, e.g., the TL1a-binding protein does not detectably inhibit the level of apoptosis of TF-1 cells cultured in the presence of the relevant TL1a, e.g. as determined using an assay described herein.

In one example, a TL1a-binding protein of the disclosure detectably inhibits interaction of TL1a from rat and DR3. For example, the TL1a-binding protein detectably inhibits the level of apoptosis of TF-1 cells cultured in the presence of the relevant TL1a, e.g. as determined using an assay described herein.

In one example, a TL1a-binding protein of the disclosure detectably binds to an isoform of TL1a consisting of amino acids 72-251 of SEQ ID NO: 123 and/or to an isoform of TL1a consisting of amino acids 84-251 of SEQ ID NO: 123.

For example, binding is assessed by an ELISA in which the isoform of TL1a is immobilized at a concentration of 1 µg/ml and the TL1a-binding protein is contacted to the isoform and the level of binding assessed.

The disclosure additionally or alternatively provides an isolated or recombinant TL1a-binding protein comprising an antigen binding domain of an antibody comprising any one or more of the following:

- (i) a V_H comprising a sequence set forth in SEQ ID NO: 94 and a V_L comprising a sequence set forth in SEQ ID NO: 95;

- (ii) a V_H comprising a sequence set forth in SEQ ID NO: 137 and a V_L comprising a sequence set forth in SEQ ID NO: 138;

- (iii) a V_H comprising a sequence set forth in SEQ ID NO: 162 and a V_L comprising a sequence set forth in SEQ ID NO: 172; or

- (iv) a V_H comprising a sequence set forth in SEQ ID NO: 173 and a V_L comprising a sequence set forth in SEQ ID NO: 174.

The disclosure additionally or alternatively provides an isolated or recombinant TL1a-binding protein comprising an antigen binding domain of an antibody comprising any one or more of the following:

- (i) a V_H comprising a sequence set forth in SEQ ID NO: 2 and a V_L comprising a sequence set forth in SEQ ID NO: 6;

- (ii) a V_H comprising a sequence set forth in SEQ ID NO: 10 and a V_L comprising a sequence set forth in SEQ ID NO: 14;

- (iii) a V_H comprising a sequence set forth in SEQ ID NO: 18 and a V_L comprising a sequence set forth in SEQ ID NO: 22;

- (iv) a V_H comprising a sequence set forth in SEQ ID NO: 26 and a V_L comprising a sequence set forth in SEQ ID NO: 30;

- (v) a V_H comprising a sequence set forth in SEQ ID NO: 34 and a V_L comprising a sequence set forth in SEQ ID NO:38;

- (vi) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 46;

- (vii) a V_H comprising a sequence set forth in SEQ ID NO: 50 and a V_L comprising a sequence set forth in SEQ ID NO: 54;

- (viii) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 46;

- (ix) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 62;

- (x) a V_H comprising a sequence set forth in SEQ ID NO: 66 and a V_L comprising a sequence set forth in SEQ ID NO: 62;

- (xi) a V_H comprising a sequence set forth in SEQ ID NO: 70 and a V_L comprising a sequence set forth in SEQ ID NO: 62;

- (xii) a V_H comprising a sequence set forth in SEQ ID NO: 74 and a V_L comprising a sequence set forth in SEQ ID NO: 62;

- (xiii) a V_H comprising a sequence set forth in SEQ ID NO: 78 and a V_L comprising a sequence set forth in SEQ ID NO: 62;

- (xiv) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 82;

- (xv) a V_H comprising a sequence set forth in SEQ ID NO: 86 and a V_L comprising a sequence set forth in SEQ ID NO: 46;

- (xvi) a V_H comprising a sequence set forth in SEQ ID NO: 90 and a V_L comprising a sequence set forth in SEQ ID NO: 46;

- (xvii) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 62;

- (xviii) a V_H comprising a sequence set forth in SEQ ID NO: 154 and a V_L comprising a sequence set forth in SEQ ID NO: 164;

- (xix) a V_H comprising a sequence set forth in SEQ ID NO: 155 and a V_L comprising a sequence set forth in SEQ ID NO: 163;
- (xx) a V_H comprising a sequence set forth in SEQ ID NO: 156 and a V_L comprising a sequence set forth in SEQ ID NO: 165;
- (xxi) a V_H comprising a sequence set forth in SEQ ID NO: 157 and a V_L comprising a sequence set forth in SEQ ID NO: 166;
- (xxii) a V_H comprising a sequence set forth in SEQ ID NO: 158 and a V_L comprising a sequence set forth in SEQ ID NO: 167;
- (xxiii) a V_H comprising a sequence set forth in SEQ ID NO: 159 and a V_L comprising a sequence set forth in SEQ ID NO: 168;
- (xxiv) a V_H comprising a sequence set forth in SEQ ID NO: 160 and a V_L comprising a sequence set forth in SEQ ID NO: 169;
- (xxv) a V_H comprising a sequence set forth in SEQ ID NO: 161 and a V_L comprising a sequence set forth in SEQ ID NO: 170;
- (xxvi) a V_H comprising a sequence set forth in SEQ ID NO: 234 and a V_L comprising a sequence set forth in SEQ ID NO: 169;
- (xxvii) a V_H comprising a sequence set forth in SEQ ID NO: 154 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxviii) a V_H comprising a sequence set forth in SEQ ID NO: 155 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxix) a V_H comprising a sequence set forth in SEQ ID NO: 156 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxx) a V_H comprising a sequence set forth in SEQ ID NO: 157 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxxi) a V_H comprising a sequence set forth in SEQ ID NO: 158 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxxii) a V_H comprising a sequence set forth in SEQ ID NO: 159 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxxiii) a V_H comprising a sequence set forth in SEQ ID NO: 160 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxxiv) a V_H comprising a sequence set forth in SEQ ID NO: 161 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxxv) a V_H comprising a sequence set forth in SEQ ID NO: 234 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxxvi) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 164;
- (xxxvii) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 165;
- (xxxviii) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 166;
- (xxxix) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 167;
- (xl) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 168;

- (xli) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 169;
- (xlii) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 170;
- (xlili) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 171;
- (xliv) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 172;
- (xlv) a V_H comprising a sequence set forth in SEQ ID NO: 175 and a V_L comprising a sequence set forth in SEQ ID NO: 188;
- (xlvi) a V_H comprising a sequence set forth in SEQ ID NO: 176 and a V_L comprising a sequence set forth in SEQ ID NO: 189;
- (xlvii) a V_H comprising a sequence set forth in SEQ ID NO: 177 and a V_L comprising a sequence set forth in SEQ ID NO: 190;
- (xlviii) a V_H comprising a sequence set forth in SEQ ID NO: 178 and a V_L comprising a sequence set forth in SEQ ID NO: 191;
- (xlix) a V_H comprising a sequence set forth in SEQ ID NO: 179 and a V_L comprising a sequence set forth in SEQ ID NO: 192;
- (l) a V_H comprising a sequence set forth in SEQ ID NO: 180 and a V_L comprising a sequence set forth in SEQ ID NO: 193;
- (li) a V_H comprising a sequence set forth in SEQ ID NO: 181 and a V_L comprising a sequence set forth in SEQ ID NO: 194;
- (lli) a V_H comprising a sequence set forth in SEQ ID NO: 182 and a V_L comprising a sequence set forth in SEQ ID NO: 195;
- (liii) a V_H comprising a sequence set forth in SEQ ID NO: 183 and a V_L comprising a sequence set forth in SEQ ID NO: 196;
- (liv) a V_H comprising a sequence set forth in SEQ ID NO: 184 and a V_L comprising a sequence set forth in SEQ ID NO: 197;
- (lv) a V_H comprising a sequence set forth in SEQ ID NO: 185 and a V_L comprising a sequence set forth in SEQ ID NO: 198;
- (lvi) a V_H comprising a sequence set forth in SEQ ID NO: 186 and a V_L comprising a sequence set forth in SEQ ID NO: 199; or
- (lvii) a V_H comprising a sequence set forth in SEQ ID NO: 187 and a V_L comprising a sequence set forth in SEQ ID NO: 200.

In a particular example, there is provided an isolated or recombinant TL1a-binding protein comprising an antigen binding domain of an antibody comprising a V_H and a V_L , wherein the V_H and V_L respectively comprise sequences selected from the group consisting of

- (i) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (ii) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 106 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence

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The disclosure additionally or alternatively provides an isolated or recombinant TL1a-binding protein comprising an antigen binding domain of an antibody comprising a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 46, wherein the V_H and/or V_L comprise one or more of the following substitutions or groups of substitutions:

- (i) the V_H comprises an alanine at position 16 of SEQ ID NO: 42;
- (ii) the V_H comprises an alanine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (iii) the V_H comprises a serine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (iv) the V_H comprises a histidine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (v) the V_H comprises a leucine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (vi) the V_H comprises an aspartic acid at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (vii) the V_H comprises a tyrosine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (viii) the V_H comprises a proline at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (ix) the V_H comprises a glutamine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (x) the V_H comprises a lysine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xi) the V_H comprises an alanine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xii) the V_H comprises a serine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xiii) the V_H comprises a histidine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xiv) the V_H comprises a leucine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xv) the V_H comprises an aspartic acid at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xvi) the V_H comprises a tyrosine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xvii) the V_H comprises a glutamine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xviii) the V_H comprises a lysine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xix) the V_H comprises an alanine at position 102 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xx) the V_H comprises a serine at position 102 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;

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- (xxi) the V_H comprises a histidine at position 102 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxii) the V_H comprises a leucine at position 102 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxiii) the V_H comprises a tyrosine at position 102 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxiv) the V_H comprises a proline at position 102 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxv) the V_H comprises a glutamine at position 102 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxvi) the V_H comprises a lysine at position 102 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxvii) the V_H comprises an alanine at position 103 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxviii) the V_H comprises a serine at position 103 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxix) the V_H comprises a histidine at position 103 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxx) the V_H comprises a leucine at position 103 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxxi) the V_H comprises an aspartic acid at position 103 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxxii) the V_H comprises a tyrosine at position 103 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxxiii) the V_H comprises a proline at position 103 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxxiv) the V_H comprises a glutamine at position 103 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxxv) the V_H comprises a lysine at position 103 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxxvi) the V_H comprises a serine at position 104 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxxvii) the V_H comprises a histidine at position 104 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxxviii) the V_H comprises a leucine at position 104 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxxix) the V_H comprises an aspartic acid at position 104 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xl) the V_H comprises a tyrosine at position 104 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xli) the V_H comprises a proline at position 104 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xlii) the V_H comprises a glutamine at position 104 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;

each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24, a threonine at position 76 and a glutamic acid at position 51 each relative to SEQ ID NO: 46;

(lxxxi) the V_H comprises a proline at position 41, a leucine at position 51, an alanine at position 72, an aspartic acid at position 73, an arginine at position 74, a threonine at position 76, a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24, a threonine at position 76 and a glutamic acid at position 51 each relative to SEQ ID NO: 46; and

(lxxxii) the V_H comprises a proline at position 41, a leucine at position 51, an alanine at position 72, an aspartic acid at position 73, an arginine at position 74, a threonine at position 76, a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24, a threonine at position 76 and a glycine at position 51 each relative to SEQ ID NO: 46.

In one example, the TL1a-binding protein comprises an antigen binding domain of an antibody comprising any one or more of the following:

- (i) a V_H comprising a sequence set forth in SEQ ID NO: 26 and a V_L comprising a sequence set forth in SEQ ID NO: 30;
- (ii) a V_H comprising a sequence set forth in SEQ ID NO: 34 and a V_L comprising a sequence set forth in SEQ ID NO: 38; and
- (iii) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 46.

In one particular example, there is provided an isolated or recombinant TL1a-binding protein comprising an antigen binding domain of an antibody comprising a V_H and a V_L , wherein the V_H and V_L respectively comprise sequences selected from the group consisting of

- (i) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (ii) a V_H comprising a sequence set forth in SEQ ID NO: 175 and a V_L comprising a sequence set forth in SEQ ID NO: 188;
- (iii) a V_H comprising a sequence set forth in SEQ ID NO: 176 and a V_L comprising a sequence set forth in SEQ ID NO: 189;
- (iv) a V_H comprising a sequence set forth in SEQ ID NO: 177 and a V_L comprising a sequence set forth in SEQ ID NO: 190;
- (v) a V_H comprising a sequence set forth in SEQ ID NO: 178 and a V_L comprising a sequence set forth in SEQ ID NO: 191;
- (vi) a V_H comprising a sequence set forth in SEQ ID NO: 179 and a V_L comprising a sequence set forth in SEQ ID NO: 192;
- (vii) a V_H comprising a sequence set forth in SEQ ID NO: 180 and a V_L comprising a sequence set forth in SEQ ID NO: 193;
- (viii) a V_H comprising a sequence set forth in SEQ ID NO: 181 and a V_L comprising a sequence set forth in SEQ ID NO: 194;
- (ix) a V_H comprising a sequence set forth in SEQ ID NO: 183 and a V_L comprising a sequence set forth in SEQ ID NO: 196;

(x) a V_H comprising a sequence set forth in SEQ ID NO: 185 and a V_L comprising a sequence set forth in SEQ ID NO: 198;

(xi) a V_H comprising a sequence set forth in SEQ ID NO: 186 and a V_L comprising a sequence set forth in SEQ ID NO: 199; and

(xii) a V_H comprising a sequence set forth in SEQ ID NO: 187 and a V_L comprising a sequence set forth in SEQ ID NO: 200.

In one example, a TL1a-binding protein of the disclosure comprises an antigen binding domain comprising a CDR3 of a variable region of an antibody recited above. For example, the CDR3 is defined according to the Kabat numbering system and comprises a sequence set forth in any one of SEQ ID NOs: 5, 9, 13, 17, 21, 25, 29, 33, 37, 41, 45, 49, 53, 57, 61, 65, 69, 73, 77, 81, 85, 89, 93 or a sequence labeled as "CDR3" and shown in bold text in FIG. 1A to 1H or in FIG. 9B or 9C or a sequence comprising amino acids 99 to 108 of any one of SEQ ID NOs: 175 to 187 or amino acids 91 to 100 of any one of SEQ ID NOs: 188 to 200 or 234.

For example, the CDR3 is defined according to the Kabat numbering system and comprises a sequence set forth in SEQ ID NO: 45 or 49 or a sequence comprising amino acids 99 to 108 of any one of SEQ ID NOs: 175 to 181, 183 or 185 to 187 or amino acids 91 to 100 of any one of SEQ ID NOs: 188 to 194, 196 or 198 to 200.

In another example, the CDR3 (e.g., a HCDR3) is defined according to the enhanced Chothia numbering system and comprises a sequence labeled as "CDR3" and shown in underlined text in any one of FIG. 1A, 1C, 1D or 1E or 9B.

In one example, the CDR3 comprises a sequence $\text{EVPX}_1\text{TAX}_2\text{FEY}$ (SEQ ID NO: 143), wherein X_1 is aspartic acid or glutamic acid and X_2 is serine or alanine.

In one example, the CDR3 comprises a sequence $\text{EX}_1\text{PX}_2\text{X}_3\text{AX}_4\text{FX}_5\text{Y}$ (SEQ ID NO: 235), wherein:

X_1 is an amino acid selected from the group consisting of valine, alanine, serine, histidine, aspartic acid, leucine, tyrosine, proline, glutamine or lysine;

X_2 is an amino acid selected from the group consisting of alanine, serine, histidine, lysine, glutamic acid or aspartic acid;

X_3 is an amino acid selected from the group consisting of alanine, serine, aspartic acid, tyrosine or threonine;

X_4 is an amino acid selected from the group consisting of serine, alanine, histidine, leucine, aspartic acid or tyrosine; and

X_5 is an amino acid selected from the group consisting of alanine, serine, histidine, leucine, aspartic acid, proline, glutamine, glutamic acid or lysine.

In one example, the CDR3 (e.g., a LCDR3) comprises a sequence set forth in SEQ ID NO: 141 (or sequence labeled as CDR3 in bold text of the sequence labeled "Consensus" in FIG. 1F or 9C).

For example, the antigen binding domain comprises three CDRs of a variable region of the antibody.

In some examples of the disclosure, the antigen binding domain is an antibody variable region comprising three CDRs of a variable region comprising an amino acid sequence set forth in any one of SEQ ID NOs: 2, 6, 10, 14, 18, 22, 26, 30, 34, 38, 42, 46, 50, 54, 58, 62, 66, 70, 74, 78, 82, 86, 90, 94, 95, 137, 138, 152 to 200 or 234.

In some examples, the antigen binding domain is an antibody variable region comprising three CDRs of a variable region comprising:

- (a) a V_H comprising a sequence set forth in SEQ ID NO: 42 and comprising one or more of the following substitutions or groups of substitutions:

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- (i) an alanine at position 16 of SEQ ID NO: 42;
- (ii) an alanine at position 100 of SEQ ID NO: 42;
- (iii) a serine at position 100 of SEQ ID NO: 42;
- (iv) a histidine at position 100 of SEQ ID NO: 42;
- (v) a leucine at position 100 of SEQ ID NO: 42;
- (vi) an aspartic acid at position 100 of SEQ ID NO: 42;
- (vii) a tyrosine at position 100 of SEQ ID NO: 42;
- (viii) a proline at position 100 of SEQ ID NO: 42;
- (ix) a glutamine at position 100 of SEQ ID NO: 42;
- (x) a lysine at position 100 of SEQ ID NO: 42;
- (xi) an alanine at position 101 of SEQ ID NO: 42;
- (xii) a serine at position 101 of SEQ ID NO: 42;
- (xiii) a histidine at position 101 of SEQ ID NO: 42;
- (xiv) a leucine at position 101 of SEQ ID NO: 42;
- (xv) an aspartic acid at position 101 of SEQ ID NO: 42;
- (xvi) a tyrosine at position 101 of SEQ ID NO: 42;
- (xvii) a glutamine at position 101 of SEQ ID NO: 42;
- (xviii) a lysine at position 101 of SEQ ID NO: 42;
- (xix) an alanine at position 102 of SEQ ID NO: 42;
- (xx) a serine at position 102 of SEQ ID NO: 42;
- (xxi) a histidine at position 102 of SEQ ID NO: 42;
- (xxii) a leucine at position 102 of SEQ ID NO: 42;
- (xxiii) a tyrosine at position 102 of SEQ ID NO: 42;
- (xxiv) a proline at position 102 of SEQ ID NO: 42;
- (xxv) a glutamine at position 102 of SEQ ID NO: 42;
- (xxvi) a lysine at position 102 of SEQ ID NO: 42;
- (xxvii) an alanine at position 103 of SEQ ID NO: 42;
- (xxviii) a serine at position 103 of SEQ ID NO: 42;
- (xxix) a histidine at position 103 of SEQ ID NO: 42;
- (xxx) a leucine at position 103 of SEQ ID NO: 42;
- (xxxi) an aspartic acid at position 103 of SEQ ID NO: 42;
- (xxxii) a tyrosine at position 103 of SEQ ID NO: 42;
- (xxxiii) a proline at position 103 of SEQ ID NO: 42;
- (xxxiv) a glutamine at position 103 of SEQ ID NO: 42;
- (xxxv) a lysine at position 103 of SEQ ID NO: 42;
- (xxxvi) a serine at position 104 of SEQ ID NO: 42;
- (xxxvii) a histidine at position 104 of SEQ ID NO: 42;
- (xxxviii) a leucine at position 104 of SEQ ID NO: 42;
- (xxxix) an aspartic acid at position 104 of SEQ ID NO: 42;
- (xl) a tyrosine at position 104 of SEQ ID NO: 42;
- (xli) a proline at position 104 of SEQ ID NO: 42;
- (xlii) a glutamine at position 104 of SEQ ID NO: 42;
- (xliii) a lysine at position 104 of SEQ ID NO: 42;
- (xliv) an alanine at position 105 of SEQ ID NO: 42;
- (xlv) a histidine at position 105 of SEQ ID NO: 42;
- (xlvi) a leucine at position 105 of SEQ ID NO: 42;
- (xlvii) an aspartic acid at position 105 of SEQ ID NO: 42;
- (xlviii) a tyrosine at position 105 of SEQ ID NO: 42;
- (xlix) a proline at position 105 of SEQ ID NO: 42;
- (l) a glutamine at position 105 of SEQ ID NO: 42;
- (li) a lysine at position 105 of SEQ ID NO: 42;
- (lii) an alanine at position 107 of SEQ ID NO: 42;
- (liii) a serine at position 107 of SEQ ID NO: 42;
- (liv) a histidine at position 107 of SEQ ID NO: 42;
- (lv) a leucine at position 107 of SEQ ID NO: 42;
- (lvi) an aspartic acid at position 107 of SEQ ID NO: 42;
- (lvii) a tyrosine at position 107 of SEQ ID NO: 42;
- (lviii) a proline at position 107 of SEQ ID NO: 42;
- (lix) a glutamine at position 107 of SEQ ID NO: 42;
- (lx) a lysine at position 107 of SEQ ID NO: 42;
- (lxi) a threonine at position 41 of SEQ ID NO: 42;
- (lxii) a serine at position 47 of SEQ ID NO: 42;
- (lxiii) a proline at position 41, an alanine at position 72, an aspartic acid at position 73 and an arginine at position 74 and a threonine at position 76 each relative to SEQ ID NO: 42;

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- (lxiv) a proline at position 41, a leucine at position 51 and a glutamic acid at position 102 each relative to SEQ ID NO: 42;
- (lxv) a proline at position 41, a leucine at position 51, a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42;
- (lxvi) a proline at position 41, a leucine at position 51, an alanine at position 72, an aspartic acid at position 73, an arginine at position 74, a threonine at position 76, a glutamic acid at position 102 and an alanine at position 105; or
- (b) a V_L comprising a sequence set forth in SEQ ID NO: 46 and comprising one or more of the following substitutions or groups of substitutions:
 - (i) a threonine at position 76 of SEQ ID NO: 46;
 - (ii) a threonine at position 23 of SEQ ID NO: 46;
 - (iii) an asparagine at position 28 of SEQ ID NO: 46;
 - (iv) a tyrosine at position 33 of SEQ ID NO: 46;
 - (v) an aspartic acid at position 34 of SEQ ID NO: 46;
 - (vi) an asparagine at position 53 of SEQ ID NO: 46;
 - (vii) a serine at position 54 of SEQ ID NO: 46;
 - (viii) an alanine at position 82 of SEQ ID NO: 46;
 - (ix) a serine at position 95 of SEQ ID NO: 46;
 - (x) a serine at position 96 of SEQ ID NO: 46;
 - (xi) a threonine at position 23, a serine at position 24 and a threonine at position 76 each relative to SEQ ID NO: 46;
 - (xii) a threonine at position 23 and a threonine at position 76 each relative to SEQ ID NO: 46;
 - (xiii) a threonine at position 23, a serine at position 24, a threonine at position 76 and a glutamic acid at position 51 each relative to SEQ ID NO: 46; or
 - (xiv) a threonine at position 23, a serine at position 24, a threonine at position 76 and a glycine at position 51 each relative to SEQ ID NO: 46.

For example, the antigen binding domain is a V_H comprising three CDRs of an amino acid sequence set forth in any one of SEQ ID NOs: 2, 10, 18, 26, 34, 42, 50, 58, 66, 70, 74, 78, 86, 90, 94, 137, 152, 154 to 162, 173, 175 to 187 or 234.

In one example, the CDRs are defined according to the Kabat numbering system.

For example, the TL1a-binding protein comprises a V_H including CDRs as follows:

- (i) a CDR1 comprising a sequence set forth in SEQ ID NO: 3, a CDR2 comprising a sequence set forth in SEQ ID NO: 4 (wherein any one or more of the five C-terminal amino acids of the CDR2 amino acid sequence are substituted with any other naturally-occurring amino acid) and a CDR3 comprising a sequence set forth in SEQ ID NO: 5 (or sequences labeled as CDRs 1, 2 and 3 in bold text of antibody C336 in FIG. 1A);
- (ii) a CDR1 comprising a sequence set forth in SEQ ID NO: 11, a CDR2 comprising a sequence set forth in SEQ ID NO: 12 (wherein any one or more of the five C-terminal amino acids of the CDR2 amino acid sequence are substituted with any other naturally-occurring amino acid) and a CDR3 comprising a sequence set forth in SEQ ID NO: 13 (or sequences labeled as CDRs 1, 2 and 3 in bold text of antibody C334 in FIG. 1A);
- (iii) a CDR1 comprising a sequence set forth in SEQ ID NO: 19, a CDR2 comprising a sequence set forth in SEQ ID NO: 20 (wherein any one or more of the five C-terminal amino acids of the CDR2 amino acid sequence are substituted with any other naturally-oc-

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185 to 187 (wherein any one or more of the five C-terminal amino acids of the CDR2 amino acid sequence are substituted with any other naturally-occurring amino acid) and amino acids 99 to 108 of any one of SEQ ID NOs: 175 to 181, 183 or 185 to 187

In one example, the TL1a-binding protein comprises a V_H including CDRs as follows:

- (i) a CDR1 comprising a sequence set forth in SEQ ID NO: 43 (or sequence labeled as CDR 1 in bold text of the sequence labeled "Consensus" in FIG. 1E);
- (ii) a CDR2 comprising a sequence $WX_1NPNSGNTGYAQKFQG$ (SEQ ID NO: 142), wherein X_1 is methionine or leucine (or sequence labeled as CDR 2 in bold text of the sequence labeled "Consensus" in FIG. 1E or FIG. 9B); and
- (iii) a CDR3 comprising a sequence EVX_1TAX_2FEY (SEQ ID NO: 143), wherein X_1 is aspartic acid or glutamic acid and X_2 is serine or alanine. (or sequence labeled as CDR3 in bold text of the sequence labeled "Consensus" in FIG. 1E or FIG. 9B) or comprising a sequence $EX_1PX_2X_3AX_4FX_5Y$ (SEQ ID NO: 235), wherein:

X_1 is an amino acid selected from the group consisting of valine, alanine, serine, histidine, aspartic acid, leucine, tyrosine, proline, glutamine or lysine;

X_2 is an amino acid selected from the group consisting of alanine, serine, histidine, lysine, glutamic acid or aspartic acid;

X_3 is an amino acid selected from the group consisting of alanine, serine, aspartic acid, tyrosine or threonine;

X_4 is an amino acid selected from the group consisting of serine, alanine, histidine, leucine, aspartic acid or tyrosine; and

X_5 is an amino acid selected from the group consisting of alanine, serine, histidine, leucine, aspartic acid, proline, glutamine, glutamic acid or lysine.

Additional residues suitable for inclusion in CDR3 are described herein and are to be taken to apply mutatis mutandis to the present example of the disclosure.

In one example, the CDRs are defined according to the enhanced Chothia numbering system. For example, the TL1a-binding protein comprises a V_H including CDRs labeled as CDRs 1, 2 and 3 in underlined text of antibody 336, 334, 333, 323, 321, 320 or 319 in FIG. 1A or of antibody C320-90, C320-103, C320-114, C320-115, C320-129 or C320-130 in FIG. 1C or of the sequence labeled "Consensus" in FIG. 1C or FIG. 1E or FIG. 9B.

In one example, the TL1a-binding protein additionally comprises the following:

- (i) a heavy chain FR1 comprising an amino acid sequence set forth in SEQ ID NO: 144;
- (ii) a heavy chain FR2 comprising an amino acid sequence set forth in SEQ ID NO: 145;
- (iii) a heavy chain FR3 comprising an amino acid sequence set forth in SEQ ID NO: 146; and
- (iv) a heavy chain FR4 comprising an amino acid sequence set forth in SEQ ID NO: 147.

In one example, the TL1a-binding protein comprises the following:

- (i) a heavy chain FR1 comprising an amino acid sequence set forth in SEQ ID NO: 144;
- (ii) a heavy chain CDR1 comprising a sequence set forth in SEQ ID NO: 43;
- (iii) a heavy chain FR2 comprising an amino acid sequence set forth in SEQ ID NO: 145;
- (iv) a heavy chain CDR2 comprising a sequence set forth in SEQ ID NO: 142;

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(v) a heavy chain FR3 comprising an amino acid sequence set forth in SEQ ID NO: 146;

(vi) a heavy chain CDR3 comprising a sequence set forth in SEQ ID NO: 143 or 235; and

(vii) a heavy chain FR4 comprising an amino acid sequence set forth in SEQ ID NO: 147.

For example, the antigen binding domain is a V_L comprising three CDRs of an amino acid sequence set forth in any one of SEQ ID NOs: 6, 14, 22, 30, 38, 46, 54, 62, 82, 95, 153, 163 to 172, 174, or 188 to 200. In one example, the CDRs are defined according to the Kabat numbering system. For example, the TL1a-binding protein comprises a V_L including CDRs as follows:

- (i) a CDR1 comprising a sequence set forth in SEQ ID NO: 7, a CDR2 comprising a sequence set forth in SEQ ID NO: 8 and a CDR3 comprising a sequence set forth in SEQ ID NO: 9 (or sequences labeled as CDRs 1, 2 and 3 in bold text of antibody C336 in FIG. 1B);
- (ii) a CDR1 comprising a sequence set forth in SEQ ID NO: 15, a CDR2 comprising a sequence set forth in SEQ ID NO: 16 and a CDR3 comprising a sequence set forth in SEQ ID NO: 17 (or sequences labeled as CDRs 1, 2 and 3 in bold text of antibody C334 in FIG. 1B);
- (iii) a CDR1 comprising a sequence set forth in SEQ ID NO: 23, a CDR2 comprising a sequence set forth in SEQ ID NO: 24 and a CDR3 comprising a sequence set forth in SEQ ID NO: 25 (or sequences labeled as CDRs 1, 2 and 3 in bold text of antibody C333 in FIG. 1B);
- (iv) a CDR1 comprising a sequence set forth in SEQ ID NO: 31, a CDR2 comprising a sequence set forth in SEQ ID NO: 32 and a CDR3 comprising a sequence set forth in SEQ ID NO: 33 (or sequences labeled as CDRs 1, 2 and 3 in bold text of antibody C323 in FIG. 1B);
- (v) a CDR1 comprising a sequence set forth in SEQ ID NO: 39, a CDR2 comprising a sequence set forth in SEQ ID NO: 40 and a CDR3 comprising a sequence set forth in SEQ ID NO: 41 (or sequences labeled as CDRs 1, 2 and 3 in bold text of antibody C321 in FIG. 1B);
- (vi) a CDR1 comprising a sequence set forth in SEQ ID NO: 47, a CDR2 comprising a sequence set forth in SEQ ID NO: 48 and a CDR3 comprising a sequence set forth in SEQ ID NO: 49 (or sequences labeled as CDRs 1, 2 and 3 in bold text of antibody C320 in FIG. 1B);
- (vii) a CDR1 comprising a sequence set forth in SEQ ID NO: 55, a CDR2 comprising a sequence set forth in SEQ ID NO: 56 and a CDR3 comprising a sequence set forth in SEQ ID NO: 57 (or sequences labeled as CDRs 1, 2 and 3 in bold text of antibody C319 in FIG. 1B);
- (viii) a CDR1 comprising a sequence set forth in SEQ ID NO: 83, a CDR2 comprising a sequence set forth in SEQ ID NO: 84 and a CDR3 comprising a sequence set forth in SEQ ID NO: 85 (or sequences labeled as CDRs 1, 2 and 3 in bold text of antibody C320-120 in FIG. 1F); or
- (ix) a CDR1 comprising amino acids 23 to 36 of any one of SEQ ID NOs: 188 to 200, a CDR2 comprising amino acids 52 to 58 of any one of SEQ ID NOs: 188 to 200 and a CDR3 comprising amino acids 91 to 100 of any one of SEQ ID NOs: 188 to 200.

In one example, the TL1a-binding protein comprises a V_L including CDRs as follows:

- (i) a CDR1 comprising a sequence $X_1X_2SSSDIG-AGLG VH$ (SEQ ID NO: 139), wherein X_1 is alanine or threonine; X_2 is glycine or serine (or sequence labeled as CDR 1 in bold text of the sequence labeled "Consensus" in FIG. 9C);

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- (ii) a CDR2 comprising a sequence set forth in SEQ ID NO: 140; and
- (iii) a CDR3 comprising a sequence set forth in SEQ ID NO: 141.

In one example, the TL1a-binding protein additionally comprises the following:

- (i) a light chain FR1 comprising an amino acid sequence set forth in SEQ ID NO: 148;
- (ii) a light chain FR2 comprising an amino acid sequence set forth in SEQ ID NO: 149;
- (iii) a light chain FR3 comprising an amino acid sequence set forth in SEQ ID NO: 150; and
- (iv) a light chain FR4 comprising an amino acid sequence set forth in SEQ ID NO: 151.

In one example, the TL1a-binding protein comprises the following:

- (i) a light chain FR1 comprising an amino acid sequence set forth in SEQ ID NO: 148;
- (ii) a light chain CDR1 comprising a sequence set forth in SEQ ID NO: 139;
- (iii) a light chain FR2 comprising an amino acid sequence set forth in SEQ ID NO: 149; (iv) a light chain CDR2 comprising a sequence set forth in SEQ ID NO: 140;
- (v) a light chain FR3 comprising an amino acid sequence set forth in SEQ ID NO: 150;
- (vi) a light chain CDR3 comprising a sequence set forth in SEQ ID NO: 141; and
- (vii) a light chain FR4 comprising an amino acid sequence set forth in SEQ ID NO: 151.

In one example, the CDRs are defined according to the enhanced Chothia numbering system. For example, the TL1a-binding protein comprises a V_L including CDRs labeled as CDRs 1, 2 and 3 in underlined text of antibody 336, 334, 333, 323, 321, 320 or 319 in FIG. 1B or of antibody C320-120 in FIG. 1F or of the sequence labeled "Consensus" in FIG. 1H and FIG. 9.

In one example, the antigen binding domain comprises six CDRs of one of the following pairs of variable regions:

- (i) a V_H comprising a sequence set forth in SEQ ID NO: 2 and a V_L comprising a sequence set forth in SEQ ID NO: 6;
- (ii) a V_H comprising a sequence set forth in SEQ ID NO: 10 and a V_L comprising a sequence set forth in SEQ ID NO: 14;
- (iii) a V_H comprising a sequence set forth in SEQ ID NO: 18 and a V_L comprising a sequence set forth in SEQ ID NO: 22;
- (iv) a V_H comprising a sequence set forth in SEQ ID NO: 26 and a V_L comprising a sequence set forth in SEQ ID NO: 30;
- (v) a V_H comprising a sequence set forth in SEQ ID NO: 34 and a V_L comprising a sequence set forth in SEQ ID NO: 38;
- (vi) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (vii) a V_H comprising a sequence set forth in SEQ ID NO: 50 and a V_L comprising a sequence set forth in SEQ ID NO: 54;
- (viii) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (ix) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 62;

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- (x) a V_H comprising a sequence set forth in SEQ ID NO: 66 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (xi) a V_H comprising a sequence set forth in SEQ ID NO: 70 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (xii) a V_H comprising a sequence set forth in SEQ ID NO: 74 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (xiii) a V_H comprising a sequence set forth in SEQ ID NO: 78 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (xiv) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 82;
- (xv) a V_H comprising a sequence set forth in SEQ ID NO: 86 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xvi) a V_H comprising a sequence set forth in SEQ ID NO: 90 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xvii) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (xviii) a V_H comprising a sequence set forth in SEQ ID NO: 154 and a V_L comprising a sequence set forth in SEQ ID NO: 164;
- (xix) a V_H comprising a sequence set forth in SEQ ID NO: 155 and a V_L comprising a sequence set forth in SEQ ID NO: 163;
- (xx) a V_H comprising a sequence set forth in SEQ ID NO: 156 and a V_L comprising a sequence set forth in SEQ ID NO: 165;
- (xxi) a V_H comprising a sequence set forth in SEQ ID NO: 157 and a V_L comprising a sequence set forth in SEQ ID NO: 166;
- (xxii) a V_H comprising a sequence set forth in SEQ ID NO: 158 and a V_L comprising a sequence set forth in SEQ ID NO: 167;
- (xxiii) a V_H comprising a sequence set forth in SEQ ID NO: 159 and a V_L comprising a sequence set forth in SEQ ID NO: 168;
- (xxiv) a V_H comprising a sequence set forth in SEQ ID NO: 160 and a V_L comprising a sequence set forth in SEQ ID NO: 169;
- (xxv) a V_H comprising a sequence set forth in SEQ ID NO: 161 and a V_L comprising a sequence set forth in SEQ ID NO: 170;
- (xxvi) a V_H comprising a sequence set forth in SEQ ID NO: 234 and a V_L comprising a sequence set forth in SEQ ID NO: 169;
- (xxvii) a V_H comprising a sequence set forth in SEQ ID NO: 154 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxviii) a V_H comprising a sequence set forth in SEQ ID NO: 155 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxix) a V_H comprising a sequence set forth in SEQ ID NO: 156 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxx) a V_H comprising a sequence set forth in SEQ ID NO: 157 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxxii) a V_H comprising a sequence set forth in SEQ ID NO: 158 and a V_L comprising a sequence set forth in SEQ ID NO: 46;

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- (xxxii) a V_H comprising a sequence set forth in SEQ ID NO: 159 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxxiii) a V_H comprising a sequence set forth in SEQ ID NO: 160 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxxiv) a V_H comprising a sequence set forth in SEQ ID NO: 161 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxxv) a V_H comprising a sequence set forth in SEQ ID NO: 234 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxxvi) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 164;
- (xxxvii) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 165;
- (xxxviii) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 166;
- (xxxix) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 167;
- (xl) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 168;
- (xli) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 169;
- (xlii) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 170;
- (xliii) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 171;
- (xliv) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 172;
- (xlv) a V_H comprising a sequence set forth in SEQ ID NO: 175 and a V_L comprising a sequence set forth in SEQ ID NO: 188;
- (xlvi) a V_H comprising a sequence set forth in SEQ ID NO: 176 and a V_L comprising a sequence set forth in SEQ ID NO: 189;
- (xlvii) a V_H comprising a sequence set forth in SEQ ID NO: 177 and a V_L comprising a sequence set forth in SEQ ID NO: 190;
- (xlviii) a V_H comprising a sequence set forth in SEQ ID NO: 178 and a V_L comprising a sequence set forth in SEQ ID NO: 191;
- (xlix) a V_H comprising a sequence set forth in SEQ ID NO: 179 and a V_L comprising a sequence set forth in SEQ ID NO: 192;
- (l) a V_H comprising a sequence set forth in SEQ ID NO: 180 and a V_L comprising a sequence set forth in SEQ ID NO: 193;
- (li) a V_H comprising a sequence set forth in SEQ ID NO: 181 and a V_L comprising a sequence set forth in SEQ ID NO: 194;
- (lii) a V_H comprising a sequence set forth in SEQ ID NO: 182 and a V_L comprising a sequence set forth in SEQ ID NO: 195;
- (liii) a V_H comprising a sequence set forth in SEQ ID NO: 183 and a V_L comprising a sequence set forth in SEQ ID NO: 196;

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- (liv) a V_H comprising a sequence set forth in SEQ ID NO: 184 and a V_L comprising a sequence set forth in SEQ ID NO: 197;
- (lv) a V_H comprising a sequence set forth in SEQ ID NO: 185 and a V_L comprising a sequence set forth in SEQ ID NO: 198;
- (lvi) a V_H comprising a sequence set forth in SEQ ID NO: 186 and a V_L comprising a sequence set forth in SEQ ID NO: 199; or
- (lvii) a V_H comprising a sequence set forth in SEQ ID NO: 187 and a V_L comprising a sequence set forth in SEQ ID NO: 200.

In one example, the antigen binding domain comprises six CDRs of an antibody comprising a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 46, wherein the V_H and/or V_L comprise one or more of the following substitutions or groups of substitutions:

- (i) the V_H comprises an alanine at position 16 of SEQ ID NO: 42;
- (ii) the V_H comprises an alanine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (iii) the V_H comprises a serine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (iv) the V_H comprises a histidine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (v) the V_H comprises a leucine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (vi) the V_H comprises an aspartic acid at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (vii) the V_H comprises a tyrosine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (viii) the V_H comprises a proline at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (ix) the V_H comprises a glutamine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (x) the V_H comprises a lysine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xi) the V_H comprises an alanine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xii) the V_H comprises a serine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xiii) the V_H comprises a histidine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xiv) the V_H comprises a leucine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xv) the V_H comprises an aspartic acid at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xvi) the V_H comprises a tyrosine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;

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- (lxi) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises an alanine at position 23 of SEQ ID NO: 46;
- (lxii) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises an aspartic acid at position 28 of SEQ ID NO: 46;
- (lxiii) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises a tyrosine at position 33 of SEQ ID NO: 46;
- (lxiv) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises an aspartic acid at position 34 of SEQ ID NO: 46;
- (lxv) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises an asparagine at position 53 of SEQ ID NO: 46;
- (lxvi) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises a serine at position 54 of SEQ ID NO: 46;
- (lxvii) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises an alanine at position 82 of SEQ ID NO: 46;
- (lxviii) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises a serine at position 95 of SEQ ID NO: 46;
- (lxix) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises a serine at position 96 of SEQ ID NO: 46;
- (lxx) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (lxxi) the V_H comprises a serine at position 47 of SEQ ID NO: 42 and the V_L comprises a threonine at position 23 of SEQ ID NO: 46;
- (lxxii) the V_H comprises a proline at position 41, an alanine at position 72, an aspartic acid at position 73 and an arginine at position 74 and a threonine at position 76 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (lxxiii) the V_H comprises a proline at position 41, a leucine at position 51 and a glutamic acid at position 102 each relative to SEQ ID NO: 42 and the V_L comprises a serine at position 24 and a threonine at position 76 each relative to SEQ ID NO: 46;
- (lxxiv) the V_H comprises a proline at position 41, a leucine at position 51 and a glutamic acid at position 102 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24 and a threonine at position 76 each relative to SEQ ID NO: 46;
- (lxxv) the V_H comprises a proline at position 41, a leucine at position 51 and a glutamic acid at position 102 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23 and a threonine at position 76 each relative to SEQ ID NO: 46;
- (lxxvi) the V_H comprises a proline at position 41, a leucine at position 51 and a glutamic acid at position 102 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (lxxvii) the V_H comprises a proline at position 41, a leucine at position 51, a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24 and a threonine at position 76 each relative to SEQ ID NO: 46;
- (lxxviii) the V_H comprises a proline at position 41, a leucine at position 51, a glutamic acid at position 102

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- and an alanine at position 105 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
 - (lxxix) the V_H comprises a proline at position 41, an alanine at position 72, an aspartic acid at position 73 an arginine at position 74 and a threonine at position 76 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24 and a threonine at position 76 each relative to SEQ ID NO: 46;
 - (lxxx) the V_H comprises a proline at position 41, an alanine at position 72, an aspartic acid at position 73 an arginine at position 74 and a threonine at position 76 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24, a threonine at position 76 and a glutamic acid at position 51 each relative to SEQ ID NO: 46;
 - (lxxxi) the V_H comprises a proline at position 41, a leucine at position 51, an alanine at position 72, an aspartic acid at position 73, an arginine at position 74, a threonine at position 76, a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24, a threonine at position 76 and a glutamic acid at position 51 each relative to SEQ ID NO: 46; or
 - (lxxxii) the V_H comprises a proline at position 41, a leucine at position 51, an alanine at position 72, an aspartic acid at position 73, an arginine at position 74, a threonine at position 76, a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24, a threonine at position 76 and a glycine at position 51 each relative to SEQ ID NO: 46.
- In one example, the antigen binding domain comprises six CDRs of one of the following pairs of variable regions:
- (i) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
 - (ii) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 106 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 107 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions;
 - (iii) a V_H comprising a sequence set forth in SEQ ID NO: 175 and a V_L comprising a sequence set forth in SEQ ID NO: 188;
 - (iv) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 222 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 228 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions;
 - (v) a V_H comprising a sequence set forth in SEQ ID NO: 176 and a V_L comprising a sequence set forth in SEQ ID NO: 189;
 - (vi) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 223 or a nucleic acid that

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- hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 228 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions; 5
- (vii) a V_H comprising a sequence set forth in SEQ ID NO: 177 and a V_L comprising a sequence set forth in SEQ ID NO: 190;
- (viii) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 224 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 229 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions; 10 15
- (ix) a V_H comprising a sequence set forth in SEQ ID NO: 178 and a V_L comprising a sequence set forth in SEQ ID NO: 191; 20
- (x) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 224 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 230 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions; 25 30
- (xi) a V_H comprising a sequence set forth in SEQ ID NO: 179 and a V_L comprising a sequence set forth in SEQ ID NO: 192;
- (xii) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 224 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 231 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions; 35 40
- (xiii) a V_H comprising a sequence set forth in SEQ ID NO: 180 and a V_L comprising a sequence set forth in SEQ ID NO: 193; 45
- (xiv) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 228 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions; 50
- (xv) a V_H comprising a sequence set forth in SEQ ID NO: 181 and a V_L comprising a sequence set forth in SEQ ID NO: 194; 55
- (xvi) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 225 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 230 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions; 60
- (xvii) a V_H comprising a sequence set forth in SEQ ID NO: 183 and a V_L comprising a sequence set forth in SEQ ID NO: 196; 65

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- (xviii) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 226 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 230 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions;
- (xix) a V_H comprising a sequence set forth in SEQ ID NO: 185 and a V_L comprising a sequence set forth in SEQ ID NO: 198;
- (xx) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 226 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 232 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions;
- (xxi) a V_H comprising a sequence set forth in SEQ ID NO: 186 and a V_L comprising a sequence set forth in SEQ ID NO: 199; and
- (xxii) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 227 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 232 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions;
- (xxiii) a V_H comprising a sequence set forth in SEQ ID NO: 187 and a V_L comprising a sequence set forth in SEQ ID NO: 200; or
- (xxiv) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 227 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 233 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions.
- In one example, the TL1a-binding protein comprises the following six CDRs:
- (i) a heavy chain CDR1 comprising a sequence set forth in SEQ ID NO: 43 (or sequence labeled as CDR 1 in bold text of the sequence labeled "Consensus" in FIG. 1E);
- (ii) a heavy chain CDR2 comprising a sequence $WX_1NPNSGNTGYAQKFQG$ (SEQ ID NO: 142), wherein X_1 is methionine or leucine (or sequence labeled as CDR 2 in bold text of the sequence labeled "Consensus" in FIG. 1E);
- (iii) a heavy chain CDR3 comprising a sequence $EVPX_1TAX_2FEY$ (SEQ ID NO: 143), wherein X_1 is aspartic acid or glutamic acid and X_2 is serine or alanine. (or the sequence labeled as CDR 3 in bold text of the sequence labeled "Consensus" in FIG. 9B) or a sequence $EX_1PX_2X_3AX_4FX_5Y$ (SEQ ID NO: 235), wherein:
- X_1 is an amino acid selected from the group consisting of valine, alanine, serine, histidine, aspartic acid, leucine, tyrosine, proline, glutamine or lysine;

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X₂ is an amino acid selected from the group consisting of alanine, serine, histidine, lysine, glutamic acid or aspartic acid;

X₃ is an amino acid selected from the group consisting of alanine, serine, aspartic acid, tyrosine or threonine;

X₄ is an amino acid selected from the group consisting of serine, alanine, histidine, leucine, aspartic acid or tyrosine; and

X₅ is an amino acid selected from the group consisting of alanine, serine, histidine, leucine, aspartic acid, proline, glutamine, glutamic acid or lysine;

(iv) a CDR1 comprising a sequence X₁X₂SSSDI-GAGLG^{VH} (SEQ ID NO: 139), wherein X₁ is alanine or threonine; X₂ is glycine or serine (or sequence labeled as CDR 1 in bold text of the sequence labeled "Consensus" in FIG. 9C);

(v) a CDR2 comprising a sequence set forth in SEQ ID NO: 48; and

(vi) a CDR3 comprising a sequence set forth in SEQ ID NO: 49

In one example, the TL1a-binding protein comprises the following:

(a) a V_H comprising:

(i) a heavy chain FR1 comprising an amino acid sequence set forth in SEQ ID NO: 144;

(ii) a heavy chain CDR1 comprising a sequence set forth in SEQ ID NO: 43;

(iii) a heavy chain FR2 comprising an amino acid sequence set forth in SEQ ID NO: 145;

(iv) a heavy chain CDR2 comprising a sequence set forth in SEQ ID NO: 142;

(v) a heavy chain FR3 comprising an amino acid sequence set forth in SEQ ID NO: 146;

(vi) a heavy chain CDR3 comprising a sequence set forth in SEQ ID NO: 143 or 235; and

(vii) a heavy chain FR4 comprising an amino acid sequence set forth in SEQ ID NO: 147; and

(b) a V_L comprising:

(i) a light chain FR1 comprising an amino acid sequence set forth in SEQ ID NO: 148;

(ii) a light chain CDR1 comprising a sequence set forth in SEQ ID NO: 139;

(iii) a light chain FR2 comprising an amino acid sequence set forth in SEQ ID NO: 149;

(iv) a light chain CDR2 comprising a sequence set forth in SEQ ID NO: 48;

(v) a light chain FR3 comprising an amino acid sequence set forth in SEQ ID NO: 150;

(vi) a light chain CDR3 comprising a sequence set forth in SEQ ID NO: 49; and

(vii) a light chain FR4 comprising an amino acid sequence set forth in SEQ ID NO: 151.

Additional residues suitable for inclusion in heavy chain CDR3 are described herein and are to be taken to apply mutatis mutandis to the present example of the disclosure.

In one example, the CDRs are defined according to the Kabat numbering system. Exemplary CDRs defined according to the Kabat numbering system are described above and/or in FIG. 1A to 1H, 9B or 9C labeled as CDRs 1 to 3 in bold text and are taken to apply mutatis mutandis to the present example of the disclosure.

In one example, the CDRs are defined according to the enhanced Chothia numbering system. Exemplary CDRs defined according to the enhanced Chothia numbering system are described above and/or in FIGS. 1A to 1H labeled as CDRs 1 to 3 in underlined text and are taken to apply mutatis mutandis to the present example of the disclosure.

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In one example, the TL1a-binding protein comprises a variable region of the antibody.

In one example, the TL1a-binding protein comprises a V_H comprising a sequence set forth in any one of SEQ ID NOs: 2, 10, 18, 26, 34, 42, 50, 58, 66, 70, 74, 78, 86, 90, 94, 137, 152, 154 to 162, 173, 175 to 187 or 234 or a sequence having at least about 80% identity to any one of the foregoing. In one example, the V_H comprises a sequence set forth in any one of SEQ ID NOs: 26, 34, 42 or 94 or a sequence having at least about 80% identity to any one of the foregoing. In one example, the V_H comprises a sequence set forth in any one of SEQ ID NOs: 42, 175 to 181, 183 or 185 to 187 or a sequence having at least about 80% identity to any one of the foregoing.

In one example, the TL1a-binding protein comprises a V_H comprising a sequence set forth in SEQ ID NO: 94, 137, 152, 162 or 173. In one example, the V_H comprises a sequence set forth in any one of SEQ ID NOs: 42, 58, 66, 70, 74, 78, 86 or 90. In one example, the V_H comprises a sequence set forth in SEQ ID NO: 42.

In one example, the TL1a-binding protein comprises a V_H comprising a sequence set forth in SEQ ID NO: 42 and comprising one or more of the following substitutions or groups of substitutions:

- (i) an alanine at position 16 of SEQ ID NO: 42;
- (ii) an alanine at position 100 of SEQ ID NO: 42;
- (iii) a serine at position 100 of SEQ ID NO: 42;
- (iv) a histidine at position 100 of SEQ ID NO: 42;
- (v) a leucine at position 100 of SEQ ID NO: 42;
- (vi) an aspartic acid at position 100 of SEQ ID NO: 42;
- (vii) a tyrosine at position 100 of SEQ ID NO: 42;
- (viii) a proline at position 100 of SEQ ID NO: 42;
- (ix) a glutamine at position 100 of SEQ ID NO: 42;
- (x) a lysine at position 100 of SEQ ID NO: 42;
- (xi) an alanine at position 101 of SEQ ID NO: 42;
- (xii) a serine at position 101 of SEQ ID NO: 42;
- (xiii) a histidine at position 101 of SEQ ID NO: 42;
- (xiv) a leucine at position 101 of SEQ ID NO: 42;
- (xv) an aspartic acid at position 101 of SEQ ID NO: 42;
- (xvi) a tyrosine at position 101 of SEQ ID NO: 42;
- (xvii) a glutamine at position 101 of SEQ ID NO: 42;
- (xviii) a lysine at position 101 of SEQ ID NO: 42;
- (xix) an alanine at position 102 of SEQ ID NO: 42;
- (xx) a serine at position 102 of SEQ ID NO: 42;
- (xxi) a histidine at position 102 of SEQ ID NO: 42;
- (xxii) a leucine at position 102 of SEQ ID NO: 42;
- (xxiii) a tyrosine at position 102 of SEQ ID NO: 42;
- (xxiv) a proline at position 102 of SEQ ID NO: 42;
- (xxv) a glutamine at position 102 of SEQ ID NO: 42;
- (xxvi) a lysine at position 102 of SEQ ID NO: 42;
- (xxvii) an alanine at position 103 of SEQ ID NO: 42;
- (xxviii) a serine at position 103 of SEQ ID NO: 42;
- (xxix) a histidine at position 103 of SEQ ID NO: 42;
- (xxx) a leucine at position 103 of SEQ ID NO: 42;
- (xxxi) an aspartic acid at position 103 of SEQ ID NO: 42;
- (xxxii) a tyrosine at position 103 of SEQ ID NO: 42;
- (xxxiii) a proline at position 103 of SEQ ID NO: 42;
- (xxxiv) a glutamine at position 103 of SEQ ID NO: 42;
- (xxxv) a lysine at position 103 of SEQ ID NO: 42;
- (xxxvi) a serine at position 104 of SEQ ID NO: 42;
- (xxxvii) a histidine at position 104 of SEQ ID NO: 42;
- (xxxviii) a leucine at position 104 of SEQ ID NO: 42;
- (xxxix) an aspartic acid at position 104 of SEQ ID NO: 42;
- (xl) a tyrosine at position 104 of SEQ ID NO: 42;
- (xli) a proline at position 104 of SEQ ID NO: 42;
- (xlii) a glutamine at position 104 of SEQ ID NO: 42;

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(xliii) a lysine at position 104 of SEQ ID NO: 42;
 (xliv) an alanine at position 105 of SEQ ID NO: 42;
 (xlv) a histidine at position 105 of SEQ ID NO: 42;
 (xlvi) a leucine at position 105 of SEQ ID NO: 42;
 (xlvii) an aspartic acid at position 105 of SEQ ID NO: 42;
 (xlviii) a tyrosine at position 105 of SEQ ID NO: 42;
 (xlix) a proline at position 105 of SEQ ID NO: 42;
 (l) a glutamine at position 105 of SEQ ID NO: 42;
 (li) a lysine at position 105 of SEQ ID NO: 42;
 (lii) an alanine at position 107 of SEQ ID NO: 42;
 (liii) a serine at position 107 of SEQ ID NO: 42;
 (liv) a histidine at position 107 of SEQ ID NO: 42;
 (lv) a leucine at position 107 of SEQ ID NO: 42;
 (lvi) an aspartic acid at position 107 of SEQ ID NO: 42;
 (lvii) a tyrosine at position 107 of SEQ ID NO: 42;
 (lviii) a proline at position 107 of SEQ ID NO: 42;
 (lix) a glutamine at position 107 of SEQ ID NO: 42;
 (lx) a lysine at position 107 of SEQ ID NO: 42;
 (lxi) a threonine at position 41 of SEQ ID NO: 42;
 (lxii) a serine at position 47 of SEQ ID NO: 42;
 (lxiii) a proline at position 41, an alanine at position 72,
 a aspartic acid at position 73 an arginine at position 74
 and a threonine at position 76 each relative to SEQ ID
 NO: 42;
 (lxiv) a proline at position 41, a leucine at position 51 and
 a glutamic acid at position 102 each relative to SEQ ID
 NO: 42;
 (lxv) a proline at position 41, a leucine at position 51, a
 glutamic acid at position 102 and an alanine at position
 105 each relative to SEQ ID NO: 42;
 (lxvi) a proline at position 41, a leucine at position 51, an
 alanine at position 72, a aspartic acid at position 73, an
 arginine at position 74, a threonine at position 76 a
 glutamic acid at position 102 and an alanine at position
 105; or
 (lxvii) a sequence having at least about 80% identity to
 any one of the foregoing.

In one example, the TL1a-binding protein comprises a V_H
 encoded by a nucleic acid comprising a sequence set forth in
 any one of SEQ ID NOs: 96, 98, 100, 102, 104, 106, 108,
 110, 112, 113, 114, 115, 117, 118 or 222 to 227 or a sequence
 at least about 80% identical thereto or a nucleic acid that
 hybridizes thereto under moderate to high stringency con-
 ditions.

In one example, the TL1a-binding protein comprises a V_H
 encoded by a nucleic acid comprising a sequence set forth in
 SEQ ID NO: 106 or 222 to 227 or a sequence at least about
 80% identical thereto or a nucleic acid that hybridizes
 thereto under moderate to high stringency conditions.

In one example, the TL1a-binding protein comprises a V_L
 comprising a sequence set forth in any one of SEQ ID NOs:
 6, 14, 22, 30, 38, 46, 54, 62, 82, 95, 138, 153, 163 or 174,
 or a sequence having at least about 80% identity to any one
 of the foregoing. In one example, the V_L comprises a
 sequence set forth in any one of SEQ ID NOs: 30, 38, 46,
 188 to 194, 196 or 198 to 200 or a sequence having at least
 about 80% identity to any one of the foregoing.

In one example, the TL1a-binding protein comprises a V_L
 comprising a sequence set forth in SEQ ID NO: 95, 138,
 153, 163 or 174. In one example, the TL1a-binding protein
 comprises a V_L comprising a sequence set forth in any one
 of SEQ ID NOs: 46, 62 or 82. In one example, the TL1a-
 binding protein comprises a V_L comprising a sequence set
 forth in SEQ ID NO: 46.

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In one example, the TL1a-binding protein comprises a V_L
 comprising a sequence set forth in SEQ ID NO: 46 and
 comprising one or more of the following substitutions or
 groups of substitutions:

- (i) a threonine at position 76 of SEQ ID NO: 46;
- (ii) an threonine at position 23 of SEQ ID NO: 46;
- (iii) an asparagine at position 28 of SEQ ID NO: 46;
- (iv) a tyrosine at position 33 of SEQ ID NO: 46;
- (v) an aspartic acid at position 34 of SEQ ID NO: 46;
- (vi) an asparagine at position 53 of SEQ ID NO: 46;
- (vii) a serine at position 54 of SEQ ID NO: 46;
- (viii) an alanine at position 82 of SEQ ID NO: 46;
- (ix) a serine at position 95 of SEQ ID NO: 46;
- (x) a serine at position 96 of SEQ ID NO: 46;
- (xi) a threonine at position 23, a serine at position 24 and
 a threonine at position 76 each relative to SEQ ID NO:
 46;
- (xii) a threonine at position 23 and a threonine at position
 76 each relative to SEQ ID NO: 46;
- (xiii) a threonine at position 23, a serine at position 24, a
 threonine at position 76 and a glutamic acid at position
 51 each relative to SEQ ID NO: 46;
- (xiv) a threonine at position 23, a serine at position 24, a
 threonine at position 76 and a glycine at position 51
 each relative to SEQ ID NO: 46; or
- (xv) a sequence having at least about 80% identity to any
 one of the foregoing.

In one example, the TL1a-binding protein comprises a V_L
 encoded by a nucleic acid comprising a sequence set forth in
 any one of SEQ ID NOs: 97, 99, 101, 103, 105, 107, 109,
 111, 116 or 228 to 233 or a sequence at least about 80%
 identical thereto or a nucleic acid that hybridizes thereto
 under moderate to high stringency conditions.

In one example, the protein comprises a V_L encoded by a
 nucleic acid comprising a sequence set forth in SEQ ID NO:
 107 or 228 to 233 or a sequence at least about 80% identical
 thereto or a nucleic acid that hybridizes thereto under
 moderate to high stringency conditions.

In one example, the TL1a-binding protein is a domain
 antibody, optionally linked to a heavy chain constant region
 or a Fc or a heavy chain constant domain (C_H) 2 and/or C_H 3
 or a protein that binds to an immune effector cell.

In one example, a TL1a-binding protein of the disclosure
 comprises at least a V_H and a V_L , wherein the V_H and V_L
 bind to form a Fv comprising the antigen binding domain.
 For example, the TL1a-binding protein comprises any one of
 the following pairs of V_H and V_L :

- (i) a V_H comprising a sequence set forth in SEQ ID NO:
 2 and a V_L comprising a sequence set forth in SEQ ID
 NO: 6;
- (ii) a V_H comprising a sequence set forth in SEQ ID NO:
 10 and a V_L comprising a sequence set forth in SEQ ID
 NO: 14;
- (iii) a V_H comprising a sequence set forth in SEQ ID NO:
 18 and a V_L comprising a sequence set forth in SEQ ID
 NO: 22;
- (iv) a V_H comprising a sequence set forth in SEQ ID NO:
 26 and a V_L comprising a sequence set forth in SEQ ID
 NO: 30;
- (v) a V_H comprising a sequence set forth in SEQ ID NO:
 34 and a V_L comprising a sequence set forth in SEQ ID
 NO:38;
- (vi) a V_H comprising a sequence set forth in SEQ ID NO:
 42 and a V_L comprising a sequence set forth in SEQ ID
 NO: 46;

- (li) a V_H comprising a sequence set forth in SEQ ID NO: 181 and a V_L comprising a sequence set forth in SEQ ID NO: 194;
- (lli) a V_H comprising a sequence set forth in SEQ ID NO: 182 and a V_L comprising a sequence set forth in SEQ ID NO: 195;
- (liii) a V_H comprising a sequence set forth in SEQ ID NO: 183 and a V_L comprising a sequence set forth in SEQ ID NO: 196;
- (liv) a V_H comprising a sequence set forth in SEQ ID NO: 184 and a V_L comprising a sequence set forth in SEQ ID NO: 197;
- (lv) a V_H comprising a sequence set forth in SEQ ID NO: 185 and a V_L comprising a sequence set forth in SEQ ID NO: 198;
- (lvi) a V_H comprising a sequence set forth in SEQ ID NO: 186 and a V_L comprising a sequence set forth in SEQ ID NO: 199; or
- (lvii) a V_H comprising a sequence set forth in SEQ ID NO: 187 and a V_L comprising a sequence set forth in SEQ ID NO: 200.

In one example, the TL1a-binding protein comprises any one of the following pairs of V_H and V_L :

- (i) a V_H comprising a sequence set forth in SEQ ID NO: 26 and a V_L comprising a sequence set forth in SEQ ID NO: 30;
- (ii) a V_H comprising a sequence set forth in SEQ ID NO: 34 and a V_L comprising a sequence set forth in SEQ ID NO: 38;
- (iii) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (iv) a V_H comprising a sequence set forth in SEQ ID NO: 175 and a V_L comprising a sequence set forth in SEQ ID NO: 188;
- (v) a V_H comprising a sequence set forth in SEQ ID NO: 176 and a V_L comprising a sequence set forth in SEQ ID NO: 189;
- (vi) a V_H comprising a sequence set forth in SEQ ID NO: 177 and a V_L comprising a sequence set forth in SEQ ID NO: 190;
- (vii) a V_H comprising a sequence set forth in SEQ ID NO: 178 and a V_L comprising a sequence set forth in SEQ ID NO: 191;
- (viii) a V_H comprising a sequence set forth in SEQ ID NO: 179 and a V_L comprising a sequence set forth in SEQ ID NO: 192;
- (ix) a V_H comprising a sequence set forth in SEQ ID NO: 180 and a V_L comprising a sequence set forth in SEQ ID NO: 193;
- (x) a V_H comprising a sequence set forth in SEQ ID NO: 181 and a V_L comprising a sequence set forth in SEQ ID NO: 194;
- (xi) a V_H comprising a sequence set forth in SEQ ID NO: 183 and a V_L comprising a sequence set forth in SEQ ID NO: 196;
- (xii) a V_H comprising a sequence set forth in SEQ ID NO: 185 and a V_L comprising a sequence set forth in SEQ ID NO: 198;
- (xiii) a V_H comprising a sequence set forth in SEQ ID NO: 186 and a V_L comprising a sequence set forth in SEQ ID NO: 199; or
- (xiv) a V_H comprising a sequence set forth in SEQ ID NO: 187 and a V_L comprising a sequence set forth in SEQ ID NO: 200;

In one example, the TL1a-binding protein comprises a V_H comprising a sequence set forth in SEQ ID NO: 94 and a V_L comprising a sequence set forth in SEQ ID NO: 95.

In one example, the TL1a-binding protein comprises a V_H comprising a sequence set forth in SEQ ID NO: 137 and a V_L comprising a sequence set forth in SEQ ID NO: 138.

In one example, the TL1a-binding protein comprises a V_H comprising a sequence set forth in SEQ ID NO: 152 and a V_L comprising a sequence set forth in SEQ ID NO: 153.

In one example, the TL1a-binding protein comprises a V_H comprising a sequence set forth in SEQ ID NO: 173 and a V_L comprising a sequence set forth in SEQ ID NO: 174.

In one example, the TL1a-binding protein comprises any one of the following pairs of V_H and V_L :

- (i) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (ii) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (iii) a V_H comprising a sequence set forth in SEQ ID NO: 66 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (iv) a V_H comprising a sequence set forth in SEQ ID NO: 70 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (v) a V_H comprising a sequence set forth in SEQ ID NO: 74 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (vi) a V_H comprising a sequence set forth in SEQ ID NO: 78 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (vii) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 82;
- (viii) a V_H comprising a sequence set forth in SEQ ID NO: 86 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (ix) a V_H comprising a sequence set forth in SEQ ID NO: 90 and a V_L comprising a sequence set forth in SEQ ID NO: 46; or
- (x) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 62.

For example, the TL1a-binding protein comprises a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 46, wherein the V_H and/or V_L comprise one or more of the following substitutions or groups of substitutions:

- (i) the V_H comprises an alanine at position 16 of SEQ ID NO: 42;
- (ii) the V_H comprises an alanine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (iii) the V_H comprises a serine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (iv) the V_H comprises a histidine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (v) the V_H comprises a leucine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (vi) the V_H comprises an aspartic acid at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;

In one example, the V_H and the V_L are in a single polypeptide chain. For example, the TL1a-binding protein is:

- (i) a single chain Fv fragment (scFv);
- (ii) a dimeric scFv (di-scFv);
- (iii) one of (i) or (ii) linked to a heavy chain constant region or a Fc or a heavy chain constant domain (C_H) 2 and/or C_H3 ; or
- (iv) one of (i) or (ii) linked to a protein that binds to an immune effector cell.

In another example, the V_L and V_H are in separate polypeptide chains. For example, the TL1a-binding protein is:

- (i) a diabody;
- (ii) a triabody;
- (iii) a tetra body;
- (iv) a Fab;
- (v) a $F(ab')_2$;
- (vi) a Fv;
- (vii) one of (i) to (vi) linked to a heavy chain constant region or a Fc or a heavy chain constant domain (C_H) 2 and/or C_H3 ; or
- (viii) one of (i) to (vi) linked to a protein that binds to an immune effector cell.

In an exemplary form of the disclosure, the TL1a-binding protein is an antibody.

Exemplary TL1a-binding proteins of the disclosure are chimeric, de-immunized, humanized, human, synhumanized or primatized.

In one example, the disclosure provides an antibody comprising an antigen binding domain, wherein the antigen binding domain specifically binds to TL1a and, wherein the antibody inhibits interaction of TL1a and DR3 and does not inhibit interaction of TL1a and DcR3, the antigen binding domain comprising any one of:

- (i) a V_H comprising a sequence set forth in SEQ ID NO: 2 and a V_L comprising a sequence set forth in SEQ ID NO: 6;
- (ii) a V_H comprising a sequence set forth in SEQ ID NO: 10 and a V_L comprising a sequence set forth in SEQ ID NO: 14;
- (iii) a V_H comprising a sequence set forth in SEQ ID NO: 18 and a V_L comprising a sequence set forth in SEQ ID NO: 22;
- (iv) a V_H comprising a sequence set forth in SEQ ID NO: 26 and a V_L comprising a sequence set forth in SEQ ID NO: 30;
- (v) a V_H comprising a sequence set forth in SEQ ID NO: 34 and a V_L comprising a sequence set forth in SEQ ID NO: 38;
- (vi) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (vii) a V_H comprising a sequence set forth in SEQ ID NO: 50 and a V_L comprising a sequence set forth in SEQ ID NO: 54;
- (viii) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (ix) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (x) a V_H comprising a sequence set forth in SEQ ID NO: 66 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xi) a V_H comprising a sequence set forth in SEQ ID NO: 70 and a V_L comprising a sequence set forth in SEQ ID NO: 62;

- (xii) a V_H comprising a sequence set forth in SEQ ID NO: 74 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (xiii) a V_H comprising a sequence set forth in SEQ ID NO: 78 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (xiv) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 82;
- (xv) a V_H comprising a sequence set forth in SEQ ID NO: 86 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xvi) a V_H comprising a sequence set forth in SEQ ID NO: 90 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xvii) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (xviii) a V_H comprising a sequence set forth in SEQ ID NO: 154 and a V_L comprising a sequence set forth in SEQ ID NO: 164;
- (xix) a V_H comprising a sequence set forth in SEQ ID NO: 155 and a V_L comprising a sequence set forth in SEQ ID NO: 163;
- (xx) a V_H comprising a sequence set forth in SEQ ID NO: 156 and a V_L comprising a sequence set forth in SEQ ID NO: 165;
- (xxi) a V_H comprising a sequence set forth in SEQ ID NO: 157 and a V_L comprising a sequence set forth in SEQ ID NO: 166;
- (xxii) a V_H comprising a sequence set forth in SEQ ID NO: 158 and a V_L comprising a sequence set forth in SEQ ID NO: 167;
- (xxiii) a V_H comprising a sequence set forth in SEQ ID NO: 159 and a V_L comprising a sequence set forth in SEQ ID NO: 168;
- (xxiv) a V_H comprising a sequence set forth in SEQ ID NO: 160 and a V_L comprising a sequence set forth in SEQ ID NO: 169;
- (xxv) a V_H comprising a sequence set forth in SEQ ID NO: 161 and a V_L comprising a sequence set forth in SEQ ID NO: 170;
- (xxvi) a V_H comprising a sequence set forth in SEQ ID NO: 234 and a V_L comprising a sequence set forth in SEQ ID NO: 169;
- (xxvii) a V_H comprising a sequence set forth in SEQ ID NO: 154 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxviii) a V_H comprising a sequence set forth in SEQ ID NO: 155 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxix) a V_H comprising a sequence set forth in SEQ ID NO: 156 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxx) a V_H comprising a sequence set forth in SEQ ID NO: 157 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxxii) a V_H comprising a sequence set forth in SEQ ID NO: 158 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxxiii) a V_H comprising a sequence set forth in SEQ ID NO: 159 and a V_L comprising a sequence set forth in SEQ ID NO: 46;

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- (ix) a V_H comprising a sequence set forth in SEQ ID NO: 178 and a V_L comprising a sequence set forth in SEQ ID NO: 191;
- (x) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 224 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 230 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions;
- (xi) a V_H comprising a sequence set forth in SEQ ID NO: 179 and a V_L comprising a sequence set forth in SEQ ID NO: 192;
- (xii) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 224 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 231 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions;
- (xiii) a V_H comprising a sequence set forth in SEQ ID NO: 180 and a V_L comprising a sequence set forth in SEQ ID NO: 193;
- (xiv) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 228 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions;
- (xv) a V_H comprising a sequence set forth in SEQ ID NO: 181 and a V_L comprising a sequence set forth in SEQ ID NO: 194;
- (xvi) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 225 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 230 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions;
- (xvii) a V_H comprising a sequence set forth in SEQ ID NO: 183 and a V_L comprising a sequence set forth in SEQ ID NO: 196;
- (xviii) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 226 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 230 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions;
- (xix) a V_H comprising a sequence set forth in SEQ ID NO: 185 and a V_L comprising a sequence set forth in SEQ ID NO: 198;
- (xx) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 226 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the

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- sequence set forth in SEQ ID NO: 232 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions;
- (xxi) a V_H comprising a sequence set forth in SEQ ID NO: 186 and a V_L comprising a sequence set forth in SEQ ID NO: 199;
- (xxii) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 227 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 232 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions;
- (xxiii) a V_H comprising a sequence set forth in SEQ ID NO: 187 and a V_L comprising a sequence set forth in SEQ ID NO: 200; or
- (xxiv) a V_H encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 227 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions and a V_L encoded by a nucleic acid comprising a sequence at least about 95% identical to the sequence set forth in SEQ ID NO: 233 or a nucleic acid that hybridizes thereto under moderate to high stringency conditions.
- In one example, the antibody comprises a V_H comprising sequence set forth in SEQ ID NO: 94 and a V_L comprising sequence set forth in SEQ ID NO: 95.
- In one example, the antibody comprises any one of the following pairs of V_H and V_L :
- (i) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (ii) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (iii) a V_H comprising a sequence set forth in SEQ ID NO: 66 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (iv) a V_H comprising a sequence set forth in SEQ ID NO: 70 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (v) a V_H comprising a sequence set forth in SEQ ID NO: 74 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (vi) a V_H comprising a sequence set forth in SEQ ID NO: 78 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (vii) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 82;
- (viii) a V_H comprising a sequence set forth in SEQ ID NO: 86 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (ix) a V_H comprising a sequence set forth in SEQ ID NO: 90 and a V_L comprising a sequence set forth in SEQ ID NO: 46; or
- (x) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 62.

The disclosure also provides a TL1a-binding antibody comprising a V_H of SEQ ID NO:42 and a V_L of SEQ ID NO:46.

The disclosure also provides a TL1a-binding antibody comprising a V_H of SEQ ID NO: 175 and a V_L of SEQ ID NO: 188.

The disclosure also provides a TL1a-binding antibody comprising a V_H of SEQ ID NO: 176 and a V_L of SEQ ID NO: 189.

The disclosure also provides a TL1a-binding antibody comprising a V_H of SEQ ID NO: 177 and a V_L of SEQ ID NO: 190.

The disclosure also provides a TL1a-binding antibody comprising a V_H of SEQ ID NO: 178 and a V_L of SEQ ID NO: 191.

The disclosure also provides a TL1a-binding antibody comprising a V_H of SEQ ID NO: 179 and a V_L of SEQ ID NO: 192.

The disclosure also provides a TL1a-binding antibody comprising a V_H of SEQ ID NO: 180 and a V_L of SEQ ID NO: 193.

The disclosure also provides a TL1a-binding antibody comprising a V_H of SEQ ID NO: 181 and a V_L of SEQ ID NO: 194.

The disclosure also provides a TL1a-binding antibody comprising a V_H of SEQ ID NO: 183 and a V_L of SEQ ID NO: 196.

The disclosure also provides a TL1a-binding antibody comprising a V_H of SEQ ID NO: 185 and a V_L of SEQ ID NO: 198.

The disclosure also provides a TL1a-binding antibody comprising a V_H of SEQ ID NO: 186 and a V_L of SEQ ID NO: 199.

The disclosure also provides a TL1a-binding antibody comprising a V_H of SEQ ID NO: 187 and a V_L of SEQ ID NO: 200.

The antibodies set out in the foregoing list provide one or more of the following advantages:

- (i) inhibit TL1a interaction with DR3 and do not inhibit interaction of TL1a and DcR3 as determined by the methods described herein;
- (ii) reduce the level of apoptosis of the TF-1 cells (e.g., about 7×10^4 cells- 8×10^4 cells (e.g., 7.5×10^4 cells)) with an EC_{50} of less than about 2 nM (such as less than about 1.5 nM or 1.2 nM or 1.1 nM; or 1 nM or less) compared to the level of apoptosis in the absence of the TL1a-binding protein; or
- (iii) bind to TL1a on the surface of a cell with an EC_{50} of less than about 10 nM, such as less than about 5 nM or 3 nM or 2 nM, e.g., as assessed using flow cytometry performed with about 2×10^5 to 3×10^5 cells (e.g., 2.5×10^5 cells).

In one example, the antibody comprises a V_H comprising a sequence set forth in SEQ ID NO: 137 and a V_L comprising a sequence set forth in SEQ ID NO: 138.

In one example, the TL1a-binding protein comprises a V_H comprising a sequence set forth in SEQ ID NO: 152 and a V_L comprising a sequence set forth in SEQ ID NO: 153.

In one example, the TL1a-binding protein comprises a V_H comprising a sequence set forth in

SEQ ID NO: 162 and a V_L comprising a sequence set forth in SEQ ID NO: 172.

In one example, the TL1a-binding protein comprises a V_H comprising a sequence set forth in SEQ ID NO: 163 and a V_L comprising a sequence set forth in SEQ ID NO: 174.

For example, the antibody comprises a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 46, wherein the V_H and/or V_L comprise one or more of the following substitutions or groups of substitutions:

- (i) the V_H comprises an alanine at position 16 of SEQ ID NO: 42;
- (ii) the V_H comprises an alanine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (iii) the V_H comprises a serine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (iv) the V_H comprises a histidine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (v) the V_H comprises a leucine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (vi) the V_H comprises an aspartic acid at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (vii) the V_H comprises a tyrosine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (viii) the V_H comprises a proline at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (ix) the V_H comprises a glutamine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (x) the V_H comprises a lysine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xi) the V_H comprises an alanine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xii) the V_H comprises a serine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xiii) the V_H comprises a histidine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xiv) the V_H comprises a leucine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xv) the V_H comprises an aspartic acid at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xvi) the V_H comprises a tyrosine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xvii) the V_H comprises a glutamine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xviii) the V_H comprises a lysine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xix) the V_H comprises an alanine at position 102 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xx) the V_H comprises a serine at position 102 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxi) the V_H comprises a histidine at position 102 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xxii) the V_H comprises a leucine at position 102 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;

- (lxvii) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises a alanine at position 82 of SEQ ID NO: 46;
- (lxviii) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises a serine at position 95 of SEQ ID NO: 46;
- (lxix) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises a serine at position 96 of SEQ ID NO: 46;
- (lxx) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (lxxi) the V_H comprises a serine at position 47 of SEQ ID NO: 42 and the V_L comprises an threonine at position 23 of SEQ ID NO: 46;
- (lxxii) the V_H comprises a proline at position 41, an alanine at position 72, an aspartic acid at position 73 an arginine at position 74 and a threonine at position 76 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (lxxiii) the V_H comprises a proline at position 41, a leucine at position 51 and a glutamic acid at position 102 each relative to SEQ ID NO: 42 and the V_L comprises a serine at position 24 and a threonine at position 76 each relative to SEQ ID NO: 46;
- (lxxiv) the V_H comprises a proline at position 41, a leucine at position 51 and a glutamic acid at position 102 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24 and a threonine at position 76 each relative to SEQ ID NO: 46;
- (lxxv) the V_H comprises a proline at position 41, a leucine at position 51 and a glutamic acid at position 102 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23 and a threonine at position 76 each relative to SEQ ID NO: 46;
- (lxxvi) the V_H comprises a proline at position 41, a leucine at position 51 and a glutamic acid at position 102 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (lxxvii) the V_H comprises a proline at position 41, a leucine at position 51, a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24 and a threonine at position 76 each relative to SEQ ID NO: 46;
- (lxxviii) the V_H comprises a proline at position 41, a leucine at position 51, a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (lxxix) the V_H comprises a proline at position 41, an alanine at position 72, an aspartic acid at position 73 an arginine at position 74 and a threonine at position 76 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24 and a threonine at position 76 each relative to SEQ ID NO: 46;
- (lxxx) the V_H comprises a proline at position 41, an alanine at position 72, an aspartic acid at position 73 an arginine at position 74 and a threonine at position 76 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24, a threonine at position 76 and a glutamic acid at position 51 each relative to SEQ ID NO: 46;
- (lxxxii) the V_H comprises a proline at position 41, a leucine at position 51, an alanine at position 72, an aspartic acid at position 73, an arginine at position 74, a threonine at position 76, a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24, a threonine at position 76 and a glycine at position 51 each relative to SEQ ID NO: 46.

- aspartic acid at position 73, an arginine at position 74, a threonine at position 76, a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24, a threonine at position 76 and a glutamic acid at position 51 each relative to SEQ ID NO: 46; or
- (lxxxii) the V_H comprises a proline at position 41, a leucine at position 51, an alanine at position 72, an aspartic acid at position 73, an arginine at position 74, a threonine at position 76, a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24, a threonine at position 76 and a glycine at position 51 each relative to SEQ ID NO: 46.

In one example, a TL1a-binding protein of the disclosure comprises a human or non-human primate heavy chain immunoglobulin constant region selected from a group consisting of IgG1, IgG2, IgG3, IgG4, IgD, IgM, IgE and IgA. An exemplary heavy chain immunoglobulin constant region is an IgG constant region, e.g., an IgG1 constant region, such as a human IgG1 constant region. For example, a human IgG1 constant region comprises a Fc region comprising a sequence set forth in SEQ ID NO: 134. In one example, the human or non-human primate heavy chain immunoglobulin constant region lacks a C-terminal lysine residue.

In another example, a TL1a-binding protein of the disclosure comprises a human or non-human primate light chain immunoglobulin constant region selected from a group consisting of kappa or lambda. In one example, the TL1a-binding protein comprises a human light chain constant region comprising a sequence set forth in SEQ ID NO: 135 (kappa) or SEQ ID NO: 136 (lambda).

The disclosure also provides an isolated or recombinant nucleic acid encoding the TL1a-binding protein of the disclosure or a sequence having at least about 80% identity thereto or a sequence that hybridizes thereto under moderate or high stringency conditions. In this regard, the disclosure is not limited to the specific exemplified nucleic acids described herein, but also encompasses any nucleic acid that encodes a TL1a-binding protein of the disclosure as a result of degeneracy of the genetic code. For example, the nucleic acid may be codon optimized for expression in a particular cell type.

In one example, the nucleic acid comprises a sequence set forth in any one of SEQ ID NOs: 96 to 118 or a sequence having at least about 80% identity thereto or a sequence that hybridizes thereto under moderate or high stringency conditions.

In one example, the nucleic acid comprises a sequence set forth in SEQ ID NO: 102 or a sequence having at least about 80% identity thereto or a sequence that hybridizes thereto under moderate or high stringency conditions.

In one example, the nucleic acid comprises a sequence set forth in SEQ ID NO: 103 or a sequence having at least about 80% identity thereto or a sequence that hybridizes thereto under moderate or high stringency conditions.

In one example, the nucleic acid comprises a sequence set forth in SEQ ID NO: 102 and a sequence set forth in SEQ ID NO: 103 or a sequence having at least about 80% identity thereto or a sequence that hybridizes thereto under moderate or high stringency conditions.

In one example, the nucleic acid comprises a sequence set forth in SEQ ID NO: 104 or a sequence having at least about

In one example, such a nucleic acid is included in an expression construct in which the nucleic acid is operably linked to a promoter. Such an expression construct can be in a vector, e.g., a plasmid.

In examples of the disclosure directed to single polypeptide TL1a-binding proteins, the expression construct may comprise a promoter linked to a nucleic acid encoding that polypeptide chain.

In examples directed to multiple polypeptides that form a TL1a-binding protein, an expression construct of the disclosure comprises a nucleic acid encoding one of the polypeptides (e.g., comprising a V_H) operably linked to a promoter and a nucleic acid encoding another of the polypeptides (e.g., comprising a V_L) operably linked to a promoter.

In another example, the expression construct is a bicistronic expression construct, e.g., comprising the following operably linked components in 5' to 3' order:

- (i) a promoter
- (ii) a nucleic acid encoding a first polypeptide;
- (iii) an internal ribosome entry site; and
- (iv) a nucleic acid encoding a second polypeptide.

For example, the first polypeptide comprises a V_H and the second polypeptide comprises a V_L , or the first polypeptide comprises a V_L and the second polypeptide comprises a V_H .

The disclosure also contemplates separate expression constructs one of which encodes a first polypeptide (e.g., comprising a V_H) and another of which encodes a second polypeptide (e.g., comprising a V_L). For example, the disclosure also provides a composition comprising:

- (i) a first expression construct comprising a nucleic acid encoding a polypeptide (e.g., comprising a V_H operably linked to a promoter); and
- (ii) a second expression construct comprising a nucleic acid encoding a polypeptide (e.g., comprising a V_L operably linked to a promoter),

wherein the first and second polypeptides associate to form a TL1a-binding protein of the disclosure.

The disclosure also provides an isolated cell expressing a TL1a-binding protein of the disclosure or a recombinant cell genetically-modified to express a TL1a-binding protein of the disclosure.

In one example, the cell comprises the expression construct of the disclosure or:

- (i) a first expression construct comprising a nucleic acid encoding a polypeptide (e.g., comprising a V_H) operably linked to a promoter; and
- (ii) a second expression construct comprising a nucleic acid encoding a polypeptide (e.g., comprising a V_L) operably linked to a promoter,

wherein the first and second polypeptides associate to form a TL1a-binding protein of the disclosure.

Examples of cells of the disclosure include bacterial cells, yeast cells, insect cells or mammalian cells. Exemplary cells are mammalian.

The disclosure additionally provides methods for producing a TL1a-binding protein of the disclosure. For example, such a method involves maintaining the expression construct(s) of the disclosure under conditions sufficient for the TL1a-binding protein to be produced.

In one example, a method for producing a TL1a-binding protein of the disclosure comprises culturing the cell of the disclosure under conditions sufficient for the protein to be produced and, optionally, secreted.

In one example, the method for producing a TL1a-binding protein of the disclosure additionally comprises isolating the protein.

The disclosure also provides a composition comprising the TL1a-binding protein, nucleic acid, expression construct or cell of the disclosure and a suitable carrier. In one example, the composition comprises the TL1a-binding protein of the disclosure and a suitable carrier. In one example, the carrier is pharmaceutically acceptable, e.g., the composition is a pharmaceutical composition.

The disclosure also provides a method for treating or preventing symptoms of a condition (e.g., a TL1a-mediated condition) in a cell, tissue, organ or subject, the method comprising administering the TL1a-binding protein, nucleic acid, expression construct, cell or composition of the disclosure to the cell, tissue, organ or subject. In one example, the disclosure provides a method for treating or preventing a condition (e.g., a TL1a-mediated condition) in a subject, the method comprising administering the TL1a-binding protein, nucleic acid, expression construct, cell or composition of the disclosure to the subject. In this regard, a method of preventing a condition can prevent a relapse of a condition having a relapsing-remitting form, such as multiple sclerosis, e.g., the protein is administered during remission to thereby prevent a relapse.

The disclosure also provides a method for inducing or enhancing angiogenesis in a subject, the method comprising administering the TL1a-binding protein, nucleic acid, expression construct, cell or composition of the disclosure to the subject.

The disclosure also provides for use of the TL1a-binding protein, nucleic acid, expression construct, cell or composition of the disclosure in medicine.

The disclosure also provides for use of the TL1a-binding protein, nucleic acid, expression construct, or cell in the manufacture of a medicament or prevention for treating or preventing symptoms of a condition (e.g., a TL1a-mediated condition) or for treating or preventing a condition (e.g., a TL1a-mediated condition) or for inducing or enhancing angiogenesis in a subject.

The disclosure also provides the TL1a-binding protein, nucleic acid, expression construct, or cell for use in treating or preventing symptoms of a condition (e.g., a TL1a-mediated condition) or for treating or preventing a condition (e.g., a TL1a-mediated condition) or for inducing or enhancing angiogenesis in a subject.

In one example, a method of treatment or prophylaxis of the disclosure additionally comprises diagnosing the condition, e.g., by performing a method described herein.

In one example, a method of treatment or prophylaxis of the disclosure additionally comprises detecting the level of TL1a in a subject and administering a further dose of the TL1a-binding protein, nucleic acid, expression construct, cell or composition of the disclosure if the level of TL1a is not significantly reduced or is not reduced to a level not associated with a condition.

The disclosure also provides a method for inhibiting interaction of TL1a and DR3 (and in one example, not inhibiting interaction of TL1a and DcR3) in a cell, tissue, organ or subject, the method comprising administering the TL1a-binding protein, nucleic acid, expression construct, cell or composition of the disclosure to the cell, tissue, organ or subject. In one example, the subject suffers from a condition (e.g., a TL1a-mediated condition).

The disclosure also provides a method for detecting TL1a in a sample, the method comprising contacting a sample with the TL1a-binding protein of the disclosure such that an antigen-protein complex forms and detecting the complex, wherein detecting the complex is indicative of TL1a in the sample.

The disclosure also provides a method for detecting TL1a in a subject, the method comprising detecting the TL1a-binding protein of the disclosure in the subject, wherein the protein is conjugated to a detectable label. In one example, the method comprises administering the protein to the subject.

The disclosure also provides a method for diagnosing a TL1a-mediated condition in a subject, the method comprising performing the method of the disclosure to detect TL1a in a sample from the subject, wherein detection of TL1a in the sample is indicative of the TL1a-mediated condition.

In one example, the method comprises determining the level of TL1a in the sample, wherein an increased or decreased level of TL1a in the sample compared to a control sample is indicative of the TL1a-mediated condition.

In one example, the results of a method to detect TL1a or diagnose/prognose a condition are provided, e.g., in paper or machine-readable form.

The disclosure also provides a method comprising obtaining the results of a method to detect TL1a or diagnose/prognose a condition of the disclosure and administering a therapeutic or prophylactic composition or recommending such administration. In one example, the composition is a composition of the disclosure.

Exemplary conditions to be treated, prevented, diagnosed or prognosed are T_H17 -mediated conditions, inflammatory conditions, autoimmune conditions or conditions associated with or caused by insufficient angiogenesis. Suitable conditions are described herein.

In one example, a condition to be treated, prevented, diagnosed or prognosed is an autoimmune disease. For example, the condition is uveitis, ulcerative colitis, Crohn's disease, irritable bowel syndrome, rheumatoid arthritis, polyarthritis, multiple sclerosis, asthma or chronic obstructive pulmonary disease.

In one example, the condition is an inflammatory bowel condition, such as, colitis, e.g., ulcerative colitis or Crohn's disease.

The disclosure also provides a method of selecting a TL1a-binding protein which binds specifically to TL1a and inhibits interaction of TL1a and DR3 and which does not inhibit interaction of TL1a and DcR3 from a plurality of TL1a-binding proteins, the method comprising:

contacting the plurality of TL1a-binding proteins to a TL1a mutein in which the arginine at amino acid position 32 of SEQ ID NO:202 has been substituted with an alanine and/or the arginine at amino acid position 85 has been substituted with alanine under conditions sufficient to allow binding of TL1a-binding proteins to the mutein to form a TL1a-binding protein-TL1a mutein complex and a depleted plurality of TL1a-binding proteins which do not bind the TL1a mutein,

and collecting TL1a-binding proteins which do not bind to the TL1a mutein from the depleted plurality of TL1a-binding proteins, wherein the collected TL1a-binding proteins bind specifically to TL1a and inhibit interaction of TL1a and DR3 and do not inhibit interaction of TL1a and DcR3.

The disclosure also provides a method of isolating a TL1a-binding protein which binds specifically to TL1a and inhibits interaction of TL1a and DR3 and which does not inhibit interaction of TL1a and DcR3 from a plurality of TL1a-binding proteins, the method comprising isolating from the plurality of TL1a-binding proteins one or more TL1a-binding proteins that do not bind to a TL1a mutein in which the arginine at amino acid position 32 of SEQ ID

NO:202 has been substituted with an alanine and/or the arginine at amino acid position 85 has been substituted with alanine.

The plurality of TL1a-binding proteins may be present in a library of antibodies or antigen binding domains, such as, a phage display, ribosome display or yeast display library. The plurality of TL1a-binding proteins may be present in an antiserum. The plurality of TL1a-binding proteins may be present in hybridoma culture supernatants.

The disclosure also provides an isolated polypeptide comprising a sequence set forth in SEQ ID NO: 202 in which the arginine at amino acid position 32 as been substituted with an alanine and/or the arginine at amino acid position 85 has been substituted with alanine.

BRIEF DESCRIPTION OF THE FIGURES

FIGS. 1A to 1H are diagrammatic representations showing sequences of variable regions of antibodies. In alignments shown in FIGS. 1D to 1H any identical amino acids are indicated by a period, i.e., ".". FIG. 1A shows sequences of V_H regions of human anti-TL1a antibodies. FIG. 1B shows sequences of V_L regions of human anti-TL1a antibodies. FIG. 1C shows sequences of V_H regions of anti-TL1a antibody C320 and some derivatives thereof. FIG. 1D shows alignment of V_H sequences of selected human antibodies into which the V_H CDRs 1, 2 and 3 of C320 were grafted.

FIG. 1E shows a consensus sequence of V_H regions of derivatives of C320. FIG. 1F shows sequences of V_L regions of anti-TL1a antibody C320 and derivatives thereof. FIG. 1G shows an alignment of human V_L sequences of antibodies into which the V_L CDRs 1, 2 and 3 of C320 were grafted. FIG. 1H shows a consensus sequence of V_L regions of derivatives of C320. Boxed regions contain CDRs (as indicated) as defined by the Kabat numbering system and the enhanced Chothia numbering system. CDRs defined by the Kabat numbering system are shown in bold. CDRs defined by the enhanced Chothia numbering system are underlined.

FIG. 2 is a graphical representation showing results of a potency assay demonstrating the ability of a range of anti-TL1a antibodies to inhibit TL1a-induced apoptosis of TF-1 cells. Five $\mu\text{g/ml}$ of anti-TL1a antibodies were screened for their ability to inhibit TL1a-induced apoptosis in TF-1 cells.

FIG. 3 is a graphical representation of results of an assay to identify highly potent anti-TL1a antibodies. Results depicted show the level of inhibition of TL1a-induced apoptosis of TF-1 cells achieved at various concentrations of test antibodies (maximum concentration tested 5 $\mu\text{g/ml}$). These levels permit determination of the EC_{50} of anti-TL1a antibodies for the inhibition of TL1a-induced apoptosis of TF-1 cells, which allows for functional comparisons based on relative potency.

FIG. 4A is a graphical representation showing the level of inhibition of interaction of TL1a with DR3 achieved at various concentrations of test antibodies (maximum concentration tested 25 $\mu\text{g/ml}$). Antibodies C320, C321 and C323 inhibited the interaction between TL1a and DR3 at numerous concentrations.

FIG. 4B is a graphical representation showing the level of inhibition of interaction of TL1a with DcR3 achieved at various concentrations of test antibodies (maximum concentration tested 25 $\mu\text{g/ml}$). Antibodies C320, C321 and C323 did not detectably inhibit the interaction between TL1a and DcR3 at antibody concentrations of 10 $\mu\text{g/ml}$ or less.

FIG. 4C is a graphical representation showing the level of inhibition of interaction of TL1a with DR3 achieved at

various concentrations of test antibodies (maximum concentration tested 10 $\mu\text{g/mL}$). Antibodies C320-0, C320-168 and C320-179 inhibited the interaction between TL1a and DR3 at numerous concentrations. Antibodies 1B4 and C300-25 are included for the purposes of comparison

FIG. 4D is a graphical representation showing the level of inhibition of interaction of TL1a and DcR3 achieved at various concentrations of test antibodies (maximum concentration tested 100 $\mu\text{g/mL}$). Antibodies C320-0, C320-168 and C320-179 did not detectably inhibit the interaction between TL1a and DcR3. Antibodies 1B4 and C300-25 inhibited the interaction between TL1a and DcR3 at various concentrations and are included for the purposes of comparison

FIG. 5A includes two a graphical representations showing the binding of various concentrations of test antibodies to soluble human TL1a (SEQ ID NO: 202; top panel) and soluble human TL1a in which arginine at residue 32 has been substituted with alanine (R32A mutein, bottom panel). Antibodies C320-0, C320-168 and C320-179 all bound TL1a at various concentrations but not R32A mutein TL1a. Anti-TL1a antibodies 1B4 and 16H2 (as described in US20090280116) bound both TL1a and R32A mutein TL1a at various concentrations. The isotype control antibody did not bind any form of TL1a

FIG. 5B is a graphical representation showing the binding of various concentrations of test antibodies to soluble human TL1a in which arginine at residue 85 has been substituted with alanine (R85A mutein, top panel) and soluble human TL1a in which arginines at residue 32 and residue 85 have been substituted with alanine (R32A+R85A mutein, bottom panel). Antibodies C320-0, C320-168 and C320-179 did not bind either R85A or R32A+R85A mutein TL1a. Anti-TL1a antibodies 1B4 and 16H2 (as described in US20090280116) bound both R85A or R32A+R85A mutein TL1a at various concentrations. The isotype control antibody did not bind any form of TL1a FIG. 5C is a graphical representation showing the binding of soluble human TL1a, R32A mutein TL1a and R85A mutein TL1a at a concentration of 1 $\mu\text{g/mL}$ to receptors DR3 and DcR3. TL1a bound both receptors equally well. R32A and R85A mutein TL1a bound DcR3 to a similar extent as did TL1a but bound DR3 at a level approximately 50% or more lower than TL1a binding to DcR3

FIG. 5D is a diagrammatic representation depicting the X-ray crystal structure of trimeric human TL1a (PDB: 3K51) (gray) with the residues R32 and R85 highlighted in black on each monomer.

FIG. 6 is a graphical representation showing the ability of antibodies C320, C321, C323 and 1B4 to bind to cell surface TL1a. Binding was assessed using flow cytometry, in which results are presented as mean fluorescence intensity (MFI). All antibodies, except the isotype control, bound TL1a

FIG. 7A is a graphical representation showing production of cell surface TL1a by mitogen (Concanavalin A)-stimulated peripheral blood mononuclear cells (PBMCs). The shaded graph represents an isotype control antibody and the line graph represents cell surface TL1a as detected by a chimeric rat/human anti-human TL1a.

FIG. 7B is a graphical representation showing increased production of secreted TL1a in PBMCs as mitogen (Concanavalin A)-stimulation is increased.

FIG. 8A is a graphical representation showing the ability of anti-TL1a antibodies (maximum concentration tested 50 $\mu\text{g/mL}$) to inhibit interferon gamma (IFN- γ) production induced by endogenous human TL1a. Endogenous TL1a enhanced cytokine production by stimulated PBMCs. Anti-

bodies C320 and C323 inhibited the production of IFN- γ . Antibody 1B4 is included for the purposes of comparison

FIG. 8B is a graphical representation showing the ability of anti-TL1a antibodies (maximum concentration tested 50 $\mu\text{g/mL}$) to inhibit IL-13 production induced by endogenous human TL1a. Endogenous TL1a enhanced cytokine production by stimulated PBMCs. Antibodies C320 and C323 inhibited the production of IL-13 Antibody 1B4 is included for the purposes of comparison.

FIG. 9A is a diagrammatic representation showing an alignment of the light chain sequence of C320 (SEQ ID NO: 46) against the germline sequence of highest homology, IGLV1-40*01 (SEQ ID NO: 201). Any identical amino acids are indicated by a period, i.e., ".". Differences in amino acid sequences in the CDR regions are identified.

FIG. 9B is a diagrammatic representation showing an alignment of V_H regions of antibodies identified herein.

FIG. 9C is a diagrammatic representation showing an alignment of V_L regions of antibodies identified herein.

FIG. 9D is a copy of a photographic representation showing results of isoelectric focusing gel comparing C320-168, C320-163 and C320-170. C320-168 has 5-6 distinct charged isoforms compared to 1-2 isoforms visualized for C320-163 and C320-170.

FIG. 9E is a graphical representation showing the ability of antibodies C320-168, C320-179 and C320-183 to neutralize TL1a-induced apoptosis of TF-1 cells at various concentrations (maximum concentration tested 10 $\mu\text{g/mL}$). Antibodies C320-0 and 1B4 are included for comparison. All antibodies inhibit TL1a-induced apoptosis of TF-1 cells at multiple concentrations.

FIG. 10A is a graphical representation showing the total area of colon ulcerated (cm^2) in rats in the days following DNBS-induced colitis. Rats were treated with antibody C320-168 (10 mg/kg), vehicle (negative control) days 0 and 4 or sulfasalazine (standard of care compound) daily from day 0 (with results for each treatment group indicated). C320-168 reduced average ulcer area compared to vehicle treated animals to a comparable extent as sulfasalazine.

FIG. 10B is a graphical representation showing the weight change (g) in rats in the days following oxazalone-induced colitis. Rats were treated with antibody C320-168 (10 mg/kg) or isotype control (10 mg/kg) on days 0 and 4 or sulfasalazine (SoC (5-ASA); standard of care compound) daily from day 0 (with results for each treatment group indicated.) C320-168 ameliorated weight loss relative to the isotype control antibody to a comparable extent as sulfasalazine.

FIG. 10C is a graphical representation showing stool consistency (DAI—Disease Activity Index) in rats in the days following oxazalone-induced colitis. Rats were treated with antibody C320-168 (10 mg/kg) or isotype control (10 mg/kg) on days 0 and 4 or sulfasalazine (SoC (5-ASA); standard of care compound) daily from day 0 (with results for each treatment group indicated). C320-168 improved the clinical signs of disease (stool consistency) relative to the isotype control antibody.

FIG. 11A is a graphical representation showing stool consistency (DAI—Disease Activity Index) in rats in the days following dextran sulphate sodium (DSS)-induced colitis. Rats were treated with antibody C320-168 (10 mg/kg), or isotype control (10 mg/kg) twice weekly from day 4 after disease induction or sulfasalazine (SoC (5-ASA); standard of care compound) daily from day 4 after disease induction (with results for each treatment group indicated). C320-168 improved the clinical signs of disease (stool

consistency) relative to the isotype control antibody to a similar extent as sulfasalazine.

FIG. 11B is a graphical representation showing the weight change (%) in rats in the days following DSS-induced colitis. Rats were treated with antibody C320-168 (10 mg/kg or isotype control (10 mg/kg) twice weekly from day 4 after disease induction or sulfasalazine (SoC (5-ASA); standard of care compound) daily from day 4 after disease induction (with results for each treatment group indicated). C320-168 ameliorated weight loss relative to the isotype control antibody to a similar extent as sulfasalazine

FIG. 12 is a graphical representation showing the results of an ELISA assay to identify the binding of antibodies to different TL1a isoforms. Both C320-168 and C320-179 bound the longer (72-251) and shorter (84-251) isoforms of soluble, cleaved TL1a.

DETAILED DESCRIPTION OF THE INVENTION

Throughout this specification, unless specifically stated otherwise or the context requires otherwise, reference to a single step, composition of matter, group of steps or group of compositions of matter shall be taken to encompass one and a plurality (i.e. one or more) of those steps, compositions of matter, groups of steps or groups of compositions of matter. Thus, as used herein, the singular forms "a", "an" and "the" include plural aspects unless the context clearly dictates otherwise. For example, reference to "a" includes a single as well as two or more; reference to "an" includes a single as well as two or more; reference to "the" includes a single as well as two or more and so forth.

Each example of the disclosure described herein is to be applied mutatis mutandis to each and every other example unless specifically stated otherwise.

Those skilled in the art will appreciate that the disclosure herein is susceptible to variations and modifications other than those specifically described. It is to be understood that the disclosure includes all such variations and modifications. The disclosure also includes all of the steps, features, compositions and compounds referred to or indicated in this specification, individually or collectively, and any and all combinations or any two or more of said steps or features.

The disclosure is not to be limited in scope by the specific examples described herein, which are intended for the purpose of exemplification only. Functionally-equivalent products, compositions and methods are clearly within the scope of the disclosure, as described herein.

The disclosure is performed without undue experimentation using, unless otherwise indicated, conventional techniques of molecular biology, microbiology, virology, recombinant DNA technology, peptide synthesis in solution, solid phase peptide synthesis, and immunology. Such procedures are described, for example, in Sambrook, Fritsch & Maniatis, *Molecular Cloning: A Laboratory Manual*, Cold Spring Harbor Laboratories, New York, Second Edition (1989), whole of Vols I, II, and III; Benny K. C. Lo, *Antibody Engineering: Methods and Protocols*, (2004) Humana Press, Vol. 248; *DNA Cloning: A Practical Approach*, Vols. I and II (D. N. Glover, ed., 1985), IRL Press, Oxford, whole of text; *Oligonucleotide Synthesis: A Practical Approach* (M. J. Gait, ed, 1984) IRL Press, Oxford, whole of text, and particularly the papers therein by Gait, pp 1-22; Atkinson et al., pp 35-81; Sproat et al., pp 83-115; and Wu et al., pp 135-151; 4. *Nucleic Acid Hybridization: A Practical Approach* (B. D. Hames & S. J. Higgins, eds., 1985) IRL Press, Oxford, whole of text; *Immobilized Cells and*

Enzymes: A Practical Approach (1986) IRL Press, Oxford, whole of text; Perbal, B., *A Practical Guide to Molecular Cloning* (1984); *Methods In Enzymology* (S. Colowick and N. Kaplan, eds., Academic Press, Inc.), whole of series; J. F. Ramalho Ortigao, "The Chemistry of Peptide Synthesis" In: *Knowledge database of Access to Virtual Laboratory website* (Interactiva, Germany); Sakakibara *Biochem. Biophys. Res. Commun.* 73: 336-342, 1976; Merrifield *J. Am. Chem. Soc.* 85: 2149-2154, 1963; Barany and Merrifield (1979) in *The Peptides* (Gross, E. and Meienhofer, J. eds.), vol. 2, pp. 1-284, Academic Press, New York. 12. Wunsch, E., ed. (1974) *Synthese von Peptiden in Houben-Weyls Methoden der Organischen Chemie* (Müller, E., ed.), vol. 15, 4th edn., Parts 1 and 2, Thieme, Stuttgart; Bodanszky, M. (1984) *Principles of Peptide Synthesis*, Springer-Verlag, Heidelberg; Bodanszky, M. & Bodanszky, A. (1984) *The Practice of Peptide Synthesis*, Springer-Verlag, Heidelberg; Bodanszky *Int. J. Peptide Protein Res.* 25: 449-474, 1985; *Handbook of Experimental Immunology*, Vols. I-IV (D. M. Weir and C. C. Blackwell, eds., 1986, Blackwell Scientific Publications); and *Animal Cell Culture: Practical Approach*, Third Edition (John R. W. Masters, ed., 2000), ISBN 0199637970, whole of text.

The term "and/or", e.g., "X and/or Y" shall be understood to mean either "X and Y" or "X or Y" and shall be taken to provide explicit support for both meanings or for either meaning.

Throughout this specification the word "comprise", or variations such as "comprises" or "comprising", will be understood to imply the inclusion of a stated element, integer or step, or group of elements, integers or steps, but not the exclusion of any other element, integer or step, or group of elements, integers or steps.

Key to Sequence Listing

SEQ ID NO: 1: amino acid sequence of human TL1a extracellular domain with N-terminal HIS and FLAG tags

SEQ ID NO 2: amino acid sequence of C336 V_H

SEQ ID NO: 3: amino acid sequence of C336 HCDR1

SEQ ID NO: 4: amino acid sequence of C336 HCDR2

SEQ ID NO: 5: amino acid sequence of C336 HCDR3

SEQ ID NO: 6: amino acid sequence of C336 V_L

SEQ ID NO: 7: amino acid sequence of C336 LCDR1

SEQ ID NO: 8: amino acid sequence of C336 LCDR2

SEQ ID NO: 9: amino acid sequence of C336 LCDR3

SEQ ID NO: 10: amino acid sequence of C334 V_H

SEQ ID NO: 11: amino acid sequence of C334 HCDR1

SEQ ID NO: 12: amino acid sequence of C334 HCDR2

SEQ ID NO: 13: amino acid sequence of C334 HCDR3

SEQ ID NO: 14: amino acid sequence of C334 V_L

SEQ ID NO: 15: amino acid sequence of C334 LCDR1

SEQ ID NO: 16: amino acid sequence of C334 LCDR2

SEQ ID NO: 17: amino acid sequence of C334 LCDR3

SEQ ID NO: 18: amino acid sequence of C333 V_H

SEQ ID NO: 19: amino acid sequence of C333 HCDR1

SEQ ID NO: 20: amino acid sequence of C333 HCDR2

SEQ ID NO: 21: amino acid sequence of C333 HCDR3

SEQ ID NO: 22: amino acid sequence of C333 V_L

SEQ ID NO: 23: amino acid sequence of C333 LCDR1

SEQ ID NO: 24: amino acid sequence of C333 LCDR2

SEQ ID NO: 25: amino acid sequence of C333 LCDR3

SEQ ID NO: 26: amino acid sequence of C323 V_H

SEQ ID NO: 27: amino acid sequence of C323 HCDR1

SEQ ID NO: 28: amino acid sequence of C323 HCDR2

SEQ ID NO: 29: amino acid sequence of C323 HCDR3

SEQ ID NO: 30: amino acid sequence of C323 V_L

SEQ ID NO: 31: amino acid sequence of C323 LCDR1

SEQ ID NO: 32: amino acid sequence of C323 LCDR2
 SEQ ID NO: 33: amino acid sequence of C323 LCDR3
 SEQ ID NO: 34: amino acid sequence of C321 V_H
 SEQ ID NO: 35: amino acid sequence of C321 HCDR1
 SEQ ID NO: 36: amino acid sequence of C321 HCDR2 5
 SEQ ID NO: 37: amino acid sequence of C321 HCDR3
 SEQ ID NO: 38: amino acid sequence of C321 V_L
 SEQ ID NO: 39: amino acid sequence of C321 LCDR1
 SEQ ID NO: 40: amino acid sequence of C321 LCDR2
 SEQ ID NO: 41: amino acid sequence of C321 LCDR3 10
 SEQ ID NO: 42: amino acid sequence of C320 V_H
 SEQ ID NO: 43: amino acid sequence of C320 HCDR1
 SEQ ID NO: 44: amino acid sequence of C320 HCDR2
 SEQ ID NO: 45: amino acid sequence of C320 HCDR3
 SEQ ID NO: 46: amino acid sequence of C320 V_L 15
 SEQ ID NO: 47: amino acid sequence of C320 LCDR1
 SEQ ID NO: 48: amino acid sequence of C320 LCDR2
 SEQ ID NO: 49: amino acid sequence of C320 LCDR3
 SEQ ID NO: 50: amino acid sequence of C319 V_H
 SEQ ID NO: 51: amino acid sequence of C319 HCDR1 20
 SEQ ID NO: 52: amino acid sequence of C319 HCDR2
 SEQ ID NO: 53: amino acid sequence of C319 HCDR3
 SEQ ID NO: 54: amino acid sequence of C319 V_L
 SEQ ID NO: 55: amino acid sequence of C319 LCDR1
 SEQ ID NO: 56: amino acid sequence of C319 LCDR2 25
 SEQ ID NO: 57: amino acid sequence of C319 LCDR3
 SEQ ID NO: 58: amino acid sequence of C320-3 V_H
 SEQ ID NO: 59: amino acid sequence of C320-3 HCDR1
 SEQ ID NO: 60: amino acid sequence of C320-3 HCDR2
 SEQ ID NO: 61: amino acid sequence of C320-3 HCDR3 30
 SEQ ID NO: 62: amino acid sequence of C320-5 V_L
 SEQ ID NO: 63: amino acid sequence of C320-5 LCDR1
 SEQ ID NO: 64: amino acid sequence of C320-5 LCDR2
 SEQ ID NO: 65: amino acid sequence of C320-5 LCDR3
 SEQ ID NO: 66: amino acid sequence of C320-90 V_H 35
 SEQ ID NO: 67: amino acid sequence of C320-90 HCDR1
 SEQ ID NO: 68: amino acid sequence of C320-90 HCDR2
 SEQ ID NO: 69: amino acid sequence of C320-90 HCDR3 40
 SEQ ID NO: 70: amino acid sequence of C320-103 V_H
 SEQ ID NO: 71: amino acid sequence of C320-103 HCDR1
 SEQ ID NO: 72: amino acid sequence of C320-103 HCDR2 45
 SEQ ID NO: 73: amino acid sequence of C320-103 HCDR3
 SEQ ID NO: 74: amino acid sequence of C320-114 V_H
 SEQ ID NO: 75: amino acid sequence of C320-114 HCDR1 50
 SEQ ID NO: 76: amino acid sequence of C320-114 HCDR2
 SEQ ID NO: 77: amino acid sequence of C320-114 HCDR3 55
 SEQ ID NO: 78: amino acid sequence of C320-115 V_H
 SEQ ID NO: 79: amino acid sequence of C320-115 HCDR1
 SEQ ID NO: 80: amino acid sequence of C320-115 HCDR2 60
 SEQ ID NO: 81: amino acid sequence of C320-115 HCDR3
 SEQ ID NO: 82: amino acid sequence of C320-120 V_L
 SEQ ID NO: 83: amino acid sequence of C320-120 LCDR1 65
 SEQ ID NO: 84: amino acid sequence of C320-120 LCDR2

SEQ ID NO: 85: amino acid sequence of C320-120 LCDR3
 SEQ ID NO: 86: amino acid sequence of C320-129 V_H
 SEQ ID NO: 87: amino acid sequence of C320-129 HCDR1
 SEQ ID NO: 88: amino acid sequence of C320-129 HCDR2
 SEQ ID NO: 89: amino acid sequence of C320-129 HCDR3
 SEQ ID NO: 90: amino acid sequence of C320-130 V_H
 SEQ ID NO: 91: amino acid sequence of C320-130 HCDR1
 SEQ ID NO: 92: amino acid sequence of C320-130 HCDR2
 SEQ ID NO: 93: amino acid sequence of C320-130 HCDR3
 SEQ ID NO: 94: amino acid sequence of V_H consensus sequence of C320 and derivatives
 SEQ ID NO: 95: amino acid sequence of V_L consensus sequence of C320 and derivatives
 SEQ ID NO: 96: nucleotide sequence encoding V_H of C336
 SEQ ID NO: 97: nucleotide sequence encoding V_L of C336
 SEQ ID NO: 98: nucleotide sequence encoding V_H of C334
 SEQ ID NO: 99: nucleotide sequence encoding V_L of C334
 SEQ ID NO: 100: nucleotide sequence encoding V_H of C333
 SEQ ID NO: 101: nucleotide sequence encoding V_L of C333
 SEQ ID NO: 102: nucleotide sequence encoding V_H of C323
 SEQ ID NO: 103: nucleotide sequence encoding V_L of C323
 SEQ ID NO: 104: nucleotide sequence encoding V_H of C321
 SEQ ID NO: 105: nucleotide sequence encoding V_L of C321
 SEQ ID NO: 106: nucleotide sequence encoding V_H of C320
 SEQ ID NO: 107: nucleotide sequence encoding V_L of C320
 SEQ ID NO: 108: nucleotide sequence encoding V_H of C319
 SEQ ID NO: 109: nucleotide sequence encoding V_L of C319
 SEQ ID NO: 110: nucleotide sequence encoding V_H of C320-3
 SEQ ID NO: 111: nucleotide sequence encoding V_L of C320-5
 SEQ ID NO: 112: nucleotide sequence encoding V_H of C320-90
 SEQ ID NO: 113: nucleotide sequence encoding V_H of C320-103
 SEQ ID NO: 114: nucleotide sequence encoding V_H of C320-114
 SEQ ID NO: 115: nucleotide sequence encoding V_H of C320-115
 SEQ ID NO: 116: nucleotide sequence encoding V_L of C320-120
 SEQ ID NO: 117: nucleotide sequence encoding V_H of C320-129
 SEQ ID NO: 118: nucleotide sequence encoding V_H of C320-130

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SEQ ID NO: 119: amino acid sequence of V_H of humanized antibody 1B4
 SEQ ID NO: 120: amino acid sequence of V_L of humanized antibody 1B4
 SEQ ID NO: 121: nucleotide sequence encoding V_H of humanized antibody 1B4 5
 SEQ ID NO: 122: nucleotide sequence encoding V_L of humanized antibody 1B4
 SEQ ID NO: 123: amino acid sequence of human TL1a (derived from GenBank Gene accession no 9966 as of 8 May 2011) 10
 SEQ ID NO: 124: amino acid sequence of mouse TL1a Extracellular Domain
 SEQ ID NO: 125: amino acid sequence of cynomolgus/Rhesus TL1a Extracellular Domain 15
 SEQ ID NO: 126: amino acid sequence of rat TL1a Extracellular Domain
 SEQ ID NO: 127: amino acid sequence of rabbit TL1a Extracellular Domain 20
 SEQ ID NO: 128: amino acid sequence of guinea Pig TL1a Extracellular Domain
 SEQ ID NO: 129: amino acid sequence of human Death Receptor 3 (derived from Genbank Gene Accession No. 8718 as of 8 May 2011) 25
 SEQ ID NO: 130: amino acid sequence of human Decoy Receptor 3 (derived from Genbank Gene Accession No. 8771 as of 8 May 2011)
 SEQ ID NO: 131: sequence of region of TL1a
 SEQ ID NO: 132: sequence of region of TL1a 30
 SEQ ID NO: 133: sequence of region of TL1a
 SEQ ID NO: 134: amino acid sequence of human IgG1 Fc region
 SEQ ID NO: 135: amino acid sequence of human kappa constant region 35
 SEQ ID NO: 136: amino acid sequence of human lambda constant region.
 SEQ ID NO: 137: amino acid sequence of V_H consensus sequence of C320 and derivatives
 SEQ ID NO: 138: amino acid sequence of V_L consensus sequence of C320 and derivatives 40
 SEQ ID NO: 139: amino acid sequence of LCDR1 consensus sequence of C320 and derivatives
 SEQ ID NO: 140: amino acid sequence of LCDR2 consensus sequence of C320 and derivatives 45
 SEQ ID NO: 141: amino acid sequence of LCDR3 consensus sequence of C320 and derivatives
 SEQ ID NO: 142: amino acid sequence of HCDR2 consensus sequence of C320 and derivatives
 SEQ ID NO: 143: amino acid sequence of HCDR3 consensus sequence of C320 and derivatives 50
 SEQ ID NO: 144: amino acid sequence of HFR1 consensus sequence of C320 and derivatives
 SEQ ID NO: 145: amino acid sequence of HFR2 consensus sequence of C320 and derivatives 55
 SEQ ID NO: 146: amino acid sequence of HFR3 consensus sequence of C320 and derivatives
 SEQ ID NO: 147: amino acid sequence of HFR4 consensus sequence of C320 and derivatives
 SEQ ID NO: 148: amino acid sequence of LFR1 consensus sequence of C320 and derivatives 60
 SEQ ID NO: 149: amino acid sequence of LFR2 consensus sequence of C320 and derivatives
 SEQ ID NO: 150: amino acid sequence of LFR3 consensus sequence of C320 and derivatives 65
 SEQ ID NO: 151: amino acid sequence of LFR4 consensus sequence of C320 and derivatives

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SEQ ID NO: 152: amino acid sequence of V_H consensus sequence of C320 and derivatives
 SEQ ID NO: 153: amino acid sequence of V_L consensus sequence of C320 and derivatives
 SEQ ID NO: 154: amino acid sequence of V_H comprising CDRs from C320 grafted onto FRs of antibody 1TZG.
 SEQ ID NO: 155: amino acid sequence of V_H comprising CDRs from C320 grafted onto FRs of antibody 1RHH.
 SEQ ID NO: 156: amino acid sequence of V_H comprising CDRs from C320 grafted onto FRs of antibody 2DD8.
 SEQ ID NO: 157: amino acid sequence of V_H comprising CDRs from C320 grafted onto FRs of antibody 2J B5.
 SEQ ID NO: 158: amino acid sequence of V_H comprising CDRs from C320 grafted onto FRs of antibody 3FKU.
 SEQ ID NO: 159: amino acid sequence of V_H comprising CDRs from C320 grafted onto FRs of antibody 3GBM.
 SEQ ID NO: 160: amino acid sequence of V_H comprising CDRs from C320 grafted onto FRs of antibody 3LMJ.
 SEQ ID NO: 161: amino acid sequence of V_H comprising CDRs from C320 grafted onto FRs of antibody 3P30.
 SEQ ID NO: 162: amino acid sequence of V_H consensus sequence of C320 and derivatives
 SEQ ID NO: 163: amino acid sequence of V_L comprising CDRs from C320 grafted onto FRs of antibody 1RHH.
 SEQ ID NO: 164: amino acid sequence of V_L comprising CDRs from C320 grafted onto FRs of antibody 1TZG L.
 SEQ ID NO: 165: amino acid sequence of V_L comprising CDRs from C320 grafted onto FRs of antibody 2DD8.
 SEQ ID NO: 166: amino acid sequence of V_L comprising CDRs from C320 grafted onto FRs of antibody 2J B5.
 SEQ ID NO: 167: amino acid sequence of V_L comprising CDRs from C320 grafted onto FRs of antibody 3FKU.
 SEQ ID NO: 168: amino acid sequence of V_L comprising CDRs from C320 grafted onto FRs of antibody 3GBM.
 SEQ ID NO: 169: amino acid sequence of V_L comprising CDRs from C320 grafted onto FRs of antibody 3LMJ.
 SEQ ID NO: 170: amino acid sequence of V_L comprising CDRs from C320 grafted onto FRs of antibody 3P30.
 SEQ ID NO: 171: amino acid sequence of V_L comprising CDRs from C320 grafted onto FRs of antibody 31YW.
 SEQ ID NO: 172: amino acid sequence of V_L consensus sequence of C320 and derivatives
 SEQ ID NO: 173: amino acid sequence of V_H consensus sequence of C320 and derivatives
 SEQ ID NO: 174: amino acid sequence of V_L consensus sequence of C320 and derivatives
 SEQ ID NO: 175: amino acid sequence of V_H of antibody C320-162
 SEQ ID NO: 176: amino acid sequence of V_H of antibody C320-163
 SEQ ID NO: 177: amino acid sequence of V_H of antibody C320-164
 SEQ ID NO: 178: amino acid sequence of V_H of antibody C320-165
 SEQ ID NO: 179: amino acid sequence of V_H of antibody C320-166
 SEQ ID NO: 180: amino acid sequence of V_H of antibody C320-167
 SEQ ID NO: 181: amino acid sequence of V_H of antibody C320-168
 SEQ ID NO: 182: amino acid sequence of V_H of antibody C320-169
 SEQ ID NO: 183: amino acid sequence of V_H of antibody C320-170
 SEQ ID NO: 184: amino acid sequence of V_H of antibody C320-171

SEQ ID NO: 185: amino acid sequence of V_H of antibody C320-172
 SEQ ID NO: 186: amino acid sequence of V_H of antibody C320-179
 SEQ ID NO: 187: amino acid sequence of V_H of antibody C320-183
 SEQ ID NO: 188: amino acid sequence of V_L of antibody C320-162
 SEQ ID NO: 189: amino acid sequence of V_L of antibody C320-163
 SEQ ID NO: 190: amino acid sequence of V_L of antibody C320-164
 SEQ ID NO: 191: amino acid sequence of V_L of antibody C320-165
 SEQ ID NO: 192: amino acid sequence of V_L of antibody C320-166
 SEQ ID NO: 193: amino acid sequence of V_L of antibody C320-167
 SEQ ID NO: 194: amino acid sequence of V_L of antibody C320-168
 SEQ ID NO: 195: amino acid sequence of V_L of antibody C320-169
 SEQ ID NO: 196: amino acid sequence of V_L of antibody C320-170
 SEQ ID NO: 197: amino acid sequence of V_L of antibody C320-171
 SEQ ID NO: 198: amino acid sequence of V_L of antibody C320-172
 SEQ ID NO: 199: amino acid sequence of V_L of antibody C320-179
 SEQ ID NO: 200: amino acid sequence of V_L of antibody C320-183
 SEQ ID NO: 201: amino acid sequence of V_L of germline sequence IGLV1-40*1
 SEQ ID NO: 202: amino acid sequence of soluble human TL1a
 SEQ ID NO: 203: amino acid sequence of N-linked glycosylation site in V_H of C320
 SEQ ID NO: 204: amino acid sequence from V_H of 1TZG corresponding to N-linked glycosylation site in V_H of C320
 SEQ ID NO: 205: amino acid sequence of peptide from V_H of C320-168
 SEQ ID NO: 206: amino acid sequence of peptide from V_L of C320-168
 SEQ ID NO: 207: amino acid sequence of influenza peptide
 SEQ ID NO: 208: amino acid sequence of mutant peptide from V_L of C320-168
 SEQ ID NO: 209: amino acid sequence of mutant peptide from V_L of C320-168
 SEQ ID NO: 210: amino acid sequence of mutant peptide from V_L of C320-168
 SEQ ID NO: 211: amino acid sequence of mutant peptide from V_L of C320-168
 SEQ ID NO: 212: amino acid sequence of mutant peptide from V_L of C320-168
 SEQ ID NO: 213: amino acid sequence of mutant peptide from V_L of C320-168
 SEQ ID NO: 214: amino acid sequence of mutant peptide from V_L of C320-168
 SEQ ID NO: 215: amino acid sequence of mutant peptide from V_L of C320-168
 SEQ ID NO: 216: amino acid sequence of mutant peptide from V_L of C320-168
 SEQ ID NO: 217: amino acid sequence of mutant peptide from V_L of C320-168

SEQ ID NO: 218: amino acid sequence of mutant peptide from V_L of C320-168
 SEQ ID NO: 219: amino acid sequence of mutant peptide from V_L of C320-168
 SEQ ID NO: 220: amino acid sequence of mutant peptide from V_L of C320-168
 SEQ ID NO: 221: amino acid sequence of mutant peptide from V_L of C320-168
 SEQ ID NO: 222: nucleotide sequence encoding V_H of antibody C320-162
 SEQ ID NO: 223: nucleotide sequence encoding V_H of antibody C320-163
 SEQ ID NO: 224: nucleotide sequence encoding V_H of antibodies C320-164, C320-165, C320-166 and C320-167
 SEQ ID NO: 225: nucleotide sequence encoding V_H of antibodies C320-168 and C320-169
 SEQ ID NO: 226: nucleotide sequence encoding V_H of antibodies C320-170 and C320-172
 SEQ ID NO: 227: nucleotide sequence encoding V_H of antibodies C320-179 and C320-183
 SEQ ID NO: 228: nucleotide sequence encoding V_L of antibodies C320-162, C320-163, C320-167 and C320-169
 SEQ ID NO: 229: nucleotide sequence encoding V_L of antibody C320-164
 SEQ ID NO: 230: nucleotide sequence encoding V_L of antibodies C320-165, C320-168 and C320-170
 SEQ ID NO: 231: nucleotide sequence encoding V_L of antibody C320-166
 SEQ ID NO: 232: nucleotide sequence encoding V_L of antibodies C320-172 and C320-179
 SEQ ID NO: 233: nucleotide sequence encoding V_L of antibody C320-183
 SEQ ID NO: 234: amino acid sequence of C320-13 and C320-22
 SEQ ID NO: 235: amino acid sequence of consensus of HCDR3 of C320 and derivatives

For the purposes of nomenclature and not limitation the amino acid sequence of human TL1a is set forth in SEQ ID NO: 123. Additional sequences of human TL1a are set out in Genbank Gene Accession No. 9966. Accordingly, in one example, the amino acid sequence of human TL1a comprises a sequence set forth in SEQ ID NO: 123. Other isoforms of TL1a have been described: 72-251 (Position 72 to 251 of SEQ ID No: 123), 84-251 (Position 84 to 251 of SEQ ID No: 123); 101-251 or VEGI-174 (Position 101 to 251 of SEQ ID No: 123) and 86-251 or VEGI-192 (Position 86 to 251 in SEQ ID No: 123). The sequences of the extracellular domain of TL1a from various species are set forth in SEQ ID NO: 1 (amino acids 16 to 184; human), SEQ ID NO: 124 (mouse), SEQ ID NO: 125 (cynomolgus/rhesus monkey), SEQ ID NO: 126 (rat), SEQ ID NO: 127 (rabbit) and SEQ ID NO: 128 (guinea pig). TL1a generally forms a homotrimer in a subject and signals through DR3. Exemplary TL1a-binding proteins of the disclosure bind to or bind specifically to human TL1a (abbreviated herein as human TL1a), including recombinant forms thereof.

For the purposes of nomenclature and not limitation, a sequence of a human DR3 is set forth in SEQ ID NO: 129. Additional sequences of human DR3 are set out in Genbank Gene Accession No. 8718. In one example, DR3 encompassed by the disclosure is human DR3 comprising a sequence set forth in SEQ ID NO: 129.

For the purposes of nomenclature and not limitation, a sequence of a human DcR3 is set forth in SEQ ID NO: 130. Additional sequences of human DcR3 are set out in Gen-

bank Gene Accession No. 8771. In one example, DcR3 encompassed by the disclosure is human DcR3 comprising a sequence set forth in SEQ ID NO: 130.

The term “isolated protein” or “isolated polypeptide” is intended to mean a protein or polypeptide that by virtue of its origin or source of derivation is not associated with naturally-associated components that accompany it in its native state; is substantially free of other proteins from the same source. A protein may be rendered substantially free of naturally associated components or substantially purified by isolation, using protein purification techniques known in the art. By “substantially purified” is meant the protein is substantially free of contaminating agents, e.g., at least about 70% or 75% or 80% or 85% or 90% or 95% or 96% or 97% or 98% or 99% free of contaminating agents.

The term “recombinant” shall be understood to mean the product of artificial genetic recombination. Accordingly, in the context of a recombinant protein comprising an antigen binding domain, this term does not encompass an antibody naturally-occurring within a subject’s body that is the product of natural recombination that occurs during B cell maturation. However, if such an antibody is isolated, it is to be considered an isolated protein comprising an antigen binding domain. Similarly, if nucleic acid encoding the protein is isolated and expressed using recombinant means, the resulting protein is a recombinant protein comprising an antibody antigen binding domain. A recombinant protein also encompasses a protein expressed by artificial recombinant means when it is within a cell, tissue or subject, e.g., in which it is expressed.

The term “TL1a-binding protein” shall be taken to include a single polypeptide chain (i.e., a series of contiguous amino acids linked by peptide bonds), or a series of polypeptide chains covalently or non-covalently linked to one another (i.e., a polypeptide complex) capable of binding to TL1a in the manner described and/or claimed herein. For example, the series of polypeptide chains can be covalently linked using a suitable chemical or a disulphide bond. Examples of non-covalent bonds include hydrogen bonds, ionic bonds, Van der Waals forces, and hydrophobic interactions.

The term “polypeptide” or “polypeptide chain” will be understood from the foregoing paragraph to mean a series of contiguous amino acids linked by peptide bonds.

As used herein, the term “antigen binding domain” shall be taken to mean a region of an antibody that is capable of specifically binding to an antigen, i.e., a V_H or a V_L or an Fv comprising both a V_H and a V_L . The antigen binding domain need not be in the context of an entire antibody, e.g., it can be in isolation (e.g., a domain antibody) or in another form, e.g., as described herein, such as a scFv.

For the purposes for the disclosure, the term “antibody” includes a protein capable of specifically binding to one or a few closely related antigens (e.g., TL1a) by virtue of an antigen binding domain contained within a Fv. This term includes four chain antibodies (e.g., two light chains and two heavy chains), recombinant or modified antibodies (e.g., chimeric antibodies, humanized antibodies, human antibodies, CDR-grafted antibodies, primatized antibodies, de-immunized antibodies, synhumanized antibodies, half-antibodies, bispecific antibodies). An antibody generally comprises constant domains, which can be arranged into a constant region or constant fragment or fragment crystallizable (Fc). Exemplary forms of antibodies comprise a four-chain structure as their basic unit. Full-length antibodies comprise two heavy chains (~50 to 70 kD) covalently linked and two light chains (~23 kDa each). A light chain generally comprises a variable region (if present) and a constant domain and in

mammals is either a K light chain or a light chain. A heavy chain generally comprises a variable region and one or two constant domain(s) linked by a hinge region to additional constant domain(s). Heavy chains of mammals are of one of the following types α , δ , ϵ , γ , or μ . Each light chain is also covalently linked to one of the heavy chains. For example, the two heavy chains and the heavy and light chains are held together by inter-chain disulfide bonds and by non-covalent interactions. The number of inter-chain disulfide bonds can vary among different types of antibodies. Each chain has an N-terminal variable region (V_H or V_L wherein each are ~110 amino acids in length) and one or more constant domains at the C-terminus. The constant domain of the light chain (C_L which is ~110 amino acids in length) is aligned with and disulfide bonded to the first constant domain of the heavy chain (C_H1 which is 330 to 440 amino acids in length). The light chain variable region is aligned with the variable region of the heavy chain. The antibody heavy chain can comprise 2 or more additional C_H domains (such as, C_H2, C_H3 and the like) and can comprise a hinge region between the C_H1 and C_H2 constant domains. Antibodies can be of any type (e.g., IgG, IgE, IgM, IgD, IgA, and IgY), class (e.g., IgG₁, IgG₂, IgG₃, IgG₄, IgA₁ and IgA₂) or subclass. In one example, the antibody is a murine (mouse or rat) antibody or a primate (such as, human) antibody. In one example the antibody heavy chain is missing a C-terminal lysine residue. In one example, the antibody is humanized, synhumanized, chimeric, CDR-grafted or deimmunized.

As used herein, “variable region” refers to the portions of the light and/or heavy chains of an antibody as defined herein that is capable of specifically binding to an antigen and, includes amino acid sequences of complementarity determining regions (CDRs); i.e., CDRI, CDR2, and CDR3, and framework regions (FRs). For example, the variable region comprises three or four FRs (e.g., FR1, FR2, FR3 and optionally FR4) together with three CDRs. V_H refers to the variable region of the heavy chain. V_L refers to the variable region of the light chain.

As used herein, the term “complementarity determining regions” (syn. CDRs; i.e., CDRI, CDR2, and CDR3) refers to the amino acid residues of an antibody variable region the presence of which are major contributors to specific antigen binding. Each variable region domain (V_H or V_L) typically has three CDR regions identified as CDRI, CDR2 and CDR3. In one example, the amino acid positions assigned to CDRs and FRs are defined according to Kabat *Sequences of Proteins of Immunological Interest*, National Institutes of Health, Bethesda, Md., 1987 and 1991 (also referred to herein as “the Kabat numbering system”). In another example, the amino acid positions assigned to CDRs and FRs are defined according to the Enhanced Chothia Numbering Scheme (www.bioinfo.org.uk/mdex.html). According to the numbering system of Kabat, V_H FRs and CDRs are positioned as follows: residues 1 to 30 (FR1), 31 to 35 (CDR1), 36 to 49 (FR2), 50 to 65 (CDR2), 66 to 94 (FR3), 95 to 102 (CDR3) and 103 to 113 (FR4). According to the numbering system of Kabat, V_L FRs and CDRs are positioned as follows: residues 1 to 23 (FR1), 24 to 34 (CDR1), 35 to 49 (FR2), 50 to 56 (CDR2), 57 to 88 (FR3), 89 to 97 (CDR3) and 98 to 107 (FR4). The disclosure is not limited to FRs and CDRs as defined by the Kabat numbering system, but includes all numbering systems, including the canonical numbering system or of Chothia and Lesk *J. Mol. Biol.* 196: 901-917, 1987; Chothia et al., *Nature* 342: 877-883, 1989; and/or Al-Lazikani et al., *J. Mol. Biol.* 273: 927-948, 1997; the numbering system of Honneger and Plückthun *J. Mol. Biol.* 309: 657-670, 2001; or the IMGT

system discussed in Giudicelli et al., *Nucleic Acids Res.* 25: 206-211 1997. In one example, the CDRs are defined according to the Kabat numbering system. Optionally, heavy chain CDR2 according to the Kabat numbering system does not comprise the five C-terminal amino acids listed herein or any one or more of those amino acids are substituted with another naturally-occurring amino acid. In an additional, or alternative, option, light chain CDR1 does not comprise the four N-terminal amino acids listed herein or any one or more of those amino acids are substituted with another naturally-occurring amino acid. In this regard, Padlan et al., *FASEB J.*, 9: 133-139, 1995 established that the five C-terminal amino acids of heavy chain CDR2 and/or the four N-terminal amino acids of light chain CDR1 are not generally involved in antigen binding.

"Framework regions" (FRs) are those variable region residues other than the CDR residues.

As used herein, the term "Fv" shall be taken to mean any protein, whether comprised of multiple polypeptides or a single polypeptide, in which a V_L and a V_H associate and form a complex having an antigen binding domain, i.e., capable of specifically binding to an antigen. The V_H and the V_L which form the antigen binding domain can be in a single polypeptide chain or in different polypeptide chains. Furthermore, an Fv of the disclosure (as well as any protein of the disclosure) may have multiple antigen binding domains which may or may not bind the same antigen. This term shall be understood to encompass fragments directly derived from an antibody as well as proteins corresponding to such a fragment produced using recombinant means. In some examples, the V_H is not linked to a heavy chain constant domain (C_H) 1 and/or the V_L is not linked to a light chain constant domain (C_L). Exemplary Fv containing polypeptides or proteins include a Fab fragment, a Fab' fragment, a F(ab') fragment, a scFv, a diabody, a triabody, a tetrabody or higher order complex, or any of the foregoing linked to a constant region or domain thereof, e.g., C_H2 or C_H3 domain, e.g., a minibody. A "Fab fragment" consists of a monovalent antigen-binding fragment of an antibody, and can be produced by digestion of a whole antibody with the enzyme papain, to yield a fragment consisting of an intact light chain and a portion of a heavy chain or can be produced using recombinant means. A "Fab' fragment" of an antibody can be obtained by treating a whole antibody with pepsin, followed by reduction, to yield a molecule consisting of an intact light chain and a portion of a heavy chain comprising a V_H and a single constant domain. Two Fab' fragments are obtained per antibody treated in this manner. A Fab' fragment can also be produced by recombinant means. A "F(ab')₂ fragment" of an antibody consists of a dimer of two Fab' fragments held together by two disulfide bonds, and is obtained by treating a whole antibody molecule with the enzyme pepsin, without subsequent reduction. A "Fab₂" fragment is a recombinant fragment comprising two Fab fragments linked using, for example a leucine zipper or a C_H3 domain. A "single chain Fv" or "scFv" is a recombinant molecule containing the variable region fragment (Fv) of an antibody in which the variable region of the light chain and the variable region of the heavy chain are covalently linked by a suitable, flexible polypeptide linker.

As used herein, the term "binds" in reference to the interaction of a TL1a-binding protein or an antigen binding domain thereof with an antigen means that the interaction is dependent upon the presence of a particular structure (e.g., an antigenic determinant or epitope) on the antigen. For example, an antibody recognizes and binds to a specific protein structure rather than to proteins generally. If an

antibody binds to epitope "A", the presence of a molecule containing epitope "A" (or free, unlabeled "A"), in a reaction containing labeled "A" and the antibody, will reduce the amount of labeled "A" bound to the antibody.

As used herein, the term "specifically binds" or "binds specifically" shall be taken to mean that a protein of the disclosure reacts or associates more frequently, more rapidly, with greater duration and/or with greater affinity with a particular antigen or cell expressing same than it does with alternative antigens or cells. For example, a protein that specifically binds to an antigen binds that antigen with greater affinity, avidity, more readily, and/or with greater duration than it binds to other antigens. For example, a protein binds to TL1a (e.g., human TL1a) with materially greater affinity than it does to other TNF superfamily ligands or to antigens commonly recognized by polyreactive natural antibodies (i.e., by naturally occurring antibodies known to bind a variety of antigens naturally found in humans). It is also understood by reading this definition that, for example, a protein that specifically binds to a first antigen may or may not specifically bind to a second antigen. As such, "specific binding" does not necessarily require exclusive binding or non-detectable binding of another antigen, this is meant by the term "selective binding". In one example, "specific binding" of a TL1a-binding protein of the disclosure to an antigen, means that the protein binds to the antigen with an equilibrium constant (K_D) of 100 nM or less, such as 50 nM or less, for example 20 nM or less, such as, 15 nM or less or 10 nM or less or 5 nM or less.

As used herein, the term "does not detectably bind" shall be understood to mean that a TL1a-binding protein, e.g., an antibody, binds to a candidate antigen at a level less than 20%, or 10% or 6% or 5% above background. The background can be the level of binding signal detected in the absence of the TL1a-binding protein and/or in the presence of a negative control protein (e.g., an isotype control antibody) and/or the level of binding detected in the presence of a negative control antigen. The level of binding is detected, for example, using ELISA in which the antigen is immobilized and contacted with a TL1a-binding protein.

As used herein, the term "epitope" (syn. "antigenic determinant") shall be understood to mean a region of TL1a to which a protein comprising an antigen binding domain of an antibody binds. This term is not necessarily limited to the specific residues or structure to which the protein makes contact. For example, this term includes the region spanning amino acids contacted by the protein and/or at least 5 to 10 or 2 to 5 or 1 to 3 amino acids outside of this region. In some examples, the epitope is a linear series amino acids. An epitope may also comprise a series of discontinuous amino acids that are positioned close to one another when TL1a is folded, i.e., a "conformational epitope". The skilled artisan will also be aware that the term "epitope" is not limited to peptides or polypeptides. For example, the term "epitope" includes chemically active surface groupings of molecules such as sugar side chains, phosphoryl side chains, or sulfonyl side chains, and, in certain examples, may have specific three dimensional structural characteristics, and/or specific charge characteristics. An epitope or peptide or polypeptide comprising same can be administered to an animal to generate antibodies against the epitope.

As used herein, the term "inhibits interaction of TL1a and DR3" shall be understood to mean that in an assay to measure binding of TL1a and DR3, a protein is capable of inhibiting 50% of binding (i.e., has an EC_{50}) of less than about 13 nM, for example, less than about 10 nM, such as

less than about 7 nM, e.g., less than about 5 nM, for example, less than about 3 nM.

As used herein, the term “does not inhibit interaction of TL1a and DcR3” will be understood to mean that a TL1a-binding protein described herein does not inhibit interaction of TL1a and DcR3 (i.e., such that interaction is no longer detectable, e.g., using an ELISA assay described herein). For example, at a concentration of 100 µg/ml, the protein does not completely inhibit interaction of TL1a and DcR3. In some examples, the protein reduces the interaction of TL1a and DcR3 by less than about 20% or 15% or 10%, e.g., when tested at a concentration of 10 µg/ml or 100 µg/ml.

As used herein, the term “does not detectably reduce” shall be understood to mean that a protein as described herein reduces binding of TL1a (or a biotinylated form thereof) to DcR3 by no more than 20% or 8% or 6% or 5% or 4% or 3% or 2% above the level of interaction in the absence of the protein or above the level of background interaction, when tested at a concentration of 10 µg/ml or 100 µg/ml. The background can be the level of binding signal detected in the absence of the protein and/or in the presence of a negative control protein (e.g., an isotype control antibody).

As used herein, the term “neutralize” shall be taken to mean that a TL1a-binding protein is capable of reducing or preventing TL1a-mediated activity in a cell. Methods for determining neutralization are known in the art and/or described herein. For example, TF-1 cells are contacted with TL1a, such as human TL1a (e.g., expressed by a mammalian cell) and cycloheximide in the presence or absence of the TL1a-binding protein. A TL1a-binding protein that reduces the level of apoptosis of the cells compared to the level in the absence of the protein is considered to “neutralize” TL1a activity.

As used herein, the term “condition” refers to a disruption of or interference with normal function, and is not to be limited to any specific condition, and will include diseases or disorders.

As used herein, a “TL1a-associated condition” refers to any condition that is caused by or associated with TL1a or a cell expressing TL1a. The skilled artisan will be readily able to determine such conditions based on the disclosure herein and/or by performing an assay to diagnose a TL1a-associated condition as described herein. In this regard, in some examples the condition is an inflammatory condition, an autoimmune condition and a condition that can be treated by enhancing angiogenesis. A description of exemplary conditions is included herein.

As used herein, the terms “preventing”, “prevent” or “prevention” include administering a protein of the disclosure to thereby stop or hinder the development of at least one symptom of a condition. This term also encompasses treatment of a subject in remission to prevent or hinder relapse. For example, a subject suffering from relapsing-remitting multiple sclerosis is treated during remission to thereby prevent a relapse.

As used herein, the terms “treating”, “treat” or “treatment” include administering a protein described herein to thereby reduce or eliminate at least one symptom of a specified disease or condition.

As used herein, the term “subject” shall be taken to mean any animal, such as, a mammal. In one example, the mammal is a human or non-human primate. In one example, the mammal is a human.

Reference herein to a “sample” should be understood as a reference to any sample derived from a subject such as, but not limited to, a body fluid (e.g., blood or blood fraction such

as serum or plasma, tears, urine, synovial fluid or cerebrospinal fluid), cellular material (e.g. tissue aspirate), tissue biopsy specimens or surgical specimens. In some examples, the “sample” is any one or more of serum, plasma, PBMCs, or a buffy coat fraction.

As used herein, the term “diagnosis”, and variants thereof such as, but not limited to, “diagnose”, “diagnosed” or “diagnosing” includes any primary diagnosis of a clinical state or diagnosis of recurrent disease.

“Prognosis”, “prognosing” and variants thereof as used herein refer to the likely outcome or course of a disease, including the chance of recovery or recurrence or the outcome of treatment.

The term “expression construct” is to be taken in its broadest context and includes a nucleic acid comprising one or more promoter sequences operably linked with one or more nucleic acids as described herein.

The term “expression vector” refers to a nucleic acid comprising an expression construct that is additionally capable of maintaining and/or replicating nucleic acid in an expressible format. For example, an expression vector may comprise a plasmid, bacteriophage, phagemid, cosmid, virus sub-genomic or genomic fragment. Selection of appropriate vectors is within the knowledge of those having skill in the art.

As used herein, the term “promoter” is to be taken in its broadest context and includes the transcriptional regulatory sequences of a genomic gene, including the TATA box or initiator element, which is required for accurate transcription initiation, with or without additional regulatory elements (e.g., upstream activating sequences, transcription factor binding sites, enhancers and silencers) that alter expression of a nucleic acid, e.g., in response to a developmental and/or external stimulus, or in a tissue specific manner. In the present context, the term “promoter” is also used to describe a recombinant, synthetic or fusion nucleic acid, or derivative which confers, activates or enhances the expression of a nucleic acid to which it is operably linked. An exemplary promoter can contain additional copies of one or more specific regulatory elements to further enhance expression and/or alter the spatial expression and/or temporal expression of said nucleic acid.

As used herein, the term “operably linked to” means positioning a promoter relative to a nucleic acid such that expression of the nucleic acid is controlled by the promoter. A promoter can be operably linked to numerous nucleic acids, e.g., through an internal ribosome entry site.

Antibodies

Immunization-Based Methods

To generate antibodies, TL1a or an epitope bearing fragment or portion thereof or a modified form thereof (e.g., a fusion protein comprising a human epitope within a mouse TL1a protein) or nucleic acid encoding same, optionally formulated with any suitable or desired adjuvant and/or pharmaceutically acceptable carrier, is administered to a subject (for example, a non-human animal subject, such as, a mouse, a rat, a chicken etc.) in the form of an injectable composition. Exemplary non-human animals are mammals, such as murine animals (e.g., rats or mice). Injection may be intranasal, intramuscular, sub-cutaneous, intravenous, intradermal, intraperitoneal, or by other known route. Optionally, the TL1a or epitope bearing fragment or portion thereof or a nucleic acid encoding same is administered numerous times. Means for preparing and characterizing antibodies are known in the art (See, e.g., *Antibodies: A Laboratory Manual*, Cold Spring Harbor Laboratory, 1988).

The production of polyclonal antibodies may be monitored by sampling blood of the immunized animal at various points following immunization. A second, booster injection, may be given, if required to achieve a desired antibody titer. The process of boosting and titering is repeated until a suitable titer is achieved. When a desired level of immunogenicity is obtained, the immunized animal is bled and the serum isolated and stored, and/or the animal is used to generate monoclonal antibodies (mAbs).

Monoclonal antibodies are exemplary antibodies contemplated by the disclosure. Generally, production of monoclonal antibodies involves, immunizing a subject (e.g., a rodent, e.g., mouse or rat) with TL1a or an epitope bearing fragment or portion thereof or a nucleic acid encoding same under conditions sufficient to stimulate antibody producing cells. In some examples, a mouse genetically-engineered to express human immunoglobulin proteins and not express murine immunoglobulin proteins, is immunized to produce an antibody (e.g., as described in PCT/US2007/008231 and/or Lonberg et al., *Nature* 368: 856-859, 1994). Following immunization, antibody producing somatic cells (e.g., B lymphocytes) are fused with immortal cells, e.g., immortal myeloma cells. Various methods for producing such fused cells (hybridomas) are known in the art and described, for example, in Kohler and Milstein, *Nature* 256: 495-497, 1975. The hybridoma cells can then be cultured under conditions sufficient for antibody production.

The disclosure contemplates other methods for producing antibodies, e.g., ABL-MYC technology (as described, for example in Largaespada et al., *Curr. Top. Microbiol. Immunol.* 166: 91-96, 1990). Suitable antibodies are then selected based on methods described herein.

Library-Based Methods

The disclosure also encompasses screening of libraries of antibodies or proteins comprising antigen binding domains thereof (e.g., comprising variable regions thereof) to identify a TL1a-binding protein of the disclosure.

Examples of libraries contemplated by this disclosure include naïve libraries (from unchallenged subjects), immunized libraries (from subjects immunized with an antigen) or synthetic libraries. Nucleic acid encoding antibodies or regions thereof (e.g., variable regions) are cloned by conventional techniques (e.g., as disclosed in Sambrook and Russell, eds, *Molecular Cloning: A Laboratory Manual*, 3rd Ed, vols. 1-3, Cold Spring Harbor Laboratory Press, 2001) and used to encode and display proteins using a method known in the art. Other techniques for producing libraries of proteins are described in, for example in U.S. Pat. No. 6,300,064 (e.g., a HuCAL library of Morphosys AG); U.S. Pat. Nos. 5,885,793, 6,204,023, 6,291,158, 6,248,516.

The TL1a-binding proteins according to the disclosure may be soluble secreted proteins or may be presented as a fusion protein on the surface of a cell, or particle (e.g., a phage or other virus, a ribosome or a spore). Various display library formats are known in the art. For example, the library is an in vitro display library (e.g., a ribosome display library, a covalent display library or a mRNA display library, e.g., as described in U.S. Pat. No. 7,270,969). In yet another example, the display library is a phage display library wherein proteins comprising antigen binding domains of antibodies are expressed on phage, e.g., as described in U.S. Pat. No. 6,300,064; 5,885,793; 6,204,023; 6,291,158; or 6,248,516. Other phage display methods are known in the art and are contemplated by the disclosure. Similarly, methods of cell display are contemplated by the disclosure, e.g., bacterial display libraries, e.g., as described in U.S. Pat. No.

5,516,637; yeast display libraries, e.g., as described in U.S. Pat. No. 6,423,538 or a mammalian display library.

Methods for screening display libraries are known in the art. In one example, a display library of the disclosure is screened using affinity purification, e.g., as described in Scopes (In: *Protein purification: principles and practice*, Third Edition, Springer Verlag, 1994). Methods of affinity purification typically involve contacting proteins comprising antigen binding domains displayed by the library with a target antigen (e.g., TL1a) and, following washing, eluting those domains that remain bound to the antigen.

Any variable regions or scFvs identified by screening are readily modified into a complete antibody, if desired. Exemplary methods for modifying or reformatting variable regions or scFvs into a complete antibody are described, for example, in Jones et al., *J. Immunol. Methods* 354: 85-90, 2010; or Jostock et al., *J. Immunol. Methods*, 289: 65-80, 2004. Alternatively, or additionally, standard cloning methods are used, e.g., as described in Ausubel et al., (In: *Current Protocols in Molecular Biology*, Wiley Interscience, ISBN 047 150338, 1987), and/or (Sambrook et al., (In: *Molecular Cloning: Molecular Cloning: A Laboratory Manual*, Cold Spring Harbor Laboratories, New York, Third Edition 2001). Deimmunized, Chimeric, CDR Grafted, Humanized, Syn-humanized, Primatized, Human and Composite TL1a-Binding Proteins

The TL1a-binding proteins of the disclosure may be CDR grafted proteins which include CDRs from an antibody from a non-human species (e.g., mouse or rat or non-human primate) grafted onto or inserted into FRs from a human antibody or which include CDRs from an antibody from one type of antibody (e.g., one type of human antibody) grafted onto or inserted into FRs from another type of antibody (e.g., another type of human antibody). This term also encompasses a composite protein comprising, for example, one or more CDR grafted variable regions and one or more, e.g., human variable regions, chimeric variable regions, synhumanized variable regions or primatized variable regions. Such proteins are exemplified herein by the antibodies designated C320-16 to C320-33.

The TL1a-binding proteins of the disclosure may be a humanized protein.

The term "humanized protein" shall be understood to refer to a protein comprising a human-like variable region, which includes CDRs from an antibody from a non-human species (e.g., mouse or rat or non-human primate) grafted onto or inserted into FRs from a human antibody (this type of antibody is also referred to as a "CDR-grafted antibody"). Humanized proteins also include proteins in which one or more residues of the human protein are modified by one or more amino acid substitutions and/or one or more FR residues of the human protein are replaced by corresponding non-human residues. Humanized proteins may also comprise residues which are found in neither the human antibody or in the non-human antibody. Any additional regions of the protein (e.g., Fc region) are generally human. Humanization can be performed using a method known in the art, e.g., U.S. Pat. No. 5,225,539, 6,054,297, 7,566,771 or 5,585,089. The term "humanized protein" also encompasses a super-humanized protein, e.g., as described in U.S. Pat. No. 7,732,578. This term also encompasses a composite protein comprising, for example, one or more humanized variable regions and one or more, e.g., human variable regions, chimeric variable regions, synhumanized variable regions or primatized variable regions.

In one example, a humanized TL1a-binding protein comprises the regions between 27d and 34, 50 and 55, and 89 and

96 in a light chain sequence disclosed herein; and 31 and 35b, 50 and 58, and 95 and 101 in a heavy chain sequence disclosed herein (numbering according to the Kabat numbering system). In this regard, Padlan et al., *FASEB J.*, 9: 133-139, 1995 presents evidence that these regions are those most likely to bind or contact antigen.

The TL1a-binding proteins of the disclosure may be human proteins. The term "human protein" as used herein refers to proteins having variable and, optionally, constant antibody regions found in humans, e.g. in the human germline or somatic cells or from libraries produced using such regions. The "human" antibodies can include amino acid residues not encoded by human sequences, e.g. mutations introduced by random or site directed mutations in vitro (in particular mutations which involve conservative substitutions or mutations in a small number of residues of the protein, e.g. in 1, 2, 3, 4 or 5 of the residues of the protein). These "human antibodies" do not necessarily need to be generated as a result of an immune response of a human, rather, they can be generated using recombinant means (e.g., screening a phage display library) and/or by a transgenic animal (e.g., a mouse) comprising nucleic acid encoding human antibody constant and/or variable regions and/or using guided selection (e.g., as described in U.S. Pat. No. 5,565,332). This term also encompasses affinity matured forms of such antibodies. For the purposes of the disclosure, a human protein will also be considered to include a protein comprising FRs from a human antibody or FRs comprising sequences from a consensus sequence of human FRs and in which one or more of the CDRs are random or semi-random, e.g., as described in U.S. Pat. Nos. 6,300,064, 6,248,516.

Exemplary human TL1a-binding proteins are antibodies comprising the following pairs of variable regions:

- (i) a V_H comprising a sequence set forth in SEQ ID NO: 2 and a V_L comprising a sequence set forth in SEQ ID NO: 6;
- (ii) a V_H comprising a sequence set forth in SEQ ID NO: 10 and a V_L comprising a sequence set forth in SEQ ID NO: 14;
- (iii) a V_H comprising a sequence set forth in SEQ ID NO: 18 and a V_L comprising a sequence set forth in SEQ ID NO: 22;
- (iv) a V_H comprising a sequence set forth in SEQ ID NO: 26 and a V_L comprising a sequence set forth in SEQ ID NO: 30;
- (v) a V_H comprising a sequence set forth in SEQ ID NO: 34 and a V_L comprising a sequence set forth in SEQ ID NO: 38;
- (vi) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (vii) a V_H comprising a sequence set forth in SEQ ID NO: 50 and a V_L comprising a sequence set forth in SEQ ID NO: 54;
- (viii) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (ix) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (x) a V_H comprising a sequence set forth in SEQ ID NO: 66 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (xi) a V_H comprising a sequence set forth in SEQ ID NO: 70 and a V_L comprising a sequence set forth in SEQ ID NO: 62;

- (xii) a V_H comprising a sequence set forth in SEQ ID NO: 74 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (xiii) a V_H comprising a sequence set forth in SEQ ID NO: 78 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (xiv) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 82;
- (xv) a V_H comprising a sequence set forth in SEQ ID NO: 86 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xvi) a V_H comprising a sequence set forth in SEQ ID NO: 90 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xvii) a V_H comprising a sequence set forth in SEQ ID NO: 58 and a V_L comprising a sequence set forth in SEQ ID NO: 62;
- (xviii) a V_H comprising a sequence set forth in SEQ ID NO: 154 and a V_L comprising a sequence set forth in SEQ ID NO: 164;
- (xix) a V_H comprising a sequence set forth in SEQ ID NO: 155 and a V_L comprising a sequence set forth in SEQ ID NO: 163;
- (xx) a V_H comprising a sequence set forth in SEQ ID NO: 156 and a V_L comprising a sequence set forth in SEQ ID NO: 165;
- (xxi) a V_H comprising a sequence set forth in SEQ ID NO: 157 and a V_L comprising a sequence set forth in SEQ ID NO: 166;
- (xxii) a V_H comprising a sequence set forth in SEQ ID NO: 158 and a V_L comprising a sequence set forth in SEQ ID NO: 167;
- (xxiii) a V_H comprising a sequence set forth in SEQ ID NO: 159 and a V_L comprising a sequence set forth in SEQ ID NO: 168;
- (xxiv) a V_H comprising a sequence set forth in SEQ ID NO: 160 and a V_L comprising a sequence set forth in SEQ ID NO: 169;
- (xxv) a V_H comprising a sequence set forth in SEQ ID NO: 161 and a V_L comprising a sequence set forth in SEQ ID NO: 170;
- (xxvi) a V_H comprising a sequence set forth in SEQ ID NO: 234 and a V_L comprising a sequence set forth in SEQ ID NO: 169;
- (xxvii) a V_H comprising a sequence set forth in SEQ ID NO: 154 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxviii) a V_H comprising a sequence set forth in SEQ ID NO: 155 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxix) a V_H comprising a sequence set forth in SEQ ID NO: 156 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxx) a V_H comprising a sequence set forth in SEQ ID NO: 157 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxxii) a V_H comprising a sequence set forth in SEQ ID NO: 158 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxxiii) a V_H comprising a sequence set forth in SEQ ID NO: 160 and a V_L comprising a sequence set forth in SEQ ID NO: 46;

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- (xxxiv) a V_H comprising a sequence set forth in SEQ ID NO: 161 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxxv) a V_H comprising a sequence set forth in SEQ ID NO: 234 and a V_L comprising a sequence set forth in SEQ ID NO: 46;
- (xxxvi) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 164;
- (xxxvii) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 165;
- (xxxviii) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 166;
- (xxxix) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 167;
- (xl) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 168;
- (xli) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 169;
- (xlii) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 170;
- (xliii) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 171;
- (xliv) a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 172;
- (xlv) a V_H comprising a sequence set forth in SEQ ID NO: 175 and a V_L comprising a sequence set forth in SEQ ID NO: 188;
- (xlvi) a V_H comprising a sequence set forth in SEQ ID NO: 176 and a V_L comprising a sequence set forth in SEQ ID NO: 189;
- (xlvii) a V_H comprising a sequence set forth in SEQ ID NO: 177 and a V_L comprising a sequence set forth in SEQ ID NO: 190;
- (xlviii) a V_H comprising a sequence set forth in SEQ ID NO: 178 and a V_L comprising a sequence set forth in SEQ ID NO: 191;
- (xlix) a V_H comprising a sequence set forth in SEQ ID NO: 179 and a V_L comprising a sequence set forth in SEQ ID NO: 192;
- (l) a V_H comprising a sequence set forth in SEQ ID NO: 180 and a V_L comprising a sequence set forth in SEQ ID NO: 193;
- (li) a V_H comprising a sequence set forth in SEQ ID NO: 181 and a V_L comprising a sequence set forth in SEQ ID NO: 194;
- (lii) a V_H comprising a sequence set forth in SEQ ID NO: 182 and a V_L comprising a sequence set forth in SEQ ID NO: 195;
- (liii) a V_H comprising a sequence set forth in SEQ ID NO: 183 and a V_L comprising a sequence set forth in SEQ ID NO: 196;
- (liv) a V_H comprising a sequence set forth in SEQ ID NO: 184 and a V_L comprising a sequence set forth in SEQ ID NO: 197;
- (lv) a V_H comprising a sequence set forth in SEQ ID NO: 185 and a V_L comprising a sequence set forth in SEQ ID NO: 198;

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- (lvi) a V_H comprising a sequence set forth in SEQ ID NO: 186 and a V_L comprising a sequence set forth in SEQ ID NO: 199; or
- (lvii) a V_H comprising a sequence set forth in SEQ ID NO: 187 and a V_L comprising a sequence set forth in SEQ ID NO: 200.

Additional exemplary human TL1a-binding proteins are antibodies comprising a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 46, wherein the V_H and/or V_L comprise one or more of the following substitutions or groups of substitutions:

- (i) the V_H comprises an alanine at position 16 of SEQ ID NO: 42;
- (ii) the V_H comprises an alanine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (iii) the V_H comprises a serine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (iv) the V_H comprises a histidine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (v) the V_H comprises a leucine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (vi) the V_H comprises an aspartic acid at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (vii) the V_H comprises a tyrosine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (viii) the V_H comprises a proline at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (ix) the V_H comprises a glutamine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (x) the V_H comprises a lysine at position 100 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xi) the V_H comprises an alanine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xii) the V_H comprises a serine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xiii) the V_H comprises a histidine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xiv) the V_H comprises a leucine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xv) the V_H comprises an aspartic acid at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xvi) the V_H comprises a tyrosine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xvii) the V_H comprises a glutamine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (xviii) the V_H comprises a lysine at position 101 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;

- (lxiii) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises a tyrosine at position 33 of SEQ ID NO: 46;
- (lxiv) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises an aspartic acid at position 34 of SEQ ID NO: 46; 5
- (lxv) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises an asparagine at position 53 of SEQ ID NO: 46;
- (lxvi) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises a serine at position 54 of SEQ ID NO: 46; 10
- (lxvii) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises an alanine at position 82 of SEQ ID NO: 46; 15
- (lxviii) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises a serine at position 95 of SEQ ID NO: 46;
- (lxix) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises a serine at position 96 of SEQ ID NO: 46; 20
- (lxx) the V_H comprises a threonine at position 41 of SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46;
- (lxxi) the V_H comprises a serine at position 47 of SEQ ID NO: 42 and the V_L comprises a threonine at position 23 of SEQ ID NO: 46; 25
- (lxxii) the V_H comprises a proline at position 41, an alanine at position 72, an aspartic acid at position 73, an arginine at position 74 and a threonine at position 76 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46; 30
- (lxxiii) the V_H comprises a proline at position 41, a leucine at position 51 and a glutamic acid at position 102 each relative to SEQ ID NO: 42 and the V_L comprises a serine at position 24 and a threonine at position 76 each relative to SEQ ID NO: 46; 35
- (lxxiv) the V_H comprises a proline at position 41, a leucine at position 51 and a glutamic acid at position 102 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24 and a threonine at position 76 each relative to SEQ ID NO: 46; 40
- (lxxv) the V_H comprises a proline at position 41, a leucine at position 51 and a glutamic acid at position 102 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23 and a threonine at position 76 each relative to SEQ ID NO: 46; 45
- (lxxvi) the V_H comprises a proline at position 41, a leucine at position 51 and a glutamic acid at position 102 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46; 50
- (lxxvii) the V_H comprises a proline at position 41, a leucine at position 51, a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24 and a threonine at position 76 each relative to SEQ ID NO: 46; 55
- (lxxviii) the V_H comprises a proline at position 41, a leucine at position 51, a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 76 of SEQ ID NO: 46; 60
- (lxxix) the V_H comprises a proline at position 41, an alanine at position 72, an aspartic acid at position 73, an arginine at position 74 and a threonine at position 76 each relative to SEQ ID NO: 42 and the V_L comprises 65

- a threonine at position 23, a serine at position 24 and a threonine at position 76 each relative to SEQ ID NO: 46;
 - (lxxx) the V_H comprises a proline at position 41, an alanine at position 72, an aspartic acid at position 73, an arginine at position 74 and a threonine at position 76 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24, a threonine at position 76 and a glutamic acid at position 51 each relative to SEQ ID NO: 46;
 - (lxxxi) the V_H comprises a proline at position 41, a leucine at position 51, an alanine at position 72, an aspartic acid at position 73, an arginine at position 74, a threonine at position 76, a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24, a threonine at position 76 and a glutamic acid at position 51 each relative to SEQ ID NO: 46; and
 - (i) the V_H comprises a proline at position 41, a leucine at position 51, an alanine at position 72, an aspartic acid at position 73, an arginine at position 74, a threonine at position 76, a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42 and the V_L comprises a threonine at position 23, a serine at position 24, a threonine at position 76 and a glycine at position 51 each relative to SEQ ID NO: 46.
- Optionally, the V_H is linked to a heavy chain constant region, e.g., an IgG1 heavy chain constant region. In one example, the heavy chain constant region lacks the c-terminal lysine residue.
- Optionally, the V_L is linked to a light chain constant region.
- The TL1a-binding proteins of the disclosure may be synhumanized proteins. The term "synhumanized protein" refers to a protein prepared by a method described in WO2007/019620. A synhumanized TL1a-binding protein includes a variable region of an antibody, wherein the variable region comprises FRs from a New World primate antibody variable region and CDRs from a non-New World primate antibody variable region. For example, a synhumanized TL1a-binding protein includes a variable region of an antibody, wherein the variable region comprises FRs from a New World primate antibody variable region and CDRs from a mouse or rat antibody. In one example, the synhumanized TL1a-binding protein is a TL1a-binding antibody in which one or both of the variable regions are synhumanized. This term also encompasses a composite protein comprising, for example, one or more synhumanized variable regions and one or more, e.g., human variable regions or humanized variable regions or chimeric variable regions.
- The TL1a-binding proteins of the disclosure may be primatized proteins. A "primatized protein" comprises variable region(s) from an antibody generated following immunization of a non-human primate (e.g., a cynomolgus macaque). Optionally, the variable regions of the non-human primate antibody are linked to human constant regions to produce a primatized antibody. Exemplary methods for producing primatized antibodies are described in U.S. Pat. No. 6,113,898. This term also encompasses a composite protein comprising, for example, one or more primatized variable regions and one or more, e.g., human variable regions or humanized variable regions or chimeric variable regions.
- In one example a TL1a-binding protein of the disclosure is a chimeric protein. The term "chimeric proteins" refers to proteins in which an antigen binding domain is from a

particular species (e.g., murine, such as mouse or rat) or belonging to a particular antibody class or subclass, while the remainder of the protein is from a protein derived from another species (such as, for example, human or non-human primate) or belonging to another antibody class or subclass. In one example, a chimeric protein is a chimeric antibody comprising a V_H and/or a V_L from a non-human antibody (e.g., a murine antibody) and the remaining regions of the antibody are from a human antibody. The production of such chimeric proteins is known in the art, and may be achieved by standard means (as described, e.g., in U.S. Pat. Nos. 6,331,415; 5,807,715; 4,816,567 and 4,816,397). This term also encompasses a composite protein comprising, for example, one or more chimeric variable regions and one or more, e.g., human variable regions or humanized variable regions or chimeric variable regions.

The disclosure also contemplates a deimmunized TL1a-binding protein, e.g., as described in WO2000/34317 and WO2004/108158. De-immunized antibodies and proteins have one or more epitopes, e.g., B cell epitopes or T cell epitopes removed (i.e., mutated) to thereby reduce the likelihood that a subject will raise an immune response against the antibody or protein. For example, a TL1a-binding protein of the disclosure is analyzed to identify one or more B or T cell epitopes and one or more amino acid residues within the epitope is mutated to thereby reduce the immunogenicity of the TL1a-binding protein. The present inventors have used such techniques to identify epitopes that are predicted to bind to MHC Class II molecules and identify TL1a binding proteins less likely to induce an immune response in a subject.

It will be apparent to the skilled artisan from the foregoing disclosure that a "composite" protein comprises one form of V_H (e.g., human) and another form of V_L (e.g., humanized). The disclosure explicitly encompasses all combinations of forms of V_H and V_L . Other TL1a-Binding Proteins Comprising an Antigen Binding Domain

The disclosure also contemplates other TL1a-binding proteins comprising a variable region or antigen binding domain of an antibody, such as:

- (i) a single-domain antibody, which is a single polypeptide chain comprising all or a portion of the V_H or a V_L of an antibody (see, e.g., U.S. Pat. No. 6,248,516);
- (ii) diabodies, triabodies and tetrabodies, e.g., as described in U.S. Pat. No. 5,844,094 and/or US2008152586;
- (iii) scFvs, e.g., as described in U.S. Pat. No. 5,260,203;
- (iv) minibodies, e.g., as described in U.S. Pat. No. 5,837,821;
- (v) "key and hole" bispecific proteins as described in U.S. Pat. No. 5,731,168;
- (vi) heteroconjugate proteins, e.g., as described in U.S. Pat. No. 4,676,980;
- (vii) heteroconjugate proteins produced using a chemical cross-linker, e.g., as described in U.S. Pat. No. 4,676,980;
- (viii) Fab'-SH fragments, e.g., as described in Shalaby et al., J. Exp. Med., 175: 217-225, 1992; or
- (ix) Fab₃ (e.g., as described in EP19930302894).

Constant Domain Fusions

The disclosure encompasses a TL1a-binding protein comprising an antigen binding domain of an antibody and a constant region or Fc or a domain thereof, e.g., C_H2 and/or C_H3 domain. Suitable constant regions and/or domains will be apparent to the skilled artisan and/or the sequences of such polypeptides are readily available from publicly avail-

able databases. Kabat et al also provide description of some suitable constant regions/domains.

Constant regions and/or domains thereof are useful for providing biological activities such as, dimerization, extended serum half life (e.g., by binding to FcRn), antibody-dependent cell cytotoxicity (ADCC), complement dependent cytotoxicity (CDC), antibody-dependent cell phagocytosis (ADCP).

The disclosure also contemplates TL1a-binding proteins comprising mutant constant regions or domains, e.g., as described in U.S. Pat. Nos. 7,217,797; 7,217,798; or US20090041770 (having increased half-life) or US2005037000 (increased ADCC).

The C-terminal lysine of the heavy chain constant region of an antibody of the disclosure or TL1a-binding protein of the disclosure comprising a constant region or Fc may be removed, for example, during production or purification of the antibody, or by recombinantly engineering the nucleic acid encoding a heavy chain of the antibody or protein. Accordingly, whole antibodies or proteins may comprise antibody or protein populations with all C-terminal lysine residues removed, antibody or protein populations with no C-terminal lysine residues removed, or antibody or protein populations having a mixture of antibodies with and without the C-terminal lysine residue. In some examples, the antibody or protein populations may additionally comprise antibodies or proteins in which the C-terminal lysine residue is removed in one of the heavy chain constant regions. Similarly, a composition of antibodies or proteins may comprise the same or a similar mix of antibody or protein populations with or without the C-terminal lysine residue. Enhancing Effector Function

In one example, a TL1a-binding protein of the disclosure may induce effector function or enhanced effector function.

In the context of the disclosure, "effector functions" refer to those biological activities mediated by cells or proteins that bind to the Fc region (a native sequence Fc region or amino acid sequence variant Fc region) of an antibody that result in killing of a cell. Examples of effector functions induced by antibodies include: complement dependent cytotoxicity; antibody-dependent-cell-mediated cytotoxicity (ADCC); antibody-dependent-cell-phagocytosis (ADCP); and B-cell activation.

In one example, a TL1a-binding protein of the disclosure binds to TL1a on the surface of a cell in such a manner that it is capable of inducing an effector function, such as, ADCC or CDC.

For example, the TL1a-binding protein remains bound to the TL1a on the surface of the cell for a time sufficient to induce an effector function, such as ADCC and/or CDC.

In one example, a TL1a-binding protein of the disclosure is capable of inducing enhanced effector function, e.g., by virtue of a modified Fc region or by virtue of comprising a region capable of binding to an immune effector cell. For example, the level of effector function is increased compared to the level induced by a human IgG1 or IgG3 Fc region. Enhancing effector function induced by a TL1a-binding protein of the disclosure may result in enhanced therapeutic or prophylactic effects, e.g., by not only blocking the action of TL1a but also by killing or depleting cells causing a condition, e.g., by killing auto-reactive T cells.

In one example, the Fc region of a TL1a-binding protein of the disclosure is modified to increase the level of effector function it is capable of inducing compared to the Fc region without the modification. Such modifications can be at the

amino acid level and/or the secondary structural level and/or the tertiary structural level and/or to the glycosylation of the Fc region.

The skilled addressee will appreciate that greater effector function may be manifested in any of a number of ways, for example as a greater level of effect, a more sustained effect or a faster rate of effect.

In one example, the Fc region comprises one or more amino acid modifications that increase its ability to induce enhanced effector function. In one example, the Fc region binds with greater affinity to one or more FcγRs, such as FcγRIII. In one example, the Fc region comprise at least one amino acid substitution at a position selected from the group consisting of: 230, 233, 234, 235, 239, 240, 243, 264, 266, 272, 274, 275, 276, 278, 302, 318, 324, 325, 326, 328, 330, 332, and 335, numbered according to the EU index of Kabat. In one example, the Fc region comprises the following amino acid substitutions 5239D/1332E, numbered according to the EU index of Kabat. This Fc region has about 14 fold increase in affinity for FcγRIIIa compared to a wild-type Fc region and about 3.3 increased ability to induce ADCC compared to a wild-type Fc region. In one example, the Fc region comprises the following amino acid substitutions 5239D/A330L/1332E, numbered according to the EU index of Kabat. This Fc region has about 138 fold increase in affinity for FcγRIIIa compared to a wild-type Fc region and about 323 fold increased ability to induce ADCC compared to a wild-type Fc region.

Additional amino acid substitutions that increase ability of a Fc region to induce effector function are known in the art and/or described, for example, in U.S. Pat. No. 6,737,056 or U.S. Pat. No. 7,317,091.

In one example, the glycosylation of the Fc region is altered to increase its ability to induce enhanced effector function. In this regard, native antibodies produced by mammalian cells typically comprise a branched, biantennary oligosaccharide that is generally attached by an N-linkage to Asn297 of the C_H2 domain of the Fc region. The oligosaccharide may include various carbohydrates, e.g., mannose, N-acetyl glucosamine (GlcNAc), galactose, and sialic acid, as well as a fucose attached to a GlcNAc in the "stem" of the biantennary oligosaccharide structure. In some examples, Fc regions according to the disclosure comprise a carbohydrate structure that lacks fucose attached (directly or indirectly) to an Fc region, i.e., the Fc region is "afucosylated". Such variants may have an improved ability to induce ADCC. Methods for producing afucosylated antibodies include, expressing the antibody or antigen binding fragment thereof in a cell line incapable of expressing α-1,6-fucosyltransferase (FUT8) (e.g., as described in Yumane-Ohnuki et al., *Biotechnol. Bioengineer.* 87: 614-622, 2004), expressing the antibody or antigen binding fragment thereof in cells expressing a small interfering RNA against FUT8 (e.g., as described in Mori et al., *Biotechnol. Bioengineer.* 88: 901-908, 2004), expressing the antibody or antigen binding fragment thereof in cells incapable of expressing guanosine diphosphate (GDP)-mannose 4,6-dehydratase (GMD) (e.g., as described in Kanda et al., *J. Biotechnol.*, 130: 300-310, 2007). The disclosure also contemplates the use of proteins having a reduced level of fucosylation, e.g., produced using a cell line modified to express β-(1,4)-N-acetylglucosaminyltransferase III (GnT-III) (e.g., as described in Umāna et al., *Nat. Biotechnol.* 17: 176-180, 1999).

Other methods include the use of cell lines which inherently produce antibodies capable of inducing enhanced Fc-mediated effector function (e.g. duck embryonic derived

stem cells for the production of viral vaccines, WO2008/129058; Recombinant protein production in avian EBX® cells, WO 2008/142124).

TL1a-binding proteins of the disclosure also include those with bisected oligosaccharides, e.g., in which a biantennary oligosaccharide attached to the Fc region is bisected by GlcNAc. Such proteins may have reduced fucosylation and/or improved ADCC function. Examples of such proteins are described, e.g., in U.S. Pat. No. 6,602,684 and US20050123546.

TL1a-binding proteins with at least one galactose residue in the oligosaccharide attached to the Fc region are also contemplated. Such proteins may have improved CDC function. Such proteins are described, e.g., in WO1997/30087 and WO1999/22764.

TL1a-binding proteins can also comprise a Fc region capable of inducing enhanced levels of CDC. For example, hybrids of IgG1 and IgG3 produce antibodies having enhanced CDC activity (Natsume et al., *Cancer Res.* 68: 3863-3872, 2008).

TL1a-binding proteins can also or alternatively be fused to or conjugated to proteins (e.g., antibody variable regions) that bind to immune effector cells, e.g., by virtue of binding to CD3 or CD16.

Methods for determining effector function are known in the art. In one example, the level of ADCC activity is assessed using a ⁵¹Cr release assay, an europium release assay or a ³⁵S release assay. In each of these assays, cells expressing TL1a are cultured with one or more of the recited compounds for a time and under conditions sufficient for the compound to be taken up by the cell. In the case of a ³⁵S release assay, the cells can be cultured with ³⁵S-labeled methionine and/or cysteine for a time sufficient for the labeled amino acids to be incorporated into newly synthesized proteins. Cells are then cultured in the presence or absence of the protein and in the presence of immune effector cells, e.g., PBMCs and/or NK cells. The amount of ⁵¹Cr, europium and/or ³⁵S in cell culture medium is then detected, and an increase in the presence of the protein compared to in the absence of protein indicates that the binding molecule/agent has effector function. Exemplary publications disclosing assays for assessing the level of ADCC induced by a protein include Hellstrom et al. *Proc. Nod Acad. Sci. USA* 83: 7059-7063, 1986 and Bruggemann et al., *J. Exp. Med.* 166: 1351-1361, 1987.

Other assays for assessing the level of ADCC induced by a protein include ACTITM nonradioactive cytotoxicity assay for flow cytometry (CellTechnology, Inc. CA, USA) or CytoTox 96® non-radioactive cytotoxicity assay (Promega, WI, USA).

Alternatively, or additionally, effector function of a TL1a-binding protein is assessed by determining its affinity for one or more FcγRs, e.g., as described in U.S. Pat. No. 7,317,091.

C1q binding assays may also be carried out to confirm that the TL1a-binding protein is able to bind C1q and may induce CDC. To assess complement activation, a CDC assay may be performed (see, for example, Gazzano-Santoro et al., *J. Immunol. Methods* 202: 163, 1996).

In another example, the TL1a-binding protein comprises one or more amino acid substitutions that increase the half-life of the protein. For example, the TL1a-binding protein comprises a constant region comprising one or more amino acid substitutions that increase the affinity of the constant region for the neonatal Fc region (FcRn). For example, the constant region has increased affinity for FcRn at lower pH, e.g., about pH 6.0, to facilitate Fc/FcRn binding in an endosome. In one example, the constant region has

increased affinity for FcRn at about pH 6 compared to its affinity at about pH 7.4, which facilitates the re-release of Fc into blood following cellular recycling. These amino acid substitutions are useful for extending the half life of a TL1a-binding protein, by reducing clearance from the blood.

Exemplary amino acid substitutions include T250Q and/or M428L or T252A, T254S and T266F or M252Y, S254T and T256E or H433K and N434F according to the EU numbering system. Additional or alternative amino acid substitutions are described, for example, in US20070135620 or U.S. Pat. No. 7,083,784.

Stabilized TL1a-Binding Proteins

Neutralizing TL1a-binding proteins of the disclosure can comprise an IgG4 constant region or a stabilized IgG4 constant region. The term “stabilized IgG4 constant region” will be understood to mean an IgG4 constant region that has been modified to reduce Fab arm exchange or the propensity to undergo Fab arm exchange or formation of a half-antibody or a propensity to form a half-antibody. “Fab arm exchange” refers to a type of protein modification for human IgG4, in which an IgG4 heavy chain and attached light chain (half-molecule) is swapped for a heavy-light chain pair from another IgG4 molecule. Thus, IgG4 molecules may acquire two distinct Fab arms recognizing two distinct antigens (resulting in bispecific molecules). Fab arm exchange occurs naturally in vivo and can be induced in vitro by purified blood cells or reducing agents such as reduced glutathione. A “half-antibody” forms when an IgG4 antibody dissociates to form two molecules each containing a single heavy chain and a single light chain.

In one example, a stabilized IgG4 constant region comprises a proline at position 241 of the hinge region according to the system of Kabat (Kabat et al., Sequences of Proteins of Immunological Interest Washington DC United States Department of Health and Human Services, 1987 and/or 1991). This position corresponds to position 228 of the hinge region according to the EU numbering system (Kabat et al., Sequences of Proteins of Immunological Interest Washington DC United States Department of Health and Human Services, 2001 and Edelman et al., *Proc. Natl. Acad. USA*, 63: 78-85, 1969). In human IgG4, this residue is generally a serine. Following substitution of the serine for proline, the IgG4 hinge region comprises a sequence CPPC. In this regard, the skilled person will be aware that the “hinge region” is a proline-rich portion of an antibody heavy chain constant region that links the Fc and Fab regions that confers mobility on the two Fab arms of an antibody. The hinge region includes cysteine residues which are involved in inter-heavy chain disulfide bonds. It is generally defined as stretching from Glu226 to Pro243 of human IgG1 according to the numbering system of Kabat. Hinge regions of other IgG isotypes may be aligned with the IgG1 sequence by placing the first and last cysteine residues forming inter-heavy chain disulphide (S—S) bonds in the same positions (see for example WO2010/080538).

Mutant TL1a-Binding Proteins

The disclosure also provides a TL1-binding protein or a nucleic acid encoding same having at least 80% identity to a sequence disclosed herein. In one example, a TL1a-binding protein or nucleic acid of the disclosure comprises sequence at least about 85% or 90% or 95% or 97% or 98% or 99% identical to a sequence disclosed herein, wherein the protein specifically binds to TL1a and inhibits interaction of TL1a and DR3 and does not inhibit interaction of TL1a and DcR3.

Alternatively, or additionally, the TL1a-binding protein comprises a CDR (e.g., three CDRs) at least about 80% or

85% or 90% or 95% or 97% or 98% or 99% identical to CDR(s) of a V_H or V_L as described herein according to any example, wherein the protein is capable of specifically binding to TL1a and inhibiting interaction of TL1a and DR3, wherein the protein does not inhibit interaction of TL1a and DcR3. In this regard, the inventors have produced numerous antibodies having diverse sequences within their CDRs. Methods for determining binding of a protein TL1a and determining the interaction of TL1a and DR3 or TL1a and DcR3 are described herein.

For example, the inventors have identified a group of TL1a-binding proteins sharing 60% identity in their HCDR1 according to the Kabat numbering system and another subgroup of proteins sharing 80% identity in their HCDR1 according to the Kabat numbering system.

The inventors have also identified a subclass of TL1a-binding proteins sharing 40% identity or 47% identity in their HCDR2 according to the Kabat numbering system.

As discussed herein, it is also known in the art that the five C-terminal residues of heavy chain CDR2 can be mutated to conservative or non-conservative amino acid substitutions (31% of residues) (Padlan et al., *FASEB J.* 9: 133-139, 1995). Thus, a protein can comprise a CDR2 having at least about 69% identity to a heavy chain CDR2 sequence disclosed herein.

The inventors have also identified a class of proteins comprising variants of the variable regions of antibody C320 described herein. These variants permit identification of sites within the variable regions that can be substituted without loss of function.

For example, the inventors have identified several residues in a V_H comprising a sequence set forth in SEQ ID NO: 42 that can be substituted without loss of function. Accordingly, the disclosure encompasses proteins comprising a V_H with at least about 86% identity to a sequence set forth in SEQ ID NO: 42. In this regard, the inventors have produced a modified form of SEQ ID NO: 42 having about 17 amino acid substitutions (i.e., about 86% identity thereto) that retains a function recited herein. In this regard, the inventors have also produced a modified form of SEQ ID NO: 42 having about 90 or 94% identity thereto that retains a function recited herein. In one example, the sequence has at least about 95% or 96% or 97% or 98% identity to a sequence set forth in SEQ ID NO: 42. In this regard, the inventors have produced proteins having about 97% or 98% or 99% identity to a sequence set forth in SEQ ID NO: 42. In one example, the sequence has at least about 99% identity to a sequence set forth in SEQ ID NO: 42. In one example, the sequence has at least about 99.2% identity to a sequence set forth in SEQ ID NO: 42.

In one example, the TL1a-binding protein comprises between 1 and 17 amino acid substitutions compared to SEQ ID NO: 42. For example, the TL1a-binding protein comprises 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 amino acid substitutions compared to SEQ ID NO: 42.

In one example, the TL1a-binding protein comprises between 1 and 17 amino acid substitutions in the FRs compared to SEQ ID NO: 42. For example, the TL1a-binding protein comprises 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 amino acid substitutions in the FRs compared to SEQ ID NO: 42. In one example, the substitution is not at position 73 of SEQ ID NO: 42.

In one example, the TL1a-binding protein comprises between 1 and 3 amino acid substitutions in CDR3 compared to SEQ ID NO: 42. For example, the TL1a-binding

protein comprises 1 or 2 or 3 amino acid substitutions in the CDR3 compared to SEQ ID NO: 42. In one example, the substitution is not at one or more of positions 99 or 101 or 104 or 108 of SEQ ID NO: 42.

In one example, TL1a-binding protein comprises one or more substitutions to prevent or reduce glycosylation of the protein, wherein the substitution(s) are between amino acids 72 to 76 of SEQ ID NO: 42. For example, the protein comprises an alanine in place of the arginine at position 71 and/or an aspartic acid in place of the asparagine at position 72 and/or an arginine in place of the threonine at position 73 and/or a threonine in place of the isoleucine at position 75 of SEQ ID NO: 42.

In one example, the V_H of the protein is not glycosylated and/or does not comprise a consensus site for N-linked glycosylation.

In one example, a TL1a-binding protein of the disclosure comprises a mutant of a sequence set forth in SEQ ID NO: 42, wherein the mutant sequence at least comprises a serine at position 16 of SEQ ID NO: 42.

In one example, a TL1a-binding protein of the disclosure comprises a mutant of a sequence set forth in SEQ ID NO: 42, wherein the mutant sequence at least comprises a proline at position 41.

In one example, a TL1a-binding protein of the disclosure comprises a mutant of a sequence set forth in SEQ ID NO: 42, wherein the mutant sequence at least comprises an arginine at position 74.

In one example, a TL1a-binding protein of the disclosure comprises a mutant of a sequence set forth in SEQ ID NO: 42, wherein the mutant sequence at least comprises a glutamic acid or glycine at position 49 of SEQ ID NO: 42.

In one example, a TL1a-binding protein of the disclosure comprises a mutant of a sequence set forth in SEQ ID NO: 42, wherein the mutant sequence at least comprises a proline at position 41 and a leucine at position 51 and a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42.

In one example, a TL1a-binding protein of the disclosure comprises a mutant of a sequence set forth in SEQ ID NO: 42, wherein the mutant sequence at least comprises a proline at position 41 and a leucine at position 51 and an alanine at position 72 and an aspartic acid at position 73 and an arginine at position 74 and a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42.

In one example, a TL1a-binding protein of the disclosure comprises a mutant of a sequence set forth in SEQ ID NO: 42, wherein the mutant sequence at least comprises a proline at position 41 and a leucine at position 51 and an alanine at position 72 and an aspartic acid at position 73 and an arginine at position 74 and a threonine at position 76 and a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42.

The inventors have identified a site within HCDR2 that can be substituted. Coupled with the observations of Padlan et al., supra, the disclosure thus provides a protein comprising a V_H comprising a CDR2 having a sequence at least about 65% identical to a sequence set forth in SEQ ID NO: 44. In one example, the percentage identity is at least about 70% or 75% or 80% or 90%. In one example, the percentage identity is at least about 94%.

The inventors have identified numerous sites within HCDR3 that can be substituted. The disclosure thus provides a protein comprising a V_H comprising a CDR3 having a sequence at least about 60% identical to a sequence set forth in SEQ ID NO: 45. In one example, the percentage identity

is at least about 70% identical to a sequence set forth in SEQ ID NO: 45, e.g., as exemplified herein. In one example, the percentage identity is at least about 75% or 80% or 90%. In one example, the percentage identity is at least about 90%.

In one example, a mutant form of SEQ ID NO: 45 comprises a glutamic acid at position 1 of SEQ ID NO: 45 and/or a proline at position 3 of SEQ ID NO: 45 and/or an alanine at position 6 of SEQ ID NO: 45. Optionally, the mutant form additionally comprises a phenylalanine at position 8 of SEQ ID NO: 45 and/or a tyrosine at position 10 of SEQ ID NO: 45.

Additional residues that can be mutated are set out in FIGS. 1C-1E and in SEQ ID NOs: 94, 137, 152, 162 and 173.

In one example, the inventors have identified several residues in a V_L comprising a sequence set forth in SEQ ID NO: 46 that can be substituted without loss of function. Accordingly, the disclosure encompasses proteins comprising a V_L with at least about 95% identity to a sequence set forth in SEQ ID NO: 46. For example, the inventors have mutated 32 residues in the V_L without loss of function, meaning that the disclosure provides a protein having at least about 71% identity to a sequence set forth in SEQ ID NO: 46. In one example, the percentage identity is at least about 75% or 80% or 90% or 95% or 97%. In one example, the percentage identity is at least about 98%. In one example, the percentage identity is at least about 99%. In one example, the percentage identity is at least about 99.1%.

In one example, the TL1a-binding protein comprises between 1 and 17 amino acid substitutions compared to SEQ ID NO: 42. For example, the TL1a-binding protein comprises 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 amino acid substitutions compared to SEQ ID NO: 46.

In one example, the TL1a-binding protein comprises between 1 and 32 amino acid substitutions in the FRs compared to SEQ ID NO: 46. For example, the TL1a-binding protein comprises 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 amino acid substitutions in the FRs compared to SEQ ID NO: 46.

In one example, the TL1a-binding protein comprises between 1 and 3 amino acid substitutions in a CDR compared to SEQ ID NO: 46. For example, the TL1a-binding protein comprises 1 or 2 or 3 amino acid substitutions in a CDR compared to SEQ ID NO: 46. In one example, the substitution is not at one or more of positions 34, 54, 94 of SEQ ID NO: 46.

In one example, a TL1a-binding protein of the disclosure comprises a mutant of a sequence set forth in SEQ ID NO: 46, wherein the mutant sequence at least comprises a serine at position 24 of SEQ ID NO: 46.

In one example, a TL1a-binding protein of the disclosure comprises a mutant of a sequence set forth in SEQ ID NO: 46, wherein the mutant sequence at least comprises a threonine at position 76 of SEQ ID NO: 46.

In one example, a TL1a-binding protein of the disclosure comprises a mutant of a sequence set forth in SEQ ID NO: 46, wherein the mutant sequence at least comprises a glutamine at position 81 of SEQ ID NO: 46.

In one example, a TL1a-binding protein of the disclosure comprises a mutant of a sequence set forth in SEQ ID NO: 46, wherein the mutant sequence at least comprises a threonine at position 23 and a serine at position 24 and a threonine at position 76 each relative to SEQ ID NO: 46.

In one example, a TL1a-binding protein of the disclosure comprises a mutant of a sequence set forth in SEQ ID NO: 46, wherein the mutant sequence at least comprises a threonine at position 23 and a serine at position 24 and a threonine at position 76 and a glutamic acid at position 51 each relative to SEQ ID NO: 46.

The inventors have identified a plurality of sites within LCDR1 that can be substituted. The disclosure thus provides a protein comprising a V_L comprising a CDR1 having a sequence at least about 60% identical to a sequence set forth in SEQ ID NO: 47. In one example, the percentage identity is at least about 70% or 75% or 80% or 90%. For example, the inventors have mutated two sites within LCDR1 comprising a sequence set forth in SEQ ID NO: 47, thus producing a protein having at least about 86 identity to the recited sequence. In one example, the percentage identity is at least about 90%.

In one example, a TL1a binding protein of the disclosure comprises a V_H comprising a sequence set forth in SEQ ID NO: 42 and a V_L comprising a sequence set forth in SEQ ID NO: 46, wherein the V_H and/or V_L comprise one or more of the following substitutions or groups of substitutions:

- (i) the V_H comprises at least a serine at position 16 of SEQ ID NO: 42;
- (ii) the V_L comprises at least a threonine at position 76 of SEQ ID NO: 46;
- (iii) the V_L comprises at least a glutamine at position 81 of SEQ ID NO: 46;
- (iv) the V_H comprises at least a proline at position 41 and a leucine at position 51 and a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42 and the V_L comprises at least a threonine at position 23 and a serine at position 24 and a threonine at position 76 each relative to SEQ ID NO: 46; or
- (v) the V_H comprises at least a proline at position 41 and a leucine at position 51 and an alanine at position 72 and an aspartic acid at position 73 and an arginine at position 74 and a glutamic acid at position 102 and an alanine at position 105 each relative to SEQ ID NO: 42 and a V_L comprises at least a threonine at position 23 and a serine at position 24 and a threonine at position 76 and a glutamic acid at position 51 each relative to SEQ ID NO: 46.

In another example, a nucleic acid of the disclosure comprises a sequence at least about 80% or 85% or 90% or 95% or 97% or 98% or 99% identical to a sequence set forth herein and encoding a TL1a-binding protein which is capable of specifically binding to TL1a and inhibiting interaction of TL1a and DR3, wherein the protein does not inhibit interaction of TL1a and DcR3. The disclosure also encompasses nucleic acids encoding a TL1a-binding protein of the disclosure, which differs from a sequence exemplified herein as a result of degeneracy of the genetic code.

The % identity of a nucleic acid or polypeptide is determined by GAP (Needleman and Wunsch. *Mol. Biol.* 48, 443-453, 1970) analysis (GCG program) with a gap creation penalty=5, and a gap extension penalty=0.3. The query sequence is at least 50 residues in length, and the GAP analysis aligns the two sequences over a region of at least 50 residues. For example, the query sequence is at least 100 residues in length and the GAP analysis aligns the two sequences over a region of at least 100 residues. For example, the two sequences are aligned over their entire length.

As discussed above, the disclosure also contemplates a nucleic acid that hybridizes under stringent hybridization conditions to a nucleic acid encoding a TL1a-binding pro-

tein described herein, e.g., nucleic acid encoding a V_H or V_L of antibody C319, C320, C321, C323, C333, C334, C336, C320-3, C320-5, C320-90, C320-103, C320-114, C320-115, C320-120, C320-129, C320-130, C320-135, C320-162, C320-163, C320-164, C320-165, C320-166, C320-167, C320-168, C320-169, C320-170, C320-171, C320-172, C320-179 or C320-183. A "moderate stringency" is defined herein as being a hybridization and/or washing carried out in 2xSSC buffer, 0.1% (w/v) SDS at a temperature in the range 45° C. to 65° C., or equivalent conditions. A "high stringency" is defined herein as being a hybridization and/or wash carried out in 0.1xSSC buffer, 0.1% (w/v) SDS, or lower salt concentration, and at a temperature of at least 65° C., or equivalent conditions. Reference herein to a particular level of stringency encompasses equivalent conditions using wash/hybridization solutions other than SSC known to those skilled in the art. For example, methods for calculating the temperature at which the strands of a double stranded nucleic acid will dissociate (also known as melting temperature, or T_m) are known in the art. A temperature that is similar to (e.g., within 5° C. or within 10° C.) or equal to the T_m of a nucleic acid is considered to be high stringency. Medium stringency is to be considered to be within 10° C. to 20° C. or 10° C. to 15° C. of the calculated T_m of the nucleic acid.

The disclosure also contemplates mutant forms of a TL1a-binding protein of the disclosure comprising one or more conservative amino acid substitutions compared to a sequence set forth herein. In some examples, the TL1a-binding protein comprises 10 or fewer, e.g., 9 or 8 or 7 or 6 or 5 or 4 or 3 or 2 or 1 conservative amino acid substitutions. A "conservative amino acid substitution" is one in which the amino acid residue is replaced with an amino acid residue having a similar side chain and/or hydropathicity and/or hydrophilicity.

Families of amino acid residues having similar side chains have been defined in the art, including basic side chains (e.g., lysine, arginine, histidine), acidic side chains (e.g., aspartic acid, glutamic acid), uncharged polar side chains (e.g., glycine, asparagine, glutamine, serine, threonine, tyrosine, cysteine), nonpolar side chains (e.g., alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, tryptophan), 6-branched side chains (e.g., threonine, valine, isoleucine) and aromatic side chains (e.g., tyrosine, phenylalanine, tryptophan, histidine). Hydropathic indices are described, for example in Kyte and Doolittle *J. Mol. Biol.*, 157: 105-132, 1982 and hydrophilic indices are described in, e.g., U.S. Pat. No. 4,554,101.

The disclosure also contemplates non-conservative amino acid changes. For example, of particular interest are substitutions of charged amino acids with another charged amino acid and with neutral or positively charged amino acids. In some examples, the TL1a-binding protein comprises 10 or fewer, e.g., 9 or 8 or 7 or 6 or 5 or 4 or 3 or 2 or 1 non-conservative amino acid substitutions.

In one example, the mutation(s) occur within a FR of an antigen binding domain of a TL1a-binding protein of the disclosure. In another example, the mutation(s) occur within a CDR of a TL1a-binding protein of the disclosure.

Exemplary methods for producing mutant forms of a TL1a-binding protein include:

mutagenesis of DNA (Thie et al., *Methods Mol. Biol.* 525: 309-322, 2009) or RNA (Kopsidas et al., *Immunol. Lett.* 107:163-168, 2006; Kopsidas et al. *BMC Biotechnology*, 7: 18, 2007; and WO1999/058661);

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introducing a nucleic acid encoding the polypeptide into a mutator cell, e.g., XL-1Red, XL-mutS and XL-mutS-Kanr bacterial cells (Stratagene); DNA shuffling, e.g., as disclosed in Stemmer, *Nature* 370: 389-91, 1994; and site directed mutagenesis, e.g., as described in Dieffenbach (ed) and Dveksler (ed) (In: PCR 5 Primer: A Laboratory Manual, Cold Spring Harbor Laboratories, N Y, 1995).

Exemplary methods for determining biological activity of the mutant TL1a-binding proteins of the disclosure will be apparent to the skilled artisan and/or described herein, e.g., antigen binding. For example, methods for determining antigen binding, competitive inhibition of binding, affinity, association, dissociation and therapeutic efficacy are described herein.

Exemplary TL1a-Binding Proteins

Exemplary variable region containing TL1a-binding proteins produced by the inventors and their encoding nucleic acids are described in Tables 1 and 2.

TABLE 1

Sequences of exemplary TL1a-binding proteins and encoding nucleic acids				
Antibody Name	V _H amino acid SEQ ID NO	V _H chain nucleotide SEQ ID NO	V _L amino acid SEQ ID NO	V _L chain nucleotide SEQ ID NO
1 C336	2	96	6	97
2 C334	10	98	14	99
3 C333	18	100	22	101
4 C323	26	102	30	103
5 C321	34	104	38	105
6 C320	42	106	46	107
7 C319	50	108	54	109
8 C320-3	58	110	46	107
9 C320-5	42	106	62	111
10 C320-90	66	112	62	111
11 C320-103	70	113	62	111
12 C320-114	74	114	62	111
13 C320-115	78	115	62	111
14 C320-120	58	110	82	116
15 C320-129	86	117	46	107
16 C320-130	90	118	46	107
17 C320-135	58	110	62	111
18 C320-7	154		164	
19 C320-8	155		163	
20 C320-9	156		165	
21 C320-10	157		166	
22 C320-11	158		167	
23 C320-12	159		168	
24 C320-13	234		169	
25 C320-14	160		170	
26 C320-15	161		171	
27 C320-16	154		46	
28 C320-17	155		46	
29 C320-18	156		46	
30 C320-19	157		46	
31 C320-20	158		46	
32 C320-21	159		46	
33 C320-22	234		46	
34 C320-23	160		46	
35 C320-24	161		46	
36 C320-25	42		164	
37 C320-26	42		163	
38 C320-27	42		165	
39 C320-28	42		166	
40 C320-29	42		167	
41 C320-30	42		168	
42 C320-31	42		169	
43 C320-32	42		170	
44 C320-33	42		171	
45 C320-162	175	222	188	228

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TABLE 1-continued

Sequences of exemplary TL1a-binding proteins and encoding nucleic acids				
Antibody Name	V _H amino acid SEQ ID NO	V _H chain nucleotide SEQ ID NO	V _L amino acid SEQ ID NO	V _L chain nucleotide SEQ ID NO
46 C320-163	176	223	189	228
47 C320-164	177	224	190	229
48 C320-165	178	224	191	230
49 C320-166	179	224	192	231
50 C320-167	180	224	193	228
51 C320-168	181	225	194	230
52 C320-169	182	225	195	228
53 C320-170	183	226	196	230
54 C320-171	184		197	
55 C320-172	185	226	198	232
56 C320-179	186	227	199	232
57 C320-183	187	227	200	233

TABLE 2

Amino acid substitutions in V _H (relative to SEQ ID NO: 42) and V _L (relative to SEQ ID NO: 46) of exemplary TL1a-binding proteins.		
Antibody name	V _H substitution ¹	V _L substitution ¹
1 C320-2	A16S	none
2 C320-53	E99S	A76T
3 C320-54	E99H	A76T
4 C320-55	E99L	A76T
5 C320-56	E99D	A76T
6 C320-57	E99Y	A76T
7 C320-58	E99P	A76T
8 C320-59	E99Q	A76T
9 C320-60	E99K	A76T
10 C320-61	V100A	A76T
11 C320-62	V100S	A76T
12 C320-63	V100H	A76T
13 C320-64	V100L	A76T
14 C320-65	V100D	A76T
15 C320-66	V100Y	A76T
16 C320-67	V100P	A76T
17 C320-68	V100Q	A76T
18 C320-69	V100K	A76T
19 C320-70	P101A	A76T
20 C320-71	P101S	A76T
21 C320-72	P101H	A76T
22 C320-73	P101L	A76T
23 C320-74	P101D	A76T
24 C320-75	P101Y	A76T
25 C320-76	P101Q	A76T
26 C320-77	P101K	A76T
27 C320-78	D102A	A76T
28 C320-79	D102S	A76T
29 C320-80	D102H	A76T
30 C320-81	D102L	A76T
31 C320-82	D102Y	A76T
32 C320-83	D102P	A76T
33 C320-84	D102Q	A76T
34 C320-85	D102K	A76T
35 C320-86	T103A	A76T
36 C320-87	T103S	A76T
37 C320-88	T103H	A76T
38 C320-89	T103L	A76T
39 C320-90	T103D	A76T
40 C320-91	T103Y	A76T
41 C320-92	T103P	A76T
42 C320-93	T103Q	A76T
43 C320-94	T103K	A76T
44 C320-95	A104S	A76T
45 C320-96	A104H	A76T
46 C320-97	A104L	A76T
47 C320-98	A104D	A76T

TABLE 2-continued

Amino acid substitutions in V_H (relative to SEQ ID NO: 42) and V_L (relative to SEQ ID NO: 46) of exemplary TL1a-binding proteins.			
Antibody name	V_H substitution ¹	V_L substitution ¹	
48 C320-99	A104Y	A76T	
49 C320-100	A104P	A76T	
50 C320-101	A104Q	A76T	
51 C320-102	A104K	A76T	
52 C320-103	S105A	A76T	
53 C320-104	S105H	A76T	
54 C320-105	S105L	A76T	
55 C320-106	S105D	A76T	
56 C320-107	S105Y	A76T	
57 C320-108	S105P	A76T	
58 C320-109	S105Q	A76T	
59 C320-110	S105K	A76T	
60 C320-111	E107A	A76T	
61 C320-112	E107S	A76T	
62 C320-113	E107H	A76T	
63 C320-114	E107L	A76T	
64 C320-115	E107D	A76T	
65 C320-116	E107Y	A76T	
66 C320-117	E107P	A76T	
67 C320-118	E107Q	A76T	
68 C320-119	E107K	A76T	
69 C320-120	T41P	A23T	
70 C320-121	T41P	D28N	
71 C320-122	T41P	L33Y	
72 C320-123	T41P	G34D	
73 C320-124	T41P	Y53N	
74 C320-125	T41P	Y54S	
75 C320-126	T41P	P82A	
76 C320-127	T41P	G95S	
77 C320-128	T41P	T96S	
78 C320-129	D102E	None	
79 C320-130	M51L	None	
80 C320-131	None	D49E	
81 C320-135	T41P	A76T	
82 C320-162	T41P + M51L + S75A + D102E	A76T	
83 C320-163	T41P + R72A + N73D + T74R + I76T	A76T	
84 C320-164	T41P + M51L + D102E	G24S + A76T	
85 C320-165	T41P + M51L + D102E	A23T + G24S + A76T	
86 C320-166	T41P + M51L + D102E	A23T + A76T	
87 C320-167	T41P + M51L + D102E	A76T	
98 C320-168	T41P + M51L + D102E + S105A	A23T + G24S + A76T	
89 C320-169	T41P + M51L + D102E + S105A	A76T	
90 C320-170	T41P + R72A + N73D + T74R + I76T	A23T + G24S + A76T	
91 C320-171	T41P + R72A + N73D + T74R + I76T	A23T + G24S + A76T + Y51P	
92 C320-172	T41P + R72A + N73D + T74R + I76T	A23T + G24S + A76T + Y51E	
93 C320-179	T41P + M51L + R72A + N73D + T74R + I76T + D102E + S105A	A23T + G24S + A76T + Y51E	
94 C320-183	T41P + M51L + R72A + N73D + T74R + I76T + D102E + S105A	A23T + G24S + A76T + Y51G	

¹substitutions are listed as: residue in SEQ ID NO: 42 or 46; position in SEQ ID NO: 42 or 46; substituted amino acid, i.e., A16S in V_H , means that at position 16 of SEQ ID NO: 42 there is an alanine that has been substituted with a serine in the TL1a-binding protein.

Recombinant Expression

As discussed herein, a nucleic acid encoding a TL1a-binding protein of the disclosure (and/or polypeptides included in such a TL1a-binding protein) is introduced into an expression construct, such that it is operably linked to a promoter to thereby facilitate its expression. Methods for producing expression constructs, e.g., cloning into expression constructs/vectors are known in the art and/or described in Ausubel et al., (In: Current Protocols in Molecular Biology. Wiley Interscience, ISBN 047 150338, 1987), and

(Sambrook et al., (In: Molecular Cloning: Molecular Cloning: A Laboratory Manual, Cold Spring Harbor Laboratories, New York, Third Edition 2001) and U.S. Pat. No. 7,270,969.

5 In one example, the TL1a-binding protein of the disclosure is expressed in a bacterial cell. Typical promoters suitable for expression in bacterial cells such as for example a bacterial cell selected from the group comprising *E. coli*, *Staphylococcus* sp., *Corynebacterium* sp., *Salmonella* sp., *Bacillus* sp., and *Pseudomonas* sp., include, but are not limited to a promoter such as lacZ, lpp, a temperature-sensitive (λ or ϕ promoters, T7, T3, SP6 or semi-artificial promoters such as the IPTG-inducible tac promoter or lacUV5 promoter.

15 In another example, the TL1a-binding protein is expressed in a yeast cell. Typical promoters suitable for expression in yeast cells such as, *Pichia pastoris*, *Saccharomyces cerevisiae* and *S. pombe*, include, but are not limited to promoters from the following genes ADH1, GAL1, GAL4, CUP1, PHO5, nmt, RPR1, or TEF1.

20 In a further example, the TL1a-binding protein is expressed in an insect cell. Typical promoters suitable for expression in insect cells, or in insects, include, but are not limited to, the OPE12 promoter, the insect actin promoter isolated from *Bombyx mori*, the *Drosophila* sp. dsh promoter (Marsh et al., *Hum. Mol. Genet.* 9: 13-25, 2000).

A TL1a-binding protein of the disclosure can also be expressed in plant cells. Promoters for expressing peptides in plant cells are known in the art, and include, but are not limited to, the *Hordeum vulgare* amylase gene promoter, the cauliflower mosaic virus 35S promoter, the nopaline synthase (NOS) gene promoter, and the auxin inducible plant promoters P1 and P2.

35 In one example, a TL1a-binding protein of the disclosure is expressed in a mammalian cell or in a mammal. Typical promoters suitable for expression in a mammalian cell include, for example a promoter selected from the group consisting of, retroviral LTR elements, the SV40 early promoter, the SV40 late promoter, the CMV IE (cytomegalovirus immediate early) promoter, the EF₁ promoter (from human elongation factor 1), the EM7 promoter, the UbC promoter (from human ubiquitin C). Examples of useful mammalian host cell lines include monkey kidney CV1 line transformed by SV40 (COS-7); human embryonic kidney line (HEK-293 cells); baby hamster kidney cells (BHK); Chinese hamster ovary cells (CHO); African green monkey kidney cells (VERO-76); or myeloma cells (e.g., NS/O cells).

Other elements of expression constructs/vectors are known in the art and include, for example, enhancers, transcriptional terminators, polyadenylation sequences, nucleic acids encoding selectable or detectable markers and origins of replication.

50 In one example, an expression construct is a bicistronic expression construct. By "bicistronic" is meant a single nucleic acid molecule that is capable of encoding two distinct polypeptides from different regions of the nucleic acid, for example, a single nucleic acid capable of encoding a V_H containing polypeptide and a V_L containing polypeptide as distinct polypeptides. Generally, the regions encoding each distinct polypeptide are separated by an internal ribosome entry site (IRES) and the region 5' of the IRES does not comprise a transcription termination sequence. Exemplary IRESs are described, for example, in US20090247455.

Following production of a suitable expression construct, it is introduced into a suitable cell using any method known in the art. Exemplary methods include microinjection, trans-

fection mediated by DEAE-dextran, transfection mediated by liposomes such as using commercially available reagents, PEG-mediated DNA uptake, electroporation and microparticle bombardment such as by using DNA-coated tungsten or gold particles (Agracetus Inc., WI, USA) amongst others.

The cells used to produce the TL1a-binding protein of this disclosure are then cultured under conditions known in the art to produce a TL1a-binding protein of the disclosure.

Cell free expression systems are also contemplated by the disclosure, e.g., the TNT T7 and TNT T3 systems (Promega), the pEXP1-DEST and pEXP2-DEST vectors (Invitrogen).

Protein Purification

Following production/expression, a TL1a-binding protein of the disclosure is purified using a method known in the art. Such purification provides the protein of the disclosure substantially free of nonspecific protein, acids, lipids, carbohydrates, and the like. In one example, the protein will be in a preparation wherein more than about 90% (e.g. 95%, 98% or 99%) of the protein in the preparation is a TL1a-binding protein of the disclosure.

Standard methods of peptide purification are employed to obtain an isolated TL1a-binding protein of the disclosure, including but not limited to various high-pressure (or performance) liquid chromatography (HPLC) and non-HPLC polypeptide isolation protocols, such as size exclusion chromatography, ion exchange chromatography, hydrophobic interaction chromatography, mixed mode chromatography, phase separation methods, electrophoretic separations, precipitation methods, salting in/out methods, immunochromatography, and/or other methods.

In one example, affinity purification is useful for isolating a fusion protein comprising a label. Methods for isolating a protein using affinity chromatography are known in the art and described, for example, in Scopes (In: Protein purification: principles and practice, Third Edition, Springer Verlag, 1994). For example, an antibody or compound that binds to the label (in the case of a polyhistidine tag this may be, for example, nickel-NTA) is immobilized on a solid support. A sample comprising a protein is then contacted to the immobilized antibody or compound for a time and under conditions sufficient for binding to occur. Following washing to remove any unbound or non-specifically bound protein, the protein is eluted.

In the case of a TL1a-binding protein comprising a Fc region of an antibody, protein A or protein G or modified forms thereof can be used for affinity purification. Protein A is useful for isolating purified proteins comprising a human $\gamma 1$, $\gamma 2$, or $\gamma 4$ heavy chain Fc region. Protein G is recommended for all mouse Fc isotypes and for human $\gamma 3$.

In one example, a TL1a-binding protein of the disclosure is conjugated to a compound. For example, the compound is selected from the group consisting of a radioisotope, a detectable label, a therapeutic compound, a colloid, a toxin, a nucleic acid, a peptide, a protein, a compound that increases the half life of the TL1a-binding protein in a subject and mixtures thereof.

The other compound can be directly or indirectly bound to the TL1a-binding protein (e.g., can comprise a linker in the case of indirect binding). Examples of compounds include, a radioisotope (e.g., iodine-131, yttrium-90 or indium-111), a detectable label (e.g., a fluorophore or a fluorescent nanocrystal or quantum dot), a therapeutic compound (e.g., a chemotherapeutic or an anti-inflammatory), a colloid (e.g., gold), a toxin (e.g., ricin or tetanus toxoid), a nucleic acid, a peptide (e.g., a serum albumin binding peptide), a protein (e.g., a protein comprising an antigen

binding domain of an antibody or serum albumin), a compound that increases the half life of the TL1a-binding protein in a subject (e.g., polyethylene glycol or other water soluble polymer having this activity) and mixtures thereof. Exemplary compounds that can be conjugated to a TL1a-binding protein of the disclosure and methods for such conjugation are known in the art and described, for example, in WO2010/059821.

The TL1a-binding protein may be conjugated to nanoparticles (for example as reviewed in Kogan et al., *Nanomedicine* (Lond). 2: 287-306, 2007). The nanoparticles may be metallic nanoparticles.

The TL1a-binding protein may be comprised in antibody-targeted bacterial minicells (for example as described in PCT/162005/000204).

Some exemplary compounds that can be conjugated to a TL1a-binding protein of the disclosure are listed in Table 3.

TABLE 3

Compounds useful in conjugation.	
Group	Detail
Radioisotopes (either directly or indirectly)	^{123}I , ^{125}I , ^{130}I , ^{133}I , ^{135}I , ^{47}Sc , ^{72}As , ^{72}Sc , ^{90}Y , ^{88}Y , ^{97}Ru , ^{100}Pd , ^{101m}Rh , ^{101m}Rh , ^{119}Sb , ^{128}Ba , ^{197}Hg , ^{211}At , ^{212}Bi , ^{153}Sm , ^{169}Eu , ^{212}Pb , ^{109}Pd , ^{111}In , ^{67}Ga , ^{68}Ga , ^{67}Cu , ^{75}Br , ^{76}Br , ^{77}Br , ^{99m}Tc , ^{11}C , ^{13}N , ^{15}O , ^{18}F , ^{188}Re , ^{203}Pb , ^{64}Cu , ^{105}Rh , ^{198}Au , ^{199}Ag or ^{177}Lu
Half life extenders	Polyethylene glycol Glycerol Glucose
Fluorescent probes	Phycoerythrin (PE) Allophycocyanin (APC) Alexa Fluor 488 Cy5.5
Biologics	fluorescent proteins such as <i>Renilla</i> luciferase, GFP immune modulators or proteins, such as cytokines, e.g., an interferon toxins an immunoglobulin or antibody or antibody variable region half life extenders such as albumin or antibody variable regions or peptides that bind to albumin
Chemo-therapeutics	Taxol 5-FU Doxorubicin Idarubicin

TL1a-binding proteins comprising antibody binding domains of the disclosure are readily screened for biological activity, e.g., as described below.

Binding Assays

One form of assay is an antigen binding assay, e.g., as described in Scopes (In: Protein purification: principles and practice, Third Edition, Springer Verlag, 1994). Such a method generally involves labeling the TL1a-binding protein and contacting it with immobilized antigen. Following washing to remove non-specific bound protein, the amount of label and, as a consequence, bound protein is detected. Of course, the TL1a-binding protein can be immobilized and the antigen labeled. Panning-type assays, e.g., as described or exemplified herein can also be used. Alternatively, or additionally, surface plasmon resonance assays can be used.

In one example, a binding assay is performed with peptide comprising an epitope of TL1a. In this way, TL1a-binding proteins that bind to a specific region of TL1a are selected. Inhibition of Interaction of TL1a and DR3

Methods for identifying TL1a-binding proteins that inhibit interaction of TL1a and DR3 will be apparent to the skilled artisan based on the description herein.

For example, DR3 (e.g., 2 $\mu\text{g/ml}$ of DR3) is immobilized on a surface and contacted with TL1a (e.g., 1 $\mu\text{g/ml}$ TL1a) and with a TL1a-binding protein to be tested (in the case of controls, an isotype matched control antibody is added). A reduced level of TL1a bound to the DR3 in the presence of the TL1a-binding protein compared to in the absence of the TL1a-binding protein indicates that the TL1a-binding protein inhibits binding of TL1a to DR3. The assay can also be performed with immobilized TL1a to which DR3 is contacted. The assay can also be performed with labeled TL1a and/or DR3 to assist with detection.

In some examples, various concentrations of the TL1a-binding protein are tested and the concentration at which 50% of the maximum inhibition of binding of TL1a to DR3 by the TL1a-binding protein is determined (this concentration is known as EC_{50}). In one example, the EC_{50} of a TL1a-binding protein of the disclosure is less than about 5 nM or 4 nM or 3.5 nM or 3 nM or 2.5 nM or 2.3 nM or 1 nM or 0.5 nM. In one example, the EC_{50} is less than 3 nM. In one example, the EC_{50} is less than 2.5 nM. In one example, the EC_{50} is less than 1 nM. In one example, the EC_{50} is less than 0.5 nM.

In some examples, the maximal inhibition of interaction of TL1a and DR3 is assessed by determining the level of interaction of TL1a and DR3 in the presence and absence of a TL1a-binding protein. The level of inhibition of interaction of TL1a and DR3 in the presence of the protein is then expressed as a percentage of the level of interaction in the absence of the protein. In one example, the TL1a-binding protein inhibits at least about 80% of interaction between TL1a and DR3. For example, the percentage inhibition is at least about 84% or 85% or 90% or 93% or 94% or 95%. In one example, the percentage inhibition is at least about 93%. In one example, the percentage inhibition is at least about 94%. In one example, the percentage inhibition is assessed using a polypeptide comprising DR3 fused to an antibody Fc region. In one example, the TL1a-binding protein is used at a concentration of about 10 $\mu\text{g/ml}$.

Selective Inhibition of Interaction of TL1a and DR3

Methods for identifying TL1a-binding proteins that inhibit interaction of TL1a and DR3 but not TL1a and DcR3 will be apparent to the skilled artisan and/or described herein.

For example, DcR3 (e.g., 2 $\mu\text{g/ml}$ DcR3) is immobilized on a surface and contacted with TL1a (e.g., 1 $\mu\text{g/ml}$ TL1a) and a TL1a-binding protein to be tested (in the case of negative controls, an isotype-matched control antibody is used). The level of TL1a bound to DcR3 is then determined. A reduced level of TL1a bound to the DcR3 in the presence of the TL1a-binding protein compared to in the absence of the TL1a-binding protein indicates that the TL1a-binding protein inhibits binding of TL1a to DcR3. A similar level of bound TL1a in the presence or absence of the TL1a-binding protein indicates that the TL1a-binding protein does not inhibit interaction of TL1a and DcR3. The assay can also be performed with immobilized TL1a to which DcR3 is contacted.

In some examples, various concentrations of the TL1a-binding protein are tested to determine the level of inhibition of TL1a interaction with DcR3 at different concentrations.

In some examples, the maximal inhibition of interaction of TL1a and DcR3 is assessed by determining the level of interaction of TL1a and DcR3 in the presence and absence of a TL1a-binding protein. The level of inhibition of interaction of TL1a and DcR3 in the presence of the protein is then expressed as a percentage of the level of interaction in the absence of the protein. In one example, the TL1a-binding

protein at a concentration of 10 $\mu\text{g/ml}$ inhibits 25% or less of the interaction between TL1a and DcR3. For example, the percentage inhibition is 20% or less or 18% or less or 15% or less or 12% or less or 10% or less or 8% or less or 5% or less. In one example, the percentage inhibition is about 18% or less. In one example, the percentage inhibition is about 7% or less. In one example, the percentage inhibition is about 5% or less. In one example, the percentage inhibition is assessed using a polypeptide comprising DcR3 fused to an antibody Fc region. In one example, the TL1a-binding protein is used at a concentration of about 10 $\mu\text{g/ml}$.

Neutralization Assays

Methods for identifying TL1a-binding proteins that neutralize TL1a activity through DR3 will also be apparent to the skilled artisan, e.g., based on the description herein.

For example, DR3-expressing cells (e.g., TF-1 cells) (e.g., about 7×10^4 cells to 8×10^4 cells (e.g., 7.5×10^4 cells) are contacted with TL1a and a protein synthesis inhibitor (e.g. cycloheximide) in the presence or absence of a TL1a-binding protein to be tested. The level of apoptosis of the cells is then assessed, e.g., by detecting activation of caspases or propidium iodide uptake or other known assays. A TL1a-binding protein that reduces the level of apoptosis compared to the level of apoptosis in the absence of the TL1a-binding protein is considered to inhibit TL1a activity or neutralize TL1a activity through DR3.

In some examples, various concentrations of the TL1a-binding protein are tested to determine the level of neutralization at different concentrations. In some examples, the concentration at which 50% of the maximum inhibition of apoptosis by the TL1a-binding protein is determined (this concentration is known as EC_{50}). In one example, the EC_{50} of a TL1a-binding protein of the disclosure is 3 nM or less, for example, about 2.5 nM or less, such as about 2.4 nM or less, for example, about 2 nM or less, such as about 1.5 nM or less, such as about 1 nM or less. In one example, the EC_{50} of a TL1a-binding protein of the disclosure about 0.99 nM or less. In one example, the EC_{50} of a TL1a-binding protein of the disclosure is about 0.6 nM or less. In one example, the EC_{50} of a TL1a-binding protein of the disclosure is about 0.4 nM or less.

The ability of a TL1a-binding protein of the disclosure to neutralize TL1a-activity can also be assessed by determining their ability to reduce cytokine secretion by immune cells. For example, PBMCs are contacted with Concanavalin A in the presence or absence of a TL1a-binding protein. The level of secretion of a TL1a-induced cytokine (e.g., interferon γ or IL-13) is then assessed, e.g., using an ELISA. Reduction of cytokine secretion in the presence of the TL1a-binding protein compared to in the absence of the protein indicates that the TL1a-binding protein neutralizes TL1a activity.

In some examples, various concentrations of the TL1a-binding protein are tested to determine the level of reduction in cytokine secretion at different concentrations. In some examples, the concentration at which 50% of the maximum inhibition of cytokine secretion by the TL1a-binding protein is determined (this concentration is known as EC_{50}). For example, the EC_{50} for inhibiting secretion of interferon- γ is 4 nM or less, such as 3 nM or less or 2.5 nM or less or 2 nM or less. In another example, the EC_{50} for inhibition of secretion of IL-13 is 15 nM or less, such as 10 nM or less, for example, 5 nM or less, such as 1 nM or less. For example, the EC_{50} for inhibition of secretion of IL-13 is 0.5 nM or less.

Other assays for determining neutralization of TL1a activity include determining the level of proliferation, migration and tube formation of endothelial cells in the presence and

absence of the TL1a-binding protein. For example, endothelial cells are cultured in an extracellular matrix, e.g., Matrigel™, and the level of migration and/or tube formation is determined, e.g., using microscopy. An increase in migration and/or tube formation in the presence of the TL1a-binding protein compared to in the absence of the TL1a-binding protein indicates that the TL1a-binding protein neutralizes TL1a activity through DR3.

In vivo Matrigel™ plug assays can also be performed, e.g., essentially as described in Bagley et al., *Cancer Res* 63: 5866, 2003.

Additional assays include assessing the ability of a TL1a-binding protein to reduce or prevent interferon γ secretion from peripheral blood T cells and/or NK cells stimulated with IL-12 and/or IL-18 or in Fc γ R activated monocytes.

In Vivo Assays

TL1a-binding proteins of the disclosure can also be assessed for therapeutic efficacy in an animal model of a condition, e.g., a TL1a-mediated condition. For example, the TL1a-binding protein is administered to a model of inflammatory bowel disease or colitis (e.g., dextran sodium sulphate (DSS)-induced colitis or CD45Rb adoptive transfer model of colitis (e.g., Kanai et al., *Inflamm. Bowel Dis.* 12: 89-99, 2006). In another example, a TL1a-binding protein is administered to a model of multiple sclerosis, e.g., EAE models in which a mouse or rat is immunized with a myelin sheath protein or peptide derived therefrom (e.g., MOG, MBP or PLP) and an immune response is generated against the protein thereby inducing a model of multiple sclerosis. Exemplary EAE models are reviewed in, for example Tsunoda and Fujinami, *J. Neuropathol. Exp. Neurol.* 55: 673-686, 1996. The TL1a-binding protein can also or alternatively be tested in a model of arthritis e.g., a SKG strain of mouse (Sakaguchi et al., *Nature* 426: 454-460, 1995), rat type II collagen arthritis model, mouse type II collagen arthritis model or antigen induced arthritis models (Bendele *J. Musculoskel. Neuron. Interact.* 1: 377-385, 2001) and/or a model of inflammatory airway disease (for example, OVA challenge or cockroach antigen challenge), or in a model of inflammatory uveitis for example interphotoreceptor retinoid binding protein immunization-induced uveoretinitis (Caspi, *Curr Protoc Immunol Chapter 15*: unit 15.6, 2003).

The ability of a TL1a-binding protein of the disclosure to neutralize TL1a activity can also or alternatively be assessed in a model of graft-versus-host-response, e.g., in which splenocytes from one animal are injected into an allogeneic animal (e.g., a MHC or HLA unmatched animal).

Epitope Mapping Assays

In another example, the epitope bound by a protein described herein is mapped. Epitope mapping methods will be apparent to the skilled artisan. For example, a series of overlapping peptides spanning the TL1a sequence or a region thereof comprising an epitope of interest, e.g., peptides comprising 10 to 15 amino acids are produced. The TL1a-binding protein is then contacted to each peptide or a combination thereof and the peptide(s) to which it binds determined. This permits determination of peptide(s) comprising the epitope to which the TL1a-binding protein binds. If multiple non-contiguous peptides are bound by the protein, the protein may bind a conformational epitope.

Alternatively, or in addition, amino acid residues within TL1a are mutated, e.g., by alanine scanning mutagenesis, and mutations that reduce or prevent protein binding are determined. Any mutation that reduces or prevents binding of the TL1a-binding protein is likely to be within the epitope bound by the protein. A form of this method is exemplified herein.

A further method involves binding TL1a or a region thereof to an immobilized TL1a-binding protein of the disclosure and digesting the resulting complex with proteases. Peptide that remains bound to the immobilized protein are then isolated and analyzed, e.g., using mass spectrometry, to determine their sequence.

Affinity Assays

Optionally, the dissociation constant (K_d) or association constant (K_a) or equilibrium constant (K_D) of a TL1a-binding protein for TL1a or a peptide comprising an epitope thereof is determined. These constants for a TL1a-binding protein are in one example measured by a radiolabeled or fluorescently-labeled TL1a binding assay. This assay equilibrates the protein with a minimal concentration of labeled TL1a in the presence of a titration series of unlabeled TL1a. Following washing to remove unbound TL1a, the amount of label is determined.

Affinity measurements can be determined by standard methodology for antibody reactions, for example, immunoassays, surface plasmon resonance (SPR) (Rich and Myszkka *Curr. Opin. Biotechnol* 11: 54, 2000; Englebienne *Analyst*. 123: 1599, 1998), isothermal titration calorimetry (ITC) or other kinetic interaction assays known in the art.

In one example, the constants are measured by using surface plasmon resonance assays, e.g., using BIAcore surface plasmon resonance (BIAcore, Inc., Piscataway, NJ) with immobilized TL1a or a region thereof. Exemplary SPR methods are described in U.S. Pat. No. 7,229,619.

In one example, a TL1a-binding protein as described herein according to any example has a K_D for TL1a of 100 nM or less, such as 50 nM or less, for example, 20 nM or less, for example, 10 nM or less or 6 nM or less. For example, a TL1a-binding protein has a K_D of 5.5 nM or less. For example, a TL1a-binding protein has a K_D of 5 nM or less. For example, a TL1a-binding protein has a K_D of 4 nM or less.

Half Life Assays

Some TL1a-binding proteins encompassed by the disclosure have an improved half-life, e.g., are modified to extend their half-life compared to TL1a-binding proteins that are unmodified. Methods for determining a TL1a-binding protein with an improved half-life will be apparent to the skilled person. For example, the ability of a TL1a-binding protein to bind to a neonatal Fc receptor (FcRn) is assessed. In this regard, increased binding affinity for FcRn increased the serum half-life of the TL1a-binding protein (see for example, Kim et al., *Eur. J. Immunol.* 24: 2429, 1994).

The half-life of a TL1a-binding protein of the disclosure can also be measured by pharmacokinetic studies, e.g., according to the method described by Kim et al, *Eur. J. of Immunol.* 24: 542, 1994. According to this method radiolabeled TL1a-binding protein is injected intravenously into mice and its plasma concentration is periodically measured as a function of time, for example at 3 minutes to 72 hours after the injection. The clearance curve thus obtained should be biphasic, that is, an alpha phase and beta phase. For the determination of the in vivo half-life of the TL1a-binding protein, the clearance rate in beta-phase is calculated and compared with that of the wild type or unmodified TL1a-binding protein.

Stability Assays

Stability of a TL1a-binding protein of the disclosure can be assessed by any of a variety of assays. For example, the TL1a-binding protein is exposed to a condition, e.g., heat or acid or stored for a period of time (e.g., 1 month) at room temperature. Aggregation of the TL1a-binding protein can then be assessed by determining turbidity (with an increase

in turbidity following exposure to the condition indicating instability), size exclusion chromatography, non-reducing gel electrophoresis or a binding or neutralization study described herein.

The TL1a-binding protein of the disclosure or nucleic acid encoding same or cell expressing same (syn. active ingredient) is useful for parenteral, topical, oral, or local administration, aerosol administration, or transdermal administration, for prophylactic or for therapeutic treatment.

Formulation of a TL1a-binding protein or nucleic acid encoding same or cell expressing same to be administered will vary according to the route of administration and formulation (e.g., solution, emulsion, capsule) selected. An appropriate pharmaceutical composition comprising TL1a-binding protein or nucleic acid encoding same or cell expressing same to be administered can be prepared in a physiologically acceptable carrier. A mixture of TL1a-binding proteins can also be used. For solutions or emulsions, suitable carriers include, for example, aqueous or alcoholic/aqueous solutions, emulsions or suspensions, including saline and buffered media. Parenteral vehicles can include sodium chloride solution, Ringer's dextrose, dextrose and sodium chloride, lactated Ringer's or fixed oils. A variety of appropriate aqueous carriers are known to the skilled artisan, including water, buffered saline, polyols (e.g., glycerol, propylene glycol, liquid polyethylene glycol), dextrose solution and glycine. Intravenous vehicles can include various additives, preservatives, or fluid, nutrient or electrolyte replenishers (See, generally, Remington's Pharmaceutical Science, 16th Edition, Mack, Ed. 1980). The compositions can optionally contain pharmaceutically acceptable auxiliary substances as required to approximate physiological conditions such as pH adjusting and buffering agents and toxicity adjusting agents, for example, sodium acetate, sodium chloride, potassium chloride, calcium chloride and sodium lactate. The TL1a-binding protein of this disclosure can be lyophilized for storage and reconstituted in a suitable carrier prior to use according to art-known lyophilization and reconstitution techniques.

The optimum concentration of the active ingredient(s) in the chosen medium can be determined empirically, according to procedures well known to the skilled artisan, and will depend on the ultimate pharmaceutical formulation desired.

The dosage ranges for the administration of the TL1a-binding protein of the disclosure are those large enough to produce the desired effect. For example, the composition comprises a therapeutically or prophylactically effective amount of the TL1a-binding protein or nucleic acid encoding same or cell expressing same.

As used herein, the term "effective amount" shall be taken to mean a sufficient quantity of the TL1a-binding protein, nucleic acid or cells to induce/increase or inhibit/reduce/prevent signaling of TL1a in a subject. The skilled artisan will be aware that such an amount will vary depending on, for example, the TL1a-binding protein, nucleic acid or cells and/or the particular subject and/or the type or severity of a condition being treated. Accordingly, this term is not to be construed to limit the disclosure to a specific quantity, e.g., weight or number of TL1a-binding proteins, nucleic acids or cells.

As used herein, the term "therapeutically effective amount" shall be taken to mean a sufficient quantity of TL1a-binding protein, nucleic acid or cells to reduce or inhibit one or more symptoms of a condition.

As used herein, the term "prophylactically effective amount" shall be taken to mean a sufficient quantity of

TL1a-binding protein, nucleic acid or cells to prevent or inhibit or delay the onset of one or more detectable symptoms of a condition.

The dosage should not be so large as to cause adverse side effects, such as hyper viscosity syndromes, pulmonary edema, congestive heart failure, and the like. Generally, the dosage will vary with the age, condition, sex and extent of the disease in the patient and can be determined by one of skill in the art. The dosage can be adjusted by the individual physician in the event of any complication. Dosage can vary from about 0.1 mg/kg to about 300 mg/kg, e.g., from about 0.2 mg/kg to about 200 mg/kg, such as, from about 0.5 mg/kg to about 20 mg/kg, in one or more dose administrations daily, for one or several days.

In one example, the TL1a-binding protein is administered at a dosage of between about 1 mg/kg to about 15 mg/kg. In one example, the TL1a-binding protein is administered at a dosage of between about 2 mg/kg to about 10 mg/kg. In one example, the TL1a-binding protein is administered subcutaneously or intravenously.

In some examples, the TL1a-binding protein or other active ingredient is administered at an initial (or loading) dose which is higher than subsequent (maintenance doses). For example, the binding molecule is administered at an initial dose of between about 1 mg/kg to about 30 mg/kg. The binding molecule is then administered at a maintenance dose of between about 0.0001 mg/kg to about 1 mg/kg. The maintenance doses may be administered every 7 to 35 days, such as, every 14 or 21 or 28 days.

In some examples, a dose escalation regime is used, in which a TL1a-binding protein or other active ingredient is initially administered at a lower dose than used in subsequent doses. This dosage regime is useful in the case of subject's initially suffering adverse events

In the case of a subject that is not adequately responding to treatment, multiple doses in a week may be administered. Alternatively, or in addition, increasing doses may be administered.

One or more TL1a-binding proteins of the disclosure can be administered to an individual by an appropriate route, either alone or in combination with (before, simultaneous with, or after) another drug or agent. For example, the TL1a-binding protein of the disclosure can also be used in combination with proteins, e.g., a TNF antagonist, an anti-IL-12/23 antibody, an anti-inflammatory, a corticosteroid, methotrexate or a painkiller. The TL1a-binding protein of the disclosure can be used as separately administered compositions given in conjunction with antibiotics and/or antimicrobial agents.

It will be appreciated by those skilled in the art that the TL1a-binding proteins of the disclosure may be introduced into a subject by administering an expression construct of the disclosure or a cell expressing a TL1a-binding protein of the disclosure. A variety of methods can be used for introducing a nucleic acid encoding the antibody into a target cell in vivo. For example, the naked nucleic acid may be injected at the target site, may be encapsulated into liposomes, or may be introduced by way of a viral vector.

The following assays can be performed with a TL1a-binding protein of the disclosure, e.g., a TL1a-binding protein conjugated to a detectable label as discussed herein. Detection of TL1a with an assay described herein is useful for diagnosing or prognosing a condition.

An immunoassay is an exemplary assay format for diagnosing a condition in a subject or detecting TL1a in a sample. The disclosure contemplates any form of immunoassay, including Western blotting, enzyme-linked immu-

nosorbent assay (ELISA), fluorescence-linked immunosorbent assay (FLISA), competition assay, radioimmunoassay, lateral flow immunoassay, flow-through immunoassay, electrochemiluminescent assay, nephelometric-based assays, turbidometric-based assay, and fluorescence activated cell sorting (FACS)-based assays.

One form of a suitable immunoassay is, for example, an ELISA or FLISA.

In one form such an assay involves immobilizing a TL1a-binding protein of the disclosure onto a solid matrix, such as, for example a polystyrene or polycarbonate microwell or dipstick, a membrane, or a glass support (e.g. a glass slide). A test sample is then brought into direct contact with the TL1a-binding protein and TL1a in the sample is bound or captured. Following washing to remove any unbound protein in the sample, a protein that binds to TL1a at a distinct epitope is brought into direct contact with the captured TL1a. This detector protein is generally labeled with a detectable reporter molecule, such as for example, an enzyme (e.g. horseradish peroxidase (HRP), alkaline phosphatase (AP) or β -galactosidase) in the case of an ELISA or a fluorophore in the case of a FLISA. Alternatively, a second labeled protein can be used that binds to the detector protein. Following washing to remove any unbound protein the detectable reporter molecule is detected by the addition of a substrate in the case of an ELISA, such as for example hydrogen peroxide, TMB, or toluidine, or 5-bromo-4-chloro-3-indol-beta-D-galactopyranoside (x-gal). Of course, the immobilized (capture) protein and the detector protein may be used in the opposite manner.

The level of the antigen in the sample is then determined using a standard curve that has been produced using known quantities of the marker or by comparison to a control sample.

The assays described above are readily modified to use chemiluminescence or electrochemiluminescence as the basis for detection.

As will be apparent to the skilled artisan, other detection methods based on an immunosorbent assay are useful in the performance of the disclosure. For example, an immunosorbent method based on the description supra using a radio-label for detection, or a gold label (e.g. colloidal gold) for detection, or a liposome, for example, encapsulating NAD⁺ for detection or an acridinium linked immunosorbent assay.

In some examples of the disclosure, the level of TL1a is determined using a surface plasmon resonance detector (e.g., BIAcore™, GE Healthcare, Piscataway, N.J.), a flow through device, for example, as described in U.S. Pat. No. 7,205,159; a micro- or nano-immunoassay device (e.g., as described in US20030124619); a lateral flow devices (e.g., as described in US20040228761 or US20040265926); a fluorescence polarization immunoassay (FPIA e.g., as described in U.S. Pat. No. 4,593,089 or U.S. Pat. No. 4,751,190); or an immunoturbidimetric assay (e.g., as described in U.S. Pat. No. 5,571,728 or U.S. Pat. No. 6,248,597).

Samples and Control Samples

As will be apparent to the skilled artisan, some of the examples described herein require some degree of quantification to determine the level of TL1a. Such quantification may be determined by the inclusion of a suitable control sample in an assay of the disclosure.

In one example, a suitable control sample is a sample that is derived from a healthy subject or a normal subject.

In the present context, the term “healthy subject” shall be taken to mean an individual who is known not to suffer from a condition associated with TL1a, e.g., an inflammatory condition.

The term “normal subject” shall be taken to mean an individual having a normal level of TL1a in a sample compared to a population of individuals.

The disclosure also contemplates the control sample as being a data set obtained from a normal and/or healthy subject or a population of normal and/or healthy subjects.

In one example, a method of the disclosure additionally comprises determining the level of TL1a in a control sample, e.g., using a method described herein.

In one example, a sample from the subject and a control sample are assayed at approximately or substantially the same time.

In one example, the sample from the subject and the control sample are assayed using the same method of the disclosure as described herein in any one or more examples to allow for comparison of results.

Exemplary conditions that can be treated/prevented/diagnosed/prognosed by performing a method of the disclosure include autoimmune diseases, inflammatory conditions, and conditions characterized by insufficient angiogenesis.

In one example, the condition is an autoimmune disease.

Exemplary conditions include inflammatory bowel disease (IBD), irritable bowel syndrome (IBS), Crohn's disease, ulcerative colitis, diverticular disease, systemic lupus erythematosus, rheumatoid arthritis, juvenile chronic arthritis, spondyloarthropathies, systemic sclerosis (scleroderma), idiopathic inflammatory myopathies (dermatomyositis, polymyositis), Sjogren's syndrome, systemic vasculitis, sarcoidosis, demyelinating diseases of the central and peripheral nervous systems such as multiple sclerosis, idiopathic polyneuropathy, autoimmune or immune-mediated eye disease such as autoimmune uveitis and uveitis associated with various vasculitides, autoimmune or immune-mediated skin diseases including bullous skin diseases, erythema multiforme and contact dermatitis, psoriasis, allergic diseases of the lung such as asthma, airway hypersensitivity, eosinophilic pneumonia, chronic obstructive pulmonary disease (COPD), idiopathic pulmonary fibrosis and hypersensitivity pneumonitis, atherosclerosis or graft versus host disease.

In one example, the condition is arthritis, e.g., rheumatoid arthritis, polyarthritis, osteoarthritis or a spondyloarthropathy. In this regard, Bull et al., *J. Exp. Med.* 205: 2457, 2008 have shown that antibodies that antagonize TL1a are useful for the treatment of rheumatoid arthritis. The selective nature of the TL1a-binding proteins of the disclosure make them useful for treating rheumatoid arthritis.

In one example, the condition is multiple sclerosis. In this regard, US20090317388 shows that mice deficient in TL1a do not develop experimental autoimmune encephalomyelitis (EAE), which is an accepted model of multiple sclerosis.

In one example, the inflammatory condition is an inflammatory mucosal condition, e.g., an inflammatory disease of the bowel (e.g., inflammatory bowel disease, Crohn's disease or ulcerative colitis), or an inflammatory disease of the lung (e.g., airway hyperreactivity or asthma).

In one example, the condition is inflammatory bowel disease and/or colitis. In this regard, Takedatsu et al., *Gastroenterology* 135: 552, 2008 have shown that antibodies that antagonize TL1a are useful for the treatment of colitis.

In one example, the condition is asthma or airway hypersensitivity or chronic obstructive pulmonary disease (COPD).

In another example, the condition is an inflammatory skin disease (e.g., an autoimmune or immune-mediated skin disease), e.g., a bullous skin diseases, erythema multiforme, contact dermatitis. Alternatively, the skin disease is psoriasis.

Exemplary conditions characterized by insufficient angiogenesis include cardiovascular disease, autoimmune conditions (e.g., rheumatoid arthritis, psoriatic arthritis, systemic lupus erythematosus (SLE) and systemic sclerosis), antineutrophil cytoplasmic antibodies (ANCA)-associated vasculitis, ischemia (including ischemia resulting from a transplant) or necrosis.

The disclosure additionally comprises a kit comprising one or more of the following:

- (i) a TL1a-binding protein of the disclosure or expression construct(s) encoding same;
- (ii) a cell of the disclosure; or
- (iii) a pharmaceutical composition of the disclosure.

In the case of a kit for detecting TL1a, the kit can additionally comprise a detection means, e.g., linked to a TL1a-binding protein of the disclosure.

In the case of a kit for therapeutic/prophylactic use, the kit can additionally comprise a pharmaceutically acceptable carrier.

Optionally a kit of the disclosure is packaged with instructions for use in a method described herein according to any example.

The disclosure includes the following non-limiting Examples.

Example 1: Materials and Methods

In the following examples, reference to a position of a residue is a reference to the position in the relevant sequence as set forth herein, unless otherwise indicated.

1.1 HEK293/pTT5 Expression System

For all transfections involving the HEK293E/pTT5 expression system (Durocher et al., *Nucl. Acids Res.*, 30: E9, 2002), HEK293E cells were cultured in complete cell growth media (1 L of F17 medium (Invitrogen), 9 ml of Pluronic F68 (Invitrogen), 2 mM Glutamine containing 20% (w/v) Tryptone NI (Organotechnie) with Geneticin™ (50 mg/ml, Invitrogen) at 50 µl/100 ml culture). At the day before transfection, the cells were harvested by centrifugation and re-suspended in fresh media without Geneticin™. The next day, DNA was mixed with a commercial transfection reagent and the DNA transfection mix added to the culture drop-wise. The culture was incubated overnight at 37° C., 5% CO₂ and 120 rpm without Geneticin™. The next day 12.5 ml of Tryptone and 250 µl of Geneticin™ were added per 500 ml culture. The culture was incubated at 37° C., 5% CO₂ and 120 rpm for seven days, then the supernatants were harvested and purified.

1.2 TL1a Protein

Human TL1a was purchased (Peprotech and Genscript: both *E. coli*-expressed) or produced in the mammalian HEK293E/pTT5 expression system, using a DNA expression construct coding for the extracellular domain (ECD) of human TL1a with an N-terminally located HIS and FLAG tag (SEQ ID NO: 1). Culture supernatant containing the secreted TL1a protein was harvested by centrifugation at 2000 g for 10 mins to remove the cells. The TL1a protein was purified from the supernatant using a HisTrap™ HP column (GE Healthcare). The eluted protein was buffer-exchanged into PBS using a HiLoad 16/60 Superdex 200 prep grade column (GE Healthcare) and ~70 kDa fraction

was separated by gel filtration on a HiLoad 26/60 Superdex 200 prep grade column (GE Healthcare).

For phage display experiments, recombinant human TL1a (Peprotech, Genscript HEK293E-derived) was biotinylated using an EZ-link Sulfo-NHS-LC-biotin kit (Pierce) at a 3:1 ratio of biotin:TL1a. Free biotin was removed from the protein preparation by dialysis against PBS using a Slide-A-Lyzer dialysis cassette with a 3.5 kDa molecular weight cut-off.

TL1a was also produced with a tag that allowed for single site biotinylation. This TL1a protein was expressed in HEK-293 cells and purified as described previously. It was then biotinylated using an enzyme that selectively incorporates a biotin on the TL1a protein.

1.3 Phage Display

Antibodies that bind specifically to TL1a were isolated from a naive phagemid library comprising more than 10×10¹⁰ individual human FAb fragments.

Anti-TL1a antibodies were isolated from the phage display library over the course of several panning 'campaigns' (i.e. discrete phage display experiments with different reagents or panning conditions). The general protocol followed the method outlined by Marks and Bradbury (*Methods Mol. Biol.* 248: 161-176, 2004)

Each phage display campaign involved three rounds of panning. For each round, 1×10¹³ phage particles were blocked by mixing 1:1 with blocking buffer (5% skim milk in phosphate buffered saline (PBS) pH 7.4) and incubating for 1 hr at room temperature. The blocked phage library was then pre-depleted for streptavidin binders by incubation for 45 mins with 100 µL of streptavidin-coupled Dynabeads (Invitrogen), which were blocked as described for the library. The beads (and streptavidin binders attached to them) were discarded after the incubation step.

Recombinant human TL1a antigen was prepared for panning by capture onto the surface of streptavidin-coupled Dynabeads (Invitrogen). To achieve this, 10 to 100 pmol of biotinylated TL1a was incubated with 100 µl of beads for 45 mins at room temperature. The resulting TL1a-bead complexes were washed with PBS to remove free TL1a and then used in the subsequent panning reaction.

Library panning was conducted by mixing the blocked and pre-depleted library with the TL1a-bead complexes in a 1.5 mL microcentrifuge tube and rotating for 2 hrs at room temperature. Non-specifically bound phage were removed using a series of washes. Each wash involved pulling the bead complexes from the solution onto the tube wall using a magnetic rack, aspirating the supernatant and then re-suspending the beads in fresh wash buffer. This was repeated a number of times with either PBS wash buffer (PBS with 0.5% skim milk) or PBS-T wash buffer (PBS with 0.05% TWEEN-20 (Sigma) and 0.5% skim milk). Phage that remained bound after the washing process were eluted from the TL1a-bead complexes by incubation with 0.5 mL of 100 mM triethylamine (TEA) (Merck) for 20 mins at room temperature. The eluted 'output' phage were neutralized by adding 0.25 mL of 1 M Tris-HCl pH 7.4 (Sigma).

At the end of the first and second rounds of panning, the output phage were added to a 10 mL culture of exponentially growing TG1 *E. coli* (yeast-tryptone (YT) growth media) and allowed to infect the cells by incubating for 30 mins at 37° C. without shaking, then with shaking at 250 rpm for 30 mins. The phagemids encoding the phage display output were then rescued as phage particles following a standard protocol (Marks and Bradbury, supra).

At the end of the third panning round TG1 cells were infected with output phage, but the cells were plated on solid

YT growth media (supplemented with 2% glucose and 100 µg/mL carbenicillin) at a sufficient dilution to produce discrete *E. coli* colonies. These colonies were used to inoculate 1 mL liquid cultures to allow expression of FAB fragments for use in screening experiments.

1.4 SPR-Based Screening of FAbs for TL1a Binding

Each individual *E. coli* colony was used to express a FAB that could be screened for TL1a binding activity. Colonies were inoculated into 1 mL YT starter cultures (supplemented with 100 µg/mL carbenicillin and 2% glucose) in 96-well deepwell plates (Costar) and incubated overnight at 30° C. with shaking at 650 rpm. These starter cultures were diluted 1:50 into a 1 mL expression culture (YT supplemented with 100 µg/mL carbenicillin only) and grown to an optical density of 0.8 to 1.0 at 600 nm. FAB expression was induced by adding isopropyl-beta-D-thiogalactopyranoside to a final concentration of 1 mM. Cultures were incubated at 20° C. for 16 hrs.

FAB samples were prepared by harvesting cells by centrifugation (2500 g, 10 mins) and performing a periplasmic extraction. The cell pellet was resuspended in 75 µL of extraction buffer (30 mM Tris-HCl, pH 8.0, 1 mM EDTA, 20% Sucrose) and shaken at 1000 rpm for 10 mins at 4° C. Extract preparation was completed by adding 225 µL of H₂O, shaking at 1000 rpm for 1 hr and clearing the extract by centrifugation at 2500 g for 10 mins. The supernatants were recovered, filtered through Acroprep 100 kDa molecular-weight cutoff plates (Pall Corporation) and stored at 4° C. until required for further experiments.

The FAB samples were screened for TL1a-binding activity using a surface plasmon resonance (SPR) assay. High-throughput SPR screening was conducted using a BIAcore 4000 Biosensor (GE Healthcare) in a single concentration analyte pass assay. Approximately 10,000 RU of antiV5 antibody (Invitrogen cat #R960CUS) was immobilized on a CM5 Series S Sensor chip, using standard amine coupling chemistry at pH 5.5 on spots 1, 2, 4 & 5 of each of the four flow cells leaving spot 3 unmodified. The running buffer used was HBS-EP+ (GE Healthcare) and all interactions measured at 25° C. and data collection rate set to 10 Hz. Crude periplasmic preparations of V5-tagged FAbs, were diluted two-fold in running buffer before capturing at a flow rate of 10 µl/min for 100 sec (typically around 200RU of FAB was captured) on spot 1 or 5 of each flow cell. Following a short stabilization period, human TL1a trimer was passed over all spots of all four flow cells simultaneously at a flow rate of 30 µl/min for 100 sec. Dissociation of the interaction was measured for 100 sec prior to regeneration back to the antiV5 antibody using a 30 sec pulse of 100 mM phosphoric acid. Generated sensorgrams were referenced against an adjacent antiV5 antibody spot for each flow cell, and fitted using a 1:1 Langmuir equation to determine k_a , k_d and K_D .

The data from the SPR screening process was used to select potential TL1a-binders. FAB k_d values were ranked and up to 200 of the strongest binders were submitted for DNA sequence analysis. FAB with unique sequences were selected for conversion to full-length human IgG 1 antibodies.

1.5 Variable Region Sequencing

DNA sequencing was conducted by the Applied Genetic Diagnostics group at the Melbourne University Department of Pathology (Melbourne, Australia). Phagemid DNA (~500 ng) was mixed with 5 pmols of the appropriate primer for sequencing either the Fab V_H or V_L chain. Sequencing reactions were conducted using a BigDye® Terminator v3.1 Cycle Sequencing Kit (Applied Biosystems) according to

manufacturer's instructions. Samples were analyzed by capillary separation on 3130xl Genetic Analyzers (Applied Biosystems). Sequence chromatogram data was analyzed using the Chromas Lite software package (Technelysium, Brisbane, QLD, Australia). Sequence text files were translated to amino acid sequences and analyzed using the SeqAgent software package (Xoma, Berkely, CA, USA).

1.6 Construction of Vectors Expressing Antibodies

V_H amino acid chains were expressed with a human constant region (human IgG1 heavy chain C_H1 , hinge, C_H2 and C_H3 domains (e.g., SwissProt No. P01857)). This was achieved by back-translation of amino acid sequences into DNA sequences which were synthesized de novo by assembly of synthetic oligonucleotides. Following gene synthesis the whole sequence was subcloned into the multiple cloning site of the pTT5 heavy chain vector (Durocher et al., *Nucl. Acids Res.* 30: E9, 2002). V_L amino acid chains were expressed with a human kappa light chain constant region (SwissProt No. P01834.1) or a human lambda light chain constant region (SwissProt No. P0CG05.1) by subcloning the sequence into the multiple cloning site of the pTT5 light chain vector.

Alternatively, FAB V_H and V_L sequences were amplified directly from phagemid DNA by PCR and subcloned into IgG expression vectors. PCR reactions were carried out using a Platinum PCR Supermix kit (Invitrogen) as per manufacturer's instructions. Approximately 50 ng of template phagemid DNA encoding a FAB was mixed with 10 pmols of the appropriate forward and reverse PCR primers and the PCR Supermix reagent to a final volume of 50 µL. Each reaction used a sequence-specific forward primer combined with a generic V_H or V_L specific reverse primer. To facilitate cloning into IgG expression vectors, primers for V_H amplification were designed to add 5' BsiWI and 3' NheI restriction enzyme sites, primers for V_L -kappa amplification added 5' BssHII and 3' BsiWI sites and primers for V_L -lambda amplification added 5' BssHII and 3' AvrII sites.

PCRs were carried out in thin-walled PCR (96-well GENEAMP® PCR system 9700, Applied Biosystems, Scoresby, Victoria, Australia). Cycling conditions comprised an initial five minute denaturation step at 94° C., followed by 30 amplification cycles (denaturation at 94° C. for 30 seconds, followed by primer annealing at 55° C. for 30 seconds, then extension at 68° C. for 30 to 90 seconds), and a final extension step at 68° C. for seven minutes. The amplified V_H genes were purified using a Minelute PCR purification kit (Qiagen), digested with BsiWI and NheI enzymes (New England Biolabs) and re-purified using a Minelute Reaction Clean-up kit (Qiagen). Amplified V_L -kappa genes were prepared similarly, but digestion was carried out using BssHII and BsiWI. Amplified V_L -lambda genes were also prepared in the same way, but digested with BssHII and AvrII. The resulting V_H and V_L gene fragments were cloned into the multicloning site of the appropriate pTT5 heavy or light chain vector (either kappa or lambda light chain), as described above.

1.7 Expression and Purification of Antibodies

Heavy and light chain DNA were co-transfected into the HEK293/pTT5 expression system and cultured for seven days. The supernatants derived from these transfections were adjusted to pH 7.4 before being loaded onto a HiTrap Protein A column (5 ml, GE Healthcare). The column was washed with 50 ml of PBS (pH 7.4). Elution was performed using 0.1M citric acid pH 2.5. The eluted antibody was desalted using Zeba Desalting columns (Pierce) into PBS (pH 7.4). The antibody was analyzed using SDS-PAGE. The concentration of the antibody was determined using a BCA

assay kit (Pierce). For conversion between antibody concentrations in $\mu\text{g/ml}$ and molar concentrations an assumed molecular weight of 150 kDa was used for all antibodies.

1.8 TF-1 Cell Line Potency Assay

To determine which anti-TL1a antibodies functionally neutralize the biological activity of TL1a, antibodies were tested for their ability to neutralize TL1a-induced apoptosis in a TF-1 cell line. The TF-1 human erythroleukemic cell line (ATCC: CRL-2003) was maintained in culture under standard conditions. TF-1 cells ($7.5 \times 10^4/\text{well}$) were incubated in black-sided 96-well plates (Greiner) with recombinant human TL1a 100 ng/ml and cycloheximide 10 $\mu\text{g/ml}$ to induce apoptosis. Test antibodies at a concentration of 10 $\mu\text{g/mL}$ (66.7 nM) or less were added to the plates and incubated for 4 to 5 hours. Induction of apoptosis was then assessed using the Homogeneous Caspases Kit (Roche) according to manufacturer's instructions. In experiments to test the ability of antibodies to neutralize function of TL1a from cynomolgus and rhesus macaque, mouse, rat, guinea pig, pig or rabbit, the appropriate species TL1a was substituted for human TL1a in this protocol.

Data were normalized by expression as a percentage of maximum apoptosis (apoptosis levels achieved by recombinant human TL1a plus cycloheximide in the absence of anti-TL1a antibody).

1.9 Receptor Selectivity of Lead Antibodies

TL1a binds both to its cognate signaling receptor, DR3, and to a decoy receptor, DcR3, which also serves as a decoy receptor for TNF family members Fas-L and LIGHT.

Antibodies were assessed for their ability to inhibit binding of TL1a to its receptors in a competition ELISA. DR3/Fc Chimera (R&D Systems) or DcR3/Fc Chimera (R&D Systems) was coated onto a 96-well plate (Maxisorp, Nunc) at a concentration of 2 $\mu\text{g/ml}$. Serially diluted test antibodies were pre-incubated with single-site biotinylated recombinant human TL1a 1 $\mu\text{g/ml}$ for 30 minutes then added to the DR3/Fc or DcR3/Fc coated wells. Bound TL1a was detected using streptavidin-horseradish peroxidase 1:2000 (BD Pharmingen). Data were normalized by expression as a percentage of maximum binding of TL1a to receptor in the absence of anti-TL1a antibody.

An antibody described to inhibit TL1a activity (1B4; which has a V_H comprising a sequence set forth in SEQ ID NO: 119 and a V_L comprising a sequence set forth in SEQ ID NO: 120, which is described in WO 2009/064854 and US patent application publication US 200900280116) was included in assays for comparison.

1.10 Epitope Mapping

Epitope mapping was performed using alanine scanning experiments. Modeling analysis was carried out to determine probable exposed residues on TL1a. TL1a constructs were then designed in which each of these theoretically exposed residues was substituted with an alanine. These constructs were then expressed and supernatant from the expression cultures tested for protein expression and binding to anti-TL1a antibody C320 using SPR. A Ni-NTA sensor chip (Biacore) was used to capture the HIS tagged TL1a mutein from the supernatant of transfected HEK-293E cells. The mutein was captured on flow cell 2 (or 4) and then C320 was passed over flow cell 1+2 (or 3+4). The mutein expression (RU) was determined as the change in RU at the end of injection of the mutein. The C320 binding (RU) was determined as the change in RU determined at the end of the C320 injection.

TL1a constructs that expressed but had apparently lost binding to C320 were re-transfected and purified for testing by ELISA. ELISA plates were coated with polyclonal rabbit

anti-human TL1a (Peprotech) (1 $\mu\text{g/ml}$), DR3/Fc chimera (R&D systems) (2 $\mu\text{g/ml}$) or DcR3/Fc chimera (R&D systems) (1 $\mu\text{g/ml}$). TL1a muteins or unsubstituted TL1a were added to the plates at a concentration of 1 $\mu\text{g/ml}$. Bound TL1a was then detected with biotinylated polyclonal rabbit anti-human TL1a (Peprotech) (250 ng/ml) and streptavidin-horseradish peroxidase (BD Pharmingen) 1:2000. Constructs that could be detected by polyclonal anti-TL1a were deemed to have expressed and folded appropriately. Appropriately expressed and folded constructs that did not bind to one or either of the receptors were considered to be important for TL1a binding to that receptor. Binding of test antibodies to TL1a muteins was tested by direct ELISA. ELISA plates were coated with different TL1a muteins (1 $\mu\text{g/ml}$). Serially diluted test antibodies were then added, and bound antibodies were detected with anti-human IgG-horseradish peroxidase (Invitrogen) 1:2000. Binding of test antibodies to different TL1a isoforms was detected using the same method.

1.11 Creation of Membrane Bound TL1a (mbTL1a)-Expressing Cell Line

HEK 293 cells were electroporated with a vector containing a sequence encoding full length TL1a (SEQ ID NO: 123) and maintained in selective media (media supplemented with blasticidin 6 $\mu\text{g/ml}$). After multiple passages, the cells were tested for cell surface expression of TL1a by flow cytometry. 2.5×10^5 cells per well were plated into 96-well round bottom plates (Corning) and incubated with biotinylated polyclonal rabbit anti-human TL1a (Peprotech) on ice for 30 minutes. Samples were washed then incubated with streptavidin-FITC for a further 30 minutes on ice. Samples were then washed, resuspended and data acquired on a flow cytometer.

1.12 Flow Cytometry Detection

After stable cell surface expression of TL1a had been confirmed (as described in Example 1.10), antibodies C320, C321 and C323 were screened using both transfected and untransfected cells (2.5×10^5 cells per well in 96-well round bottom plates (Corning)) at decreasing concentrations starting from 10 $\mu\text{g/ml}$. Polyclonal goat anti-human IgG-FITC (Sigma) at a dilution of 1:200 was used as detection antibody. Anti-TL1a antibody 1B4 was included for comparison.

1.13 Inhibition of Cytokine Production

To further characterize antibodies described herein, their function on primary human cells was tested. To confirm endogenous TL1a production, PBMCs were isolated from buffy coats over a lymphoprep (Nycomed) gradient and cultured in 96-well tissue culture plates (Corning) with Concanavalin A at decreasing concentrations starting from 2 $\mu\text{g/ml}$. Plates were incubated overnight then supernatants were harvested and assayed for TL1a using the human TL1a ELISA kit (Peprotech).

Additionally, adherent cells were harvested and assessed for TL1a expression by flow cytometry using biotinylated anti-human TL1a (Peprotech) 1:100 and streptavidin-FITC (Zymed) 1:200.

1.14 Isoelectric focusing gel experiments

Isoelectric focusing gels were performed using the NOVEX® Xcell Surelock™ system (Life Technologies) according to manufacturer's instructions.

1.15 Protein A HPLC

Supernatants from HEK-293E cells transfected to transiently express antibodies were analysed by Protein A HPLC using a POROS® A/20 2.1x30 mm Id column (Applied Biosystems) connected to an Agilent 1100 chromatography system. The column was equilibrated with phosphate buff-

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ered saline (PBS) pH7.4, 0.2 ml of HEK-293E supernatant containing protein was loaded and the protein eluted with PBS adjusted to pH 2.2. The chromatograms at the wavelengths of 215 nm or 280 nm were integrated using the manufacturer's software and the area under the curve (AUC) reported.

1.16 Antibody Expression and Antigen Binding as Determined by SPR

Using a CM5 sensor chip (Biacore™) Protein A was coupled to the chip surface using amine coupling. Protein A was coupled on flow cell 1 and 2 (or alternatively 3 and 4) using a Biacore 3000. Cell supernatant for HEK-293 cells containing antibody were passed over the surface of flow cell 2, while buffer (HBS-EP) was passed over flow cell 1. At the end of injection of the supernatant the change in response units was measured. This value is reported as Protein A capture (SPR). The % expression is the Protein A capture (SPR) of the tested antibody as a percentage of the Protein A capture (SPR) of C320 in the same experiment. To determine if the antibody binds TL1a, the TL1a was then passed over the flow cell 1 and 2. The sensorgrams were double referenced (Flow cell 2 is subtracted from flow cell 1 and a buffer blank).

Example 2: Results of Phage Display

Phage display campaigns were conducted against recombinant human TL1a. Six discreet campaigns were conducted using recombinant bacterial-expressed recombinant TL1a. A low number of FAbs were isolated that bound to TL1a, with no neutralizing antibodies isolated.

Subsequent campaigns were conducted with mammalian-expressed TL1a. The percentage of FAbs isolated that bound TL1a was substantially higher using mammalian derived TL1a than bacterially derived TL1a. The number of FAbs that were shown to be positive for TL1a binding across the total phage display campaigns was in excess of 200. From these, 55 FAbs with unique sequences were identified, of which 29 were selected for conversion to full length IgG 1 antibody.

Example 3: Neutralization of TL1a Activity

The ability of full length IgG1 antibodies comprising FAbs isolated using phage display to inhibit or reduce TL1a-induced apoptosis was assessed as described above (Example 1.8). Under these conditions, 15/29 of the antibodies tested showed better than 50% inhibition of TL1a-induced apoptosis (FIG. 2). These antibodies included: C319, C320, C321, C323, C333, C334, C335, C336.

These data demonstrate that while antibodies can be isolated that bind to TL1a, only a limited subset of these antibodies have the specificity required to functionally inhibit TL1a activity.

Within the group of antibodies that demonstrated inhibition of recombinant human TL1a, the relative inhibition profile of each antibody was assessed using the EC₅₀ value (Table 4 and FIG. 3). Only antibodies C320, C321 and C323 had an inhibitory EC₅₀ value of 1 nM or below.

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TABLE 4

Antibody EC ₅₀ values for inhibition of TL1a-induced apoptosis			
Antibody Designation	Variable Heavy Chain (SEQ ID NO)	Variable Light Chain (SEQ ID NO)	EC ₅₀ (nM)
C336	2	6	4.05
C334	10	14	22.14
C333	18	22	2.34
C323	26	30	0.99
C321	34	38	0.56
C320	42	46	0.31
C319	50	54	11.17
1B4	74	75	3.37

Example 4: Receptor Selectivity

Antibodies C320, C321 and C323 inhibit interaction of TL1a with DR3 (FIG. 4A) but do not inhibit interaction of TL1a with DcR3 (FIG. 4B). Using the same assay, antibodies C319, C333, C334 and C336 were shown to demonstrate the same selective neutralization.

Using a similar assay the binding of antibodies C320 and derivatives thereof described 5 below, particularly C320-168 and C320-179, were shown to inhibit interaction of TL1a with DR3 (FIG. 4C) with an EC₅₀ 1 nM or less. These antibodies also did not inhibit interaction of TL1a with DcR3 (FIG. 4D) when tested at concentrations ranging from 0.1 µg/mL to 100 µg/mL. These results contrast with those of antibodies C300-25 and 1B4. In this regard, antibody C300-25 is a rat monoclonal antibody produced by the inventors.

TABLE 5

Antibody EC ₅₀ values for inhibition of TL1a interaction with receptors DR3 and DcR3		
Antibody	EC ₅₀ (nM)	
	DR3	DcR3
C320	0.97	DNI
C320-179	0.72	DNI
C320-168	0.97	DNI
C300-25	13.5	9.84
1B4	17.8	26.5

DNI—did not inhibit

DR3 and DcR3 have been shown to compete for binding to TL1a. Therefore it was unexpected that antibodies have been isolated with specificity, targeting an epitope on TL1a that is neutralizing for DR3, but not DcR3. These results indicate that antibodies described herein are capable of preventing TL1a activity through its cognate receptor DR3 without disturbing either the natural antagonistic function of DcR3 or a homeostatic balance of DcR3 binding. Without being bound by any theory or mode of action, such selective inhibition may be biologically relevant because DcR3 regulates the amount of free Fas-L and LIGHT available for binding to their cognate signaling receptors (Fas and H-VEEM, respectively). Consequently, inhibiting the interaction between TL1a and DcR3 could increase the amount of Fas-L and LIGHT bound by DcR3 thus decreasing the amount available for signaling through their cognate receptors. As Fas-mediated killing plays a role in cancer surveillance, potential downstream consequences of increasing the amount of DcR3 to bind to Fas-L could include increased susceptibility to cancer. Again, without being bound by

theory or mode of action, isolating antibodies that specifically inhibit interaction of TL1a and DR3, but not DcR3, could be advantageous in treating disease but without compromising safety.

Example 5: Epitope Mapping

Amino acid substitutions were introduced into the sequence of soluble TL1a (SEQ ID NO: 202) to generate a series of TL1a muteins.

Using the SPR-based mutein analysis assay described in Example 1.10, TL1a muteins in supernatants from transfected HEK-293E cells were tested for expression levels and binding to C320. The results are presented in Table 6.

TABLE 6

Expression and binding of soluble human TL1a and muteins thereof to antibody C320					
Amino acid substitution	Mutein Expression (RU)	C320 Binding (RU)	Amino acid substitution	Mutein Expression (RU)	C320 Binding (RU)
Wild Type	3244.9	125.6	A55S	2933.1	33.2
L1A	3264.3	115	A55L	2746.5	-7.6
K2A	3094.4	152	A55R	2307.7	-12.3
Q4A	3366.2	120	A55G	1991.2	-5.8
E5A	3086.5	360.4	A55D	2386.5	-21.1
F6A	2948.3	240.2	T57A	3335.6	42.7
P8A	3455.9	124.3	K58A	3241.5	4.3
S9A	3594.2	94.5	N59A	3382.9	73.8
H10A	3372.5	122.7	R60A	3405.8	116.5
Q11A	3572.9	71.4	N62A	3264.9	117.4
Q12A	3098.5	193.8	T64A	3083.6	289.8
V13A	3213.9	167.8	N65A	3226.8	156.6
Y14A	2883.5	401	K66A	3183.3	27.1
P16A	2938	225	F67A	3216.1	304.7
L17A	2904.1	194.5	L69A	3389.8	212.6
R18A	3102.5	132.2	E72A	3171.5	136.2
D20A	3247.6	154.6	S73A	3129.3	146
G21S	2915.3	184.8	R85A	3047.1	-33.5
G21L	3424.4	72.8	M87A	3161.2	46.1
G21R	3197	116.2	S89A	3600.3	10.8
G21A	3483	85	E90A	3129.7	1089.1
G21D	3032	158.6	E93A	3181.6	343.9
D22A	3185.6	169.8	I94A	3125.1	352.3
R32A	3103.7	-36.5	R95A	3634.8	28.8
T34A	3278.7	389.2	Q96A	3256.5	149.4
P35A	3385	-28.6	R99A	3351.3	105.2
T36A	2771.9	486.1	P100A	2898.5	199.5
Q37A	2789.3	319.2	K102A	3420.8	155.7
H38A	3133	147.4	D104A	2982.9	533.2
F39A	2509.8	445.5	S105A	3425.8	127.1
K40A	3213.5	55.5	D115A	3648.5	110.1
N41A	2967.5	248.5	S116A	2962.4	282.1
Q42A	3073	175.7	Y117A	3323.8	189.8
F43A	3245.8	109.3	P118A	3254.9	161.7
P44A	3180.1	200.4	E119A	3263.7	278.5
A45S	3407.3	61.5	P120A	3143	164.8
A45L	3087.7	35.6	Q122A	3189	148.5
A45R	3167	107.2	S135A	3109.4	95.4
A45G	3385.9	77.1	F138A	3496.7	21
A45D	3395.1	38.9	S148A	2883.9	228.7
H47A	3466.4	159.1	Q150A	3151.1	193.5
H50A	3249.3	132.4	E151A	3796.5	90.8
E51A	3136.5	33.8	K154A	3610.3	86.7
L52A	3105.5	52.7	S160A	1272.2	901.6
G53S	3238.5	19	D161A	3391.4	223.4
G53L	1299.7	-3.9	I162A	3139.2	72.4
G53R	3796.7	7.2	S163A	3054	481
G53A	3652.4	-4.8	Y167A	2942.2	70.7
G53D	2130.3	-6.1	T168A	3335.1	2.6
			K169A	3005	69.7
			E170A	3418.7	16

Muteins that expressed well (Mutein expression greater than 2000 RU) but failed to bind antibody C320 (TL1a binding less than 17 RU) were selected for further analysis.

Muteins with substitutions at amino acids G53 and A55 expressed poorly and had reduced binding to polyclonal anti-TL1a, suggesting that these are important residues for TL1a structural integrity. The other muteins all expressed adequately, bound polyclonal anti-TL1a and ran equivalently to wild-type TL1a on both reducing and non-reducing SDS-PAGE gels. These characteristics all suggest that the muteins were appropriately expressed and folded into TL1a proteins.

The muteins exhibited differential binding patterns to the different monoclonal anti-TL1a antibodies described herein. Antibodies C320, C320-168 and C320-179 did not detectably bind to the R32A and R85A muteins or to a mutein containing both of these substitutions (FIGS. 5A and B). C320 also showed negligible binding to the P35A, T168A and E170A muteins while C320-168 and C320-179 had reduced binding to these muteins but not to the same extent as C320. In contrast, TL1a-binding antibodies 1B4 and 16H2 showed binding to the muteins R32A, R85A, P35A, T168A at greater than 50% of the wild-type TL1a, indicating that amino acids at these positions are not crucial to the binding of 1B4 and 16H2 to TL1a.

Given that the antibodies C320, C320-168 and C320-179 inhibit the binding of TL1a to DR3 but not DcR3, the antibodies are likely bind to amino acids involved in the binding of DR3 to TL1a and disrupt this interaction. DR3, but not DcR3, had significantly decreased binding to the R32A and R85A muteins, suggesting that these residues are involved with TL1a binding to DR3 (FIG. 5C).

The binding site of the antibodies C320, C320-168 and C320-179 on TL1a is illustrated in FIG. 5D. As can be seen from the diagram, the residues R32 and R85, while being distant in primary amino acid sequence are located adjacent to each other in the protein's tertiary structure. Hence, these can be considered key residues within the epitope on TL1a to which the antibodies C320, C320-168 and C320-179 bind.

Example 6—Characterization of Antibodies Binding to Cell-Surface TL1a

TL1a, like most TNF-ligand superfamily members, is present as a cell surface form that is cleaved to generate soluble TL1a and as for other members the membrane form of TL1 (mbTL1a) has been shown to have biological function. Therefore to neutralize the activity of TL1a it would be advantageous to have antibodies that bind strongly to both membrane and soluble TL1a. Antibodies that displayed functional inhibition of TL1a were screened by flow cytometry on a human cell line transfected stably with a membrane anchored TL1a (produced as described in Example 1.11).

Antibodies C320, C321, C323 and 1B4 all bound to the TL1a transfected cell line (FIG. 6) with EC₅₀s of 0.5 to 2 nM, but not normal untransfected cells.

To further characterize antibodies described herein, their function on primary human cells was tested. PBMCs were isolated as described in Example 1.12. Using the assay conditions described in Example 1.13 these cells were shown to produce both mbTL1a (FIG. 7A) and soluble TL1a (FIG. 7B).

PBMCs were stimulated as described in Example 1.13 and assayed for cytokine production in the presence of anti-TL1a antibodies. Since no exogenous TL1a was added to the cell culture, the effects of anti-TL1a antibodies were

due to neutralization of endogenous TL1a. Antibodies, C320 and C323 demonstrated good inhibition of endogenous TL1a-induced cytokine production by PBMCs (FIG. 8). This inhibition was apparent on cytokines typically produced by Th1 cells (IFN- γ FIG. 8A) and Th2 cells (IL-13; FIG. 8B). These data demonstrate that these antibodies are likely to be capable of inhibiting TL1a produced from primary human cells, and that these antibodies could be used as diagnostic reagents to detect TL1a on cells or in sera.

Example 7: Generating Improved Variants of C320

The C320 antibody was further optimized through alterations to the antibody's sequence with the aim of yielding a positive effect on the antibody's biophysical properties whilst having minimal impact on the potency of C320.

For example, alterations that enhance the expression level of the antibody with concomitantly increased production levels are desirable. Removal of a N-linked glycosylation site in the V_H through amino acid substitution to reduce product heterogeneity may further enhance C320. Substitution of amino acid residues with the potential to impact the stability of the antibody through oxidation or isomerization during purification or storage with amino acids that do not undergo such transitions (Wang et al., *Journal of Pharmaceutical Sciences* 96:1-26, 2006) may further improve C320. Substitution of rare or non-germ-line C320 sequences which may potentially contribute to immunogenicity with those of lower predicted immunogenicity could further improve C320. Such changes are described in more detail as follows. Enhancing the Expression of C320

A database of human antibody germline amino acid sequences was interrogated by Basic Local Alignment Search Tool searches (BLAST: Altschul et al., *J Mol Biol* 215: 403-410, 1990) using the C320 V_H and V_L amino acid sequences. This identified twenty of the most homologous human antibody germline sequences to those of the C320 V_H and C320 V_L , respectively. Aligning the human germline sequences with those of C320 enabled the identification of residues present in C320 but not common in the majority of the germlines. The most common amino acid from the germline sequences was substituted into the C320 amino acid sequence. Table 7 lists the amino acid substitutions and the resultant impact on the expression level of each antibody from a transient transfection in HEK-293E cells.

TABLE 7

Effect of amino acid substitutions on expression and potency			
Antibody Designation	Amino Acid Substitution (chain)	% Expression Level relative to C320	Potency TF-1 EC-50 (pM)
C320	N/A	100	233
C320-2	A16S (Heavy)	156	159
C320-3	T41P (Heavy)	338	428
C320-4	N73D (Heavy)	DNE	N/A
C320-5	A76T (Light)	231	89
C320-6	L81Q (Light)	98	67
C320-135	T41P (Heavy) & A76T (Light)	288	462

Amino acid substitutions are relative to SEQ ID NO: 42 for heavy chain and SEQ ID NO: 46 for light chain.

Two substitutions, present in C320-3 and C320-5 raised the expression level of the antibody greater than two-fold and whilst minimally impacting the potency of the antibody in a TF-1 potency assay. A further antibody incorporating both substitutions, C320-135, had largely equivalent potency and maintained the improved expression level.

C320-4 contained a N73D (N72D according to the numbering system of Kabat) substitution which aimed to

improve expression while also attempted to remove an N-linked glycosylation motif. However, the N73D substitution abolished antibody expression, suggesting N73 is desirable for expression of the antibody.

To further improve C320 expression the CDRs of C320 were grafted onto other frameworks of antibodies possessing a known crystal structure. This unusual approach was adopted as antibodies of known crystal structure usually possess a matched V_H : V_L pair. To select antibodies with suitable V_H and V_L frameworks BLAST searches of a database of crystal structures were performed to identify antibodies of similar amino acid sequence to C320. BLAST searches using the C320 V_H were used to identify the 100 most homologous antibody sequences with known crystal structures. Similarly, the 100 most homologous antibody sequences to the C320 V_L with known crystal structures were identified. Crystal structures appearing in both heavy- and light chain lists were deemed suitable acceptor frameworks for CDR grafting. These were 1TZG, 1RHH, 2DD8, 2JB5, 3FKU, 3GBM, 3IYW, 3LMJ, and 3P30. The CDRs of C320 were then used to replace the CDR sequences present in each of the above antibody sequences. An alignment of each of the sequence containing the C320 CDR regions is given in FIG. 1D (heavy chain) and FIG. 1G (light chain). The antibody heavy and light chains were then paired as follows and assessed for protein expression and TL1a binding and the results listed in Table 8.

TABLE 8

Results of CDR grafting on expression and TL1a binding					
Antibody Designation	Heavy Chain	Light Chain	Protein A capture (RU)	AUC Protein A HPLC (215 nm)	TL1a binding (RU)
C320	C320	C320	779	2764	282
mock	N/A	N/A	25	376	N/D
C320-7	1TZG	1TZG	187	912	N/D
C320-8	1RHH	1RHH	437	1718	29
C320-9	2DD8	2DD8	2235	8687	305
C320-10	2JB5	2JB5	2098	11406	368
C320-11	3FKU	3FKU	129	637	N/D
C320-12	3GBM	3GBM	100	697	N/D
C320-13	3IYW	3IYW	1566	4488	524
C320-14	3LMJ	3LMJ	570	1125	171
C320-15	3P30	3P30	1063	2192	318
C320-16	1TZG	C320	2527	11604	653
C320-17	1RHH	C320	2241	11980	683
C320-18	2DD8	C320	1528	5786	408
C320-19	2JB5	C320	1805	8018	465
C320-20	3FKU	C320	986	1909	310
C320-21	3GBM	C320	39	428	N/D
C320-22	3IYW	C320	2107	8622	665
C320-23	3LMJ	C320	1119	2044	324
C320-24	3P30	C320	1023	2353	311
C320-25	C320	1TZG	376	1062	42
C320-26	C320	1RHH	438	1012	81
C320-27	C320	2DD8	2272	9795	578
C320-28	C320	2JB5	2176	11771	623
C320-29	C320	3FKU	1549	3195	355
C320-30	C320	3GBM	1288	3356	455
C320-31	C320	3IYW	1061	2806	381
C320-32	C320	3LMJ	892	1888	331
C320-33	C320	3P30	1527	4473	521

Note:

The sequences of heavy and light chains of these antibodies are listed in FIG. 1D and 1G. RU is response units—a measure of binding to the surface using SPR and AEC is area under the curve.

When the CDRs of C320 were grafted onto different antibody frameworks a large number of the resultant antibodies expressed at a level above that of the C320 antibody. The antibody C320-16, in which the heavy chain C320 CDRs were grafted on the antibody 1TZG framework and was paired with the C320 light chain, expressed 3-fold better

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than C320. This experiment also demonstrated that it is possible to change the isotype of the antibody and retain protein expression and binding to TL1a using this approach, as some of the light chain antibody frameworks into which the C320 light chain CDRs were grafted were of the kappa isotype as opposed to the lambda isotype present in C320.

To further improve the expression of the antibody, amino acid substitutions were introduced into the CDR3 region of the variable heavy chain of C320 and the antibodies transfected into HEK-293E cells and screened for expression levels using Protein A HPLC and Protein A capture using surface plasmon resonance (SPR). The results are described in Table 9.

TABLE 9

Effect of HCDR3 substitutions on expression				
Antibody Designation	Heavy Chain Substitution (relative to SEQ ID NO: 42)	Light Chain Substitution (relative to SEQ ID NO: 46)	Protein A capture (RU)	AUC (Protein A HPLC) (280 nm)
C320-0	N/A	N/A	6413	487
mock	N/A	N/A	198	N/D
C320-53	E99S	A76T	6831	581
C320-54	E99H	A76T	6594	444
C320-55	E99L	A76T	6797	540
C320-56	E99D	A76T	5823	343
C320-57	E99Y	A76T	7789	670
C320-58	E99P	A76T	5679	335
C320-59	E99Q	A76T	8534	1149
C320-60	E99K	A76T	8293	839
C320-61	V100A	A76T	8381	863
C320-62	V100S	A76T	8423	803
C320-63	V100H	A76T	8828	751
C320-64	V100L	A76T	8288	910
C320-65	V100D	A76T	8816	1003
C320-66	V100Y	A76T	7750	570
C320-67	V100P	A76T	7831	660
C320-68	V100Q	A76T	8170	715
C320-69	V100K	A76T	8741	706
C320-70	P101A	A76T	5022	234
C320-71	P101S	A76T	5083	268
C320-72	P101H	A76T	5403	230
C320-73	P101L	A76T	5518	283
C320-74	P101D	A76T	5756	297
C320-75	P101Y	A76T	5084	212
C320-76	P101Q	A76T	5036	263
C320-77	P101K	A76T	5702	309
C320-78	D102A	A76T	8249	684
C320-79	D102S	A76T	8964	1102
C320-80	D102H	A76T	8957	830
C320-81	D102L	A76T	7969	515
C320-82	D102Y	A76T	8135	558
C320-83	D102P	A76T	7765	462
C320-84	D102Q	A76T	7826	635
C320-85	D102K	A76T	8487	890
C320-86	T103A	A76T	10209	1882
C320-87	T103S	A76T	8236	567
C320-88	T103H	A76T	4934	227
C320-89	T103L	A76T	7580	822
C320-90	T103D	A76T	8694	1106
C320-91	T103Y	A76T	3838	130
C320-92	T103P	A76T	4792	219
C320-93	T103Q	A76T	4104	164
C320-94	T103K	A76T	3592	149
C320-95	A104S	A76T	4811	240
C320-96	A104H	A76T	4882	251
C320-97	A104L	A76T	4557	189
C320-98	A104D	A76T	5371	279
C320-99	A104Y	A76T	6305	410
C320-100	A104P	A76T	5678	339
C320-101	A104Q	A76T	6634	508
C320-102	A104K	A76T	5826	438
C320-103	S105A	A76T	8159	825
C320-104	S105H	A76T	6664	426
C320-105	S105L	A76T	6193	357

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TABLE 9-continued

Effect of HCDR3 substitutions on expression				
Antibody Designation	Heavy Chain Substitution (relative to SEQ ID NO: 42)	Light Chain Substitution (relative to SEQ ID NO: 46)	Protein A capture (RU)	AUC (Protein A HPLC) (280 nm)
C320-106	S105D	A76T	7752	992
C320-107	S105Y	A76T	8209	1072
C320-108	S105P	A76T	6132	483
C320-109	S105Q	A76T	6767	465
C320-110	S105L	A76T	6999	452
C320-111	E105K	A76T	6919	500
C320-112	E107S	A76T	7713	631
C320-113	E107H	A76T	6723	459
C320-114	E107L	A76T	7739	839
C320-115	E107D	A76T	8505	1034
C320-116	E107Y	A76T	6465	375
C320-117	E107P	A76T	6699	400
C320-118	E107Q	A76T	8109	704
C320-119	E107K	A76T	8776	952

Note:

Substitutions in this table refer to those made into the C320 antibody

Several antibodies were generated that had significant expression levels above that of C320 as determined by capture on a Protein A surface. Several of the high expressing antibodies were re-transfected, purified by Protein A chromatography and their potency measured in a TF-1 assay. The results are listed in Table 10.

TABLE 10

Effect of HCDR3 substitutions on TL1a inhibition			
Antibody Designation	Heavy Chain Substitution (relative to SEQ ID NO: 42)	Light Chain Substitution (relative to SEQ ID NO: 46)	Potency TF-1 EC-50 (pM)
C320-0	N/A	N/A	233
C320-61	V100A	A76T	272
C320-63	V100H	A76T	1067
C320-65	V100D	A76T	546
C320-68	V100Q	A76T	535
C320-86	T103A	A76T	303
C320-87	T103S	A76T	356
C320-90	T103D	A76T	370
C320-103	S105A	A76T	455
C320-104	S105H	A76T	349
C320-106	S105D	A76T	1587
C320-112	E107S	A76T	242
C320-114	E107L	A76T	514
C320-115	E107D	A76T	321
C320-117	E107P	A76T	251
C320-119	E107K	A76T	4142

Note:

Substitutions in this table refer to those made into the C320 antibody

The potency of the resulting antibodies ranged from reduced compared to C320, such as C320-119, to levels equivalent to that of C320.

The light chain sequence of C320 was aligned against the germline sequence of highest homology, IGLV1-40*01. Differences in amino acid sequences in the CDR regions were identified, as shown in FIG. 9A. At a position in which the sequences differed a back substitution of the amino acid from the germline sequence into C320 was performed. The antibodies were expressed in HEK-293 cells, purified and the protein yield relative to C320 determined. A potency assay using TF-1 cells was used to characterize the inhibition profile of the antibodies. The results are displayed in Table 11.

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TABLE 11

Effect of germlining C320 on TF1a inhibition				
Antibody Designation	Heavy Chain Substitution (relative to SEQ ID NO: 42)	Light Chain Substitution (relative to SEQ ID NO: 46)	% Expression Level relative to C320	Potency TF-1 EC-50 (pM)
C320	N/A	N/A	100	233
C320-120	T41P	A23T	279	343
C320-121	T41P	D28N	239	1433
C320-122	T41P	L33Y	247	5480
C320-123	T41P	G34D	207	DNI
C320-124	T41P	Y53N	128	1613
C320-125	T41P	Y54S	259	13393
C320-126	T41P	P82A	187	539
C320-127	T41P	G95S	140	541
C320-128	T41P	T96S	134	328

Note:

Substitutions in this table refer to those made into the C320 antibody

The antibody C320-120 demonstrated a good level of expression and the highest potency of all the antibodies tested in this experiment. Some of the substitutions, such as those in C320-123 and C320-125 resulted in antibodies that expressed but no longer inhibited or weakly inhibited TL1a induced apoptosis on TF-1 cells. This suggested that the amino acids at the positions G34 (G32 according to the numbering system of Kabat) and Y54 (Y52 according to the numbering system of Kabat) of the variable light chain may be important in enabling the antibody to inhibit the activity of TL1a.

Removal of Putative Oxidation and Isomerization Sites from C320

Amino acid analysis of the variable heavy and light chain sequences identified several amino acids that may undergo oxidation or isomerization. Particular emphasis was placed on amino acids present in the CDRs of the antibody. Changes to these amino acids may, over time, alter the binding profile of the antibody. In the variable heavy chain M51 (M51 according to the numbering system of Kabat) was identified as a potential oxidation site along with the potential aspartate isomerization site D102 (D98 according to the numbering system of Kabat). In the light chain, D94 (D92 according to the numbering system of Kabat) was identified as a potential isomerization site. To reduce the potential impact of these predicted issues variants of C320 were produced containing conservative or semi-conservative amino acid substitutions in these positions. These substitutions and their impact on the potency of the resulting C320 variants are listed in Table 12.

TABLE 12

Effect of amino acid substitutions on expression and TL1a inhibition			
Antibody Designation	Amino Acid Substitution from C320 (chain)	% Expression Level relative to C-320	Potency TF-1 EC-50 (pM)
C320	N/A	100	233
C320-129	D102E (Heavy)	201	433
C320-130	M51L (Heavy)	185	528
C320-131	D94E (Light)	103	10680

Note:

Substitutions in this table refer to those made into the C320 antibody, with heavy chain substitutions relative to SEQ ID NO: 42 and light chain substitutions relative to SEQ ID NO: 46

The potential oxidation and isomerization site present in the heavy chain were successfully substituted with conservative amino acids which resulted in improved expression

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and retention of potency. The substitution of D94E in the light chain CDR region resulted in an antibody that expressed at similar levels to C320, but had significantly reduced potency, indicating that D94 of the light chain may be important in mediating the functional activity of C320. Removal of the N-Glycosylation Site from the C320 Heavy Chain

As an earlier attempt to remove the V_H glycosylation site NTS by substitution to DTS (N73D; N72D according to the numbering system of Kabat) was not successful, further attempts to disrupt the N-linked glycosylation motif, NX(S/T) (where X≠P) using a more comprehensive range of substitutions were made. Firstly, further residues were tested in place of the asparagine at position 73. Secondly the substitution T74P (T73P according to the numbering system of Kabat) was tested, since proline in this position is known to prevent N-linked glycosylation. A range of amino acids were also tested in place of S75 (S74 according to the numbering system of Kabat). The constructs tested in attempts to remove this glycosylation site and their impact on antibody expression levels are listed in Table 13.

TABLE 13

Effect of amino acid substitutions on expression levels			
Antibody Designation	Heavy Chain Substitution relative to SEQ ID NO: 42	Light Chain Substitution relative to SEQ ID NO: 46	AUC (Protein A HPLC)
C320-0	N/A	N/A	162
C320-120	T41P	A23T	710
mock	N/A	N/A	N/D
C320-138	N73S	A23T	87
C320-139	N73K	A23T	58
C320-140	N73H	A23T	N/D
C320-141	N73T	A23T	66
C320-142	N73Q	A23T	70
C320-143	N73G	A23T	N/D
C320-144	N73P	A23T	N/D
C320-145	N73L	A23T	54
C320-146	N73Y	A23T	N/D
C320-147	T74P	A23T	110
C320-148	S75L	A23T	115
C320-149	S75I	A23T	96
C320-150	S75A	A23T	194
C320-151	S75Y	A23T	88
C320-152	S75K	A23T	115
C320-153	S75E	A23T	110
C320-154	S75F	A23T	78
C320-155	S75H	A23T	118

Note:

Substitutions in this table refer to those made into the C320 antibody

The majority of the C320 variants tested exhibited lower levels of expression compared to C320, suggesting that substituting individual amino acids present in the NTS motif, in C320, reduces the expression of the antibody.

A result of earlier attempts to enhance the expression of C320 by grafting the C320 CDRs onto different antibody frameworks was the antibody designated C320-16, which used the frameworks of 1TZG. This antibody expressed well and bound TL1a (Table 7). The frameworks of 1TZG and C320-16 were devoid of the N-linked glycosylation sequence motif, as the RNTSI (SEQ ID NO: 203) at positions 72 through 76 of SEQ ID NO: 42 (71 through 75 according to the numbering system of Kabat) of C320 V_H sequence is replaced by ADRST (SEQ ID NO: 204) in the corresponding V_H sequence of 1TZG.

Variant C320-135 comprising expression enhancing substitutions V_H T41P and V_L A73T was subjected to further

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heavy chain engineering. C320 V_H sequence RNTSI (SEQ ID NO: 203) at positions 71 through 75 was replaced by ADRST (SEQ ID NO: 204) from the corresponding V_H sequence of 1TZG to produce antibody C320-163. This antibody, now devoid of an N-linked glycosylation site, expressed when transfected into HEK-293E cells and, when purified, was capable of functional inhibition of TL1a mediated apoptosis of TF-1 cells (Table 14).

Improved Variants of C320

The amino acid substitutions identified in previous experiments as conferring advantageous properties over C320 were subsequently combined and the resultant antibodies expressed, purified and assayed for potency. The sequences of these are listed in FIG. 9B (variable heavy) and FIG. 9C (variable light). Comparative potency and expression data of each antibody is listed in Table 14.

TABLE 14

Effect of amino acid substitutions on expression and TL1a inhibition		
Antibody Designation	% expression compared to C320	Potency TF-1 EC-50 (pM)
C320	100	233
C320-162	74	1053
C320-163	170	1160
C320-164	299	338
C320-165	380	240
C320-166	339	793
C320-167	344	239
C320-168	219	46
C320-169	N/A	477
C320-170	232	287
C320-171	102	DNI
C320-172	140	269
C320-179	161	96
C320-183	129	439

* Note:

DNI = did not inhibit

Comparing antibodies C320-168 (containing a N-linked glycan motif) with C320-163 and C320-170 (lacking a

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N-linked glycan motif) on an isoelectric focusing (IEF) gel demonstrated removal of charge heterogeneity from the antibody profile on removing the N-linked glycosylation site (FIG. 9D). C320-168 has 5 to 6 distinct charged isoforms compared to 1-2 isoforms visualized for C320-163 and C320-170. This reduction in charge heterogeneity is an advantageous property providing improved batch to batch consistency in large scale manufacturing processes (Wang et al., *Journal of Pharmaceutical Sciences* 96:1-26, 2006).

Of the antibodies tested, two were more potent than the parental C320 antibody—C320-168 and C320-179 (FIG. 9E). Both of these C320 variants include the light chain variable region substitution G24S (G25S according to the numbering system of Kabat) that was found to improve potency in TF-1 assays in combination with the other advantageous amino acids substitutions described previously.

Removal of Predicted MHC Class II Binding Peptides from Antibodies

The amino acid sequence of C320-168 variable heavy and light chain regions were analysed in silico using the software package Epibase™ (LONZA). This software predicts the propensity of peptide sequences present in C320-168 to bind to MHC Class II alleles. Such binding of peptides to MHC Class II alleles is an important step in a host immune response against the administered antibody.

A peptide sequence in the heavy chain of C320-168, GLEWMGWLNPNNGNT (SEQ ID NO: 205), was predicted to strongly bind to DRB1*0401. A peptide sequence in the light chain of C320-168, LLIYGYNRPSPGVPD (SEQ ID NO: 206), was predicted to strongly bind to DRB1*1101, DRB1*1104 and DRB1*1501.

The above listed peptide sequences and modified versions of these peptide sequences were synthesized and incubated with the corresponding MHC Class II protein and an ELISA performed to determine if the peptide formed a complex with the MHC Class II protein. The influenza peptide, PKYVKQNTLKLAT (SEQ ID NO: 207), was used as positive control in these assays to indicated the formation of a peptide:MHC Class II complex. The binding of each peptide to the MHC Class II allele is listed in Table 14.

TABLE 14

Binding of peptides to MHC Class II			
Peptide	MHC Class II Allele	Binding (% of positive control)	
GLEWMGWLNPNNGNT (SEQ ID NO: 205)	DRB1*0401	6	
LLIYGYNRPSPGVPD (SEQ ID NO: 206)	DRB1*1501	123	
LLIEGYNRPSPGVPD (SEQ ID NO: 208)	DRB1*1501	41	
LLIGGYNRPSPGVPD (SEQ ID NO: 209)	DRB1*1501	42	
LLIPGYNRPSPGVPD (SEQ ID NO: 210)	DRB1*1501	0	
LLIKGYNRPSPGVPD (SEQ ID NO: 211)	DRB1*1501	74	
LLIYGYNRPSPGVPD (SEQ ID NO: 212)	DRB1*1101	1	
LLIEGYNRPSPGVPD (SEQ ID NO: 213)	DRB1*1101	0	
LLIGGYNRPSPGVPD (SEQ ID NO: 214)	DRB1*1101	0	
LLIPGYNRPSPGVPD (SEQ ID NO: 215)	DRB1*1101	0	
LLIKGYNRPSPGVPD (SEQ ID NO: 216)	DRB1*1101	0	

TABLE 14-continued

Binding of peptides to MHC Class II		
Peptide	MHC Class II Allele	Binding (% of positive control)
LLIYGYNRP SGVPD (SEQ ID NO: 217)	DRB1*1104	1
LLIEGYNRP SGVPD (SEQ ID NO: 218)	DRB1*1104	0
LLIGGYNRP SGVPD (SEQ ID NO: 219)	DRB1*1104	1
LLIPGYNRP SGVPD (SEQ ID NO: 220)	DRB1*1104	0
LLIKGYNRP SGVPD (SEQ ID NO: 221)	DRB1*1104	0

Peptides that may be immunologically significant or warrant further investigation as good binders are considered to be those peptides with scores 15% of the positive control. Therefore the peptide GLEWMGWLNPNSGNT (SEQ ID NO: 205) has a low likelihood of being immunogenic and no attempt was made to modify this motif in C320 related antibodies.

The peptide LLIYGYNRP SGVPD (SEQ ID NO: 206) was shown to form a complex with MHC Class II protein DRB1*1501 (123% binding) but not DRB1*1101 or DRB1*1104. An attempt to reduce the formation of the peptide:MHC Class II complex was made by modifying the tyrosine residue at the fourth position of the peptide (corresponding to position 51 (position 49 according to the numbering system of Kabat) in the C320-168 variable light chain sequence) to one of either glutamic acid, glycine, proline, or lysine. The introduction of a glutamic acid or a glycine residue at this position reduced the complex formation by over 50% (Table 14). Introducing proline at this position prevented complex formation. Antibodies C320-172 and C320-179 incorporated substitution Y51E (Y49E according to the numbering system of Kabat), in their V_L region, whilst C320-183 incorporated the change Y51G (Y49G according to the numbering system of Kabat) in the V_L region. All three antibodies expressed and functionally inhibited TL1a, as illustrated in Table 13. Expression of antibody C320-171 incorporating substitution Y51E (Y49P according to the numbering system of Kabat) was detected, however, this antibody failed to functionally inhibit TL1a, indicating that not all amino acids are tolerated at position 49 in the light chain (Table 13).

Example 8—Generation of High Potency DR3-Selective Antibodies

New antibodies capable of neutralizing TL1a activity with high potency and capable of selective inhibition of TL1a-DR3-mediated activity are generated by focusing on the C320 antibody epitope on TL1a. For example, mutant variable regions and/or individual substitutions showing improved expression and/or TL1a-neutralization (e.g., as described in Example 7) are combined and re-tested to determine if the improvement is additive.

In other approaches antibodies that bind to the epitope bound by C320 are selected by any of a variety of techniques.

8.1 Selection from an Antibody Library Using the C320 Antibody

A phage display protocol is used where a first panning round is conducted using an antigen density (i.e. biotinylated

TL1a) of about 100 pmol and a TEA-based elution step as described previously. The second and third rounds use a reduced antigen density (e.g., about 50 pmol). Phage are eluted by adding the C320 IgG at a 10-fold molar excess and incubating the reactions at room temperature for 2, 5, 10 or 20 mins. The IgG is expected to specifically displace and elute phage expressing FAbs that bound to the C320 epitope. Non-specific binders and phage bound to other regions on the TL1a surface are less likely to elute under these conditions.

The washing regimen comprises six washes with M-PBS for round 1 and 2. For round 3 the washes are three washes with PBST and then three washes with PBS.

Eluted phage are used to infect TG1 *E. coli* for phagemid rescue or generation of colonies for screening as described for other phage display experiments.

8.2 Selection/Production of Antibodies Using Mutant TL1a

Using mutated versions of TL1a as panning reagents for phage display, antibodies that recognize an epitope similar to that of C320 may be obtained. A phage display library may be depleted of antibodies that recognize TL1a with an amino acid substitution of R32A and/or R85A. The library may then be panned against TL1a. The resultant isolated antibodies will likely bind to the residues R32 and/or R85.

Example 9: Affinity Maturation of C320

Antibody C320 is already a potent inhibitor of TL1a activity. However, potency may be enhanced using affinity maturation approaches. Enhanced potency frequently confers dosing and efficacy advantages.

Numerous methods for affinity maturation of antibodies are known in the art. Many of these are based on the general strategy of generating panels or libraries of variant proteins by mutagenesis followed by selection and/or screening for improved affinity. Mutagenesis is often performed at the DNA level, for example by error prone PCR (Thie et al., *Methods Mol. Biol.* 525: 309-322, xv, 2009), by gene shuffling (Kolkman and Stemmer, *Nat. Biotechnol.* 19: 423-428, 2001), by use of mutagenic chemicals or irradiation, by use of 'mutator' strains with error prone replication machinery (Greener et al., *In Vitro Mutagenesis Protocols*. (Humana press, NJ), 1996) or by somatic hypermutation approaches that harness natural affinity maturation machinery (Peled et al., *Annu. Rev. Immunol.* 26: 481-511, 2008). Mutagenesis is also performed at the RNA level, for example by use of Q β replicase (Kopsidas et al., *Immunol.*

Lett. 107: 163-168, 2006). Library-based methods allowing screening for improved variant proteins are based on various display technologies such as phage, yeast, ribosome, bacterial or mammalian cells, and are known in the art (Benhar, *Expert Opin. Biol. Ther.* 7: 763-779, 2007). Affinity maturation is also achieved by more directed/predictive methods for example by site-directed mutagenesis or gene synthesis guided by findings from 3D protein modeling (see for example U.S. Pat. No. 6,180,370 or 5,225,539).

Affinity maturation using ribosome display (Kopsidas et al., *BMC Biotechnol.* 7: 18, 2007) is performed using RNA encoding the V_L and V_H domains of a C320-related antibody in scFv format. A library of variants of this RNA is generated using Q β replicase (typically 1 to 3 changes per molecule) and this library is displayed and selected on ribosomes for binding to TL1a. Phenotype-genotype linkage is achieved because the RNA constructs used remain attached to the ribosome translating the functional scFv protein. C320-related scFv-RNA-ribosome complexes are panned against TL1a and isolated. The RNA is converted into DNA and incorporated into a bacterial expression system. Individual bacteria encoding a single scFv are then isolated and induced to express the scFv. Using a competition ELISA, scFvs with a higher affinity for TL1a than a C320-related ScFv are identified. These scFvs are converted into full length antibodies which are screened in the TF-1 apoptosis assay as described in Example 1.8 to determine improvement in inhibition of TL1a biological activity over C320.

Example 11: Efficacy of Anti-TL1a Antibodies in Animal Models of Colitis

Rodent cross-reactive anti-TL1a antibodies described herein were tested in rodent models of acute colitis induced by intrarectal administration of di- or tri-nitrobenzenesulfonic acid (D/TNBS) or oxazolone, and chronic colitis induced by administration of DSS in drinking water (as described in Wirtz et al., *Nat. Protoc.* 2: 541-546, 2007). DNBS and

oxazolone induce localized ulceration and inflammation. DSS administration induces robust generalized inflammation of the intestinal tract characterized by erosive lesions and inflammatory infiltrate. Symptoms of all these models usually include diarrhoea, occult blood, weight loss and occasionally rectal prolapse.

In a prophylactic model, antibody treatment was commenced at the start of administration of the colitis-inducing compound. In a therapeutic model, antibody treatment was commenced several days after commencement of induction. The effect of the treatment on weight, stool consistency and occult blood, as well as microscopic effects on epithelial integrity and degree of inflammatory infiltrate was determined. Daily clinical scoring was performed based on stool consistency and presence of occult blood giving a disease activity index (DAI) score.

Antibody C320-168 showed comparable efficacy to a standard of care compound when dosed prophylactically in DNBS- or oxazolone-induced colitis (FIGS. 10A-10C), and when dosed therapeutically in DSS-induced colitis (FIGS. 11A and 11B).

Example 12: Efficacy of Anti-TL1a Antibodies in Animal Models of Disease

Antibodies are additionally tested in a rodent model of multiple sclerosis (Racke, *Curr. Protoc. Neurosci. Chapter 9, Unit 9 7*, 2001). In this model either acute or chronic central demyelination is induced by administration of spinal cord homogenate or purified myelin peptides in adjuvant. This administration causes autoimmune destruction of the myelin sheath around the spinal cord neurons leading to lower limb weakness which may develop to paralysis, and is characterized by significant inflammatory infiltrate into the spinal cord. The acute form is monophasic and animals recover spontaneously. The chronic form resembles relapsing-remitting multiple sclerosis and consists of two or more episodes of hind limb weakness/paralysis.

SEQUENCE LISTING

Sequence total quantity: 235

SEQ ID NO: 1 moltype = AA length = 207
FEATURE Location/Qualifiers
source 1..207
mol_type = protein
organism = Homo sapiens

SEQUENCE: 1
DDDDKGSHHH HHHHHGSGSL VPRGSGSLKG QEFAPSHQQV YAPLRADGDK PRAHLTVVRQ 60
TPTQHFKNQF PALHWEHELG LAFTKNRMNY TNKFLIPES GDYFIYSQVT FRGMTSECSE 120
IRQAGRPNKP DSITVVITKV TDSYPEPTQL LMGTKSVCEV GSNWFQPIYL GAMFSLQEGD 180
KLMVNVSDIS LVDYTKEDKT FFGAFL 207

SEQ ID NO: 2 moltype = AA length = 117
FEATURE Location/Qualifiers
source 1..117
mol_type = protein
note = amino acid sequence of C336 VH
organism = synthetic construct

SEQUENCE: 2
EVQLVQSGAE VKKPGASVKV SCKASGYTFT GYYMHWVRQA PGQGLEWMGW INPNSGGTNY 60
AQKFQGRVTM TRDTSISTAY MELSLRLSDD TAVYYCASGG QTHLDVWGQG TTVTVSS 117

SEQ ID NO: 3 moltype = AA length = 5
FEATURE Location/Qualifiers
source 1..5
mol_type = protein
note = amino acid sequence of C336 HCDRI
organism = synthetic construct

SEQUENCE: 3
GYMH

-continued

SEQ ID NO: 4	moltype = AA length = 17	
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	mol_type = protein	
	note = amino acid sequence of C336 HCDR2	
	organism = synthetic construct	
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SEQ ID NO: 5	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
source	1..8	
	mol_type = protein	
	note = amino acid sequence of C336 HCDR3	
	organism = synthetic construct	
SEQUENCE: 5		
GGQTHLDV		8
SEQ ID NO: 6	moltype = AA length = 108	
FEATURE	Location/Qualifiers	
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	mol_type = protein	
	note = amino acid sequence of C336 VL	
	organism = synthetic construct	
SEQUENCE: 6		
DIQMTQSPSS LSASVGDVRT ITCQASQDIT DYLNWYQQR GKAPKLLIYD ASNLETGVPS		60
RPSGSGSGTY FTFTISLQP EDPATYYCQ YDNLPIFGQ GTRLEIKR		108
SEQ ID NO: 7	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	note = amino acid sequence of C336 LCDR1	
	organism = synthetic construct	
SEQUENCE: 7		
QASQDITDYL N		11
SEQ ID NO: 8	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	note = amino acid sequence of C336 LCDR2	
	organism = synthetic construct	
SEQUENCE: 8		
DASNLET		7
SEQ ID NO: 9	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	note = amino acid sequence of C336 LCDR3	
	organism = synthetic construct	
SEQUENCE: 9		
QQYDNLPIIT		9
SEQ ID NO: 10	moltype = AA length = 122	
FEATURE	Location/Qualifiers	
source	1..122	
	mol_type = protein	
	note = amino acid sequence of C334 VH	
	organism = synthetic construct	
SEQUENCE: 10		
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AQSPQGRVTI TADESTSTAY MELSSLRSED TAVYYCSTNS YSSSWYDAFD IWGQGMVTV		120
SS		122
SEQ ID NO: 11	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	note = amino acid sequence of C334 HCDR1	
	organism = synthetic construct	
SEQUENCE: 11		
SYTIS		5
SEQ ID NO: 12	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	

-continued

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SEQ ID NO: 13	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
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	mol_type = protein	
	note = amino acid sequence of C334 HCDR3	
	organism = synthetic construct	13
SEQUENCE: 13		
NSYSSSWYDA FDI		
SEQ ID NO: 14	moltype = AA length = 108	
FEATURE	Location/Qualifiers	
source	1..108	
	mol_type = protein	
	note = amino acid sequence of C334 VL	
	organism = synthetic construct	60
SEQUENCE: 14		
DVMTQSPAP LSVSPGERAT LSCRASQSSIS>NNLAWYQQMP GQAPRLLLYD ASTRATDIPA		108
RFSGSGSGSE FTLTISGLQS ADFAVYCCQ YNNWPLTFGG GTKLEIKR		
SEQ ID NO: 15	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	note = amino acid sequence of C334 LCDR1	
	organism = synthetic construct	11
SEQUENCE: 15		
RASQSSNNL A		
SEQ ID NO: 16	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	note = amino acid sequence of C334 LCDR2	
	organism = synthetic construct	7
SEQUENCE: 16		
DASTRAT		
SEQ ID NO: 17	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	note = amino acid sequence of C334 LCDR3	
	organism = synthetic construct	9
SEQUENCE: 17		
QQYNNWPLT		
SEQ ID NO: 18	moltype = AA length = 122	
FEATURE	Location/Qualifiers	
source	1..122	
	mol_type = protein	
	note = amino acid sequence of C333 VH	
	organism = synthetic construct	60
SEQUENCE: 18		
QVQLVQSGAE VKKPGSSVKV SCKASGGTFS SYTISWVRQA PGQGLEWMGG IIPFGTTNY		120
AQRFGQGRVTI TADESTSTAY MELSSLRSED TAIYYCSTNS YSSSWYDAFD IWGQGTMTVT		122
SS		
SEQ ID NO: 19	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	note = amino acid sequence of C333 HCDR1	
	organism = synthetic construct	5
SEQUENCE: 19		
SYTIS		
SEQ ID NO: 20	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	note = amino acid sequence of C333 HCDR2	
	organism = synthetic construct	
SEQUENCE: 20		

-continued

GIIPFGTTN YAQRFQG	17
SEQ ID NO: 21	moltype = AA length = 13
FEATURE	Location/Qualifiers
source	1..13
	mol_type = protein
	note = amino acid sequence of C333 HCDR3
	organism = synthetic construct
SEQUENCE: 21	
NSYSSSWYDA FDI	13
SEQ ID NO: 22	moltype = AA length = 108
FEATURE	Location/Qualifiers
source	1..108
	mol_type = protein
	note = amino acid sequence of C333 VL
	organism = synthetic construct
SEQUENCE: 22	
EIVLTQSPAT LSVSPGERAT LSCRASQSIT>NNLAWYQQLP GQAPRLLIYD ASTRATDIPA	60
RFSGTGSGSE FTLTISGLQS ADFAVYCCQ YNNWPLTFGG GTKVDIKR	108
SEQ ID NO: 23	moltype = AA length = 11
FEATURE	Location/Qualifiers
source	1..11
	mol_type = protein
	note = amino acid sequence of C333 LCDR1
	organism = synthetic construct
SEQUENCE: 23	
RASQSITNNL A	11
SEQ ID NO: 24	moltype = AA length = 7
FEATURE	Location/Qualifiers
source	1..7
	mol_type = protein
	note = amino acid sequence of C333 LCDR2
	organism = synthetic construct
SEQUENCE: 24	
DASTRAT	7
SEQ ID NO: 25	moltype = AA length = 9
FEATURE	Location/Qualifiers
source	1..9
	mol_type = protein
	note = amino acid sequence of C333 LCDR3
	organism = synthetic construct
SEQUENCE: 25	
QQYNNWPLT	9
SEQ ID NO: 26	moltype = AA length = 122
FEATURE	Location/Qualifiers
source	1..122
	mol_type = protein
	note = amino acid sequence of C323 VH
	organism = synthetic construct
SEQUENCE: 26	
QMQLVQSGAE VKKPGSSVKV SCKASGGTFS SYTISWVRQA PGQGLEWMGG IIPFGTTNY	60
AQRFQGRVTI TADESTSTAY MELSSLRSED TAIYYCSTNS YSSSWYDAFD IWGQGTMTV	120
SS	122
SEQ ID NO: 27	moltype = AA length = 5
FEATURE	Location/Qualifiers
source	1..5
	mol_type = protein
	note = amino acid sequence of C323 HCDR1
	organism = synthetic construct
SEQUENCE: 27	
SYTIS	5
SEQ ID NO: 28	moltype = AA length = 17
FEATURE	Location/Qualifiers
source	1..17
	mol_type = protein
	note = amino acid sequence of C323 HCDR2
	organism = synthetic construct
SEQUENCE: 28	
GIIPFGTTN YAQRFQG	17
SEQ ID NO: 29	moltype = AA length = 13
FEATURE	Location/Qualifiers

-continued

source	1..13 mol_type = protein note = amino acid sequence of C323 HCDR3 organism = synthetic construct	
SEQUENCE: 29		
NSYSSSWYDA FDI		13
SEQ ID NO: 30	moltype = AA length = 108	
FEATURE	Location/Qualifiers	
source	1..108 mol_type = protein note = amino acid sequence of C323 VL organism = synthetic construct	
SEQUENCE: 30		
AIQLTQSPSS LSASVGDRVT ITCRASQGIG SALAWYQQKP GKAPKLLIYD ASSLQSGVPS		60
RYSGSGSGTD FTLTISGLQP EDPATYYCQQ FSYYPLTFGG GTKLEIKR		108
SEQ ID NO: 31	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11 mol_type = protein note = amino acid sequence of C323 LCDR1 organism = synthetic construct	
SEQUENCE: 31		
RASQGIGSAL A		11
SEQ ID NO: 32	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7 mol_type = protein note = amino acid sequence of C323 LCDR2 organism = synthetic construct	
SEQUENCE: 32		
DASSLQS		7
SEQ ID NO: 33	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9 mol_type = protein note = amino acid sequence of C323 LCDR3 organism = synthetic construct	
SEQUENCE: 33		
QQFSYYPLT		9
SEQ ID NO: 34	moltype = AA length = 122	
FEATURE	Location/Qualifiers	
source	1..122 mol_type = protein note = amino acid sequence of C321 VH organism = synthetic construct	
SEQUENCE: 34		
QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYGISWVRQA PGQGLEWMGW ISAYNGNTNY		60
AQKLQGRVTM TTDSTSTAY MELRSLRSD TAVYICARDS HIYDILTGYD YWGQGLVTV		120
SS		122
SEQ ID NO: 35	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5 mol_type = protein note = amino acid sequence of C321 HCDR1 organism = synthetic construct	
SEQUENCE: 35		
SYGIS		5
SEQ ID NO: 36	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17 mol_type = protein note = amino acid sequence of C321 HCDR2 organism = synthetic construct	
SEQUENCE: 36		
WISAYNGNTN YAQLQG		17
SEQ ID NO: 37	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
source	1..13 mol_type = protein note = amino acid sequence of C321 HCDR3 organism = synthetic construct	

-continued

SEQUENCE: 37 DSHIYDILTG YDY		13
SEQ ID NO: 38 FEATURE source	moltype = AA length = 113 Location/Qualifiers 1..113 mol_type = protein note = amino acid sequence of C321 VL organism = synthetic construct	
SEQUENCE: 38 DVVMTQTPLS SPVTLGQPAS ISCHSSQSLV HSDGNTYLSW LQQRPGQPPR LLIYKISNRF SGVPDRFSGS GAGTDFTLKI SRVEAEDVG V Y Y C M Q D T Q F P Q T F G Q G T K V E I K R		60 113
SEQ ID NO: 39 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein note = amino acid sequence of C321 LCDR1 organism = synthetic construct	
SEQUENCE: 39 HSSQSLVHSD GNTYLS		16
SEQ ID NO: 40 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein note = amino acid sequence of C321 LCDR2 organism = synthetic construct	
SEQUENCE: 40 KISNRFS		7
SEQ ID NO: 41 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein note = amino acid sequence of C321 LCDR3 organism = synthetic construct	
SEQUENCE: 41 MQDTQFPQT		9
SEQ ID NO: 42 FEATURE source	moltype = AA length = 119 Location/Qualifiers 1..119 mol_type = protein note = amino acid sequence of C320 VH organism = synthetic construct	
SEQUENCE: 42 QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA TGQGLEWMGW MNPNSGNTGY AQKFPQGRVTM TRNTSISTAY MELSSLRSED TAVYYCAREV PDTASFEYWG QGTLVTVSS		60 119
SEQ ID NO: 43 FEATURE source	moltype = AA length = 5 Location/Qualifiers 1..5 mol_type = protein note = amino acid sequence of C320 HCDR1 organism = synthetic construct	
SEQUENCE: 43 SYDIN		5
SEQ ID NO: 44 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein note = amino acid sequence of C320 HCDR2 organism = synthetic construct	
SEQUENCE: 44 WMNPNSGNTG YAQKFQG		17
SEQ ID NO: 45 FEATURE source	moltype = AA length = 10 Location/Qualifiers 1..10 mol_type = protein note = amino acid sequence of C320 HCDR3 organism = synthetic construct	
SEQUENCE: 45 EVPDTASFEY		10
SEQ ID NO: 46 FEATURE	moltype = AA length = 111 Location/Qualifiers	

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source                1..111
                      mol_type = protein
                      note = amino acid sequence of C320 VL
                      organism = synthetic construct

SEQUENCE: 46
QSVLTQPPSV SGAPGQRVTI SCAGSSSDIG AGLGVHWYQQ LPGTAPKLLI YGYNRPSGV 60
PDRFSGSKSG TSASLAITGL LPEDEGDYYC QSYDGTLSAL FGGGTKLTVL G      111

SEQ ID NO: 47          moltype = AA length = 14
FEATURE               Location/Qualifiers
source                1..14
                      mol_type = protein
                      note = amino acid sequence of C320 LCDR1
                      organism = synthetic construct

SEQUENCE: 47
AGSSSDIGAG LGVH                                     14

SEQ ID NO: 48          moltype = AA length = 7
FEATURE               Location/Qualifiers
source                1..7
                      mol_type = protein
                      note = amino acid sequence of C320 LCDR2
                      organism = synthetic construct

SEQUENCE: 48
GYNRPS                                             7

SEQ ID NO: 49          moltype = AA length = 10
FEATURE               Location/Qualifiers
source                1..10
                      mol_type = protein
                      note = amino acid sequence of C320 LCDR3
                      organism = synthetic construct

SEQUENCE: 49
QSYDGTLSAL                                         10

SEQ ID NO: 50          moltype = AA length = 123
FEATURE               Location/Qualifiers
source                1..123
                      mol_type = protein
                      note = amino acid sequence of C319 VH
                      organism = synthetic construct

SEQUENCE: 50
QLQLQESGPG LVKPSGTLSTL TCAVSGGSIS SRNWSWVRQ SPKGLEWIG EIYHSDITNY 60
NPSLKSRVTI SVDKSKNQFS LKLSSVTAAD TAVYYCAKDG EAGGTIYIDAF DVWGQGTMTV 120
VSS                                              123

SEQ ID NO: 51          moltype = AA length = 6
FEATURE               Location/Qualifiers
source                1..6
                      mol_type = protein
                      note = amino acid sequence of C319 HCDR1
                      organism = synthetic construct

SEQUENCE: 51
SRNWS                                             6

SEQ ID NO: 52          moltype = AA length = 16
FEATURE               Location/Qualifiers
source                1..16
                      mol_type = protein
                      note = amino acid sequence of C319 HCDR2
                      organism = synthetic construct

SEQUENCE: 52
EIYHSDITNY NPSLKS                                     16

SEQ ID NO: 53          moltype = AA length = 14
FEATURE               Location/Qualifiers
source                1..14
                      mol_type = protein
                      note = amino acid sequence of C319 HCDR3
                      organism = synthetic construct

SEQUENCE: 53
DGEAGGTIYID AFDV                                     14

SEQ ID NO: 54          moltype = AA length = 111
FEATURE               Location/Qualifiers
source                1..111
                      mol_type = protein
                      note = amino acid sequence of C319 VL
                      organism = synthetic construct

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FEATURE	Location/Qualifiers	
source	1..14	
	mol_type = protein	
	note = amino acid sequence of C320-5 LCDR1	
	organism = synthetic construct	
SEQUENCE: 63		
AGSSSDIGAG LGVH		14
SEQ ID NO: 64	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	note = amino acid sequence of C320-5 LCDR2	
	organism = synthetic construct	
SEQUENCE: 64		
GYYNRPS		7
SEQ ID NO: 65	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	note = amino acid sequence of C320-5 LCDR3	
	organism = synthetic construct	
SEQUENCE: 65		
QSYDGTLSAL		10
SEQ ID NO: 66	moltype = AA length = 119	
FEATURE	Location/Qualifiers	
source	1..119	
	mol_type = protein	
	note = amino acid sequence of C320-90 VH	
	organism = synthetic construct	
SEQUENCE: 66		
QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA TQGGLEWMGW MNPNSGNTGY		60
AQKFQGRVTM TRNTSISTAY MELSSLRSED TAVYYCAREV PDDASFEYWG QGTLVTVSS		119
SEQ ID NO: 67	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	note = amino acid sequence of C320-90 HCDR1	
	organism = synthetic construct	
SEQUENCE: 67		
SYDIN		5
SEQ ID NO: 68	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	note = amino acid sequence of C320-90 HCDR2	
	organism = synthetic construct	
SEQUENCE: 68		
WMNPNSGNTG YAQKFQG		17
SEQ ID NO: 69	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	note = amino acid sequence of C320-90 HCDR3	
	organism = synthetic construct	
SEQUENCE: 69		
EVPDDASFEY		10
SEQ ID NO: 70	moltype = AA length = 119	
FEATURE	Location/Qualifiers	
source	1..119	
	mol_type = protein	
	note = amino acid sequence of C320-103 VH	
	organism = synthetic construct	
SEQUENCE: 70		
QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA TQGGLEWMGW MNPNSGNTGY		60
AQKFQGRVTM TRNTSISTAY MELSSLRSED TAVYYCAREV PDAAFEYWG QGTLVTVSS		119
SEQ ID NO: 71	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	note = amino acid sequence of C320-103 HCDR1	
	organism = synthetic construct	

-continued

SEQUENCE: 71
SYDIN 5

SEQ ID NO: 72
FEATURE
source
moltype = AA length = 17
Location/Qualifiers
1..17
mol_type = protein
note = amino acid sequence of C320-103 HCDR2
organism = synthetic construct

SEQUENCE: 72
WMNPNSGNTG YAQKFQG 17

SEQ ID NO: 73
FEATURE
source
moltype = AA length = 10
Location/Qualifiers
1..10
mol_type = protein
note = amino acid sequence of C320-103 HCDR3
organism = synthetic construct

SEQUENCE: 73
EVPDTAAFEY 10

SEQ ID NO: 74
FEATURE
source
moltype = AA length = 119
Location/Qualifiers
1..119
mol_type = protein
note = amino acid sequence of C320-114 VH
organism = synthetic construct

SEQUENCE: 74
QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA TGGGLEWMGW MNPNSGNTGY 60
AQKFQGRVTM TRNTSISTAY MELSSLRSED TAVYYCAREV PDTASFLYWG QGTLVTVSS 119

SEQ ID NO: 75
FEATURE
source
moltype = AA length = 5
Location/Qualifiers
1..5
mol_type = protein
note = amino acid sequence of C320-114 HCDR1
organism = synthetic construct

SEQUENCE: 75
SYDIN 5

SEQ ID NO: 76
FEATURE
source
moltype = AA length = 17
Location/Qualifiers
1..17
mol_type = protein
note = amino acid sequence of C320-114 HCDR2
organism = synthetic construct

SEQUENCE: 76
WMNPNSGNTG YAQKFQG 17

SEQ ID NO: 77
FEATURE
source
moltype = AA length = 10
Location/Qualifiers
1..10
mol_type = protein
note = amino acid sequence of C320-114 HCDR3
organism = synthetic construct

SEQUENCE: 77
EVPDTASFLY 10

SEQ ID NO: 78
FEATURE
source
moltype = AA length = 119
Location/Qualifiers
1..119
mol_type = protein
note = amino acid sequence of C320-115 VH
organism = synthetic construct

SEQUENCE: 78
QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA TGGGLEWMGW MNPNSGNTGY 60
AQKFQGRVTM TRNTSISTAY MELSSLRSED TAVYYCAREV PDTASFDYWG QGTLVTVSS 119

SEQ ID NO: 79
FEATURE
source
moltype = AA length = 5
Location/Qualifiers
1..5
mol_type = protein
note = amino acid sequence of C320-115 HCDR1
organism = synthetic construct

SEQUENCE: 79
SYDIN 5

SEQ ID NO: 80
FEATURE
Location/Qualifiers
moltype = AA length = 17

-continued

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source                1..17
                      mol_type = protein
                      note = amino acid sequence of C320-115 HCDR2
                      organism = synthetic construct

SEQUENCE: 80
WMNPNSGNTG YAQKFQG                                     17

SEQ ID NO: 81        moltype = AA  length = 10
FEATURE              Location/Qualifiers
source                1..10
                      mol_type = protein
                      note = amino acid sequence of C320-115 HCDR3
                      organism = synthetic construct

SEQUENCE: 81
EVPDTASFDY                                               10

SEQ ID NO: 82        moltype = AA  length = 111
FEATURE              Location/Qualifiers
source                1..111
                      mol_type = protein
                      note = amino acid sequence of C320-120 VL
                      organism = synthetic construct

SEQUENCE: 82
QSVLTQPPSV SGAPGQRVTI SCTGSSSDIG AGLGVHWYQQ LPGTAPKLLI YGYNRPSGV  60
PDRFSGSKSG TSASLAITGL LPDEGDYYC QSYDGTLSAL FGGGTKLTVL G          111

SEQ ID NO: 83        moltype = AA  length = 14
FEATURE              Location/Qualifiers
source                1..14
                      mol_type = protein
                      note = amino acid sequence of C320-120 LCDR1
                      organism = synthetic construct

SEQUENCE: 83
TGSSSDIGAG LGVH                                         14

SEQ ID NO: 84        moltype = AA  length = 7
FEATURE              Location/Qualifiers
source                1..7
                      mol_type = protein
                      note = amino acid sequence of C320-120 LCDR2
                      organism = synthetic construct

SEQUENCE: 84
GYNRPS                                                  7

SEQ ID NO: 85        moltype = AA  length = 10
FEATURE              Location/Qualifiers
source                1..10
                      mol_type = protein
                      note = amino acid sequence of C320-120 LCDR3
                      organism = synthetic construct

SEQUENCE: 85
QSYDGTLSAL                                             10

SEQ ID NO: 86        moltype = AA  length = 119
FEATURE              Location/Qualifiers
source                1..119
                      mol_type = protein
                      note = amino acid sequence of C320-129 VH
                      organism = synthetic construct

SEQUENCE: 86
QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA TGQGLEWMGW MNPNSGNTGY  60
AQKFGQGRVTM TRNTSISTAY MELSSLRSED TAVYYCAREV PETASFEYWG QGTLVTVSS  119

SEQ ID NO: 87        moltype = AA  length = 5
FEATURE              Location/Qualifiers
source                1..5
                      mol_type = protein
                      note = amino acid sequence of C320-129 HCDR1
                      organism = synthetic construct

SEQUENCE: 87
SYDIN                                                  5

SEQ ID NO: 88        moltype = AA  length = 17
FEATURE              Location/Qualifiers
source                1..17
                      mol_type = protein
                      note = amino acid sequence of C320-129 HCDR2
                      organism = synthetic construct

SEQUENCE: 88

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-continued

WMNPNSGNTG YAQKFQG 17

SEQ ID NO: 89 moltype = AA length = 10
 FEATURE Location/Qualifiers
 source 1..10
 mol_type = protein
 note = amino acid sequence of C320-129 HCDR3
 organism = synthetic construct

SEQUENCE: 89
 EVPETASFEY 10

SEQ ID NO: 90 moltype = AA length = 119
 FEATURE Location/Qualifiers
 source 1..119
 mol_type = protein
 note = amino acid sequence of C320-130 VH
 organism = synthetic construct

SEQUENCE: 90
 QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA TQGLEWMGW LNPNSGNTGY 60
 AQKFQGRVTM TRNTSISTAY MELSSLRSED TAVYYCAREV PDTASFEYWG QGTLVTVSS 119

SEQ ID NO: 91 moltype = AA length = 5
 FEATURE Location/Qualifiers
 source 1..5
 mol_type = protein
 note = amino acid sequence of C320-130 HCDR1
 organism = synthetic construct

SEQUENCE: 91
 SYDIN 5

SEQ ID NO: 92 moltype = AA length = 17
 FEATURE Location/Qualifiers
 source 1..17
 mol_type = protein
 note = amino acid sequence of C320-130 HCDR2
 organism = synthetic construct

SEQUENCE: 92
 WLNPNNSGNTG YAQKFQG 17

SEQ ID NO: 93 moltype = AA length = 10
 FEATURE Location/Qualifiers
 source 1..10
 mol_type = protein
 note = amino acid sequence of C320-130 HCDR3
 organism = synthetic construct

SEQUENCE: 93
 EVPDASFEY 10

SEQ ID NO: 94 moltype = AA length = 119
 FEATURE Location/Qualifiers
 VARIANT 41
 note = X=T or P
 VARIANT 51
 note = X=M or L
 VARIANT 102
 note = X=E or D
 VARIANT 103
 note = X=T or D
 VARIANT 105
 note = X=S or A
 VARIANT 107
 note = X=L or D or E
 source 1..119
 mol_type = protein
 note = amino acid sequence of VH consensus sequence of C320
 and derivatives
 organism = synthetic construct

SEQUENCE: 94
 QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA XQGLEWMGW XNPNSGNTGY 60
 AQKFQGRVTM TRNTSISTAY MELSSLRSED TAVYYCAREV PXXAXFPXYWG QGTLVTVSS 119

SEQ ID NO: 95 moltype = AA length = 111
 FEATURE Location/Qualifiers
 VARIANT 23
 note = X=A or T
 VARIANT 76
 note = X=A or T
 source 1..111
 mol_type = protein

-continued

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note = amino acid sequence of VL consensus sequence of C320
      and derivatives
organism = synthetic construct

```

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SEQUENCE: 95
QSVLTQPPSV SGAPGQRTVI SCXGSSSDIG AGLGVHWYQQ LPGTAPKLLI YGYNRPSGV 60
PDRFSGSKSG TSASLXITGL LPEDEGDYYC QSYDGTLSAL FGGGTKLTVL G 111

```

```

SEQ ID NO: 96      moltype = DNA length = 351
FEATURE           Location/Qualifiers
source            1..351
                  mol_type = other DNA
                  note = nucleotide sequence encoding VH of C336
                  organism = synthetic construct

```

```

SEQUENCE: 96
gaggtccagc tggtagcagtc tggggctgag gtgaagaagc ctggggcctc agtgaaggtc 60
tcctgcaagg cttctggata caccttcacc ggctactata tgcactgggt gcgacaggcc 120
cctggacaag ggcttgagtg gatgggatgg atcaacccta acagtgggtg cacaaactat 180
gcacagaagt ttcagggcag ggtcaccatg accagggaca cgtccatcag cacagcctac 240
atggagctga gcaggctgag atctgacgac acggccgtgt attactgtgc gagtgggggc 300
cagacccacc tggacgtctg gggccaaggg accacgggtc cgtctctc a 351

```

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SEQ ID NO: 97      moltype = DNA length = 324
FEATURE           Location/Qualifiers
source            1..324
                  mol_type = other DNA
                  note = nucleotide sequence encoding VL of C336
                  organism = synthetic construct

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```

SEQUENCE: 97
gacatccaga tgaccagtc tccatcctcc ctgtctgcat ctgtgggaga cagagtcacc 60
atcacttgcc aggcgagtcg ggacattacc gactatttaa attgggtatca gcagagacca 120
gggaaagccc ctaagctcct gatctacgat gcattccaatt tggaaacagg ggttccatca 180
agggttcagt gaagtggatc tgggacatat ttactttca ccatcagcag cctgcagcct 240
gaagattttg cgacatatta ctgtcaacag tatgataatc tcccgatcac cttcggccag 300
gggacacgac tggagattaa acgt 324

```

```

SEQ ID NO: 98      moltype = DNA length = 366
FEATURE           Location/Qualifiers
source            1..366
                  mol_type = other DNA
                  note = nucleotide sequence encoding VH of C334
                  organism = synthetic construct

```

```

SEQUENCE: 98
caaatgcagc tgggtcagtc tggggctgag gtgaagaagc ctgggtcctc ggtgaaggtc 60
tcctgcaagg cttctggagg caccttcacc agctatacta tcagctgggt gcgacaggcc 120
cctggacaag ggcttgagtg gatgggaggg atcatcccta tctttgttac agcaaaccac 180
gcacagagct tccagggcag agtcacgatt accgcggacg aatccacgag cacagcctac 240
atggagctga gcagcctgag atctgaggac acggccgtat attattgttc aaccaactcg 300
tatagcagca gctggtatga tgcttttgat atctggggcc aagggacaat ggtcacccgtc 360
tcttca 366

```

```

SEQ ID NO: 99      moltype = DNA length = 324
FEATURE           Location/Qualifiers
source            1..324
                  mol_type = other DNA
                  note = nucleotide sequence encoding VL of C334
                  organism = synthetic construct

```

```

SEQUENCE: 99
gatgttgtga tgacacagtc tccagcttcc ctctctgtgt ctctgggga aagagccacc 60
ctctcctgca gggccagtcg gagtattagc aacaacttag cctgggtatca gcaaatgcct 120
ggccaggctc ccaggctcct tctttatgat gcattccaca gggccactga tatccagcc 180
agggtcagtg gcagtgggtc tgggtcagag ttcactctca ccatcagcgg cctgcagtct 240
gcagattttg cagtttatta ctgtcaacaa tacaataact ggctctcac ttctggcgga 300
gggaccaagc tggagatcaa acgt 324

```

```

SEQ ID NO: 100     moltype = DNA length = 366
FEATURE           Location/Qualifiers
source            1..366
                  mol_type = other DNA
                  note = nucleotide sequence encoding VH of C333
                  organism = synthetic construct

```

```

SEQUENCE: 100
caggtccagc ttgtgcagtc tggggctgag gtgaagaagc ctgggtcctc ggtgaaggtc 60
tcctgcaagg cttctggagg caccttcacc agctatacta tcagctgggt gcgacaggcc 120
cctggacaag ggcttgagtg gatgggaggg atcatcccta tctttgttac acaaaactac 180
gcacagaggt tccagggcag agtcacgatt accgcggacg aatccacgag cacagcctac 240
atggagctga gcagcctgag atctgaggac acggccatat attactgttc aaccaactcc 300
tatagcagca gctggtatga tgcttttgat atctggggcc aagggacaat ggtcacccgtc 360
tcttca 366

```

-continued

SEQ ID NO: 101 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = other DNA
 note = nucleotide sequence encoding VL of C333
 organism = synthetic construct

SEQUENCE: 101
 gaaattgtgt tgacgcagtc tccagccacc ctgtctgtgt ctctgggga aagagccacc 60
 ctctcctgca gggccagtc gagtattacc aacaacttag cctgggtatca acaactgcct 120
 ggccaggctc ccaggctcct tatttacgat gcattccacca gggccactga tatccagcc 180
 aggttcagtg gcaactgggtc tgggtcagag ttcactctca ccatcagcgg cctgcagtct 240
 ggggattttg cagtttatta ctgtcaacaa tacaataact ggctctcac tttcgcgga 300
 gggaccaaag tggatatcaa acgt 324

SEQ ID NO: 102 moltype = DNA length = 366
 FEATURE Location/Qualifiers
 source 1..366
 mol_type = other DNA
 note = nucleotide sequence encoding VH of C323
 organism = synthetic construct

SEQUENCE: 102
 caaatgcagc tgggtgcagtc tggggctgag gtgaagaagc ctgggtcttc ggtgaaggtc 60
 tcctgcaagg cttctggagg caccttcagc agctatacta tcagctgggt gcgacaggcc 120
 cctggacaag ggcttgagtg gatgggagg atcatcccta tctttgttac aacaaactac 180
 gcacagaggt tccagggcag agtcacgatt accgcggacg aatccacgag cacagcctac 240
 atggagctga gcagcctgag atctgaggac acggccatat attactgttc aaccaactcc 300
 tatagcagca gctggtatga tgctttgat atctggggcc aagggacaat ggtcaccgtc 360
 tcttca 366

SEQ ID NO: 103 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = other DNA
 note = nucleotide sequence encoding VL of C323
 organism = synthetic construct

SEQUENCE: 103
 gccatccagt tgaccagtc tccatcctcc ctgtctgcat ctgtaggaga cagagtcacc 60
 atcacttgcc gcgcaagtca gggcattggc agtgctttag cctgggtatca gcagaaacca 120
 gggaagctc ctaagctcct gatctatgat gcctccagtt tgcaaaagtg ggtcccatca 180
 aggtacagcg gcagtggatg tgggacagat ttcactctca ccatcagcgg cctgcagcct 240
 gaagattttg caacttatta ctgtcaacag tttagtatt acccgctcac tttcgcgga 300
 gggaccaagc tggagatcaa acgt 324

SEQ ID NO: 104 moltype = DNA length = 366
 FEATURE Location/Qualifiers
 source 1..366
 mol_type = other DNA
 note = nucleotide sequence encoding VH of C321
 organism = synthetic construct

SEQUENCE: 104
 cagggtgcagc tgggtgcagtc tggggctgag gtgaagaagc ctggggcctc agtgaaggtc 60
 tcctgcaagg cttctgggta cacctttacc agctatggta tcagctgggt gcgacaggcc 120
 cctggacaag ggcttgagtg gatgggatgg atcagcgctt acaatggtaa cacaaactat 180
 gcacagaagc tccagggcag agtcaccatg accacagaca catccacgag cacagcctac 240
 atggagctga ggagcctgag atctgacgac acggccgtgt attactgtgc gagagactcc 300
 catatttacg atattttgac tggttatgac tactggggcc agggaaacct ggtcaccgtc 360
 tcctca 366

SEQ ID NO: 105 moltype = DNA length = 339
 FEATURE Location/Qualifiers
 source 1..339
 mol_type = other DNA
 note = nucleotide sequence encoding VL of C321
 organism = synthetic construct

SEQUENCE: 105
 gatgttgta tgacacagac tccactctcc tcacctgtca cccttgaca gccggcctcc 60
 atctcctgcc actctagtc aagcctcgta cacagtgatg gaaacaccta cttgagttgg 120
 cttcagcaga gggccggcca gcctccaaga ctctaatat ataagatttc taaccgggtc 180
 tctggggtcc cagacagatt cagtggcagt ggggcaggga cagatttcac actgaaaatc 240
 agcaggggtg aagctgagga tgtcggggtt tattactgca tgcaagatac acaatttcct 300
 cagacgttcg gccaaaggac caagtgga atcaaacgt 339

SEQ ID NO: 106 moltype = DNA length = 357
 FEATURE Location/Qualifiers
 source 1..357
 mol_type = other DNA
 note = nucleotide sequence encoding VH of C320
 organism = synthetic construct

SEQUENCE: 106

-continued

```

caggtgcagc tgggtgcagtc tggggctgag gtgaagaagc ctggggcctc agtgaaggtc 60
tcctgcaagg cttctggata caccttcacc agttatgata tcaactgggt gcgacaggcc 120
actggacaag ggcttgagtg gatgggatgg atgaacccta acagtggtaa cacaggctat 180
gcacagaagt tccagggcag agtcaccatg accaggaaca cctccataag cacagcctac 240
atggagctga gcagcctgag atctgaggac acggccgtgt attactgtgc gagagaagtg 300
cctgatacgg cctcctttga gtactggggc cagggaaacc tggtcaccgt ctctca 357

```

```

SEQ ID NO: 107      moltype = DNA length = 333
FEATURE            Location/Qualifiers
source             1..333
                   mol_type = other DNA
                   note = nucleotide sequence encoding VL of C320
                   organism = synthetic construct

```

```

SEQUENCE: 107
cagctctgtgc tgcagcagcc gccctcagtg tctggggccc cagggcagag ggtcaccatc 60
tcctgcgctg ggagtagttc cgacatcggg gcaggtcttg gcgtgcactg gtatcagcag 120
cttcaggaa cagcccccac actcctcatc tatggttact acaatcggcc ctccaggggtc 180
cctgaccgat tctctggctc caagtctggc acctcagcct ccctggccat cactggggctc 240
ctgcctgagg atgagggtga ttattactgc cagtcctatg acggcactct gagtgcctta 300
ttcggcggag ggaccaagct gaccgtccta ggt 333

```

```

SEQ ID NO: 108      moltype = DNA length = 369
FEATURE            Location/Qualifiers
source             1..369
                   mol_type = other DNA
                   note = nucleotide sequence encoding VH of C319
                   organism = synthetic construct

```

```

SEQUENCE: 108
cagctgcagc tgcaggagtc gggcccagga ctggtgaagc cttcggggac cctgtccctc 60
acctgcgctg tctctggtgg ctccatcagc agtagaaact ggtggagttg ggtccgccag 120
tcccaggga aggggctgga gtggattggg gaaatctatc acagtgatat aaccaactat 180
aaccgctccc tcaagagtcg agtcaccatt tcagtagaca agtccaagaa ccagttcttc 240
ctgaagctga gctctgtgac cgcgcgggac acggccgtgt attactgtgc gaaggacggg 300
gaggcgggag ggacctacat tgatgctttt gatgtctggg gccaaaggac aatggtcacc 360
gtctcttca 369

```

```

SEQ ID NO: 109      moltype = DNA length = 333
FEATURE            Location/Qualifiers
source             1..333
                   mol_type = other DNA
                   note = nucleotide sequence encoding VL of C319
                   organism = synthetic construct

```

```

SEQUENCE: 109
cagctgccc tgactcagcc tcgctcagtg tccgggtctc ctggacagtc agtcaccatc 60
tcctgcactg gaaccagcag tgatgttgg atttataact atgtotcctg gtaccaacaa 120
caccaggca aagcccccac actcataatt tatgatgtca gtgagcggcc ctccaggggtc 180
cctgatcgct tctctggctc caagtctgac aacacggcct ccctgacctc ctctggggctc 240
caggctgagg atgagggtga ttattactgc tactcatatg caggcaccta cacttcctta 300
ttcggcggag ggaccaaggt caccgtccta ggt 333

```

```

SEQ ID NO: 110      moltype = DNA length = 357
FEATURE            Location/Qualifiers
source             1..357
                   mol_type = other DNA
                   note = nucleotide sequence encoding VH of C320-3
                   organism = synthetic construct

```

```

SEQUENCE: 110
caggtgcagc tgggtgcagtc tggggctgag gtgaagaagc ctggggcctc agtgaaggtc 60
tcctgcaagg cttctggata caccttcacc agttatgata tcaactgggt gcgacaggcc 120
cccgacaag ggcttgagtg gatgggatgg atgaacccta acagtggtaa cacaggctat 180
gcacagaagt tccagggcag agtcaccatg accaggaaca cctccataag cacagcctac 240
atggagctga gcagcctgag atctgaggac acggccgtgt attactgtgc gagagaagtg 300
cctgatacgg cctcctttga gtactggggc cagggaaacc tggtcaccgt ctctca 357

```

```

SEQ ID NO: 111      moltype = DNA length = 333
FEATURE            Location/Qualifiers
source             1..333
                   mol_type = other DNA
                   note = nucleotide sequence encoding VL of C320-5
                   organism = synthetic construct

```

```

SEQUENCE: 111
cagctctgtgc tgcagcagcc gccctcagtg tctggggccc cagggcagag ggtcaccatc 60
tcctgcgctg ggagtagttc cgacatcggg gcaggtcttg gcgtgcactg gtatcagcag 120
cttcaggaa cagcccccac actcctcatc tatggttact acaatcggcc ctccaggggtc 180
cctgaccgat tctctggctc caagtctggc acctcagcct ccctgacctc cactggggctc 240
ctgcctgagg atgagggtga ttattactgc cagtcctatg acggcactct gagtgcctta 300
ttcggcggag ggaccaagct gaccgtccta ggt 333

```

```

SEQ ID NO: 112      moltype = DNA length = 357

```

-continued

FEATURE Location/Qualifiers
 source 1..357
 mol_type = other DNA
 note = nucleotide sequence encoding VH of C320-90
 organism = synthetic construct

SEQUENCE: 112

```

caggtgcagc tgggtgcagtc tggcgccgaa gtgaagaaac ctggcgccag cgtgaagggtg 60
tcctgcaagg ccagcgggcta caccttcacc agctacgaca tcaactgggt cgcacaggcc 120
accggccagg gcctgggaatg gatgggctgg atgaacccca acagcggcaa caccggctac 180
gcccagaaat tccagggcag agtgaccatg acccggaaca ccagcatcag caccgcctac 240
atggaactga gcagcctgcg gagcgaggac accgccgtgt actactgcgc cagagagggtg 300
ccgcacacgc ccagcttcga gtattggggc cagggcacc c tggtcaccgt gtctagc 357
  
```

SEQ ID NO: 113 moltype = DNA length = 357
 FEATURE Location/Qualifiers
 source 1..357
 mol_type = other DNA
 note = nucleotide sequence encoding VH of C320-103
 organism = synthetic construct

SEQUENCE: 113

```

caggtgcagc tgggtgcagtc tggcgccgaa gtgaagaaac ctggcgccag cgtgaagggtg 60
tcctgcaagg ccagcgggcta caccttcacc agctacgaca tcaactgggt cgcacaggcc 120
accggccagg gcctgggaatg gatgggctgg atgaacccca acagcggcaa caccggctac 180
gcccagaaat tccagggcag agtgaccatg acccggaaca ccagcatcag caccgcctac 240
atggaactga gcagcctgcg gagcgaggac accgccgtgt actactgcgc cagagagggtg 300
ccgcacacgc ccagcttcga gtattggggc cagggcacc c tggtcaccgt gtctagc 357
  
```

SEQ ID NO: 114 moltype = DNA length = 357
 FEATURE Location/Qualifiers
 source 1..357
 mol_type = other DNA
 note = nucleotide sequence encoding VH of C320-114
 organism = synthetic construct

SEQUENCE: 114

```

caggtgcagc tgggtgcagtc tggcgccgaa gtgaagaaac ctggcgccag cgtgaagggtg 60
tcctgcaagg ccagcgggcta caccttcacc agctacgaca tcaactgggt cgcacaggcc 120
accggccagg gcctgggaatg gatgggctgg atgaacccca acagcggcaa caccggctac 180
gcccagaaat tccagggcag agtgaccatg acccggaaca ccagcatcag caccgcctac 240
atggaactga gcagcctgcg gagcgaggac accgccgtgt actactgcgc cagagagggtg 300
ccgcacacgc ccagcttcct gtattggggc cagggcacc c tggtcaccgt gtctagc 357
  
```

SEQ ID NO: 115 moltype = DNA length = 357
 FEATURE Location/Qualifiers
 source 1..357
 mol_type = other DNA
 note = nucleotide sequence encoding VH of C320-115
 organism = synthetic construct

SEQUENCE: 115

```

caggtgcagc tgggtgcagtc tggcgccgaa gtgaagaaac ctggcgccag cgtgaagggtg 60
tcctgcaagg ccagcgggcta caccttcacc agctacgaca tcaactgggt cgcacaggcc 120
accggccagg gcctgggaatg gatgggctgg atgaacccca acagcggcaa caccggctac 180
gcccagaaat tccagggcag agtgaccatg acccggaaca ccagcatcag caccgcctac 240
atggaactga gcagcctgcg gagcgaggac accgccgtgt actactgcgc cagagagggtg 300
ccgcacacgc ccagcttcga ctattggggc cagggcacc c tggtcaccgt gtctagc 357
  
```

SEQ ID NO: 116 moltype = DNA length = 333
 FEATURE Location/Qualifiers
 source 1..333
 mol_type = other DNA
 note = nucleotide sequence encoding VL of C320-120
 organism = synthetic construct

SEQUENCE: 116

```

cagagcgtgc tgacacagcc ccctagcgtg tcaggcgccc ctggccagag agtgaccatc 60
tcctgcacgc gcagcagcag cgacatcgga gctggactgg gcgtgcactg gtatcagcag 120
ctgcttgcca ccgcccccaa gctgctgac tacggctact acaaccggcc cagcggcgtg 180
cccgacagat tcagcggcag caagagcggc accagcgcca gcctggccat cactggactg 240
ctgcccaggg acgagggcga ctactactgc cagagctacg acggcaccct gagcgccctg 300
tttgccggag gcaccaagct gacgcctcta ggt 333
  
```

SEQ ID NO: 117 moltype = DNA length = 357
 FEATURE Location/Qualifiers
 source 1..357
 mol_type = other DNA
 note = nucleotide sequence encoding VH of C320-129
 organism = synthetic construct

SEQUENCE: 117

```

caggtgcagc tgggtgcagtc tggggctgag gtgaagaagc ctggggcctc agtgaaggtc 60
tcctgcaagg cttctggata caccttcacc agttatgata tcaactgggt gcgacaggcc 120
actggacaag ggcttgagtg gatgggatgg atgaacccta acagtggtaa cacaggctat 180
  
```

-continued

```
gcacagaagt tccagggcag agtcaccatg accaggaaca cctccataag cacagcctac 240
atggagctga gcagcctgag atctgaggac acggccgtgt attactgtgc gagagaagtg 300
cctgagacgg cctcctttga gtactggggc cagggaaacc tggtcaccgt ctectca 357
```

```
SEQ ID NO: 118      moltype = DNA length = 357
FEATURE            Location/Qualifiers
source              1..357
                    mol_type = other DNA
                    note = nucleotide sequence encoding VH of C320-130
                    organism = synthetic construct
```

```
SEQUENCE: 118
caggtgcagc tgggtgcagtc tggggctgag gtgaagaagc ctggggcctc agtgaaggtc 60
tcctgcaagg ctcttgata caccctcacc agttatgata tcaactgggt gcgacaggcc 120
actggacaag ggcttgatg gatgggatgg ctgaacccta acagtggtaa cacaggctat 180
gcacagaagt tccagggcag agtcaccatg accaggaaca cctccataag cacagcctac 240
atggagctga gcagcctgag atctgaggac acggccgtgt attactgtgc gagagaagtg 300
cctgatacgg cctcctttga gtactggggc cagggaaacc tggtcaccgt ctectca 357
```

```
SEQ ID NO: 119      moltype = AA length = 124
FEATURE            Location/Qualifiers
source              1..124
                    mol_type = protein
                    note = amino acid sequence of VH of humanized antibody 1B4
                    organism = synthetic construct
```

```
SEQUENCE: 119
QVTLKESGPA LVKPTQTLTL TCTFSGFSLT TSNMGVVWIR QPPGKALEWL AHILWDDREY 60
SNPALKSRLT ISKDTSKNQV VLTMTNMDPV DTATYYCARM SRNYYGSSYV MDYWGQGTIV 120
TVSS 124
```

```
SEQ ID NO: 120      moltype = AA length = 106
FEATURE            Location/Qualifiers
source              1..106
                    mol_type = protein
                    note = amino acid sequence of VL of humanized antibody 1B4
                    organism = synthetic construct
```

```
SEQUENCE: 120
DIQLTQSPSF LSASVGDRVT ITCSASSSVN YMHWYQOKPG KAPKLLIYST SNLASGVPSR 60
FSGSGSGTEF LTISSSLQPE DFATYYCHQW NNYGTFGQGT KVEIKR 106
```

```
SEQ ID NO: 121      moltype = DNA length = 372
FEATURE            Location/Qualifiers
source              1..372
                    mol_type = other DNA
                    note = nucleotide sequence encoding VH of humanized
                    antibody 1B4
                    organism = synthetic construct
```

```
SEQUENCE: 121
caggtcacac tgaagagtc cggccctgcc ctggccaagc ccaccagac cctgaccctg 60
acatgcacct tcagcggctt cagcctgagc accagcaaca tgggcgtcgt gtggatcaga 120
cagccccctg gcaaggccct ggaatggctg gccacatcc tgtgggacga cagagagtac 180
agcaaccccg cctgaagtc cgggctgacc atcagcaagg acaccagcaa gaaccagggtg 240
gtgctgacca tgaccaacat ggaccccggt gacaccgcca cctactactg cgcccggatg 300
agccgggaact actacggcag cagctactgt atggactact ggggccaggg caccctgggtc 360
accgtgtcct ca 372
```

```
SEQ ID NO: 122      moltype = DNA length = 318
FEATURE            Location/Qualifiers
source              1..318
                    mol_type = other DNA
                    note = nucleotide sequence encoding VL of humanized
                    antibody 1B4
                    organism = synthetic construct
```

```
SEQUENCE: 122
gacatccagc tgaccagag cccagcttc ctgagcgcca gcgtggcgca cagagtgacc 60
atcacctgta gcgccagcag cagcgtgaac tacatgcact ggtatcagca gaagcccgcc 120
aaggccccca agctgctgat ctacagcacc agcaacctgg ccagcggcgt gccagcaga 180
ttttctggca gcggcagcgg caccgagttc accctgacca tcagcagcct gcagcccgag 240
gacttcgcca cctactactg ccaccagtgg aacaactacg gcaccttcgg ccagggcacc 300
aaggtggaaa tcaagcgt 318
```

```
SEQ ID NO: 123      moltype = AA length = 251
FEATURE            Location/Qualifiers
source              1..251
                    mol_type = protein
                    organism = Homo sapiens
```

```
SEQUENCE: 123
MAEDLGLSPG ETASVEMLPE HGSCRPKARS SSARWALTCC LVLLPFLAGL TTYLLVSQRL 60
AQGEACVQFQ ALKGQEFAPS HQQVYAPLRA DGDKPRAHLT VVRQTPTQHF KNQFPALHWE 120
HELGLAFTKN RMNYTNKFLI IPESGDYFIY SQVTPRGMTS ECSEIRQAGR PNKPDSITVV 180
```

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ITKVTDTYPE PTQLLMGTSK VCEVGSNWFQ PIYLGAMFSL QEGDKLMVNV SDISLVDYTK 240
EDKTFFGAFL L 251

SEQ ID NO: 124 moltype = AA length = 177
FEATURE Location/Qualifiers
source 1..177
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 124
ITEERSESPS QQVYSPPRGK PRAHLTIKKQ TPAPHLKNQL SALHWEHDLG MAFTKNGMKY 60
INKSLVIPES GDYFIYSQIT FRGTTSVCGD ISRGRRPNKP DSITMVITKV ADSYPEPARL 120
LTGSKSVCEI SNNWFQSLYL GATFSLEEGD RLMVNVSDIS LVDYTKEDKT FFGAFL 177

SEQ ID NO: 125 moltype = AA length = 180
FEATURE Location/Qualifiers
source 1..180
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 125
LKGQEFAPSH QQVYAPLRAD GDKPRAHLTV VRQTPHQHLK NQFPALHWEH ELGLAFTKNR 60
MNYTNKFLLI PESGDYFVYS QVTFRGMTSE CSEIRQAGRP NKPDSTVVI TKVTDTYPEP 120
TQLLMGTSKV CEVGSNWFQP IYLGAMFSLQ EGDKLMVNVSDIS LVDYDYTEKE DKTFFGAFL 180

SEQ ID NO: 126 moltype = AA length = 177
FEATURE Location/Qualifiers
source 1..177
 mol_type = protein
 organism = Rattus rattus

SEQUENCE: 126
VTEERSAPSA QPVYTPSRDK PKAHLTIMRQ TPVPHLKNEL AALHWENNLG MAFTKNRMNY 60
TNKFLVIPES GDYFIYSQIT FRGTTSECGD ISRVRRPKKP DSITVVITKV ADSYPEPAHL 120
LTGTSKSVCEI SSNWFQPIYL GAMFSLEEGD RLMVNVSDIS LVDYTKEDKT FFGAFLI 177

SEQ ID NO: 127 moltype = AA length = 179
FEATURE Location/Qualifiers
source 1..179
 mol_type = protein
 organism = Oryctolagus cuniculus

SEQUENCE: 127
LKGREFGPSQ QRAYMPLRAD GNKPRAHLTA VKQTPHQPLR NHFPALHWEH ELGLAFTKNR 60
MNYTNKFLVI PESGDYFVYS QVTFRGTTSE CGVINQRRRQ TKPDSIVVVI TKVTDNYPEP 120
AQLLTGTSKV CEMGNWFQPI YLGAMFSLEE GDKLMVNVSD VSLVDYTEKE KTFFGAFL 179

SEQ ID NO: 128 moltype = AA length = 180
FEATURE Location/Qualifiers
source 1..180
 mol_type = protein
 organism = Cavia porcellus

SEQUENCE: 128
TINEQRFGPS YQRVYTPLRD DRDKPRAHLT VVRQTPHQHL KNQFPALHWE HELGLAFTKN 60
RMNYTNKFLV IPETGDYFVY SQITFRGTTSE ECGISPGRQQ NKPDSTVVI TKVTDTYPEP 120
SQLLTGTSKV CEISSNWFQP LYLAMFSLQ EGDKLMVNVSDIS LVDYDYTEKE DKTFFGAFL 180

SEQ ID NO: 129 moltype = AA length = 417
FEATURE Location/Qualifiers
source 1..417
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 129
MEQRPRGCAA VAAALLLVLL GARAQGGTRS PRCDGAGDFH KKIGLFCCRG CPAGHYLKAP 60
CTEPCGNSTC LVCPQDTFLA WENHNHSECA RCQACDEQAS QVALENCASV ADTRCGCKPG 120
WFVECVQSQC VSSSPFYCQP CLDCGALHRH TRLLCSRRTG DCGTCLPGFY EHGDGCVSCP 180
TSTLGSCPER CAAVCGWRQM FWVQVLLAGL VVPLLLGATL TYTYRHCWPH KPLVTADAEAG 240
MEALTPPPAT HLSPLDSAHT LLAPPDSSSEK ICTVQLVGNS WTPGYPETQE ALCPQVTWSW 300
DQLPSRALGP AAAPTLSPEP PAGSPAMMLQ PGPQLYDVM DAVPARRWKEF VRTLGLREAE 360
IEAVEVEIGR FRDQYEMLK RWRQQQPAGL GAVYAALERM GLDGCVEDLR SRLQRGP 417

SEQ ID NO: 130 moltype = AA length = 300
FEATURE Location/Qualifiers
source 1..300
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 130
MRALEGPGLS LLCLVLALPA LLPVPAVRGV AETPTYPRWD AETGERLVCA QCPPGTFVQR 60
PCRRDSPTTC GPCPPRHYTQ FWNYLERCYR CNVLCGEREE EARACHATHN RACRCRTGFF 120
AHAGFCLEHA SCPPGAGVIA PGTPSQNTQC QCPPGTFSA SSSSEQCQP HRNCTALGLA 180
LNVPGSSSHD TLCTSCCTGFP LSTRVPGAE E CERAVIDFVA FQDISIKRLQ RLLQALEAPE 240
GWGPTPRAGR AALQLKLRRR LTELGAQDG ALLVRLQLAL RVARMPGLER SVRERFLPVH 300

-continued

SEQ ID NO: 131	moltype =	length =	
SEQUENCE: 131			
000			
SEQ ID NO: 132	moltype = AA	length = 4	
FEATURE	Location/Qualifiers		
source	1..4		
	mol_type = protein		
	note = sequence from TL1a		
	organism = synthetic construct		
SEQUENCE: 132			
QPIY			4
SEQ ID NO: 133	moltype = AA	length = 7	
FEATURE	Location/Qualifiers		
source	1..7		
	mol_type = protein		
	note = sequence from TL1a		
	organism = synthetic construct		
SEQUENCE: 133			
NVSDISL			7
SEQ ID NO: 134	moltype = AA	length = 330	
FEATURE	Location/Qualifiers		
source	1..330		
	mol_type = protein		
	organism = Homo sapiens		
SEQUENCE: 134			
ASTKGPSVFP LAPSSKSTSG	GTAALGCLVK DYPPEPVTVS	WNSGALTSGV HTPPAVLQSS	60
GLYSLSSVVT VPSSSLGTQT	YICNVNHKPS NTKVDKKVEP	KSCDKHTTCP PCPAPELLGG	120
PSVFLFPPKP KDTLMISRT	EVTCVVVDVS HEDPEVKFNW	YVDGVEVHNA KTKPREEQYN	180
STYRVVSVLT VLNQDNLNGK	EYCKKVSNKA LPAPIEKTIS	KAKGQPREPQ VYTLPPSRDE	240
LTKNQVSLTC LVKGFYPSDI	AVEWESNGQP ENNYKTTTPV	LDSDGSFFLY SKLTVDKSRW	300
QQGNVFSCSV MHEALHNHYT	QKSLSLSPGK		330
SEQ ID NO: 135	moltype = AA	length = 106	
FEATURE	Location/Qualifiers		
source	1..106		
	mol_type = protein		
	organism = Homo sapiens		
SEQUENCE: 135			
TVAAPSVFIF PPSDEQLKSG	TASVVCLLNN FYPREAKVQW	KVDNALQSGN SQESVTEQDS	60
KDSTYLSST LTLKADYEK	HKVYACEVTH QGLSSPVTKS	FNRGEC	106
SEQ ID NO: 136	moltype = AA	length = 105	
FEATURE	Location/Qualifiers		
source	1..105		
	mol_type = protein		
	organism = Homo sapiens		
SEQUENCE: 136			
QPKAAPSVTL FPPSSEELQA	NKATLVCLIS DFYPGAVTVA	WKADSSPVKA GVETTTPSKQ	60
SNNKYAASSY LSLTPEQWKS	HRSYSCQVTH EGSTVEKTVA	PTECS	105
SEQ ID NO: 137	moltype = AA	length = 120	
FEATURE	Location/Qualifiers		
VARIANT	1		
VARIANT	4		
	note = X=L or V		
VARIANT	5		
	note = X=V or L or Q		
VARIANT	6		
	note = X=L or absent		
VARIANT	7		
	note = X=Q or E		
VARIANT	13		
	note = X=K or R		
VARIANT	14		
	note = X=K or R or A		
VARIANT	17		
	note = X=A or S		
VARIANT	20		
	note = X=K or T or Q or R		
VARIANT	24		
	note = X=K or T		
VARIANT	25		
	note = X=A or S or V		
VARIANT	42		
	note = X=T or P		
VARIANT	44		

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VARIANT	note = X=Q or R or H or K 46
VARIANT	note = X=L or P 47
VARIANT	note = X=E or Q 49
VARIANT	note = X=M or L 52
VARIANT	note = X=M or L 69
VARIANT	note = X=V or I or L 70
VARIANT	note = X=T or S 71
VARIANT	note = X=M or I or F 73
VARIANT	note = X=R or A or T or E 74
VARIANT	note = X=N or D 75
VARIANT	note = X=T or R or Q or E or D 76
VARIANT	note = X=S or A or F 77
VARIANT	note = X=I or T or A 78
VARIANT	note = X=S or R or G or N 80
VARIANT	note = X=A or V 82
VARIANT	note = X=M or L 83
VARIANT	note = X=E or D 85
VARIANT	note = X=S or N or T or R 86
VARIANT	note = X=S or N 89
VARIANT	note = X=S or P or Y 90
VARIANT	note = X=E or D 94
VARIANT	note = X=V or M 98
VARIANT	note = X=A or T 99
VARIANT	note = X=R or K or T 101
VARIANT	note = X=V or A or S or H or L or D or Y or P or Q or K 103
VARIANT	note = X=E or D or A or S or H or K 104
VARIANT	note = X=T or D or A or S or D or Y 106
VARIANT	note = X=S or A or H or L or D or Y 108
VARIANT	note = X=L or D or E or A or S or H or P or Q or K 112
VARIANT	note = X=Q or K 115
source	note = X=L or T 1..120 mol_type = protein note = VH consensus sequence of C320 and derivatives organism = synthetic construct

SEQUENCE: 137

XVQXXXSGA EVXXPGXSVX VSCXXSGYTF TSYDINWVRQ AXGXGXWVG WXNPNSGNTG 60
YAQKPGGRXX XTXXXXXXTX YXXLXXLRXX DTAXYYCXHE XPXXAXFXYW GXGTXTVTSS 120

SEQ ID NO: 138

FEATURE	moltype = AA length = 112
VARIANT	Location/Qualifiers
VARIANT	1
VARIANT	note = X=Q or S or D or E 2
VARIANT	note = X=S or I or Y or P or L 3
VARIANT	note = X=V or A or G 7
VARIANT	note = X=P or S

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VARIANT	8 note = X=P or S or A
VARIANT	9 note = X=absent or G
VARIANT	10 note = X=S or T
VARIANT	11 note = X=V or Q or L
VARIANT	13 note = X=G or V or K or A or L
VARIANT	14 note = X=A or G or T or S
VARIANT	15 note = X=P or L or A
VARIANT	16 note = X=G or R
VARIANT	17 note = X=Q or K or E
VARIANT	18 note = X=R or S or K or T
VARIANT	19 note = X=V or I or A
VARIANT	20 note = X=T or R
VARIANT	21 note = X=I or S
VARIANT	22 note = X=S
VARIANT	24 note = X=A or T
VARIANT	25 note = X=G or S
VARIANT	39 note = X=Y or L
VARIANT	42 note = X=L or K or R or H
VARIANT	43 note = X=P or Q
VARIANT	45 note = X=T or Q or K or H
VARIANT	46 note = X=A or P
VARIANT	48 note = X=K or R or V
VARIANT	50 note = X=L or V or M
VARIANT	51 note = X=I or V or S
VARIANT	52 note = X=P or E or G or Y
VARIANT	54 note = X=Y
VARIANT	55 note = X=Y
VARIANT	60 note = X=G or I
VARIANT	61 note = X=V or A or S
VARIANT	62 note = X=P or E or N
VARIANT	66 note = X=S or A
VARIANT	68 note = X=S or G or N or R
VARIANT	71 note = X=G or N
VARIANT	72 note = X=T or D
VARIANT	73 note = X=S or F
VARIANT	74 note = X=A or T
VARIANT	76 note = X=L or T or G
VARIANT	77 note = X=A or T
VARIANT	78 note = X=I or G or S
VARIANT	79

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VARIANT      note = X=T or R
              80
              note = X=G or V
VARIANT      81
              note = X=L or E or Q or R
VARIANT      82
              note = X=L or A or T or S or Q
VARIANT      83
              note = X=P or G or A
VARIANT      85
              note = X=D or L or F
VARIANT      86
              note = X=E or A
VARIANT      87
              note = X=G or V or N or A
VARIANT      96
              note = X=G
VARIANT      97
              note = X=T
VARIANT      103
              note = X=G or Q or T or A
VARIANT      107
              note = X=K or V
VARIANT      108
              note = X=L or E
VARIANT      109
              note = X=T or I
VARIANT      110
              note = X=V or K
VARIANT      111
              note = X=L or R
source        1..112
              mol_type = protein
              note = consensus sequence of C320 and derivatives
              organism = synthetic construct

SEQUENCE: 138
XXXLTQXXXXX XSXXXXXXXXX XXCXXSSSDI GAGLGVHWXQ QXXGXXPXLX XXGXXNRPSX 60
XXDRFPXGKXS XXXXSXXXXXX XXXEXXXDYD CQSYDXXLSA LFXGGTXXXX XG 112

SEQ ID NO: 139      moltype = AA length = 14
FEATURE            Location/Qualifiers
VARIANT            1
VARIANT            2
                    note = X at position 2 is G or S
source             1..14
                    mol_type = protein
                    note = LCDR1 consensus sequence of C320 and derivatives
                    organism = synthetic construct

SEQUENCE: 139
XXSSSDIGAG LGVH 14

SEQ ID NO: 140      moltype = AA length = 7
FEATURE            Location/Qualifiers
source             1..7
                    mol_type = protein
                    note = LCDR2 consensus sequence of C320 and derivatives
                    organism = synthetic construct

SEQUENCE: 140
GYYNRPS 7

SEQ ID NO: 141      moltype = AA length = 10
FEATURE            Location/Qualifiers
source             1..10
                    mol_type = protein
                    note = LCDR3 consensus sequence of C320 and derivatives
                    organism = synthetic construct

SEQUENCE: 141
QSYDGTLSAL 10

SEQ ID NO: 142      moltype = AA length = 17
FEATURE            Location/Qualifiers
VARIANT            2
                    note = X at position 2 is methionine or leucine
source             1..17
                    mol_type = protein
                    note = HCDR2 consensus sequence of C320 and derivatives
                    organism = synthetic construct

SEQUENCE: 142
WXPNSGNTG YAQKFQG 17

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SEQ ID NO: 143	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
SITE	4	
	note = MISC_FEATURE - X = A or T	
SITE	7	
	note = MISC_FEATURE - X = G or S	
source	1..10	
	mol_type = protein	
	note = HCDR3 consensus sequence of C320 and derivatives	
	organism = synthetic construct	
SEQUENCE: 143		
EVPXTAXFEY		10
SEQ ID NO: 144	moltype = AA length = 30	
FEATURE	Location/Qualifiers	
source	1..30	
	mol_type = protein	
	note = amino acid sequence of HRF1 consesnys seq od C320	
	and der	
	organism = synthetic construct	
SEQUENCE: 144		
QVQLVQSGAE VKKPGASVKV SCKASGYTFT		30
SEQ ID NO: 145	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
VARIANT	6	
	note = X= T or P	
source	1..14	
	mol_type = protein	
	note = HFR2 consensus sequence of C320 and derivatives	
	organism = synthetic construct	
SEQUENCE: 145		
WVRQAXGQGL EWMG		14
SEQ ID NO: 146	moltype = AA length = 32	
FEATURE	Location/Qualifiers	
VARIANT	6	
	note = X = R or A	
VARIANT	7	
	note = X = N or D	
VARIANT	8	
	note = X = T or R	
VARIANT	9	
	note = X = S or A	
VARIANT	10	
	note = X = I or T	
source	1..32	
	mol_type = protein	
	note = HFR3 consensus sequence of C320 and derivatives	
	organism = synthetic construct	
SEQUENCE: 146		
RVTMTXXXXX STAYMELSSL RSED TAVYYC AR		32
SEQ ID NO: 147	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	note = HFR4 consensus sequence of C320 and derivatives	
	organism = synthetic construct	
SEQUENCE: 147		
WGQGLVTVS S		11
SEQ ID NO: 148	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = protein	
	note = LFR1 consensus sequence of C320 and derivatives	
	organism = synthetic construct	
SEQUENCE: 148		
QSVLTQPPSV SGAPGQRVTI SC		22
SEQ ID NO: 149	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
VARIANT	15	
	note = X = Y or E or G	
source	1..15	
	mol_type = protein	
	note = LFR2 consensus sequence of C320 and derivatives	

-continued

organism = synthetic construct
 SEQUENCE: 149
 WYQQLPGTAP KLLIX 15

SEQ ID NO: 150 moltype = AA length = 32
 FEATURE Location/Qualifiers
 VARIANT 18
 note = X=A or T
 source 1..32
 mol_type = protein
 note = LFR3 consensus sequence of C320 and derivatives
 organism = synthetic construct

SEQUENCE: 150
 GVPDRFSGSK SGTSASLXIT GLLPEDEGDY YC 32

SEQ ID NO: 151 moltype = AA length = 11
 FEATURE Location/Qualifiers
 VARIANT 3
 note = X=Q or K
 VARIANT 3
 note = X=L or T
 source 1..11
 mol_type = protein
 note = LFR4 consensus sequence of C320 and derivatives
 organism = synthetic construct

SEQUENCE: 151
 FGGGKLTVL G 11

SEQ ID NO: 152 moltype = AA length = 119
 FEATURE Location/Qualifiers
 VARIANT 41
 note = X=T or P
 VARIANT 51
 note = X=M or L
 VARIANT 102
 note = X=E or D
 VARIANT 103
 note = X=T or D
 VARIANT 105
 note = X=S or A
 VARIANT 107
 note = X=L or D or E
 source 1..119
 mol_type = protein
 note = VH consensus sequence of C320 and derivatives
 organism = synthetic construct

SEQUENCE: 152
 QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA XGQGLEWMGW XNPNSGNTGY 60
 AQKFQGRVTM TRNTSISTAY MELSSLRSED TAVYYCAREV PXXAXFPYWG QGTLVTVSS 119

SEQ ID NO: 153 moltype = AA length = 111
 FEATURE Location/Qualifiers
 VARIANT 23
 note = X=A or T
 VARIANT 76
 note = X=A or T
 source 1..111
 mol_type = protein
 note = V1 consensus sequence of C320 and derivatives
 organism = synthetic construct

SEQUENCE: 153
 QSVLTQPPSV SGAPGQRVTI SCXGSSSDIG AGLGVHWYQQ LPGTAPKLLI YGYNRPSPGV 60
 PDRFSGSGSKG TSASLXITGL LPEDEGDYYC QSYDGTLSAL FGGGKLTVL G 111

SEQ ID NO: 154 moltype = AA length = 119
 FEATURE Location/Qualifiers
 source 1..119
 mol_type = protein
 note = VH comprising CDRs from C320 grafted onto FRs of
 antibody 1TZG
 organism = synthetic construct

SEQUENCE: 154
 QVQLVQSGAE VKRPGSSVTV SCKASGYTFT SYDINWVRQA PGRGLEWMGW MNPNSGNTGY 60
 AQKFQGRITI TADRSTSTAY LELNSLRPED TAVYYCAREV PDTASFEYWG QGTLVTVSS 119

SEQ ID NO: 155 moltype = AA length = 120
 FEATURE Location/Qualifiers
 source 1..120
 mol_type = protein

-continued

note = VH comprising CDRs from C320 grafted onto FRs of
 antibody 1RHH
 organism = synthetic construct

SEQUENCE: 155
 EVQLLEQSGA EVKKPGSSVQ VSCASGYTF TSYDINWVRQ APGHGLEWMG WMNPNSGNTG 60
 YAQKFQGRVT FTADQATSTA YMELTNLRSD DTAVYYCARE VPDASFEYW QGTLVTVSS 120

SEQ ID NO: 156 moltype = AA length = 119
 FEATURE Location/Qualifiers
 source 1..119
 mol_type = protein
 note = VH comprising CDRs from C320 grafted onto FRs of
 antibody 2DD8
 organism = synthetic construct

SEQUENCE: 156
 QVQLQQSGAE VKKPGSSVKV SCKASGYTFT SYDINWVRQA PGQGLEWMGW MNPNSGNTGY 60
 AQKFQGRVTI TTDESTSTAY MELSSLRSED TAVYYCAREV PDTASFEYW QGTTVTVSS 119

SEQ ID NO: 157 moltype = AA length = 119
 FEATURE Location/Qualifiers
 source 1..119
 mol_type = protein
 note = VH comprising CDRs from C320 grafted onto FRs of
 antibody 2JB5
 organism = synthetic construct

SEQUENCE: 157
 QVQLVQSGAE VKKPGSSVKV SCKASGYTFT SYDINWVRQA PGQGLEWMGW MNPNSGNTGY 60
 AQKFQGRVTI TADESTSTAY MELSSLRSED TAVYYCAREV PDTASFEYW QGTLVTVSS 119

SEQ ID NO: 158 moltype = AA length = 119
 FEATURE Location/Qualifiers
 source 1..119
 mol_type = protein
 note = VH comprising CDRs from C320 grafted onto FRs of
 antibody 3FKU
 organism = synthetic construct

SEQUENCE: 158
 QVQLVQSGAE VKKPGSSVKV SCTSSGYTFT SYDINWVRQA PGQGLEWLGW MNPNSGNTGY 60
 AQKFQGRVTI TADQSTRTAY MDLRLRSED TAVYYCAREV PDTASFEYW QGTLVTVSS 119

SEQ ID NO: 159 moltype = AA length = 119
 FEATURE Location/Qualifiers
 source 1..119
 mol_type = protein
 note = VH comprising CDRs from C320 grafted onto FRs of
 antibody 3GBM
 organism = synthetic construct

SEQUENCE: 159
 EVQLVESGAE VKKPGSSVKV SCKASGYTFT SYDINWVRQA PGQGPWMGW MNPNSGNTGY 60
 AQKFQGRVTI TADDFAGTVY MELSSLRSED TAMYYCAKEV PDTASFEYW KGTTVTVSS 119

SEQ ID NO: 160 moltype = AA length = 119
 FEATURE Location/Qualifiers
 source 1..119
 mol_type = protein
 note = VH comprising CDRs from C320 grafted onto FRs of
 antibody 3LMJ
 organism = synthetic construct

SEQUENCE: 160
 QVQVQSGAE VRKPGASVKV SCKVSGYTFT SYDINWVRQA PGKGLEWMGW MNPNSGNTGY 60
 AQKFQKRVSM TEDTSTNTAY MELSSLRSED TAVYYCATEV PDTASFEYW QGTLVTVSS 119

SEQ ID NO: 161 moltype = AA length = 119
 FEATURE Location/Qualifiers
 source 1..119
 mol_type = protein
 note = VH comprising CDRs from C320 grafted onto FRs of
 antibody 3P30
 organism = synthetic construct

SEQUENCE: 161
 QVQLVQSGAE VKKAGSSVRV SCKASGYTFT SYDINWVRQA PGQGPQWMGW MNPNSGNTGY 60
 AQKFQGRITI TADTSTNTAY LELSSLRYDD TAVYYCTREV PDTASFEYW QGTLVTVSS 119

SEQ ID NO: 162 moltype = AA length = 120
 FEATURE Location/Qualifiers
 VARIANT 1
 note = X=Q or E
 VARIANT 4
 note = X=L or V

-continued

VARIANT	5	
	note = X=V or L or Q	
VARIANT	6	
	note = X=L or absent	
VARIANT	7	
	note = X=Q or E	
VARIANT	13	
	note = X=K or R	
VARIANT	14	
	note = X=K or R or A	
VARIANT	17	
	note = X=A or S	
VARIANT	20	
	note = X=K or T or Q or R	
VARIANT	24	
	note = X=K or T	
VARIANT	25	
	note = X=A or S or V	
VARIANT	42	
	note = X=T or P	
VARIANT	44	
	note = X=Q or R or H or K	
VARIANT	46	
	note = X=L or P	
VARIANT	47	
	note = X=E or Q	
VARIANT	49	
	note = X=M or L	
VARIANT	69	
	note = X=V or I or L	
VARIANT	70	
	note = X=T or S	
VARIANT	71	
	note = X=M or I or F	
VARIANT	73	
	note = X=R or A or T or E	
VARIANT	74	
	note = X=N or D	
VARIANT	75	
	note = X=T or R or Q or E or D	
VARIANT	76	
	note = X=S or A or F	
VARIANT	77	
	note = X=I or T or A	
VARIANT	78	
	note = X=S or R or G or N	
VARIANT	80	
	note = X=A or V	
VARIANT	82	
	note = X=M or L	
VARIANT	83	
	note = X=E or D	
VARIANT	85	
	note = X=S or N or T or R	
VARIANT	86	
	note = X=S or N	
VARIANT	89	
	note = X=S or P or Y	
VARIANT	90	
	note = X=E or D	
VARIANT	94	
	note = X=V or M	
VARIANT	98	
	note = X=A or T	
VARIANT	99	
	note = X=R or K or T	
VARIANT	112	
	note = X=Q or K	
VARIANT	115	
	note = X=L or T	
source	1..120	
	mol_type = protein	
	note = amino acid sequence of VH consensus sequence of C320	
	and derivatives	
	organism = synthetic construct	

SEQUENCE: 162

XVQXXXSGA EVXXPGXSVX VSCXXSGYTF TSYDINWVRQ AXGXGXXWVG WMNPNSGNTG 60
YAQKFGGRXX XTXXXXXXTX YXXLXXLRXX DTAXYYCXXE VPDTASFYEW GXGTXTVSS 120

-continued

SEQ ID NO: 163 moltype = AA length = 112
 FEATURE Location/Qualifiers
 source 1..112
 mol_type = protein
 note = VL comprising CDRs from C320 grafted onto FRs of
 antibody 1RHH
 organism = synthetic construct

SEQUENCE: 163
 ELVLTQSPGT LSLSAGERAT LSCAGSSSDI GAGLGVHWYQ QKPGQAPRLI IYGYNRPSG 60
 IPDRFSGSGS GTDFTLTIGR LEPEDLAVYY CQSYDGTLSA LFGQGTKLEI KR 112

SEQ ID NO: 164 moltype = AA length = 112
 FEATURE Location/Qualifiers
 source 1..112
 mol_type = protein
 note = VL comprising CDRs from C320 grafted onto FRs of
 antibody 1TZGL
 organism = synthetic construct

SEQUENCE: 164
 EIVLTQSPGT QSLSPGERAT LSCAGSSSDI GAGLGVHWYQ QRPQAPRLI IYGYNRPSG 60
 VADRFGSGSGS GTDFTLTISR LEPEDFAVYY CQSYDGTLSA LFGQGTKVEV KR 112

SEQ ID NO: 165 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein
 note = VL comprising CDRs from C320 grafted onto FRs of
 antibody 2DD8
 organism = synthetic construct

SEQUENCE: 165
 SYELTQPPSV SVAPGKTARI TCAGSSSDIG AGLGVHWYQQ KPGQAPVLVV YGYNRPSGI 60
 PERFGSGNSG NTATLTISRV EAGDEADYYC QSYDGTLSAL FGTGKTVTL G 111

SEQ ID NO: 166 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 166
 DIALTQPASV SGSPGQSITI SCAGSSSDIG AGLGVHWYQQ HPGKAPKLM I YGYNRPSGV 60
 SNRFGSGSKG NTASLTISGL QAEDEADYYC QSYDGTLSAL FGGGTKLTVL G 111

SEQ ID NO: 167 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein
 note = VL comprising CDRs from C320 grafted onto FRs of
 antibody 3FKU
 organism = synthetic construct

SEQUENCE: 167
 QPGLTQPPSV SKGLRQTATL TCAGSSSDIG AGLGVHWLQQ HQGHPPKLLS YGYNRPSGI 60
 SERFSASRSG NTASLTITGL QPEDEADYYC QSYDGTLSAL FGGGTKLTVL G 111

SEQ ID NO: 168 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein
 note = VL comprising CDRs from C320 grafted onto FRs of
 antibody 3GBM
 organism = synthetic construct

SEQUENCE: 168
 QSVLTQPPSV SAAPGQKVTI SCAGSSSDIG AGLGVHWYQQ LPGTAPKLLI YGYNRPSGI 60
 PDRFGSGSKG TSATLGITGL QTGDEADYYC QSYDGTLSAL FGGGTKLTVL G 111

SEQ ID NO: 169 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein
 note = VL comprising CDRs from C320 grafted onto FRs of
 antibody 3LMJ
 organism = synthetic construct

SEQUENCE: 169
 QSVLTQPPSV SAAPGQKVTI SCAGSSSDIG AGLGVHWYQQ LPGAAPKLLI FGYYNRPSGI 60
 PDRFGSGSKG TSATLAITGL QTGDEADYYC QSYDGTLSAL FGGGTKLTVL G 111

SEQ ID NO: 170 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein

-continued

note = VL comprising CDRs from C320 grafted onto FRs of
 antibody 3P30
 organism = synthetic construct

SEQUENCE: 170
 QSVLTQPPSV SGTPGQRTI SCAGSSSDIG AGLGVHWYQQ LPGTAPKLLI YGYNRPSGV 60
 PDRFSGSKSG TSASLAISGL RSEDEADYYC QSYDGTLSAL FGAGTKLTVL G 111

SEQ ID NO: 171 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein
 note = VL comprising CDRs from C320 grafted onto FRs of
 antibody 3IYW
 organism = synthetic construct

SEQUENCE: 171
 QSVLTQPPSV SGTPGQRTI SCAGSSSDIG AGLGVHWYQQ LPGTAPKLLI YGYNRPSGV 60
 PDRFSGSKSG TSASLAISGL QSEDEADYYC QSYDGTLSAL FGGGKLTVL G 111

SEQ ID NO: 172 moltype = AA length = 112
 FEATURE Location/Qualifiers
 VARIANT 1
 note = X=Q or E or S or D
 VARIANT 2
 note = X=S or L or I or Y
 VARIANT 3
 note = X=V or E or A or G
 VARIANT 7
 note = X=P or S
 VARIANT 8
 note = X=P or A or S
 VARIANT 9
 note = X=G or absent
 VARIANT 10
 note = X=S or T
 VARIANT 11
 note = X=V or L or Q
 VARIANT 13
 note = X=G or L or V or K or A
 VARIANT 14
 note = X=A or S or G or T
 VARIANT 15
 note = X=P or A or L
 VARIANT 16
 note = X=G or R
 VARIANT 17
 note = X=Q or E or K
 VARIANT 18
 note = X=R or T or S or K
 VARIANT 19
 note = X=V or A or I
 VARIANT 20
 note = X=T or R
 VARIANT 21
 note = X=I or L
 VARIANT 22
 note = X=S or T
 VARIANT 39
 note = X=Y or L
 VARIANT 42
 note = X=L or K or R or H
 VARIANT 43
 note = X=P or Q
 VARIANT 45
 note = X=T or Q or K or H
 VARIANT 46
 note = X=A or P
 VARIANT 48
 note = X=K or R or V
 VARIANT 50
 note = X=L or V or M
 VARIANT 51
 note = X=I or V or S
 VARIANT 60
 note = X=G or I
 VARIANT 61
 note = X=V or A or S
 VARIANT 62
 note = X=P or E or N
 VARIANT 66

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VARIANT          note = X=S or A
                  68
VARIANT          note = X=S or G or N or R
                  71
VARIANT          note = X=G or N
                  72
VARIANT          note = X=T or D
                  73
VARIANT          note = X=S or F
                  74
VARIANT          note = X=A or T
                  76
VARIANT          note = X=L or T or G
                  78
VARIANT          note = X=I or G or S
                  79
VARIANT          note = X=T or R
                  80
VARIANT          note = X=G or V
                  81
VARIANT          note = X=L or E or Q or R
                  82
VARIANT          note = X=L or A or T or S
                  83
VARIANT          note = X=P or G
                  85
VARIANT          note = X=D or L or F
                  86
VARIANT          note = X=E or A
                  87
VARIANT          note = X=G or V or N
                  103
VARIANT          note = X=G or Q or T or A
                  107
VARIANT          note = X=K or V
                  108
VARIANT          note = X=L or E
                  109
VARIANT          note = X=T or I
                  110
VARIANT          note = X=V or K
                  111
source           note = X=L or R
                  1..112
                  mol_type = protein
                  note = VL consensus sequence of C320 and derivatives
                  organism = synthetic construct

SEQUENCE: 172
XXXLTQXXXXX XSXXXXXXXXX XXCAGSSSDI GAGLGVHWXQ QXXGXXPXLX XYGYYNRPSX 60
XPDRFXGXKS XXXXSXAXXX XXXEXXXDY CQSYDGTLSA LFXGGTXXXX XG 112

SEQ ID NO: 173      moltype = AA length = 119
FEATURE            Location/Qualifiers
VARIANT            41
                   note = X= T or P
VARIANT            51
                   note = X= M or L
VARIANT            72
                   note = X= R or A
VARIANT            73
                   note = X= N or D
VARIANT            74
                   note = X= T or R
VARIANT            75
                   note = X= S or A
VARIANT            76
                   note = X=I or T
VARIANT            102
                   note = X= D or E
VARIANT            105
                   note = X= S or A
source             1..119
                   mol_type = protein
                   note = VH consensus sequence of C320 and derivatives
                   organism = synthetic construct

SEQUENCE: 173
QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA XGQGLEWMGW XNPNSGNTGY 60
AQKFGQGRVTM TXXXXXSTAY MELSSLRSED TAVYYCAREV PXTAXFEYWG QGTLVTVSS 119

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SEQ ID NO: 174      moltype = AA  length = 111
FEATURE            Location/Qualifiers
SITE               23
                   note = MISC_FEATURE - X = A or T
SITE               24
                   note = MISC_FEATURE - X = G or S
SITE               51
                   note = MISC_FEATURE - X = Y or P or E or G
SITE               76
                   note = MISC_FEATURE - X = A or T
source             1..111
                   mol_type = protein
                   note = VL consensus sequence of C320 and derivatives
                   organism = synthetic construct

SEQUENCE: 174
QSVLTQPPSV SGAPQQRVTI SCXXSSSDIG AGLGVHWYQQ LPGTAPKLLI XGYNRPSPGV  60
PDRFSGSKSG TSASLXITGL LPEDEGDYYC QSYDGTLSAL FGGGKLTVL  G      111

SEQ ID NO: 175      moltype = AA  length = 119
FEATURE            Location/Qualifiers
source             1..119
                   mol_type = protein
                   note = amino acid sequence of VH of antibody C320-162
                   organism = synthetic construct

SEQUENCE: 175
QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA PGQGLEWMGW LNPNSGNTGY  60
AQKFQGRVTM TRNTAISTAY MELSSLRSED TAVYYCAREV PETASFEYWG QGTLVTVSS  119

SEQ ID NO: 176      moltype = AA  length = 119
FEATURE            Location/Qualifiers
source             1..119
                   mol_type = protein
                   note = amino acid sequence of VH of antibody C320-163
                   organism = synthetic construct

SEQUENCE: 176
QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA PGQGLEWMGW MNPNSGNTGY  60
AQKFQGRVTM TADRSTSTAY MELSSLRSED TAVYYCAREV PDTASFEYWG QGTLVTVSS  119

SEQ ID NO: 177      moltype = AA  length = 119
FEATURE            Location/Qualifiers
source             1..119
                   mol_type = protein
                   note = amino acid sequence of VH of antibody C320-164
                   organism = synthetic construct

SEQUENCE: 177
QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA PGQGLEWMGW LNPNSGNTGY  60
AQKFQGRVTM TRNTSISTAY MELSSLRSED TAVYYCAREV PETASFEYWG QGTLVTVSS  119

SEQ ID NO: 178      moltype = AA  length = 119
FEATURE            Location/Qualifiers
source             1..119
                   mol_type = protein
                   note = amino acid sequence of VH of antibody C320-165
                   organism = synthetic construct

SEQUENCE: 178
QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA PGQGLEWMGW LNPNSGNTGY  60
AQKFQGRVTM TRNTSISTAY MELSSLRSED TAVYYCAREV PETASFEYWG QGTLVTVSS  119

SEQ ID NO: 179      moltype = AA  length = 119
FEATURE            Location/Qualifiers
source             1..119
                   mol_type = protein
                   note = amino acid sequence of VH of antibody C320-166
                   organism = synthetic construct

SEQUENCE: 179
QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA PGQGLEWMGW LNPNSGNTGY  60
AQKFQGRVTM TRNTSISTAY MELSSLRSED TAVYYCAREV PETASFEYWG QGTLVTVSS  119

SEQ ID NO: 180      moltype = AA  length = 119
FEATURE            Location/Qualifiers
source             1..119
                   mol_type = protein
                   note = amino acid sequence of VH of antibody C320-167
                   organism = synthetic construct

SEQUENCE: 180
QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA PGQGLEWMGW LNPNSGNTGY  60
AQKFQGRVTM TRNTSISTAY MELSSLRSED TAVYYCAREV PETASFEYWG QGTLVTVSS  119

SEQ ID NO: 181      moltype = AA  length = 119

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FEATURE Location/Qualifiers
 source 1..119
 mol_type = protein
 note = amino acid sequence of CH of antibody C320-168
 organism = synthetic construct

SEQUENCE: 181
 QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA PGQGLEWMGW LNPNSGNTGY 60
 AQKFQGRVTM TRNTSISTAY MELSSLRSED TAVYYCAREV PETAAFEYWG QGTLVTVSS 119

SEQ ID NO: 182 moltype = AA length = 119
 FEATURE Location/Qualifiers
 source 1..119
 mol_type = protein
 note = amino acid sequence of VH of antibody C320-169
 organism = synthetic construct

SEQUENCE: 182
 QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA PGQGLEWMGW LNPNSGNTGY 60
 AQKFQGRVTM TRNTSISTAY MELSSLRSED TAVYYCAREV PETAAFEYWG QGTLVTVSS 119

SEQ ID NO: 183 moltype = AA length = 119
 FEATURE Location/Qualifiers
 source 1..119
 mol_type = protein
 note = amino acid sequence of VH of antibody C320-170
 organism = synthetic construct

SEQUENCE: 183
 QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA PGQGLEWMGW MNPNSGNTGY 60
 AQKFQGRVTM TADRSTSTAY MELSSLRSED TAVYYCAREV PDTASFEYWG QGTLVTVSS 119

SEQ ID NO: 184 moltype = AA length = 119
 FEATURE Location/Qualifiers
 source 1..119
 mol_type = protein
 note = amino acid sequence of VH of antibody C320-171
 organism = synthetic construct

SEQUENCE: 184
 QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA PGQGLEWMGW MNPNSGNTGY 60
 AQKFQGRVTM TADRSTSTAY MELSSLRSED TAVYYCAREV PDTASFEYWG QGTLVTVSS 119

SEQ ID NO: 185 moltype = AA length = 119
 FEATURE Location/Qualifiers
 source 1..119
 mol_type = protein
 note = amino acid sequence of VH of antibody C320-172
 organism = synthetic construct

SEQUENCE: 185
 QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA PGQGLEWMGW MNPNSGNTGY 60
 AQKFQGRVTM TADRSTSTAY MELSSLRSED TAVYYCAREV PDTASFEYWG QGTLVTVSS 119

SEQ ID NO: 186 moltype = AA length = 119
 FEATURE Location/Qualifiers
 source 1..119
 mol_type = protein
 note = amino acid sequence of VH of antibody C320-179
 organism = synthetic construct

SEQUENCE: 186
 QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA PGQGLEWMGW LNPNSGNTGY 60
 AQKFQGRVTM TADRSTSTAY MELSSLRSED TAVYYCAREV PETAAFEYWG QGTLVTVSS 119

SEQ ID NO: 187 moltype = AA length = 119
 FEATURE Location/Qualifiers
 source 1..119
 mol_type = protein
 note = amino acid sequence of VH of antibody C320-183
 organism = synthetic construct

SEQUENCE: 187
 QVQLVQSGAE VKKPGASVKV SCKASGYTFT SYDINWVRQA PGQGLEWMGW LNPNSGNTGY 60
 AQKFQGRVTM TADRSTSTAY MELSSLRSED TAVYYCAREV PETAAFEYWG QGTLVTVSS 119

SEQ ID NO: 188 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein
 note = amino acid sequence of VL of antibody C320-162
 organism = synthetic construct

SEQUENCE: 188
 QSVLTQPPSV SGAPQQRVTI SCAGSSSDIG AGLGVHWYQQ LPGTAPKLLI YGYNRPSPGV 60
 PDRFSGSKSG TSASLTITGL LPEDEGDYYC QSYDGTLSAL FGGGTKLTVL G 111

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SEQ ID NO: 189 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein
 note = amino acid sequence of VL of antibody C320-163
 organism = synthetic construct

SEQUENCE: 189
 QSVLTQPPSV SGAPGQRTI SCAGSSSDIG AGLGVHWYQQ LPGTAPKLLI YGYNRPSGV 60
 PDRFSGSKSG TSASLTITGL LPEDEGDYYC QSYDGTLSAL FGGGTKLTVL G 111

SEQ ID NO: 190 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein
 note = amino acid sequence of VL of antibody C320-164
 organism = synthetic construct

SEQUENCE: 190
 QSVLTQPPSV SGAPGQRTI SCASSSSDIG AGLGVHWYQQ LPGTAPKLLI YGYNRPSGV 60
 PDRFSGSKSG TSASLTITGL LPEDEGDYYC QSYDGTLSAL FGGGTKLTVL G 111

SEQ ID NO: 191 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein
 note = amino acid sequence of VL of antibody C320-165
 organism = synthetic construct

SEQUENCE: 191
 QSVLTQPPSV SGAPGQRTI SCTSSSSDIG AGLGVHWYQQ LPGTAPKLLI YGYNRPSGV 60
 PDRFSGSKSG TSASLTITGL LPEDEGDYYC QSYDGTLSAL FGGGTKLTVL G 111

SEQ ID NO: 192 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein
 note = amino acid sequence of VL of antibody C320-166
 organism = synthetic construct

SEQUENCE: 192
 QSVLTQPPSV SGAPGQRTI SCTGSSSDIG AGLGVHWYQQ LPGTAPKLLI YGYNRPSGV 60
 PDRFSGSKSG TSASLTITGL LPEDEGDYYC QSYDGTLSAL FGGGTKLTVL G 111

SEQ ID NO: 193 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein
 note = amino acid sequence of VL of antibody C320-167
 organism = synthetic construct

SEQUENCE: 193
 QSVLTQPPSV SGAPGQRTI SCAGSSSDIG AGLGVHWYQQ LPGTAPKLLI YGYNRPSGV 60
 PDRFSGSKSG TSASLTITGL LPEDEGDYYC QSYDGTLSAL FGGGTKLTVL G 111

SEQ ID NO: 194 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein
 note = amino acid sequence of VL of antibody C320-168
 organism = synthetic construct

SEQUENCE: 194
 QSVLTQPPSV SGAPGQRTI SCTSSSSDIG AGLGVHWYQQ LPGTAPKLLI YGYNRPSGV 60
 PDRFSGSKSG TSASLTITGL LPEDEGDYYC QSYDGTLSAL FGGGTKLTVL G 111

SEQ ID NO: 195 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein
 note = amino acid sequence of VL of antibody C320-169
 organism = synthetic construct

SEQUENCE: 195
 QSVLTQPPSV SGAPGQRTI SCAGSSSDIG AGLGVHWYQQ LPGTAPKLLI YGYNRPSGV 60
 PDRFSGSKSG TSASLTITGL LPEDEGDYYC QSYDGTLSAL FGGGTKLTVL G 111

SEQ ID NO: 196 moltype = AA length = 111
 FEATURE Location/Qualifiers
 source 1..111
 mol_type = protein
 note = amino acid sequence of VL of antibody C320-170
 organism = synthetic construct

SEQUENCE: 196
 QSVLTQPPSV SGAPGQRTI SCTSSSSDIG AGLGVHWYQQ LPGTAPKLLI YGYNRPSGV 60
 PDRFSGSKSG TSASLTITGL LPEDEGDYYC QSYDGTLSAL FGGGTKLTVL G 111

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SEQUENCE: 204	mol_type = protein	
ADRS	note = amino acid sequence from VH of 1TZG corresponding to	
	N-linked glycosylation site in VH of C320	
	organism = synthetic construct	
		5
SEQ ID NO: 205	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	note = amino acid sequence of peptide from VH of C320-168	
	organism = synthetic construct	
SEQUENCE: 205		
GLEWMGWLNP NSGNT		15
SEQ ID NO: 206	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	note = amino acid sequence of peptide from VL of C320-168	
	organism = synthetic construct	
SEQUENCE: 206		
LLIYGYYNRP SGVPD		15
SEQ ID NO: 207	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
source	1..13	
	mol_type = protein	
	note = amino acid sequence of influenza peptide	
	organism = synthetic construct	
SEQUENCE: 207		
PKYVKQNTLK LAT		13
SEQ ID NO: 208	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	note = amino acid sequence of mutant peptide from VL of	
	C320-168	
	organism = synthetic construct	
SEQUENCE: 208		
LLIEGYNRP SGVPD		15
SEQ ID NO: 209	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	note = amino acid sequence of mutant peptide from VL of	
	C320-168	
	organism = synthetic construct	
SEQUENCE: 209		
LLIGGYNRP SGVPD		15
SEQ ID NO: 210	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	note = amino acid sequence of mutant peptide from VL of	
	C320-168	
	organism = synthetic construct	
SEQUENCE: 210		
LLIPGYNRP SGVPD		15
SEQ ID NO: 211	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	note = amino acid sequence of mutant peptide from VL of	
	C320-168	
	organism = synthetic construct	
SEQUENCE: 211		
LLIKGYNRP SGVPD		15
SEQ ID NO: 212	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	note = amino acid sequence of mutant peptide from VL of	

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SEQUENCE: 212	C320-168	
LLIYGYYNRP SGVPD	organism = synthetic construct	15
SEQ ID NO: 213	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	note = amino acid sequence of mutant peptide from VL of	
	C320-168	
	organism = synthetic construct	
SEQUENCE: 213		
LLIEGYYNRP SGVPD		15
SEQ ID NO: 214	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	note = amino acid sequence of mutant peptide from VL of	
	C320-168	
	organism = synthetic construct	
SEQUENCE: 214		
LLIGGYYNRP SGVPD		15
SEQ ID NO: 215	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	note = amino acid sequence of mutant peptide from VL of	
	C320-168	
	organism = synthetic construct	
SEQUENCE: 215		
LLIPGYYNRP SGVPD		15
SEQ ID NO: 216	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	note = amino acid sequence of mutant peptide from VL of	
	C320-168	
	organism = synthetic construct	
SEQUENCE: 216		
LLIKGYYNRP SGVPD		15
SEQ ID NO: 217	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	note = amino acid sequence of mutant peptide from VL of	
	C320-168	
	organism = synthetic construct	
SEQUENCE: 217		
LLIYGYYNRP SGVPD		15
SEQ ID NO: 218	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	note = amino acid sequence of mutant peptide from VL of	
	C320-168	
	organism = synthetic construct	
SEQUENCE: 218		
LLIEGYYNRP SGVPD		15
SEQ ID NO: 219	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	
	note = amino acid sequence of mutant peptide from VL of	
	C320-168	
	organism = synthetic construct	
SEQUENCE: 219		
LLIGGYYNRP SGVPD		15
SEQ ID NO: 220	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = protein	

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note = amino acid sequence of mutant peptide from VL of
C320-168
organism = synthetic construct

SEQUENCE: 220
LLIPGYNRP SGVPD 15

SEQ ID NO: 221
FEATURE
source
moltype = AA length = 15
Location/Qualifiers
1..15
mol_type = protein
note = amino acid sequence of mutant peptide from VL of
C320-168
organism = synthetic construct

SEQUENCE: 221
LLIKGYNRP SGVPD 15

SEQ ID NO: 222
FEATURE
source
moltype = DNA length = 357
Location/Qualifiers
1..357
mol_type = other DNA
note = nucleotide sequence encoding VH of antibody C320-162
organism = synthetic construct

SEQUENCE: 222
caggtgcagc tgggtgcagtc tggggccgag gtgaagaagc ctggggcctc agtgaaggtc 60
tcctgcaagg cttctggata caccttcacc agttatgata tcaactgggt gcgacaggcc 120
cccgacaag ggcttgagtg gatgggatgg ctgaacccta acagtggtaa cacaggctat 180
gcacagaagt tccagggcag agtcaccatg accaggaaca ccgccataag cacagcctac 240
atggagctga gcagcctgag atctgaggac acggccgtgt attactgtgc gagagaagtg 300
ctgagacgg cctcctttga gtactggggc caggaaccc tggtagacagt gtcctca 357

SEQ ID NO: 223
FEATURE
source
moltype = DNA length = 357
Location/Qualifiers
1..357
mol_type = other DNA
note = nucleotide sequence encoding VH of antibody C320-163
organism = synthetic construct

SEQUENCE: 223
caggtgcagc tgggtgcagtc tggggccgag gtgaagaagc ctggggcctc agtgaaggtc 60
tcctgcaagg cttctggata caccttcacc agttatgata tcaactgggt gcgacaggcc 120
cccgacaag ggcttgagtg gatgggatgg atgaacccta acagtggtaa cacaggctat 180
gcacagaagt tccagggcag agtcaccatg accgcagatc gttccaccag cacagcctac 240
atggagctga gcagcctgag atctgaggac acggccgtgt attactgtgc gagagaagtg 300
cctgatacgg cctcctttga gtactggggc caggaaccc tggtagacagt gtcctca 357

SEQ ID NO: 224
FEATURE
source
moltype = DNA length = 357
Location/Qualifiers
1..357
mol_type = other DNA
note = nucleotide sequence encoding VH of antibodies
C320-164, C320-165, C320-166 and C320-167
organism = synthetic construct

SEQUENCE: 224
caggtgcagc tgggtgcagtc tggggccgag gtgaagaagc ctggggcctc agtgaaggtc 60
tcctgcaagg cttctggata caccttcacc agttatgata tcaactgggt gcgacaggcc 120
cccgacaag ggcttgagtg gatgggatgg ctgaacccta acagtggtaa cacaggctat 180
gcacagaagt tccagggcag agtcaccatg accaggaaca cctccataag cacagcctac 240
atggagctga gcagcctgag atctgaggac acggccgtgt attactgtgc gagagaagtg 300
cctgagacgg cctcctttga gtactggggc caggaaccc tggtagacagt gtcctca 357

SEQ ID NO: 225
FEATURE
source
moltype = DNA length = 357
Location/Qualifiers
1..357
mol_type = other DNA
note = nucleotide sequence encoding VH of antibodies
C320-168 and C320-169
organism = synthetic construct

SEQUENCE: 225
caggtgcagc tgggtgcagtc tggggccgag gtgaagaagc ctggggcctc agtgaaggtc 60
tcctgcaagg cttctggata caccttcacc agttatgata tcaactgggt gcgacaggcc 120
cccgacaag ggcttgagtg gatgggatgg ctgaacccta acagtggtaa cacaggctat 180
gcacagaagt tccagggcag agtcaccatg accaggaaca cctccataag cacagcctac 240
atggagctga gcagcctgag atctgaggac acggccgtgt attactgtgc gagagaagtg 300
cctgagacgg ccgcctttga gtactggggc caggaaccc tggtagacagt gtcctca 357

SEQ ID NO: 226
FEATURE
source
moltype = DNA length = 357
Location/Qualifiers
1..357
mol_type = other DNA
note = nucleotide sequence encoding VH of antibodies

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C320-170 and C320-172
organism = synthetic construct

SEQUENCE: 226

caggtgcagc	tggtgcagtc	tggggcggag	gtgaagaagc	ctggggcctc	agtgaaggtc	60
tcctgcaagg	cttctggata	caccttcacc	agttatgata	tcaactgggt	gcgacaggcc	120
cccgacaag	ggcttgagtg	gatgggatgg	atgaacccta	acagtggtaa	cacaggtat	180
gcacagaagt	tccagggcag	agtcaccatg	accgcagatc	gttccaccag	cacagcctac	240
atggagctga	gcagcctgag	atctgaggac	acggccgtgt	attactgtgc	gagagaagtg	300
cctgatacgg	cctcctttga	gtactggggc	caggaaccc	tggtcaccgt	ctcctca	357

SEQ ID NO: 227 moltype = DNA length = 357
FEATURE Location/Qualifiers
source 1..357
 mol_type = other DNA
 note = nucleotide sequence encoding VH of antibodies
 C320-179 and C320-183
 organism = synthetic construct

SEQUENCE: 227

caggtgcagc	tggtgcagtc	tggggcggag	gtgaagaagc	ctggggcctc	agtgaaggtc	60
tcctgcaagg	cttctggata	caccttcacc	agttatgata	tcaactgggt	gcgacaggcc	120
cccgacaag	ggcttgagtg	gatgggatgg	ctgaacccta	acagtggtaa	cacaggtat	180
gcacagaagt	tccagggcag	agtcaccatg	accgcagatc	gttccaccag	cacagcctac	240
atggagctga	gcagcctgag	atctgaggac	acggccgtgt	attactgtgc	gagagaagtg	300
cctgagacgg	cgcctttga	gtactggggc	caggaaccc	tggtcaccgt	ctcctca	357

SEQ ID NO: 228 moltype = DNA length = 333
FEATURE Location/Qualifiers
source 1..333
 mol_type = other DNA
 note = nucleotide sequence encoding VL of antibodies
 C320-162, C320-163, C320-167 and C320-169
 organism = synthetic construct

SEQUENCE: 228

cagttctgtgc	tgacgcagcc	gccctcagtg	tctggggccc	cagggcagag	ggtcaccatc	60
tcctgcgctg	ggagtagttc	cgacatcggg	gcaggtcttg	gcgtgcactg	gtatcagcag	120
cttccaggaa	cagcccccaa	actcctcatc	tatggttact	acaatcggcc	ctcaggggtc	180
cctgaccgat	tctctggctc	caagtctggc	acctcagcct	ccctgaccat	cactggggtc	240
ctgcctgagg	atgaggggtga	ttattactgc	cagtcctatg	acggcactct	gagtgcccta	300
ttcgcgagg	ggaccaagct	gaccgtccta	ggt			333

SEQ ID NO: 229 moltype = DNA length = 333
FEATURE Location/Qualifiers
source 1..333
 mol_type = other DNA
 note = nucleotide sequence encoding VL of antibody C320-164
 organism = synthetic construct

SEQUENCE: 229

cagttctgtgc	tgacgcagcc	gccctcagtg	tctggggccc	cagggcagag	ggtcaccatc	60
tcctgcgcta	gcagtagttc	cgacatcggg	gcaggtcttg	gcgtgcactg	gtatcagcag	120
cttccaggaa	cagcccccaa	actcctcatc	tatggttact	acaatcggcc	ctcaggggtc	180
cctgaccgat	tctctggctc	caagtctggc	acctcagcct	ccctgaccat	cactggggtc	240
ctgcctgagg	atgaggggtga	ttattactgc	cagtcctatg	acggcactct	gagtgcccta	300
ttcgcgagg	ggaccaagct	gaccgtccta	ggt			333

SEQ ID NO: 230 moltype = DNA length = 333
FEATURE Location/Qualifiers
source 1..333
 mol_type = other DNA
 note = nucleotide sequence encoding VL of antibodies
 C320-165, C320-168 and C320-170
 organism = synthetic construct

SEQUENCE: 230

cagttctgtgc	tgacgcagcc	gccctcagtg	tctggggccc	cagggcagag	ggtcaccatc	60
tcctgcgcta	gcagtagttc	cgacatcggg	gcaggtcttg	gcgtgcactg	gtatcagcag	120
cttccaggaa	cagcccccaa	actcctcatc	tatggttact	acaatcggcc	ctcaggggtc	180
cctgaccgat	tctctggctc	caagtctggc	acctcagcct	ccctgaccat	cactggggtc	240
ctgcctgagg	atgaggggtga	ttattactgc	cagtcctatg	acggcactct	gagtgcccta	300
ttcgcgagg	ggaccaagct	gaccgtccta	ggt			333

SEQ ID NO: 231 moltype = DNA length = 333
FEATURE Location/Qualifiers
source 1..333
 mol_type = other DNA
 note = nucleotide sequence encoding VL of antibody C320-166
 organism = synthetic construct

SEQUENCE: 231

cagttctgtgc	tgacgcagcc	gccctcagtg	tctggggccc	cagggcagag	ggtcaccatc	60
tcctgcactg	ggagtagttc	cgacatcggg	gcaggtcttg	gcgtgcactg	gtatcagcag	120
cttccaggaa	cagcccccaa	actcctcatc	tatggttact	acaatcggcc	ctcaggggtc	180

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cctgaccgat tctctggctc caagtctggc acctcagcct ccttgaccat cactggggctc 240
ctgcctgagg atgaggggtga ttattactgc cagtcctatg acggcactct gagtgcctta 300
ttcgcgaggag ggaccaagct gaccgtccta ggt 333

SEQ ID NO: 232      moltype = DNA length = 333
FEATURE            Location/Qualifiers
source              1..333
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 232
cagtctgtgc tgacgcagcc gccctcagtg tctggggccc cagggcagag ggtcaccatc 60
tcttgcacta gcagtagttc cgacatcggg gcaggtcttg gcgtgcactg gtatcagcag 120
cttccaggaa cagcccccaa actcctcctc gaaggttact acaatcggcc ctcagggggtc 180
cctgaccgat tctctggctc caagtctggc acctcagcct ccttgaccat cactggggctc 240
ctgcctgagg atgaggggtga ttattactgc cagtcctatg acggcactct gagtgcctta 300
ttcgcgaggag ggaccaagct gaccgtccta ggt 333

SEQ ID NO: 233      moltype = DNA length = 333
FEATURE            Location/Qualifiers
source              1..333
                    mol_type = other DNA
                    note = nucleotide sequence encoding VL of antibody C320-183
                    organism = synthetic construct

SEQUENCE: 233
cagtctgtgc tgacgcagcc gccctcagtg tctggggccc cagggcagag ggtcaccatc 60
tcttgcacta gcagtagttc cgacatcggg gcaggtcttg gcgtgcactg gtatcagcag 120
cttccaggaa cagcccccaa actcctcctc ggaggttact acaatcggcc ctcagggggtc 180
cctgaccgat tctctggctc caagtctggc acctcagcct ccttgaccat cactgggggtc 240
ctgcctgagg atgaggggtga ttattactgc cagtcctatg acggcactct gagtgcctta 300
ttcgcgaggag ggaccaagct gaccgtccta ggt 333

SEQ ID NO: 234      moltype = AA length = 119
FEATURE            Location/Qualifiers
source              1..119
                    mol_type = protein
                    note = amino acid sequence of VH of antibody C320-12 and
                          C320-22
                    organism = synthetic construct

SEQUENCE: 234
EVQLVQSGAE VRKPGASTKV SCKASGYTFT SYDINWVRQA PGQGLEWMGW MNPNSGNTGY 60
AQQFQGRVTM TRDTSTSTVY MELSSLRSED TAVYYCAREV PDTASFEYWG QGTLVTVSS 119

SEQ ID NO: 235      moltype = AA length = 10
FEATURE            Location/Qualifiers
VARIANT            2
                    note = X = V or A or S or H or D or L or Y or P or Q or K
VARIANT            4
                    note = X = A or S or H or K or E or D
VARIANT            5
                    note = X = A or S or D or Y or T
VARIANT            7
                    note = X = A or S or H or L or D or Y
VARIANT            9
                    note = X = A or S or H or L or D or P or Q or E or K
source              1..10
                    mol_type = protein
                    note = consensus of HCDR3 of C320 and derivatives
                    organism = synthetic construct

SEQUENCE: 235
EXPXXAXFXV

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We claim:

1. A method for treating a disease or disorder selected from inflammatory bowel disease or irritable bowel syndrome, comprising administering to a subject in need thereof a recombinant antibody that binds to TNF-like ligand 1A (TL1A), wherein the antibody comprises a heavy chain variable region CDR1 comprising amino acids 31 to 35 of SEQ ID NO: 186, a heavy chain variable region CDR2 comprising amino acids 50 to 66 of SEQ ID NO: 186, a heavy chain variable region CDR3 comprising amino acids 99 to 108 of SEQ ID NO: 186, a light chain variable region CDR1 comprising amino acids 23 to 36 of SEQ ID NO: 199, a light chain variable region CDR2 comprising amino acids

52 to 58 of SEQ ID NO: 199, and a light chain variable region CDR3 comprising amino acids 91 to 100 of SEQ ID NO: 199.

2. The method according to claim 1, wherein the recombinant antibody comprises a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 186.

3. The method according to claim 1, wherein the recombinant antibody comprises a light chain variable region comprising the amino acid sequence of SEQ ID NO: 199.

4. The method according to claim 1, wherein the recombinant antibody comprises a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 186 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 199.

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5. The method according to claim 1, wherein the recombinant antibody comprises a human IgG1 heavy chain constant region.

6. The method according to claim 1, wherein the recombinant antibody comprises a human light chain lambda constant region.

7. The method according to claim 1, wherein the recombinant antibody comprises a human IgG1 heavy chain constant region and a human light chain lambda constant region.

8. The method according to claim 1, wherein the disease or disorder is inflammatory bowel disease.

9. The method according to claim 8, wherein the inflammatory bowel disease is Crohn's disease or ulcerative colitis.

10. The method according to claim 8, wherein the inflammatory bowel disease is Crohn's disease.

11. The method according to claim 1, wherein the recombinant antibody is capable of inhibiting the interaction of TL1A with the death receptor 3 (DR3).

12. The method according to claim 1, wherein the recombinant antibody is capable of inhibiting the interaction of TL1A with the death receptor 3 (DR3) and is not capable of inhibiting the interaction of TL1A with the decoy receptor 3 (DcR3).

13. A method for treating a lung disease or disorder, comprising administering to a subject in need thereof a recombinant antibody thereof that binds to TNF-like ligand 1A (TL1A), wherein the antibody comprises a heavy chain variable region CDR1 comprising amino acids 31 to 35 of SEQ ID NO: 186, a heavy chain variable region CDR2 comprising amino acids 50 to 66 of SEQ ID NO: 186, a heavy chain variable region CDR3 comprising amino acids 99 to 108 of SEQ ID NO: 186, a light chain variable region

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CDR1 comprising amino acids 23 to 36 of SEQ ID NO: 199, a light chain variable region CDR2 comprising amino acids 52 to 58 of SEQ ID NO: 199, and a light chain variable region CDR3 comprising amino acids 91 to 100 of SEQ ID NO: 199, wherein the lung disease or disorder is allergic lung inflammation, asthma, chronic obstructive pulmonary disease (COPD), or pulmonary fibrosis.

14. The method according to claim 13, wherein the recombinant antibody comprises a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 186.

15. The method according to claim 13, wherein the recombinant antibody comprises a light chain variable region comprising the amino acid sequence of SEQ ID NO: 199.

16. The method according to claim 13, wherein the recombinant antibody comprises a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 186 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 199.

17. The method according to claim 13, wherein the recombinant antibody comprises a human IgG1 heavy chain constant region and a human light chain lambda constant region.

18. The method according to claim 13, wherein the antibody is capable of inhibiting the interaction of TL1A with the death receptor 3 (DR3).

19. The method according to claim 13, wherein the recombinant antibody is capable of inhibiting the interaction of TL1A with the death receptor 3 (DR3) and is not capable of inhibiting the interaction of TL1A with the decoy receptor 3 (DcR3).

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